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Detlef Müller

An Invitation to Fourier Analysis and Distribution Theory

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Detlef Müller

An Invitation to Fourier Analysis and Distribution Theory

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Preface

This textbook originated from a lecture series I repeatedly delivered at the University of Kiel. The aim was to offer a robust introduction to the fundamentals of Fourier analysis and distribution theory (often called the "theory of generalized functions" in Russian schools) without depending on advanced knowledge of functional analysis or vector-valued integration theory. This approach was chosen because participants often had varying levels of prior knowledge and were not always majoring in mathematics, but rather in physics or engineering disciplines.

Therefore, the prerequisites for this textbook, in addition to the usual knowledge from standard lectures on analysis (preferably including also analysis of functions of a single complex variable) and linear algebra, include only a familiarity with basic concepts and results from general measure theory (especially the theory of the Lebesgue measure) and set-theoretic topology (particularly in metric spaces). No deeper knowledge of functional analysis is required – only a familiarity with basic concepts such as the definitions of normed spaces, Banach spaces, linear operators between normed spaces and their operator norms, as well as basic concepts in Hilbert space theory, including orthonormal systems and Hilbert space bases.

All other concepts and notions necessary for distribution theory will be developed in the book, including relevant facts about locally convex topological vector spaces.

The focus will not be on developing a general abstract theory of such spaces (as there are many excellent monographs on this), but rather on their applications in distribution theory.

To this end, we shall always describe the topology of the relevant vector spaces of functions and distributions by concretely specifying families of seminorms; such families will therefore play a paramount role, especially since analysis often involves estimates using appropriate norms or seminorms. This approach also has the didactic advantage of making these aspects of the theory natural generalizations of the theory of normed spaces.

The approach in this textbook differs from many other books on these topics in several additional ways, some of which will be detailed below. My guiding principle has always been to develop approaches that are as elementary and straightforward as possible, without compromising on substance.

Let me illustrate this with a few examples. In the introductory Chapter 1, I begin by briefly explaining some fundamental concepts concerning Fourier series and the expansion of periodic functions on the real line into their Fourier series, focusing on functions that are square-integrable over one period. This section requires only basic knowledge of Hilbert space theory and emphasizes the importance of orthogonality in Fourier analysis.

More advanced topics, such as the study of pointwise convergence of Fourier series, and the usual identification of 2π-periodic functions on $\mathbb{R}$ with functions on the 1-torus $\mathbb{T}$, are deferred to Chapter 3. These topics are addressed after the foundational aspects of Fourier analysis for non-periodic functions on $\mathbb{R}^n$ are developed in Chapter 2.

This somewhat unconventional "scenery change" has several advantages, in my view. The space $\mathbb{R}^n$ has a much larger automorphism group compared to the torus, a fact which is highly useful in Fourier analysis on $\mathbb{R}^n$. In particular, this group includes all scaling transformations by factors $r > 0$ (which do not exist on $\mathbb{T}$ unless $r = 1$). These scaling transformations enable, for instance, the simple construction of "approximate identities," which play a fundamental role in Fourier analysis and physics by providing a straightforward way to approximate the "Dirac delta function" (which is actually a distribution, specifically a point measure) using genuine functions. Additionally, many aspects of Fourier analysis are more elegantly formulated on $\mathbb{R}^n$, since the Fourier transform of a function on $\mathbb{R}^n$ is again a function on $\mathbb{R}^n$, whereas the Fourier transform of a function on $\mathbb{T}$ is a function on $\mathbb{Z}$.

In this chapter, I also introduce and use the Schwartz space $\mathcal{S} = \mathcal{S}(\mathbb{R}^n)$—the space of all smooth functions on $\mathbb{R}^n$ that decay rapidly along with all their derivatives—and indicate the relevance of the Fourier transform for the theory of linear differential operators.

Finally, in Section 2.8, I construct an explicit countable Hilbert space basis of $L^2(\mathbb{R}^n)$ consisting of eigenfunctions of the Fourier transform. These eigenfunctions are tensor products of one-dimensional Hermite functions and are, in particular, Schwartz functions.

One reason for including this section is that the derivation of these results relies on Heisenberg's commutation relations (see (2.34)) between the "position and momentum operators of quantum mechanics"—specifically, the multiplication operators $X_k : f \mapsto x_k f$ and the differential operators $D_j = \frac{1}{i}\frac{\partial}{\partial x_j}$. These operators play a central role, at least implicitly, in the theory of partial differential equations. By emphasizing this connection, my goal is to show that even in commutative harmonic analysis, such as Fourier analysis on the abelian groups $\mathbb{R}^n$ or $\mathbb{T}^n$, non-commutative structures (and thus also non-commutative harmonic analysis) can play an important role.

Another reason is that expanding arbitrary Schwartz functions with respect to this Hilbert space basis of Hermite functions allows for a conceptually simple proof of Schwartz's kernel theorem in Chapter 9. Since each basis function is a tensor product of one-dimensional Hermite functions, this approach at the same time allows for a straightforward definition of the tensor product of two distributions and facilitates the study of its most important properties (see Chapter 9).

The underlying core idea is not new and should rather be regarded as "mathematical folklore." Such an approach had been worked out, for example, in the bachelor's thesis of my former student Nico Hartig [19], and a similar approach can already be found in the appendix of [5], as I learned from Alessio Martini after completing Chapter 9.

Subsequently, transferring many key ideas and concepts from Chapter 2 to the theory of Fourier series in Chapter 3 becomes relatively straightforward, even though some constructions, such as the construction of "approximate identities," are technically more demanding.

Chapter 4 introduces distribution theory, starting with tempered distributions on $\mathbb{R}^n$. These are, by definition, continuous linear functionals on the Schwartz space $\mathcal{S}$, which serves as the space of "test functions."

Tempered distributions are particularly important in Fourier analysis because they not only support standard operations like differentiation, multiplication by suitable functions, and convolution with Schwartz functions, but also allow for the definition of their Fourier transform.

Another advantage of tempered distributions, compared to general distributions on $\mathbb{R}^n$, is that the topology of $\mathcal{S}$, generated by the usual natural family of norms $\{\|\cdot\|_{(N)}\}_{N\in\mathbb{N}}$, is metrizable, since only countably many seminorms are required. We can thus leverage the theory of metric spaces to address topological questions in $\mathcal{S}$, such as its completeness, with relative ease.

Distributions on arbitrary open subsets Ω of $\mathbb{R}^n$ are introduced in Chapter 5. We define them (somewhat unconventionally) as those linear functionals on the test function space $\mathcal{D}(\Omega) := C_0^\infty(\Omega)$ that, in a well-defined sense, "locally resemble" tempered distributions. Our definition is closely aligned with the original approach used by Schwartz (see Remark 5.2).

This approach offers several advantages over other common definitions of distributions:

- It immediately clarifies that distributions are characterized by local properties. Other common characterizations, such as by continuous linear functionals on the space $\mathcal{D}(\Omega)$ (endowed with its natural locally convex topology), or through $\mathcal{D}$-null sequence continuity, follow quickly (see Theorem 5.3, Proposition 5.4).
  In functional analysis books, the locally convex topology on $\mathcal{D}(\Omega)$ is typically introduced first, which may initially seem somewhat opaque to beginners in this field.
- Many operations on distributions in Ω, such as differentiation and multiplication by suitable functions, can be easily traced back to their counterparts for tempered distributions, avoiding the need for repetitive, adapted proofs, which one finds in many other approaches.

Further details on the locally convex topology of $\mathcal{D}(\Omega)$ are provided in Section 5.4, including a proof that the topology of $\mathcal{D}(\Omega)$ is not metrizable (relying on the Baire category theorem, proven for completeness in Appendix A). Additional insights are found in Section 5.5.

*Sections like these are marked with an asterisk *. This indicates that they provide supplementary information, which is not essential for a first reading or lecture course but may enhance understanding for more advanced study.*

Section 2.8 is also marked with an asterisk (*) (in parentheses), because, while it is indispensable later for the proof of Schwartz's kernel theorem in Chapter 9, it can be skipped initially without significant loss.

In my lecture courses, I placed greater emphasis on practical and important applications of Fourier analysis and distribution theory. Applications to the theory of partial differential equations are covered in Chapters 7 and 8. Chapter 7 includes explicit computations of fundamental solutions for all powers Δ^m of the Laplace operator Δ using analytic families of distributions. It also contains a proof of the Malgrange–Ehrenpreis theorem on the local solvability of linear differential operators with constant coefficients. Chapter 8 provides a brief introduction to the regularity theory of linear differential operators.

The *exercises* at the end of each chapter, varying in degree of difficulty, aim to deepen the understanding and highlight further applications. Particularly challenging exercises are marked with an asterisk *, and for these (as well as for especially important exercises like Exercise 9.1), I have included, hopefully helpful, hints.

A am grateful to many who have suggested corrections, large or small, before publication, or have helped me with creating some of the figures in the book, including Stefan Buschenhenke, Lars Niedorf and Martin Paulat. My thanks also go to Alessio Martini for calling my attention to the monograph [5] by Corwin and Greenleaf.

Kiel *Detlef Müller*

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Chapter 1
The Basic Idea of Fourier Analysis: Expansion of Periodic Functions into Trigonometric Series

Already in high school, one typically learns that the sound of a vibrating string (for instance, a violin, or a guitar string), which is fixed at both ends, can be decomposed into the "fundamental tone" and "overtones" (or "harmonics"), which are responsible for the particular sound image of the given instrument.

More specifically, this means the following: Let us imagine that the two ends of our string correspond to the points 0 and $L > 0$ on the real axis $\mathbb{R}$, and denote by $u(x,t)$ the deflection of the string at time t and position $x \in [0, L]$. Then the idea is that, for fixed time t, we should be able to decompose $u(x,t)$ as a function of x into **standing waves** of the form

$$u_m(x,t) = a_m(t)\sin(m\tfrac{\pi}{L}x), \quad m = 1, 2, 3, \ldots, \tag{1.1}$$

i.e., that we can represent $u(x,t)$ as a **superposition** of standing waves

$$u(x,t) = \sum_{m=1}^{\infty} a_m(t)\sin(m\tfrac{\pi}{L}x). \tag{1.2}$$

Mathematically, such one-dimensional waves are modeled by means of the "homogeneous wave equation." Under suitable assumptions on the material properties of the string, we may assume that the wave speed $c > 0$ is constant, and that the wave equation (in its simplest form) is given by

$$\frac{\partial^2 u}{\partial t^2} - c^2\frac{\partial^2 u}{\partial x^2} = 0. \tag{1.3}$$

For our mathematical discussion of the solutions of this differential equation, we may and shall assume (after a scaling in time by the factor $1/c$) that $c = 1$.

It is easily seen that the function u_m in (1.1) satisfies this wave equation (with $c = 1$) if and only if a_m satisfies the ordinary differential equation

$$a_m''(t) + (m\tfrac{\pi}{L})^2 a_m(t) = 0,$$

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i.e., if and only if $a_m(t)$ in (1.1) has the form

$$a_m(t) = A_m \cos(m\tfrac{\pi}{L}t) + B_m \sin(m\tfrac{\pi}{L}t),$$

with complex coefficients A_m, B_m, which, however, need to be real if $u(x,t)$ is assumed to be real. (For a more comprehensive mathematical discussion of the vibrating string, including a derivation of the wave equation from physical considerations, we refer, e.g., to Section 1.1 in Stein and Shakarchi [40]).

The term with $m = 1$ in (1.2) corresponds to the **fundamental tone** or **first harmonic** of the vibrating string, the term with $m = 2$ to the **first overtone** or **second harmonic**, etc.

Our assumption on the decomposition of $u(x,t)$ into an infinite series of standing waves would imply in particular that, for any fixed time t, it should be possible to represent the deflection of the string $f(x) = u(x,t)$ as a sine series in the variable x

$$f(x) = \sum_{m=1}^{\infty} c_m \sin(\tfrac{\pi}{L}mx),$$

with suitable coefficients c_m. Observe also that the function f can be extended in a unique way to an odd function on the interval $[-L, L]$, which would then also be represented by this sine series.

For more general functions f on the interval $[-L, L]$, which are not necessarily vanishing at the endpoints $-L$ and L, a similar heuristic analysis of the corresponding boundary value problem for the wave equation would suggest that it should be possible to expand such functions into series of the form

$$f(x) = \frac{a_0}{2} + \sum_{m=1}^{\infty} \big(a_m \cos(\tfrac{\pi}{L}mx) + b_m \sin(\tfrac{\pi}{L}mx)\big). \tag{1.4}$$

The sine terms would then correspond to the odd part $x \mapsto ((f(x) - f(-x))/2$ of f, and the cosine terms to the even part $x \mapsto ((f(x) + f(-x))/2$ of f.

Historical note. As explained in [18], the differential equation (1.3) appeared (in implicit form) for the first time in a paper from 1749 by Jean le Rond d'Alembert (1717–1783), who also studied solutions of the form $u(x,t) = \varphi(t)\psi(x)$. Leonhard Euler (1707–1783), who began to develop the theory of differential equations around that time, studied the wave equation (1.3) explicitly as an important example. He, as well as his contemporary Daniel Bernoulli (1700–1782), observed that superpositions of solutions are solutions as well.

The representation (1.2) of the deflection $u(x,t)$ of the vibrating string as a superposition of standing waves had basically already been proposed in 1753 by D. Bernoulli, and in the eighteenth century leading mathematicians such as Euler and Joseph-Louis Lagrange (1736–1813), and the Bernoulli brothers, had already been aware of expansions of particular functions into series of the form (1.4), but there were serious doubts that this would be possible for wider classes of functions.

Indeed, it is a priori not clear at all whether the physical considerations which had led to (1.4) could really be made mathematically rigorous.

In contrast to this general belief, in his seminal mémoire "Théorie analytique de la chaleur" from 1807 on heat conduction in solids, Joseph Fourier (1768–1830) expressed his conviction that indeed quite general functions f could be expanded into a series of the form (1.4). As mentioned before, leading mathematicians of his time were first quite skeptical, but Fourier would turn out to be right.

In numerous subsequent mathematical studies, beginning with a fundamental paper by Johann Peter Gustav Lejeune Dirichlet (1808–1859), Fourier's conviction has eventually been impressively justified – we shall come back to this in more detail at various places.

The ideas and visions of Fourier have had a deep impact on the development of physics, engineering and many other fields, and have created in mathematics what is now called the field of Fourier analysis, or, more generally, Harmonic Analysis,

Let us come back to (1.4). We can interpret (1.4) as the claim that any sufficiently general $2L$-periodic function f on the real axis can be expanded on $\mathbb{R}$ into a series of the form (1.4). The question of in what sense convergence should hold (pointwise, uniformly, etc.) will be left open at this point.

Let us recall: A function f on $\mathbb{R}$ is called **T-periodic** ($T > 0$) if

$$f(x+T) = f(x) \qquad \text{for all } x \in \mathbb{R}.$$

By passing by means of the dilation by the factor $T/2\pi$ from f to the 2π-periodic function $x \mapsto f(\frac{T}{2\pi}x)$, we may assume without loss of generality that the period of f is $T = 2\pi$. This is what we shall always assume in what follows, unless explicitly stated otherwise.

Then we are dealing in (1.4) with a **trigonometric series** of the form

$$\frac{a_0}{2} + \sum_{m=1}^{\infty} (a_m \cos mx + b_m \sin mx). \tag{1.5}$$

The **coefficients** a_m, b_m of such a trigonometric series can be any complex numbers. We shall therefore regard such a series a priori as a formal function series, i.e., as the sequence of the partial sums

$$\frac{a_0}{2} + \sum_{m=1}^{n} (a_m \cos mx + b_m \sin mx), \qquad n \in \mathbb{N} := \{0, 1, 2, \dots\},$$

without requiring convergence in even a single point x (in a similar way as a power series is a priori only viewed as a formal series).

By means of Euler's identity $e^{ix} = \cos x + i \sin x$, we can formally also write such a series in the form of a so-called **Fourier series**

$$\sum_{k=-\infty}^{\infty} c_k e^{ikx}. \tag{1.6}$$

Again, a Fourier series will a priori only be viewed as a formal series, for which we choose as partial sums the symmetric finite sums

$$\sum_{k=-n}^{n} c_k e^{ikx}, \qquad n \in \mathbb{N}.$$

The complex number c_k is then called the **k-th Fourier coefficient** of this series.

Suppose that (1.5) and (1.6) represent the same formal series. Then it is easily seen that the coefficients a_m, b_m in (1.5) and the coefficients c_k in (1.6) are related as follows:

$$a_m = c_m + c_{-m}, \text{ if } m \geq 0, \tag{1.7}$$

$$b_m = i(c_m - c_{-m}), \text{ if } m \geq 1. \tag{1.8}$$

From a mathematical perspective, the representation as a series of the form (1.6) is in general preferable, due to the special properties of the so-called **characters**

$$e_k(x) := e^{ikx}, \qquad k \in \mathbb{Z}.$$

Indeed, in contrast to the functional identities which are satisfied by the trigonometric functions sin and cos, these characters satisfy the following simpler identities:

$$e_k(x + y) = e_k(x)e_k(y), \quad x, y \in \mathbb{R}, \tag{1.9}$$

as well as

$$\overline{e_k} = e_{-k} \tag{1.10}$$

and

$$e_j e_k = e_{j+k}. \tag{1.11}$$

Definitions. The Fourier series (1.6) **converges at the point** $x \in \mathbb{R}$ towards $\lambda \in \mathbb{C}$ if the sequence of symmetric partial sums $\sum_{k=-n}^{n} c_k e^{ikx}$ converges to λ as $n \to \infty$. In this case, λ will be called the **value** of this series, and we shall write

$$\lambda = \sum_{k=-\infty}^{\infty} c_k e^{ikx} \quad (\text{or simply } \lambda = \sum_{k\in\mathbb{Z}} c_k e^{ikx}).$$

The series converges **pointwise (resp. uniformly)** towards the function $f : \mathbb{R} \to \mathbb{C}$ if the sequence of symmetric partial sums $\sum_{k=-n}^{n} c_k e^{ikx}$ converges pointwise (resp. uniformly) towards f. The latter just means that

$$\|f - \sum_{k=-n}^{n} c_k e_k\|_{\ell^\infty(\mathbb{R})} \to 0 \qquad \text{for } n \to \infty,$$

where

$$\|f\|_{\ell^\infty(\mathbb{R})} := \sup_{x\in\mathbb{R}} |f(x)|$$

denotes the sup norm of f. The limit function f is then continuous, and 2π- periodic as well (see Proposition 1.3).

Suppose we knew already that a given 2π-periodic function f can be expanded into a Fourier series. How, then, should its Fourier coefficients be determined? The key to answering this question lies in the identities (1.13) below for the characters e_k, which are of fundamental importance to Fourier analysis:

If a function f on $\mathbb{R}$ is 2π-periodic, then it is uniquely determined by its restriction to any semi-open interval of length 2π, for instance the interval $[0, 2\pi[$. Under the assumption that f is Lebesgue integrable over this interval, we then define its integral by

$$\fint f(x)\,\mathrm{d}x := \frac{1}{2\pi}\int_0^{2\pi} f(x)\,\mathrm{d}x.$$

Note that by means of the factor $1/2\pi$ the integral is normalized in such a way that for the constant function 1 we have $\fint 1\,\mathrm{d}x = 1$.

It is then easily seen (cf. Exercise 1.3) that for any interval I of length 2π we have

$$\fint f(x)\,\mathrm{d}x = \frac{1}{2\pi}\int_I f(x)\,\mathrm{d}x. \tag{1.12}$$

With this notion of integral the following holds true:

Proposition 1.1 *For all $j, k \in \mathbb{Z}$ we have*

$$\fint e_j(x)\overline{e_k(x)}\,\mathrm{d}x = \delta_{jk}, \tag{1.13}$$

where δ_{jk} denotes Kronecker's symbol, i.e.,

$$\delta_{jk} := \begin{cases} 1, & \text{if} \quad j = k, \\ 0, & \text{if} \quad j \neq k. \end{cases}$$

Proof. For $\ell \neq 0$ we have

$$\int_0^{2\pi} \mathrm{e}^{i\ell x}\mathrm{d}x = \frac{1}{i\ell}\mathrm{e}^{i\ell x}\Big|_0^{2\pi} = 0,$$

and for $\ell = 0$ we get $\int_0^{2\pi} 1\mathrm{d}x = 2\pi$. The identities (1.13) now follow immediately by means of (1.10) and (1.11). □

Remark 1.2 The identity (1.13) admits the following *geometric interpretation*: By

$$(f, g) := \fint f(x)\overline{g(x)}\,\mathrm{d}x$$

we introduce a hermitian inner product on the space L^2_{per} of 2π-periodic complex-valued functions which are square integrable over the interval $[0, 2\pi[$. (1.13) then just says that the characters $\{e_k\}_{k\in\mathbb{Z}}$ form an orthonormal system in L^2_{per} with respect to this inner product.

Proposition 1.3 *Assume that the Fourier series $\sum_{k=-\infty}^{\infty} c_k e_k$ converges uniformly on $\mathbb{R}$ towards the function f. Then f is continuous and 2π-periodic, and we have*

$$c_k = \frac{1}{2\pi}\int_0^{2\pi} f(x)\mathrm{e}^{-ikx}\mathrm{d}x.$$

Proof. Let $f_n := \sum_{j=-n}^{n} c_j e_j$. Then f_n is continuous and 2π-periodic. The uniform convergence of the f_n towards f then implies that f is continuous too. Moreover, for any $x \in \mathbb{R}$ we have $f(x) = \lim_{n\to\infty} f_n(x)$, and thus $f(x + 2\pi) = \lim_{n\to\infty} f_n(x + 2\pi) = \lim_{n\to\infty} f_n(x) = f(x)$, i.e., f is 2π-periodic too.

Fix now $k \in \mathbb{Z}$. Since $|\overline{e_k(x)}| = |\mathrm{e}^{-ikx}| = 1$ for $x \in \mathbb{R}$, we see that

$$\|f\overline{e_k} - \sum_{j=-n}^{n} c_j e_j \overline{e_k}\|_{\ell^\infty(\mathbb{R})} = \|f - \sum_{j=-n}^{n} c_j e_j\|_{\ell^\infty(\mathbb{R})}.$$

This shows that the sequence of functions $\{\sum_{j=-n}^{n} c_j e_j \overline{e_k}\}_n$ converges uniformly towards $f\overline{e_k}$, and thus

$$\oint f(x)\overline{e_k}(x)\mathrm{d}x = \lim_{n\to\infty} \oint \sum_{j=-n}^{n} c_j e_j(x)\overline{e_k(x)}\,\mathrm{d}x = \lim_{n\to\infty} \sum_{j=-n}^{n} c_j \oint e_j(x)\overline{e_k(x)}\,\mathrm{d}x.$$

The asserted identity now follows immediately from (1.13). □

Definition. Let $f : \mathbb{R} \to \mathbb{C}$ be a 2π-periodic function, which is Lebesgue-integrable over the interval $[0, 2\pi[$, i.e., a **2π-periodic locally integrable function.** For $k \in \mathbb{Z}$ we then denote by

$$\widehat{f}(k) := \oint f(x)\overline{e_k(x)}\,\mathrm{d}x = \frac{1}{2\pi}\int_0^{2\pi} f(x)\mathrm{e}^{-ikx}\,\mathrm{d}x$$

the k-th **Fourier coefficient** of f. The (formal) series

$$Sf(x) := \sum_{k=-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{ikx}$$

is called the **Fourier series** of f. A standard shorthand notation for this is

$$f(x) \sim \sum_{k=-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{ikx}. \tag{1.14}$$

By

$$S_n f(x) := \sum_{k=-n}^{n} \widehat{f}(k) \mathrm{e}^{ikx}$$

we shall denote the n-th symmetric partial sum of this Fourier series. If we re-write the Fourier series of f as a (formal) trigonometric series

$$f(x) \sim \frac{a_0}{2} + \sum_{m=1}^{\infty} (a_m \cos mx + b_m \sin mx) \tag{1.15}$$

as in (1.5), then from (1.7), (1.8) and (1.12) one easily deduces the following formulas for the corresponding **Fourier-cosine coefficients** a_m and the **Fourier-sine coefficients** b_m :

$$a_m = 2 \fint f(x) \cos mx \, \mathrm{d}x = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos mx \, \mathrm{d}x, \quad m = 0, 1, 2, \ldots; \tag{1.16}$$

$$b_m = 2 \fint f(x) \sin mx \, \mathrm{d}x = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin mx \, \mathrm{d}x, \quad m = 1, 2, \ldots. \tag{1.17}$$

Here we have used the symmetric interval of integration $]-\pi, \pi[$ in place of $]0, 2\pi[$, since the expansion into a trigonometric series (1.15) can have advantages compared to a Fourier series expansion (1.14), in particular when the function f exhibits symmetry properties with respect to the origin, i.e., for functions which are even, or odd.

For instance, if f is *even*, then the integrand in the identity (1.17) for b_m is odd, so that $b_m = 0$, and then the series (1.15) is a pure **cosine series**

$$f(x) \sim \frac{a_0}{2} + \sum_{m=1}^{\infty} a_m \cos mx.$$

Moreover, since the integrand (1.16) is even, we then have

$$a_m = \frac{2}{\pi} \int_0^{\pi} f(x) \cos mx \, \mathrm{d}x, \qquad m = 0, 1, 2, \ldots. \tag{1.18}$$

Analogously, if f is *odd*, then the series (1.15) is a pure **sine series**

$$f(x) \sim \sum_{m=1}^{\infty} b_m \sin mx,$$

and we have that

$$b_m = \frac{2}{\pi} \int_0^{\pi} f(x) \sin mx \, \mathrm{d}x, \qquad m = 1, 2, \ldots. \tag{1.19}$$

Finite linear combinations $\sum c_k e_k$ of the characters e_k, such as $S_n f$, are called **trigonometric polynomials**.

Example 1.4 Let $-\pi \le a \le b \le \pi$, and let f be the 2π-periodic function which agrees on the interval $[-\pi, \pi[$ with the indicator function $\mathbf{1}_{[a,b[}$ of the interval $[a, b[$. With a slight abuse of notation, this 2π-periodic function will again be simply denoted by $\mathbf{1}_{[a,b[}$. For the k-th Fourier coefficient

$$\hat{f}(k) = \frac{1}{2\pi} \int_a^b 1 \mathrm{e}^{-ikx} \, \mathrm{d}x$$

of f we then obtain

$$\hat{f}(k) = \begin{cases} \frac{b-a}{2\pi}, & \text{if } k = 0, \\ \frac{i}{2\pi k}(\mathrm{e}^{-ibk} - \mathrm{e}^{-iak}), & \text{if } k \neq 0. \end{cases} \tag{1.20}$$

In particular, for $0 < a < \pi$, we get

$$\widehat{\mathbf{1}_{[-a,a]}}(k) = \begin{cases} a/\pi, & \text{if } k = 0, \\ \frac{\sin(ak)}{\pi k}, & \text{if } k \neq 0. \end{cases} \tag{1.21}$$

For the latter function we have $\hat{f}(k) = \hat{f}(-k)$, so that the Fourier series becomes a pure cosine series (compare Figure 1.1):

$$\mathbf{1}_{[-a,a]}(x) \sim \frac{a}{\pi} + \sum_{m=1}^{\infty} \frac{2}{\pi} \frac{\sin(am)}{m} \cos(mx), \qquad -\pi \le x < \pi.$$

Since here the function f is even, we could have computed the coefficients of this cosine series more quickly by means of (1.18).

Note also that this series is typically not absolutely convergent, for instance for $a = \pi/2$ at $x = 0$.

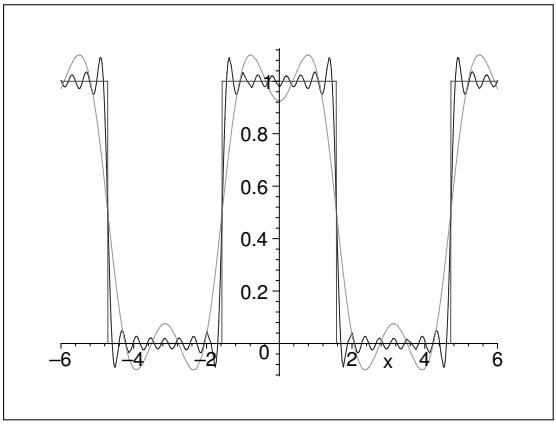


Fig. 1.1 Graphs of f, $S_3 f$ and $S_{16} f$ for $a = \pi/2$

So far, we have associated to the function f only its formal Fourier series (1.14). *The question of whether, and in what sense, this formal Fourier series converges – and, if so, whether it indeed converges to $f(x)$ – has been left open so far.* As already mentioned, in his mémoire on heat conduction, Fourier had asserted that such a kind of convergence should essentially hold true for any "reasonable" function f, but many leading mathematicians of his time had initially been skeptical about this claim.

The first who succeeded in giving a rigorous convergence proof for Fourier series for a larger class of functions, namely for all piecewise continuously differentiable functions, was Dirichlet (cf. Corollary 3.14). After this first breakthrough, step by step a better and increasingly deeper understanding of this convergence question has been gained, and by now we know that indeed remarkably many functions can be expanded into their Fourier series.

Fourier's mémoire was thus the starting point of what is nowadays called the field of "Fourier analysis," which had initially been developed in order to solve problems in physics. However, this field has turned out to be of fundamental importance for numerous other fields, in mathematics as well as in other sciences.

In this introductory chapter, I shall not yet enter into a discussion of the problem of convergence of Fourier series. Since, conceptually as well as technically, many of the aspects of these questions can be understood more easily in the analogous setting of Fourier analysis for non-periodic functions on $\mathbb{R}$, we shall indeed first look at the analogous questions on $\mathbb{R}$, more generally even on $\mathbb{R}^n$, in the next chapter, before returning to the theory of Fourier series in Chapter 3.

At least for locally square integrable 2π-periodic functions f, by means of tools from Hilbert space theory it is nowadays quite easy to make it plausible that the Fourier series of f should converge towards f (in a certain sense). But note that Hilbert spaces, named after David Hilbert (1862–1943) by his doctoral student Erhard Schmidt (1876–1959), were introduced only about a hundred years after the appearance of Fourier's mémoire (itself also motivated by Fourier's work). So let us conclude this chapter with a corresponding discussion.

We have already seen that the characters $\{e_k\}_k$ form an orthonormal system in the space L^2_{per} of 2π-periodic functions f which are square integrable over the interval $[0, 2\pi[$ (compare Remark 1.2).

Suppose we could show that this system is complete in the Hilbert space L^2_{per}, i.e., that it forms a Hilbert space basis of L^2_{per}. (We shall prove this later in Corollary 3.12, but one can also easily derive it by means of the Stone–Weierstrass approximation theorem. For yet another, elementary proof we refer to [6].)

Then abstract Hilbert space theory shows that the Fourier series of $f \in L^2_{\mathrm{per}}$ will converge towards f, at least in the sense of the norm topology induced by the L^2_{per} -norm

$$\|f\|_2 := (f, f)^{1/2} = \left(\fint |f(x)|^2 \, \mathrm{d}x\right)^{1/2},$$

i.e.,

$$\lim_{n\to\infty} \|S_n f - f\|_2 = 0. \tag{1.22}$$

Indeed, just note that for L^2_{per},

$$\hat{f}(k) = (f, e_k),$$

so that

$$S_n(f) = \sum_{k=-n}^{n} (f, e_k) e_k.$$

Note finally that abstract Hilbert space theory also tells us that the following "Plancherel identity" holds true:

$$\|f\|_2^2 = \sum_{k=-\infty}^{\infty} |(f, e_k)|^2 = \sum_{k=-\infty}^{\infty} |\hat{f}(k)|^2. \tag{1.23}$$

(We shall later return to this property and give an alternative proof in Chapter 3.) Finally, observe that (1.22) is an immediate consequence of this identity – just apply it to $S_n f - f$ in place of f :

$$\|S_n f - f\|_2^2 = \sum_{|k|>n} |\hat{f}(k)|^2 \to 0 \qquad \text{as} \quad n \to \infty.$$

1.1 Exercises

1.1 Compute the formal Fourier series of the following 2π-periodic functions on $\mathbb{R}$:

a) $f(x) := x \cos x \quad \text{on }]-\pi, \pi]$.
b) $g(x) := |\sin x| \quad \text{on }]-\pi, \pi]$.

1.2 Let f and g be the 2π-periodic functions on $\mathbb{R}$ defined on the interval $[-\pi, \pi[$ by $f(x) := \operatorname{sign} x$ and $g(x) = |x|$, respectively. Furthermore, let h be the 2π-periodic odd function defined on the interval $[0, \pi]$ by $h(x) := x(\pi - x)$.

Draw the graphs of the functions f, g and h, and determine their corresponding Fourier series. Utilize symmetry properties of these functions where applicable.

1.3 Let the function $f : \mathbb{R} \to \mathbb{C}$ be 2π-periodic and (Lebesgue-) integrable over the interval $[0, 2\pi[$. Show that:

a) For every half-open interval A of length 2π, the following holds:

$$\int_0^{2\pi} f(x)\,\mathrm{d}x = \int_A f(x)\,\mathrm{d}x.$$

More generally, this formula holds for any measurable subset $A \subset \mathbb{R}$ such that $\mathbb{R}$ is the disjoint union of the translated sets $2\pi k + A$, $k \in \mathbb{Z}$, i.e.,

$$\mathbb{R} = \dot{\bigcup_{k\in\mathbb{Z}}} 2\pi k + A.$$

Hint: Note that $[0, 2\pi[= \dot{\cup}_{k\in\mathbb{Z}} A_k$, with $A_k := (2\pi k + A) \cap [0, 2\pi[$, and exploit the 2π-periodicity of f.

b) $$\oint f(x + a)\,\mathrm{d}x = \oint f(x)\,\mathrm{d}x \quad \text{for every } a \in \mathbb{R}.$$

c) $$\oint f(nx)\,\mathrm{d}x = \oint f(x)\,\mathrm{d}x \quad \text{for every } n \in \mathbb{N}_{>0}.$$

1.4 Let $\{a_k\}_{k=1}^{\infty}$ and $\{b_k\}_{k=1}^{\infty}$ be two sequences of complex numbers. Denote by $B_n := \sum_{k=1}^{n} b_k$ the n-th partial sum of the series $\sum_{k=1}^{\infty} b_k$; here, by definition we put $B_0 := 0$.

a) Prove the following *summation by parts formula:*

$$\sum_{k=M}^{N} a_k b_k = a_N B_N - a_M B_{M-1} - \sum_{k=M}^{N-1} (a_{k+1} - a_k) B_k.$$

b) Use this identity to derive Dirichlet's test for the convergence of infinite series:
If the partial sums of the series $\sum_{k=1}^{\infty} b_k$ form a bounded sequence and if the sequence $\{a_k\}_k$ is a monotonically decreasing sequence of real numbers converging to zero, then the series $\sum_{k=1}^{\infty} a_k b_k$ is convergent.

c) Let $\alpha \geq 0$. Use this test to investigate for which $\theta \in [0, 2\pi[$ the Fourier series

$$\sum_{n=1}^{\infty} n^{-\alpha} \mathrm{e}^{in\theta}$$

is convergent.

1.5 Let $f : \mathbb{R} \to \mathbb{C}$ be a 2π-periodic function that is integrable over the interval $[0, 2\pi[$. Prove that

$$\lim_{|k|\to\infty} \hat{f}(k) = 0.$$

Hint: First, show this for 2π-periodic functions that coincide with a step function on the interval $[0, 2\pi[$.

Chapter 2
Fourier Transform and Convolution on $\mathbb{R}^n$

In a similar way as we did for 2π-periodic functions on $\mathbb{R}$, a Fourier transform can also be defined for non-periodic integrable functions on $\mathbb{R}$, more generally on $\mathbb{R}^n$.

A helpful and significant feature for studying this Fourier transform is that on $\mathbb{R}^n$, for any $r > 0$, the "scaling" transformation $x \mapsto rx$, $x \in \mathbb{R}^n$, is available as an automorphism, referred to as **dilation by the factor** r.

However, such scalings typically change the period of a periodic function on $\mathbb{R}$, making them unsuitable for the theory of Fourier series.

Using these dilations (along with other automorphisms from the general linear group $\mathrm{GL}(n, \mathbb{R})$), many key concepts in the theory can be studied and understood more easily on $\mathbb{R}^n$ than on $\mathbb{R}$ for periodic functions.

For instance, constructing approximate (convolution-) identities, as discussed in Section 2.3, becomes simpler.

There are thus many good reasons why we will first focus on the theory of Fourier integrals on $\mathbb{R}^n$, which plays a fundamental role also in the study of linear differential operators on $\mathbb{R}^n$, before returning to the theory of Fourier series in Chapter 3 (for a brief overview of the theory of higher-dimensional Fourier series, I refer to Section 3.5).

2.1 Definition and Elementary Properties of the Fourier Transform

We denote by D_j the differential operator

$$D_j := \frac{1}{i}\,\frac{\partial}{\partial x_j}, \qquad j = 1, \ldots, n,$$

and put

$$D^\alpha := D_1^{\alpha_1} \ldots D_n^{\alpha_n}, \quad \alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n.$$

Given any $\xi \in \mathbb{R}^n$, we introduce the function e_ξ:

$$e_\xi(x) := \mathrm{e}^{i\xi \cdot x} = \exp[i(\xi_1 x_1 + \cdots + \xi_n x_n)], \qquad x \in \mathbb{R}^n.$$

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D. Müller, *An Invitation to Fourier Analysis and Distribution Theory*,
Universitext, https://doi.org/10.1007/978-3-032-31550-2_2

By the properties of the exponential function, it is obvious that these so-called **characters** e_ξ form continuous homomorphisms from the additive group $\mathbb{R}^n$ into the multiplicative group $S^1 := \{z \in \mathbb{C} : |z| = 1\}$ of all complex numbers of modulus 1 (compare also Exercise 2.6) , i.e.,

$$e_\xi(x+y) = e_\xi(x)e_\xi(y) \qquad \text{for all } \xi, x, y \in \mathbb{R}^n. \tag{2.1}$$

At the same time, these characters form a family of simultaneous eigenfunctions for all differential operators D^α, since

$$D^\alpha e_\xi = \xi^\alpha e_\xi. \tag{2.2}$$

For $1 \le p \le \infty$ we denote by $L^p(\mathbb{R}^n) = L^p(\mathbb{R}^n, \mathrm{d}x)$ Lebesgue's L^p-space on $\mathbb{R}^n$.

We recall: A Lebesgue-measurable function $f : \mathbb{R} \to \mathbb{C} \cup \{\infty\}$ lies in the space $\mathcal{L}^p(\mathbb{R}^n)$ if its L^p-seminorm $\|f\|_p$ is finite. For $1 \le p < \infty$, this seminorm is given by

$$\|f\|_p = \|f\|_{L^p} := \left(\int_{\mathbb{R}^n} |f(x)|^p \, \mathrm{d}x \right)^{1/p},$$

whereas, by definition, for $p = \infty$ the $\|f\|_\infty$ -seminorm (for the notion of seminorm, see Chapter 4) is defined as the essential supremum of $|f|$. $\mathcal{L}^p(\mathbb{R}^n)$, endowed with the L^p- seminorm, then forms a seminormed vector space over $\mathbb{C}$.

If we denote by $\mathcal{N}$ the subspace of all functions f that vanish almost everywhere in the Lebesgue sense, i.e., outside a set of measure zero, then the quotient space

$$L^p(\mathbb{R}^n) := \mathcal{L}^p(\mathbb{R}^n)/\mathcal{N},$$

endowed with the induced L^p-norm $\|\cdot\|_p = \|\cdot\|_{L^p(\mathbb{R}^n)}$, becomes a complex Banach space. As usual, we shall identify functions $f \in \mathcal{L}^p(\mathbb{R}^n)$ with their corresponding equivalence classes $[f] := f + \mathcal{N} \in L^p(\mathbb{R}^n)$, unless expressly stated otherwise.

We shall leave it to the reader to verify that the respective definitions or statements about $\mathcal{L}^p$ functions will not depend on the choice of the representative from $[f]$.

Definitions. The **Fourier transform** $\hat{f} : \mathbb{R}^n \to \mathbb{C}$ of a function $f \in L^1(\mathbb{R}^n)$ is defined by

$$\hat{f}(\xi) := \int_{\mathbb{R}^n} f(x) \mathrm{e}^{-i\xi \cdot x} \mathrm{d}x, \qquad \xi \in \mathbb{R}^n.$$

In the sequel, when the domain of integration of an integral is the whole underlying space, like $\mathbb{R}^n$ in the preceding Fourier integral, we shall often drop it from the integral sign, as well as sometimes also the integration variable from the integrand, in order to defray formulas containing integrals.

For instance, according to this convention, we can then write

$$\hat{f}(\xi) = \int f \overline{e}_\xi \, \mathrm{d}x.$$

For $y \in \mathbb{R}^n$ we define the **translation operator** λ_y by

$$(\lambda_y f)(x) := f(x-y), \qquad x \in \mathbb{R}^n.$$

Moreover, for any function $f : \mathbb{R} \to \mathbb{C} \cup \{\infty\}$ we put

$$f^*(x) := \overline{f(-x)}, \qquad x \in \mathbb{R}^n.$$

Proposition 2.1 (a) *The Fourier transform $\hat{f}$ of any function $f \in L^1(\mathbb{R}^n)$ is continuous.*

(b) *The* **Fourier transform** $\mathcal{F} : f \mapsto \hat{f}$ *is a continuous linear operator*

$$\mathcal{F} : L^1(\mathbb{R}^n) \to L^\infty(\mathbb{R}^n),$$

and we have the norm bound

$$\|\mathcal{F} f\|_\infty \le \|f\|_1 \qquad \text{for every } f \in L^1(\mathbb{R}^n).$$

(c) *Let $f \in L^1(\mathbb{R}^n)$, $y \in \mathbb{R}^n$ and $A \in GL(n,\mathbb{R})$. Then the following hold true:*

(i) $(\lambda_y f)^\wedge = e_{-y}\hat{f}$;

(ii) $(e_y f)^\wedge = \lambda_y \hat{f}$;

(iii) $(f \circ A)^\wedge = \frac{1}{|\det A|}\hat{f} \circ {}^tA^{-1}$, *where ${}^tA$ denotes the transpose of A.*

(iv) $\widehat{f^*} = \overline{\hat{f}}$.

In particular, under the Fourier transform, the translation $\lambda_y f$ of f by y corresponds to the ***modulation*** $e_{-y}\hat{f}(\xi) = \mathrm{e}^{-i\xi\cdot y}\hat{f}(\xi)$ *of the Fourier transform of f by the "frequency" y.*

Proof. (a) follows by the theorem on the continuous dependence of parameter-dependent Lebesgue integrals, and (b) is immediate, since

$$|\hat{f}(\xi)| \le \int |f(x)\mathrm{e}^{-i\xi\cdot x}|\,\mathrm{d}x = \|f\|_1.$$

Next, we note that

$$\begin{aligned}
(\lambda_y f)^\wedge(\xi) &= \int f(x-y)\mathrm{e}^{-i\xi\cdot x}\mathrm{d}x \\
&= \int f(x)\mathrm{e}^{-i\xi\cdot(x+y)}\mathrm{d}x \\
&= \mathrm{e}^{-i\xi\cdot y}\int f(x)\mathrm{e}^{-i\xi\cdot x}\mathrm{d}x \\
&= \mathrm{e}^{-i\xi\cdot y}\hat{f}(\xi), \\
(e_y f)^\wedge(\xi) &= \int f(x)\mathrm{e}^{iy\cdot x}\mathrm{e}^{-i\xi\cdot x}\mathrm{d}x = \hat{f}(\xi-y),
\end{aligned}$$

which gives (c)(i) and (ii). Finally, by the change of variables formula,

$$\begin{aligned}(f\circ A)^\wedge(\xi) &= \int f(Ax)\mathrm{e}^{-i\xi\cdot x}\mathrm{d}x = \int f(x)\mathrm{e}^{-i\xi\cdot(A^{-1}x)}|\det A^{-1}|\mathrm{d}x\\ &= \frac{1}{|\det A|}\int f(x)\mathrm{e}^{-i({}^tA^{-1}\xi)\cdot x}\mathrm{d}x = \frac{1}{|\det A|}\hat{f}({}^tA^{-1}\xi),\end{aligned}$$

and clearly

$$\begin{aligned}\widehat{f^*}(\xi) &= \int \overline{f(-x)}\mathrm{e}^{-i\xi\cdot x}\mathrm{d}x = \int \overline{f}(x)\mathrm{e}^{i\xi\cdot x}\mathrm{d}x\\ &= \Big(\int f(x)\mathrm{e}^{-i\xi\cdot x}\,\mathrm{d}x\Big)^- = \overline{\hat{f}(\xi)},\end{aligned}$$

which also proves (c)(iii) and (iv). □

2.2 Convolution

In the following, we will always understand a measurable (or integrable) function to mean a Lebesgue-measurable (or integrable) function, unless expressly stated otherwise.

Definitions. Let f, g be measurable functions on $\mathbb{R}^n$. If $x \in \mathbb{R}^n$ is such that

$$\int_{\mathbb{R}^n} |f(y)g(x-y)|\,\mathrm{d}y < \infty,$$

then we set

$$\begin{aligned}f * g(x) :&= \int f(y)\,\lambda_y g(x)\,\mathrm{d}y\\ &= \int f(y)g(x-y)\,\mathrm{d}y\end{aligned}$$

and denote by $f * g(x)$ the **convolution of** f **with** g (at the point x). The substitution $y = x - t$ then shows that

$$f * g(x) = \int f(x-t)g(t)\,\mathrm{d}t = g * f(x),$$

so that this convolution is commutative (wherever it is defined).

Let us next recall that if $p \in [1,\infty]$, then $p' \in [1,\infty]$ is the **conjugate** Lebesgue exponent to p if

$$\frac{1}{p} + \frac{1}{p'} = 1$$

(where we use the standard convention that $1/\infty := 0$ and $1/0 := \infty$ in this context).

Then, if p and p' are conjugate exponents in $]1,\infty[$, by Hölder's inequality we have

$$\begin{aligned}\int |f(y)g(x-y)|\,\mathrm{d}y &\le \left(\int |f(y)|^p\,\mathrm{d}y\right)^{1/p}\left(\int |g(x-y)|^{p'}\,\mathrm{d}y\right)^{1/p'},\\ &= \left(\int |f(y)|^p\,\mathrm{d}y\right)^{1/p}\left(\int |g(y)|^{p'}\,\mathrm{d}y\right)^{1/p'},\end{aligned}$$

In particular, we see that $f * g(x)$ is defined for every $x \in \mathbb{R}^n$, if $f \in L^p(\mathbb{R}^n)$ and $g \in L^{p'}(\mathbb{R}^n)$, and then

$$\|f * g\|_\infty \le \|f\|_p \|g\|_{p'}. \tag{2.3}$$

Clearly, this remains valid when $p = 1,\ p' = \infty$, and when $p = \infty,\ p' = 1$.

Theorem 2.2 (i) *Let $f, g \in L^1(\mathbb{R}^n)$. Then $f * g(x)$ is defined for almost every (abbreviated as a.e.) $x \in \mathbb{R}^n$, and $f * g \in L^1(\mathbb{R}^n)$. Moreover:*

$$\|f * g\|_1 \le \|f\|_1\ \|g\|_1.$$

(ii) *Let $f, g, h \in L^1(\mathbb{R}^n)$, and let $\alpha, \beta \in \mathbb{C}$. Then the following hold true:*

$$\begin{aligned}(f * g) * h &= f * (g * h),\\ (\alpha f + \beta g) * h &= \alpha(f * h) + \beta(g * h),\\ (f * g)^* &= g^* * f^* = f^* * g^*,\\ (\alpha f + \beta g)^* &= \overline{\alpha} f^* + \overline{\beta} g^*,\\ \|f^*\|_1 &= \|f\|_1.\end{aligned}$$

This shows that convolution defines on $L^1(\mathbb{R}^n)$ an associative and commutative product, and that the involution $f \mapsto f^*$ is isometric on $L^1(\mathbb{R}^n)$. If we endow the Banach space $L^1(\mathbb{R}^n)$ with this convolution product and this isometric involution, we thus obtain an involutive commutative Banach algebra, the so-called **group algebra** $(L^1(\mathbb{R}^n), +, *, {}^*; \|\cdot\|_1)$ *of* $\mathbb{R}^n$.

Proof. (i) Let $f, g \in \mathcal{L}^1(\mathbb{R}^n)$. Then the functions $(x, y) \mapsto f(y)$ and $(x, y) \mapsto g(x-y)$ are measurable on $\mathbb{R}^n \times \mathbb{R}^n$, and thus so is their product

$$F(x, y) := f(y)g(x - y).$$

Moreover,

$$\begin{aligned}\int \Big(\int |F(x,y)|\,\mathrm{d}x\Big)\mathrm{d}y &= \int |f(y)|\Big(\int |g(x-y)|\,\mathrm{d}x\Big)\mathrm{d}y\\ &= \int |f(y)| \int |g(x)|\,\mathrm{d}x\mathrm{d}y = \|f\|_1\ \|g\|_1.\end{aligned} \tag{2.4}$$

By Tonelli's theorem, the function F is thus also integrable. Fubini's theorem then shows that the mapping $y \mapsto F(x, y) = f(y)g(x - y)$ is integrable for a.e. x, i.e., $f * g$ exists almost everywhere, and the mapping $x \mapsto \int F(x, y)\,\mathrm{d}y = f * g(x)$ is integrable

over $\mathbb{R}^n$. Finally, (2.4) and Fubini's theorem yield

$$\begin{aligned}\|f * g\|_1 &= \int |f * g(x)|\,\mathrm{d}x = \int \left|\int F(x,y)\,\mathrm{d}y\right|\mathrm{d}x \\ &\le \int\int |F(x,y)|\,\mathrm{d}y\mathrm{d}x = \|f\|_1\,\|g\|_1.\end{aligned}$$

(ii) can be proven by similar arguments and will be left to the interested reader (see Exercise 2.2). □

The significance of the convolution product in connection with the Fourier transform will become evident through the following proposition.

Proposition 2.3 *Let $f, g \in L^1(\mathbb{R}^n)$. Then:*

$$\widehat{f * g} = \hat{f}\,\hat{g}.$$

Proof. By Fubini's theorem and translation invariance of the Lebesgue measure, we have

$$\begin{aligned}\widehat{f * g}(\xi) &= \int \left(\int f(y)g(x-y)\,\mathrm{d}y\right)\mathrm{e}^{-i\xi\cdot x}\mathrm{d}x \\ &= \int\int f(y)\mathrm{e}^{-i\xi\cdot y}g(x-y)\mathrm{e}^{-i\xi\cdot(x-y)}\mathrm{d}y\mathrm{d}x \\ &= \int f(y)\mathrm{e}^{-i\xi\cdot y}\left(\int g(x-y)\mathrm{e}^{-i\xi\cdot(x-y)}\mathrm{d}x\right)\mathrm{d}y \\ &= \int f(y)\mathrm{e}^{-i\xi\cdot y}\left(\int g(x)\mathrm{e}^{-i\xi\cdot x}\mathrm{d}x\right)\mathrm{d}y \\ &= \hat{g}(\xi)\hat{f}(\xi).\end{aligned}$$

□

The next proposition will show that by convolution smoothness properties of one of the two convolution factors is inherited by the convolution product. We still need a bit of notation.

Assume $\Omega \subset \mathbb{R}^n$ is an open subset. By $L^1_{\mathrm{loc}}(\Omega)$ we denote the space of all (equivalence classes of) measurable functions $f : \Omega \to \mathbb{C}$ which are **locally integrable** in the following sense:

for every $x \in \Omega$ there is an open neighborhood U_x of x in Ω, so that f is integrable over U_x, i.e., $f|_{U_x} \in L^1(U_x)$.

This is clearly equivalent to the property that f is integrable over every compact subset in Ω. Note also that, by Hölder's inequality,

$$L^p(\Omega) \subset L^1_{\mathrm{loc}}(\Omega), \qquad \text{for } 1 \le p \le \infty. \tag{2.5}$$

For $k \in \mathbb{N} \cup \{\infty\}$, we shall denote by $C^k_0(\Omega)$ the space of all k-times continuously differentiable functions $f : \Omega \to \mathbb{C}$ with compact support $\operatorname{supp} f$ (recall that $\operatorname{supp} f$ is the closure in Ω of the set of all $x \in \Omega$ such that $f(x) \neq 0$).

For $x_0 \in \mathbb{R}^n$ and $r > 0$ we denote by

$$B_r(x_0) := \{x \in \mathbb{R}^n : |x - x_0| < r\}$$

the open Euclidean ball of radius r and center x_0 in $\mathbb{R}^n$, and for $r \geq 0$ let

$$\overline{B}_r(x_0) := \{x \in \mathbb{R}^n : |x - x_0| \leq r\}$$

be the corresponding closed ball.

Remark 2.4 The spaces $C_0^k(\mathbb{R}^n)$ are non-trivial. For example, the function

$$\varphi(x) := \begin{cases} \mathrm{e}^{-\frac{1}{1-|x|^2}}, & |x| < 1, \\ 0, & |x| \geq 1, \end{cases}$$

whose support is the closed unit ball $\operatorname{supp} \varphi = \overline{B_1}(0)$, belongs to $C_0^\infty(\mathbb{R}^n)$.

We leave the proof as an exercise to anyone not yet familiar with this result.

If A and B are subsets of $\mathbb{R}^n$ (or, more generally, of a vector space E over some field $\mathbb{K}$), then we denote by

$$A + B := \{a + b : a \in A \text{ and } b \in B\} \quad \text{resp.} \quad A - B := \{a - b : a \in A \text{ and } b \in B\}$$

the Minkowski sum respectively Minkowski difference of A and B, and put

$$\lambda A := \{\lambda a : a \in A\} \quad \text{and} \quad x + A := \{x\} + A$$

for any scalar $\lambda \in \mathbb{K}$ and any $x \in E$.

Proposition 2.5 *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$, $g \in C_0^k(\mathbb{R}^n)$ ($k \in \mathbb{N} \cup \{\infty\}$).*

(i) *Then $f * g \in C^k(\mathbb{R}^n)$, and*

$$D^\alpha(f * g) = f * D^\alpha g, \quad |\alpha| \leq k. \tag{2.6}$$

(ii) *If A is a closed subset of $\mathbb{R}^n$ so that f vanishes a.e. outside A, then*

$$\operatorname{supp} f * g \subset A + \operatorname{supp} g.$$

In particular,

$$\operatorname{supp} f * g \subset \operatorname{supp} f + \operatorname{supp} g,$$

if f is in $C_0(\mathbb{R}^n)$ too.

Proof. (i) Without loss of generality we may assume that $k \in \mathbb{N}$. Let $K := \operatorname{supp} g$ be the compact support of g, and let $x_0 \in \mathbb{R}^n$ be fixed. Let $\varepsilon > 0$, and assume that $x \in B_\varepsilon(x_0) =: B$. Then $g(x - y) = 0$, unless $x - y \in K$, i.e., $y \in x - K \subset B - K$. This shows that

$$f * g(x) = \int_{B-K} f(y) g(x - y)\, \mathrm{d}y.$$

Let us first assume that $k = 0$. Then $\|g\|_\infty < \infty$, and

$$|f(y)g(x-y)| \le \|g\|_\infty (\mathbf{1}_{B-K}|f|)(y)$$

for all $x \in B$. The right-hand side thus provides an x-independent integrable function which dominates the integrand $f(y)g(x-y)$, which in turn is continuous in x for every fixed y. By the theorem on continuous dependence of parameter-dependent integrals we thus see that $f * g$ is continuous at the point x_0. Thus, $f * g$ is continuous.

For $k \ge 1$, in a similar way one can prove the corresponding statement in (i) and the identity (2.6) by means of induction over k and the theorem on differentiability of parameter-dependent integrals.

(ii) Assume that $f*g(x) \neq 0$. Then there is at least one $y \in A$ such that $f(y)g(x-y) \neq 0$ (why?). For such a y, we then have $x-y \in \operatorname{supp} g$, and thus $x \in y+\operatorname{supp} g \subset A+\operatorname{supp} g$. Our claim in (ii) follows, since $A + \operatorname{supp} g$ is closed. □

2.3 Approximate Identities and Mollification of Functions in L^p

We now want to show how convolutions with suitable functions can be used in order to "smoothen" or "mollify" a given function and approximate it arbitrarily closely by C^∞-functions. This shall be examined in particular with respect to L^p-norms. To this end, we need some further tools from the theory of L^p-spaces.

We recall: By a **step function** we mean any finite linear combination of characteristic functions of cuboids, i.e., any function of the form $\sum a_j \mathbf{1}_{Q_j}$, with coefficients $a_j \in \mathbb{C}$ and cuboids $Q_j \subset \mathbb{R}^n$. A cuboid is, by definition, any direct product $Q = I_1 \times \cdots \times I_n$ of bounded intervals $I_1, \ldots, I_n$.

Lemma 2.6 (Continuity of translation in L^p) *Let $1 \le p \le \infty$, and let $f \in L^p(\mathbb{R}^n)$. Then*

$$\|\lambda_y f\|_p = \|f\|_p \qquad \text{for all} \quad y \in \mathbb{R}^n. \tag{2.7}$$

Moreover, if $1 \le p < \infty$, then

$$\lim_{y \to 0} \|\lambda_y f - f\|_p = 0. \tag{2.8}$$

Proof. The identity $\|\lambda_y f\|_p = \|f\|_p$ is an immediate consequence of the translation invariance of the Lebesgue measure.

In order to prove the continuity of translation, we make use of the fact that the space $\mathcal{T}$ of all step functions on $\mathbb{R}^n$ (compare Exercise 2.5) is dense in $L^p(\mathbb{R}^n)$, if $1 \le p < \infty$: For $p = 1$ this is well-known and follows from the usual definitions of $L^1(\mathbb{R}^n)$. For $1 < p < \infty$, we leave this an exercise to the interested reader (see Exercise 2.5).

But, if f is the indicator function of a cuboid, then (2.8) is obvious, and consequently (2.8) is also true for every step function f.

Let now $f \in L^p(\mathbb{R}^n)$ be arbitrary, and let $\varepsilon > 0$. We can then choose a step function $\varphi \in \mathcal{T}$ such that $\|f - \varphi\|_p < \varepsilon/4$. By what has been said, there is a $\delta > 0$ so that $\|\lambda_y \varphi - \varphi\|_p < \varepsilon/2$ for all y with $|y| < \delta$. For every such y we can then estimate

$$\begin{aligned}\|\lambda_y f - f\|_p &= \|\lambda_y(f-\varphi) + \lambda_y\varphi - \varphi + \varphi - f\|_p \\ &\le \|\lambda_y(f-\varphi)\|_p + \|\lambda_y\varphi - \varphi\|_p + \|f-\varphi\|_p \\ &= 2\|f-\varphi\|_p + \|\lambda_y\varphi - \varphi\|_p \\ &< \varepsilon.\end{aligned}$$

Thus, (2.8) also holds for f. □

Recall next that, for $1 \le p < \infty$, one can identify in a natural way the topological dual space of the Banach space $L^p(\mathbb{R}^n)$ with the space $L^{p'}(\mathbb{R}^n)$, where p' is the conjugate exponent to p. The next lemma is thus essentially a special case of more general duality principles from functional analysis. We prefer, however, to give a direct proof here.

Lemma 2.7 ("Dualization principle") *Let $1 \le p \le \infty$, let p' be the conjugate exponent to p, and let $f \in L^p(\mathbb{R}^n)$. Then*

$$\|f\|_p = \sup_{g \in L^{p'}(\mathbb{R}^n):\|g\|_{p'}=1} \left| \int f(x)g(x)\,\mathrm{d}x \right|. \tag{2.9}$$

Proof. Let $S^{p'} := \{g \in L^{p'}(\mathbb{R}^n) : \|g\|_{p'} = 1\}$. For $g \in S^{p'}$, by Hölder's inequality we may estimate

$$\left| \int f(x)g(x)\mathrm{d}x \right| \le \|f\|_p \|g\|_{p'} = \|f\|_p,$$

so that $\|f\|_p$ dominates the right-hand side of (2.9).

To prove the reverse inequality, we may clearly assume that $f \neq 0$, and then, without loss of generality, that $\|f\|_p = 1$. Regarding f as a concrete function (by choosing a representative from $\mathcal{L}^p(\mathbb{R}^n)$), let us then put, if $p < \infty$,

$$h(x) := \begin{cases} 0, & \text{if } f(x) = 0, \\ \frac{\overline{f(x)}}{|f(x)|}|f(x)|^{p-1}, & \text{if } f(x) \neq 0. \end{cases}$$

The function h is then clearly measurable, and since $p' = p/(p-1)$, we have

$$\int |h(x)|^{p'}\mathrm{d}x = \int |f(x)|^p \mathrm{d}x = \|f\|_p^p = 1,$$

so that $h \in S^{p'}$. Moreover, one computes that

$$\int f(x)h(x)\mathrm{d}x = \int |f(x)|^p \mathrm{d}x = 1 = \|f\|_p.$$

This shows that $\|f\|_p$ is bounded by the right-hand side of (2.9). Consequently, (2.9) holds when $1 \le p < \infty$.

Assume finally that $p = \infty$, and that, without loss of generality, $\|f\|_\infty > 0$. If $0 < \varepsilon < \|f\|_\infty$, then there is a measurable set $A \subset \mathbb{R}^n$ of strictly positive measure $|A| > 0$ such that $|f(x)| \ge \|f\|_\infty - \varepsilon$ for all $x \in A$.

We then put

$$h(x) := \begin{cases} \frac{1}{|A|} \, \frac{\overline{f(x)}}{|f(x)|} & , \text{ if } x \in A, \\ 0 & , \text{ if } x \in A^c. \end{cases}$$

Then $h \in L^1(\mathbb{R}^n)$, $\|h\|_1 = 1$, and

$$\begin{aligned} \int f(x)h(x)\,\mathrm{d}x &= \frac{1}{|A|} \int\limits_A |f(x)|\mathrm{d}x \\ &\geq \|f\|_\infty - \varepsilon. \end{aligned}$$

This shows that (2.9) also holds when $p = \infty$. □

Remarks 2.8 (a) Evidently, one can replace in (2.9) the condition $\|g\|_{p'} = 1$ by $\|g\|_{p'} \leq 1$. Sometimes this turns out to be useful.

(b) If $f : \mathbb{R} \to \mathbb{C} \cup \{\infty\}$ is measurable, and if

$$\sup_{\|g\|_{p'} \leq 1} \int |f(x)g(x)|\,\mathrm{d}x < \infty,$$

where here the supremum is taken over all measurable bounded functions g with bounded support and $\|g\|_{p'} \leq 1$, then one concludes in a similar way that f will lie in $L^p(\mathbb{R}^n)$.

Indeed, just replace the function h from the preceding proof by $h_N := \frac{\widetilde{h_N}}{\|\widetilde{h_N}\|_{p'}}$, with

$$\widetilde{h_N}(x) := \begin{cases} h(x), & \text{if } |x| \leq N \text{ and } |h(x)| \leq N, \\ 0, & \text{otherwise,} \end{cases}$$

and let N tend to infinity. The claim then follows by the monotone convergence theorem.

Theorem 2.9 (Minkowski's integral inequality) *Let $X := \mathbb{R}^m, Y := \mathbb{R}^n$, let $F : X \times Y \to \mathbb{C}$ be measurable, and let $1 \leq p \leq \infty$. Then the mapping $Y \ni y \mapsto \|F(\cdot, y)\|_p$ is measurable too. And, if we assume that $\int\limits_Y \|F(\cdot, y)\|_p \,\mathrm{d}y < \infty$, then the function $y \mapsto F(x, y)$ is integrable over Y for a.e. $x \in X$. Furthermore, the function $X \ni x \mapsto \int\limits_Y F(x, y)\,\mathrm{d}y$ is measurable, and we have that*

$$\left\| \int\limits_Y F(\cdot, y)\,\mathrm{d}y \right\|_p \leq \int\limits_Y \|F(\cdot, y)\|_p \,\mathrm{d}y. \tag{2.10}$$

Proof. If one imagines an approximation of the integral $\int F(\cdot, y)\,\mathrm{d}y$ by Riemann sums, then the inequality (2.10) becomes very plausible, in view of the triangle inequality.

As a matter of fact, it is possible to regard F as a function f on Y taking values in the Banach space $L^p(X)$, by putting $f(y)(x) := F(x,y)$. Then the integral $\int_Y F(\cdot,y)\,\mathrm{d}y$ can be regarded as a vector-valued "Bochner integral" $\int_Y f(y)\mathrm{d}y$, with values in the Banach space $L^p(X)$, and (2.10) becomes nothing but the (well-known) triangle inequality for Banach space-valued integrals (here for $f \in L^1(Y, L^p(X))$).

However, let us avoid using such results from more advanced integration theory here and give an elementary proof, based on the dualization principle.

Once again, we shall only consider the case $1 \le p < \infty$ and leave the case $p = \infty$ as an exercise to the interested reader.

Observe first that, by Tonelli's theorem, the mapping $y \mapsto \int |F(x,y)|^p\,\mathrm{d}x$, and hence also $y \mapsto \|F(\cdot,y)\|_p$, is measurable, since $(x,y) \mapsto |F(x,y)|^p$ is measurable.

Suppose that $\int \|F(\cdot,y)\|_p\,\mathrm{d}y < \infty$, and choose any $g \in S^{p'} := \{\varphi \in L^{p'}(X) : \|\varphi\|_{p'} = 1\}$. By Hölder's inequality, we then have

$$\begin{aligned}\int_Y \int_X |F(x,y)||g(x)|\,\mathrm{d}x\mathrm{d}y &= \int_Y \Big(\int_X |g(x)||F(x,y)|\mathrm{d}x\Big)\mathrm{d}y \\ &\le \int_Y \|g\|_{p'}\|F(\cdot,y)\|_p\,\mathrm{d}y = \int_Y \|F(\cdot,y)\|_p\,\mathrm{d}y < \infty.\end{aligned}$$

This shows that the function $(x,y) \mapsto g(x)F(x,y)$ is integrable for every $g \in S^{p'}$. In particular we may conclude that the function $y \mapsto F(x,y)$ is integrable for a.e. x, and by means of Fubini's theorem we can then conclude in a similar way as above that

$$\begin{aligned}&\left|\int_X \Big(\int_Y F(x,y)\,\mathrm{d}y\Big)g(x)\,\mathrm{d}x\right| = \left|\int_Y \Big(\int_X g(x)F(x,y)\,\mathrm{d}x\Big)\,\mathrm{d}y\right| \\ &\le \int_Y \int_X |g(x)||F(x,y)|\mathrm{d}x\mathrm{d}y \le \int_Y \|F(\cdot,y)\|_p\,\mathrm{d}y.\end{aligned}$$

Thus (2.10) follows from (2.9) in combination with Remark 2.8 (b). □

Corollary 2.10 (Young's inequality) *Let $f \in L^p(\mathbb{R}^n), g \in L^1(\mathbb{R}^n)$, and $1 \le p \le \infty$. Then the convolution product $g * f$ is defined almost everywhere and lies in $L^p(\mathbb{R}^n)$. Moreover, we have*

$$\|g * f\|_p \le \|g\|_1 \|f\|_p. \tag{2.11}$$

Proof. Apply Minkowski's integral inequality to the function $F(x,y) := g(y)f(x-y)$ and make use of the translation invariance (2.7). □

Remark 2.11 More generally, Lemma 2.7 and Theorem 2.9 analogously hold true for arbitrary σ-finite measure spaces $(X, \mathrm{d}\mu)$ and $(Y, d\nu)$ in place of $(\mathbb{R}^m, \mathrm{d}x)$ and $(\mathbb{R}^n, \mathrm{d}y)$. The given proofs can be copied almost verbatim.

Definition. By an **approximate (convolution-) identity** we shall understand any family $\{\varphi_\varepsilon\}_{0<\varepsilon<\varepsilon_0}$ of integrable functions on $\mathbb{R}^n$ with the following properties: there is a constant $C > 0$ such that

(i) $\int \varphi_\varepsilon(x)\,\mathrm{d}x = 1$ for every $\varepsilon \in]0, \varepsilon_0[$;
(ii) $\int |\varphi_\varepsilon(x)|\,\mathrm{d}x \le C$ for every $\varepsilon \in]0, \varepsilon_0[$;
(iii) for every $\delta > 0$ we have $\lim_{\varepsilon\to 0} \int_{|x|>\delta} |\varphi_\varepsilon(x)|\,\mathrm{d}x = 0$.

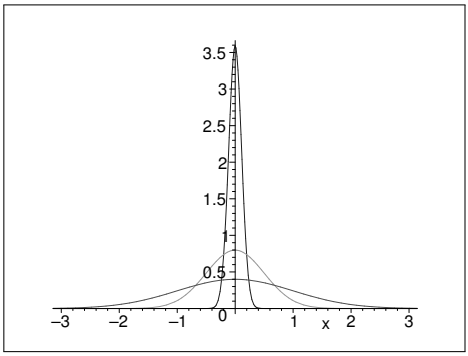


Fig. 2.1 Dirac-sequence

Example 2.12 Let $\varphi \in L^1(\mathbb{R}^n)$ such that $\int \varphi(x)\,\mathrm{d}x = 1$, and put

$$\varphi_\varepsilon(x) := \varepsilon^{-n}\varphi\left(\frac{x}{\varepsilon}\right),\ \varepsilon > 0.$$

Then the family $\{\varphi_\varepsilon\}_{\varepsilon>0}$ is an approximate identity.

Indeed, (i) follows immediately by means of the change of variables $x \mapsto \varepsilon x$. Similarly, (ii) holds true, since

$$\int |\varphi_\varepsilon(x)|\,\mathrm{d}x = \varepsilon^{-n}\int \left|\varphi\left(\frac{x}{\varepsilon}\right)\right|\mathrm{d}x = \int |\varphi(x)|\,\mathrm{d}x$$

for every $\varepsilon > 0$. Finally, given $\delta > 0$, we have

$$\int_{|x|>\delta} |\varphi_\varepsilon(x)|\,\mathrm{d}x = \int_{|y|>\delta/\varepsilon} |\varphi(y)|\,\mathrm{d}y,$$

so that (iii) is a consequence of Lebesgue's theorem on dominated convergence. □

This example demonstrates how easy it is to construct approximate identities on $\mathbb{R}^n$ by means of the dilations $x \mapsto \varepsilon^{-1}x$.

If we choose $\varphi \in C_0^\infty(\mathbb{R}^n)$ so that $\int \varphi(x)\,\mathrm{d}x = 1$, then the approximate identity constructed in Example 2.12 is also called a **Dirac family**. Moreover, if $\varepsilon_j \searrow 0$ is any decreasing **null-sequence** (i.e., $\lim_{j\to\infty} \varepsilon_j = 0$), then the sequence $\{\varphi_j\}_j := \{\varphi_{\varepsilon_j}\}_j$ will be called a **Dirac sequence**).

Theorem 2.13 *Let $\{\varphi_\varepsilon\}_{0<\varepsilon<\varepsilon_0}$ be an approximate identity.*

(a) *If $f \in L^p(\mathbb{R}^n)$, $1 \le p < \infty$, then*

$$\lim_{\varepsilon\to 0} \|f * \varphi_\varepsilon - f\|_p = 0.$$

(b) *Let $f \in \mathcal{L}^\infty(\mathbb{R}^n)$ be uniformly continuous at the points of a subset $V \subset \mathbb{R}^n$, in the following sense: for every $\kappa > 0$ there is a $\delta > 0$ such that $|f(x) - f(z)| \le \kappa$ for all $x \in V$ and all $z \in \mathbb{R}^n$ such that $|x - z| \le \delta$.*
Then $f * \varphi_\varepsilon$ converges uniformly on V towards f as $\varepsilon \to 0$.

Proof. (a) Let $f \in \mathcal{L}^p(\mathbb{R}^n)$, and let $\kappa > 0$. By (2.8) we can choose a $\delta > 0$ so that

$$\|\lambda_y f - f\|_p < \kappa \quad \text{for all } y \in \mathbb{R}^n \text{ with } |y| < \delta. \tag{2.12}$$

Observe next that by property (i) of an approximate identity

$$(f * \varphi_\varepsilon - f)(x) = \int \varphi_\varepsilon(y)[f(x-y) - f(x)]\,\mathrm{d}y.$$

Minkowski's integral inequality then implies that

$$\begin{aligned}\|f * \varphi_\varepsilon - f\|_p &\le \int |\varphi_\varepsilon(y)|\,\|\lambda_y f - f\|_p\,\mathrm{d}y \\ &= I + II,\end{aligned}$$

where

$$I := \int_{|y|<\delta} |\varphi_\varepsilon(y)|\,\|\lambda_y f - f\|_p\,\mathrm{d}y,$$

$$II := \int_{|y|>\delta} |\varphi_\varepsilon(y)|\,\|\lambda_y f - f\|_p\,\mathrm{d}y.$$

Using (2.12) and property (ii) of an approximate identity, we find that for every $\varepsilon \in]0, \varepsilon_0[$

$$I \le \int_{|y|<\delta} |\varphi_\varepsilon(y)|\kappa\,\mathrm{d}y \le \kappa \int_{\mathbb{R}^n} |\varphi_\varepsilon(y)|\mathrm{d}y \le C\kappa.$$

Moreover we may estimate

$$II \le \int_{|y|>\delta} |\varphi_\varepsilon(y)|\,2\|f\|_p\,\mathrm{d}y.$$

Thus, by property (iii) of an approximate identity, there is an $\varepsilon_1 > 0$ such that $II < \kappa$ for all ε with $0 < \varepsilon < \varepsilon_1$. For such ε we thus finally conclude that $\|f * \varphi_\varepsilon - f\|_p \le (C+1)\kappa$. This proves (a).

(b) Let $f \in \mathcal{L}^\infty(\mathbb{R}^n)$ be uniformly continuous at the points of V, and let $\kappa > 0$. By definition, there is then a $\delta > 0$ such that the following analogue of (2.12) holds true:

$$\|\lambda_y f - f\|_{\infty,V} < \kappa \quad \text{for all } y \in \mathbb{R}^n \text{ with } |y| < \delta.$$

Here, we have set $\|g\|_{\infty,V} := \sup_{x\in V} |g(x)|$.

From here on, the arguments from part (a) can directly be transferred to (b), by replacing the L^p-norm $\|\cdot\|_p$ by $\|\cdot\|_{\infty,V}$. □

Corollary 2.14 *For every $1 \le p < \infty$, the space $C_0^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$.*

Proof. Let $f \in L^p(\mathbb{R}^n)$, and let $f_N := \mathbf{1}_{B_N(0)} f$. Lebesgue's theorem on dominated convergence implies that $f = \lim_{N\to\infty} f_N$ in $L^p(\mathbb{R}^n)$.

Thus, given any $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that $\|f - f_N\|_p < \varepsilon/2$.

Choose next a Dirac-sequence $\{\varphi_j\}_j$. By Theorem 2.13 we can find a $j \in \mathbb{N}$ such that $\|f_N - f_N * \varphi_j\|_p < \varepsilon/2$, hence $\|f - f_N * \varphi_j\|_p < \varepsilon$. But, Proposition 2.5 shows that $f_N * \varphi_j$ is smooth and has compact support, so that $f_N * \varphi_j \in C_0^\infty(\mathbb{R}^n)$. □

2.4 Smooth Partitions of Unity

We will prove here a result that will later be needed, but which is relevant in many other contexts as well.

We shall use the following standard notation: if A, B are two subsets of $\mathbb{R}^n$, then the distance between A and B is defined by

$$\operatorname{dist}(A, B) := \inf\{|a - b| : a \in A, b \in B\}.$$

For any $x \in \mathbb{R}^n$, the distance of x to B is then given by $\operatorname{dist}(x, B) := \operatorname{dist}(\{x\}, B)$.

Lemma 2.15 (Smooth version of Urysohn's lemma) *Let K be a compact subset of the open subset Ω of $\mathbb{R}^n$. Then there exists a function $\psi \in C_0^\infty(\Omega)$ such that $0 \le \psi \le 1$ and $\psi \equiv 1$ on a neighborhood of K.*

Proof. Choose $0 < \varepsilon < \delta < \varepsilon + \delta < \operatorname{dist}(K, \Omega^c)$, and set $f := \mathbf{1}_{K_\delta}$, where K_δ denotes the **δ-thickening**

$$K_\delta := \{x \in \mathbb{R}^n : \operatorname{dist}(x, K) \le \delta\} = K + \overline{B}_\delta(0)$$

of the set K.

Moreover, let $\{\varphi_{\varepsilon'}\}_{\varepsilon'>0}$ be a Dirac-family of non-negative functions with $\operatorname{supp} \varphi_{\varepsilon'} \subset \overline{B}_{\varepsilon'}(0)$, and put $\tilde{\psi} := f * \varphi_\varepsilon$. Then, by Proposition 2.5, $\tilde{\psi} \in C_0^\infty(\mathbb{R}^n)$, and

$$\operatorname{supp} \tilde{\psi} \subset K_\delta + \overline{B}_\varepsilon(0) = K_{\delta+\varepsilon} \subset \Omega.$$

Moreover, if $x \in K_{\delta-\varepsilon}$ and $y \in \overline{B}_\varepsilon(0)$, then $x - y \in K_\delta$, hence $f(x-y) = 1$, and thus

$$\tilde{\psi}(x) = \int f(x-y)\varphi_\varepsilon(y)\,\mathrm{d}y = \int \varphi_\varepsilon(y)\mathrm{d}y = 1.$$

This shows that $\tilde{\psi} \equiv 1$ on the neighborhood $K_{\delta-\varepsilon}$ of K. The function $\psi := \tilde{\psi}|_\Omega$ thus has all the required properties. □

Suppose that $\Omega \subset \mathbb{R}^n$, and that $f : \Omega \to \mathbb{C} \cup \{\infty\}$ is a given function. Then in the sequel we shall denote by $\tilde{f} : \mathbb{R}^n \to \mathbb{C} \cup \{\infty\}$ the **trivial extension**

$$\tilde{f}(x) := \begin{cases} f(x), & \text{if } x \in \Omega, \\ 0, & \text{if } x \in \mathbb{R}^n \setminus \Omega, \end{cases}$$

of f to the whole of $\mathbb{R}^n$.

Remark 2.16 If $\Omega \subset \mathbb{R}^n$ is open, and if $\varphi \in C_0^k(\Omega)$, then the trivial extension $\tilde{\varphi}$ of φ obviously lies in $C_0^k(\mathbb{R}^n)$.

Conversely, if $\psi \in C_0^k(\mathbb{R}^n)$, and if $\operatorname{supp}\psi \subset \Omega$, then the restriction $\psi|_\Omega$ of ψ to Ω lies in $C_0^k(\Omega)$, and we have $\widetilde{\psi|_\Omega} = \psi$.

In view of this remark, we shall in the sequel usually identify a function $\varphi \in C_0^k(\Omega)$ with its trivial extension $\tilde{\varphi}$ whenever it is clear from the context whether we mean φ, or $\tilde{\varphi}$, i.e., we shall identify the space $C_0^k(\Omega)$ with the subspace $\{\psi \in C_0^k(\mathbb{R}^n) : \operatorname{supp}\psi \subset \Omega\}$ of $C_0^k(\mathbb{R}^n)$. For instance, in the next lemma we shall apply this to the functions φ_j.

If A is a subset of a topological space, then A^0 will as usual denote the open interior of A.

Lemma 2.17 (Existence of smooth partitions of unity) *Let $\Omega_1, \ldots, \Omega_k$ be open subsets of $\mathbb{R}^n$, and let K be a compact subset such that $K \subset \bigcup_{j=1}^k \Omega_j$.*

(a) *For every $j = 1, \ldots, k$, there exists a compact subset $K_j \subset \Omega_j$ such that $K \subset \bigcup_{j=1}^k K_j$. We may even assume that $K_j = \overline{K_j^0}$.*

(b) *There are functions $\varphi_j \in C_0^\infty(\Omega_j)$, $j = 1, \ldots, k$, such that $\varphi_j \geq 0$, $\sum_{j=1}^k \varphi_j \leq 1$ and $\sum_{j=1}^k \varphi_j \equiv 1$ on a neighborhood of K.*

Proof. (a) For $\varepsilon > 0$ let us denote by Ω_j^ε the **ε-narrowing**

$$\Omega_j^\varepsilon := \{x \in \Omega_j : \operatorname{dist}(x, \Omega_j^c) > \varepsilon\}$$

of Ω. Then Ω_j^ε is open, and $\overline{\Omega_j^\varepsilon} \subset \Omega_j$.

We shall show that for sufficiently small $\varepsilon > 0$ the inclusion

$$K \subset \bigcup_{j=1}^{k} \Omega_j^{\varepsilon}$$

holds true. If we then choose $\varepsilon > 0$ in addition so small that $K_\varepsilon := K + \overline{B}_\varepsilon(0) \subset \bigcup_{j=1}^{k} \Omega_j$, then we obtain (a), by setting $K_j := \overline{(K_\varepsilon)^0 \cap \Omega_j^{\varepsilon}}$.

Assume that the above inclusion fails to hold for every $\varepsilon > 0$. Then for every $\varepsilon > 0$ there exists some point $x_\varepsilon \in K \setminus \bigcup_{j=1}^{k} \Omega_j^{\varepsilon}$. Since K is compact, we can extract a sequence $\varepsilon_l \to 0$ and some point $x \in K$ such that $x_{\varepsilon_l} \to x$ as $l \to \infty$. But then one checks that $x \in K \setminus \bigcup_{j=1}^{k} \Omega_j$, which contradicts our assumption on K.

Indeed, if we had $x \in \Omega_j$ for some j, then there would be some $\varepsilon' > 0$ so that $x \in \Omega_j^{\varepsilon'}$. As $x_{\varepsilon_l} \to x$, this would imply that $x_{\varepsilon_l} \in \Omega_j^{\varepsilon'} \subset \Omega_j^{\varepsilon_l}$ for l sufficiently large, contradicting our selection of the points x_ε.

(b) Suppose we have chosen compact sets $K_j \subset \Omega_j$ as in (a). By Lemma 2.15, we may then choose functions $\psi_j \in C_0^\infty(\Omega_j)$ so that $0 \le \psi_j \le 1$ and $\psi_j \equiv 1$ on a neighborhood U_j of K_j.

Then clearly $(1-\psi_1)\cdots(1-\psi_k) \equiv 0$ on the neighborhood $U := \bigcup_{j=1}^{k} U_j$ of K, i.e.,

$$1 - (1-\psi_1)\cdots(1-\psi_k) \equiv 1 \qquad \text{on } U.$$

If we set $\varphi_1 := 1 - (1-\psi_1) = \psi_1$, and for $j = 2, \ldots, k$

$$\varphi_j := (1-\psi_1)\cdots(1-\psi_{j-1}) - (1-\psi_1)\cdots(1-\psi_j) = \psi_j(1-\psi_1)\cdots(1-\psi_{j-1}),$$

then by summing a telescopic series we immediately see that

$$\sum_{j=1}^{k} \varphi_j = 1 - (1-\psi_1)\cdots(1-\psi_k).$$

This implies in particular that $\sum_{j=1}^{k} \varphi_j(x) = 1$ for every $x \in U$. Moreover, evidently we have that φ_j in $C_0^\infty(\Omega_j)$, that $\varphi_j \ge 0$ and that $\sum_{j=1}^{k} \varphi_j \le 1$. □

2.5 Rapidly Decaying Functions, and the Riemann–Lebesgue Lemma

Proposition 2.18 *Let* $1 \le j \le n$.

(a) *Suppose that* $f \in L^1(\mathbb{R}^n)$ *is continuously differentiable, that* $D_j f \in L^1(\mathbb{R}^n)$, *and that* f *vanishes at infinity. Then*

$$\widehat{D_j f}(\xi) = \xi_j \hat{f}(\xi).$$

(b) *Suppose that* $f \in L^1(\mathbb{R}^n)$ *is such that the function* $x_j f : x \mapsto x_j f(x)$ *lies in* $L^1(\mathbb{R}^n)$ *too. Then the Fourier transform* $\hat{f}$ *of* f *is partially differentiable with respect to* ξ_j, *and*

$$D_j \hat{f}(\xi) = -(x_j f)^\wedge(\xi).$$

Proof. (a) Since f vanishes at infinity, an integration by parts with respect to x_j, in combination with Fubini's theorem, easily shows that

$$\begin{aligned}\widehat{D_j f}(\xi) &= \frac{1}{i}\int \partial_{x_j} f(x) \mathrm{e}^{-i\xi\cdot x}\mathrm{d}x = -\frac{1}{i}\int f(x)\partial_{x_j}(\mathrm{e}^{-i\xi\cdot x})\,\mathrm{d}x \\ &= -\frac{1}{i}(-i\xi_j)\int f(x)\mathrm{e}^{-i\xi\cdot x}\mathrm{d}x = \xi_j \hat{f}(\xi).\end{aligned}$$

(b) By viewing

$$\hat{f}(\xi) = \int f(x)\mathrm{e}^{-i\xi\cdot x}\mathrm{d}x$$

as a Lebesgue integral depending on the parameter ξ_j, (b) follows in a similar way by means of Fubini's theorem, and the theorem on differentiation of parameter-dependent Lebesgue-integrals. □

This proposition motivates the introduction of the following function space.

Definition. A function $\varphi \in C^\infty(\mathbb{R}^n)$ will be called a **Schwartz function**, if $x^\alpha D^\beta \varphi$ is bounded for all multi-indices $\alpha, \beta \in \mathbb{N}^n$.

Clearly the set of all such Schwartz functions will form a subspace of the $\mathbb{C}$-vector space $C^\infty(\mathbb{R}^n)$, called the **Schwartz space** $\mathcal{S}(\mathbb{R}^n)$. Often we shall also denote it simply by $\mathcal{S}$.

Remark 2.19 By means of Leibniz' formula (cf. Exercise 2.12), one easily shows that a function $\varphi \in C^\infty(\mathbb{R}^n)$ is a Schwartz function if and only if

$$D^\alpha(x^\beta \varphi) \text{ is bounded for all } \alpha, \beta \in \mathbb{N}^n.$$

The proof will be left as an exercise (compare also Exercise 2.13).

Moreover, if $\varphi \in \mathcal{S}(\mathbb{R}^n)$ is a Schwartz function, then clearly also

$$(1 + |\cdot|^2)^\ell D^\alpha \varphi \in \mathcal{S} \quad \text{for every } \ell \in \mathbb{N},\ \alpha \in \mathbb{N}^n,$$

and thus in particular

$$|D^\alpha \varphi(x)| \le C_\ell (1+|x|^2)^{-\ell} \quad \text{for every } \ell \in \mathbb{N}. \tag{2.13}$$

This shows that $\mathcal{S} = \mathcal{S}(\mathbb{R}^n)$ consists of all C^∞-functions which, jointly with all their derivatives, vanish of arbitrary negative order $O(|x|^{-N})$ at infinity.

For this reason, Schwartz functions are also called **rapidly decaying** functions.

Clearly, we have

$$C_0^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n). \tag{2.14}$$

An important example of a Schwartz function which has no compact support is the Gaussian $\Phi_n(x) := \mathrm{e}^{-|x|^2/2}$ (compare Lemma 2.24).

Note also that for $\ell > n/2$ the right-hand side of (2.13) lies in every Lebesgue space $L^p(\mathbb{R}^n)$. This implies that for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$

$$D^\alpha \varphi \in L^p(\mathbb{R}^n) \quad \text{for all } \alpha \in \mathbb{N}^n,\ p \in [1,\infty].$$

In particular, we see that

$$\mathcal{S}(\mathbb{R}^n) \subset L^p(\mathbb{R}^n).$$

Since, by Corollary 2.14, $C_0^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$ for $1 \le p < \infty$, (2.14) shows in particular that the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$ for this range of Lebesgue exponents as well.

By means of induction on the orders $|\alpha|$ and $|\beta|$ of α and β in combination with Leibniz' rule one easily derives the following from Proposition 2.18.

Corollary 2.20 *The Fourier transform leaves the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ invariant, i.e., $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$. Moreover, for every $\alpha, \beta \in \mathbb{N}^n$ and $\varphi \in \mathcal{S}(\mathbb{R}^n)$,*

$$\xi^\alpha D_\xi^\beta (\mathcal{F}\varphi)(\xi) = (-1)^{|\beta|} \mathcal{F}(D_x^\alpha (x^\beta \varphi))(\xi)\,. \tag{2.15}$$

Remark 2.21 A consequence of this corollary which is of fundamental importance to the theory of partial differential equations is the following: if

$$P(D) = \sum_{|\alpha| \le m} a_\alpha D^\alpha$$

is any linear differential operator with constant coefficients $a_\alpha \in \mathbb{C}$, then one calls the associated polynomial

$$P(\xi) := \sum_{|\alpha| \le m} a_\alpha \xi^\alpha, \qquad \xi \in \mathbb{R}^n,$$

the **(complete) symbol** of $P(D)$. By (2.15), we then have that

$$(P(D)\varphi)^\wedge(\xi) = P(\xi)\hat{\varphi}(\xi), \qquad \xi \in \mathbb{R}^n, \tag{2.16}$$

for all $\varphi \in \mathcal{S}(\mathbb{R}^n)$.

Note also that the Fourier transform converts the differential operator $P(D)$ into the **multiplication operator**

$$M_P : \varphi \mapsto P\varphi, \qquad \varphi \in \mathcal{S},$$

where P denotes the symbol of $P(D)$.

For example, the symbol of the **Laplace operator** (or the **Laplacian**, for short)

$$\Delta := \sum_{j=1}^{n} \frac{\partial^2}{\partial x_j^2}$$

on $\mathbb{R}^n$ is given by $P(\xi) = -|\xi|^2$.

As an easy consequence of Corollary 2.20 we obtain

Theorem 2.22 (Riemann–Lebesgue lemma) *If $f \in L^1(\mathbb{R}^n)$, then its Fourier transform $\hat{f}$ lies in the space $C_\infty(\mathbb{R}^n)$ of all continuous functions on $\mathbb{R}^n$ which vanish at infinity.*

Proof. We have seen that the space $\mathcal{S}$ is dense in $L^1(\mathbb{R}^n)$. Thus, if $f \in L^1(\mathbb{R}^n)$, then there is a sequence $\{\varphi_j\}_j$ in $\mathcal{S}$ such that $\lim_{j\to\infty} \|f - \varphi_j\|_1 = 0$. By Proposition 2.1 (b), we see in particular that $\lim_{j\to\infty} \|\hat{f} - \widehat{\varphi_j}\|_\infty = 0$, i.e., $\{\widehat{\varphi_j}\}_j$ converges uniformly towards $\hat{f}$. Since $\widehat{\varphi_j} \in \mathcal{S} \subset C_\infty(\mathbb{R}^n)$, we conclude that also $\hat{f} \in C_\infty(\mathbb{R}^n)$ (recall that $C_\infty(\mathbb{R}^n)$, endowed with the sup-norm $\|\cdot\|_{\ell^\infty}$, is a Banach space, and that the sup-norm agrees with $\|\cdot\|_\infty$ on $C_\infty(\mathbb{R}^n)$). □

Notice that the proof shows at the same time that the Fourier transform of an L^1-function is continuous, a fact that we had already seen before by means of a different argument.

Propositions 2.1 and 2.3 in combination with Theorem 2.22 immediately give the following:

Corollary 2.23 *The Fourier transform $\mathcal{F}$ is a continuous linear homomorphism from the involutive Banach algebra $(L^1(\mathbb{R}^n), +, *, {}^*; \|\cdot\|_1)$ into the involutive Banach algebra $(C_\infty(\mathbb{R}^n), +, \cdot, \overline{\ }; \|\cdot\|_\infty)$.*

By means of the Fourier inversion formula, which we shall derive next, we will show that the Fourier transform indeed maps $\mathcal{S}$ onto $\mathcal{S}$. This implies in particular that $\mathcal{S} \subset \mathcal{F}(L^1(\mathbb{R}^n))$. Since, as we have just seen, $\mathcal{S}$ is dense in C_∞, we also conclude that the so-called **Fourier algebra**

$$A(\mathbb{R}^n) := \mathcal{F}(L^1(\mathbb{R}^n)) \subset C_\infty(\mathbb{R}^n)$$

of the abelian group $\mathbb{R}^n$ is dense in $C_\infty(\mathbb{R}^n)$.

2.6 Inversion of the Fourier Transform

Lemma 2.24 (Fourier transform of the Gaussian) *Let Φ_n on $\mathbb{R}^n$ denote the* **Gaussian**

$$\Phi_n(x) := \mathrm{e}^{-|x|^2/2},\ x \in \mathbb{R}^n.$$

Then Φ_n in $\mathcal{S}(\mathbb{R}^n)$, and

$$\widehat{\Phi_n} = (2\pi)^{n/2}\Phi_n. \tag{2.17}$$

In particular, we have

$$1 = \Phi_n(0) = (2\pi)^{-n}\int_{\mathbb{R}^n} \widehat{\Phi_n}(\xi)\,\mathrm{d}\xi. \tag{2.18}$$

Proof. By induction on the order of differentiation, one easily shows that for any $\alpha \in \mathbb{N}^n$ there is a polynomial p_α such that $D^\alpha \mathrm{e}^{-|\cdot|^2/2} = p_\alpha \mathrm{e}^{-|\cdot|^2/2}$. Moreover, $|\cdot|^{2N}\mathrm{e}^{-|\cdot|^2/2}$ is bounded for every $N \in \mathbb{N}$, and consequently we see that $\Phi_n \in \mathcal{S}(\mathbb{R}^n)$.

To prove (2.17), assume first that $n = 1$. Note that the function $y = \Phi_1$ satisfies the ordinary differential equation (ODE)

$$y' + xy = 0. \tag{2.19}$$

Moreover, by (2.15) we have

$$(\widehat{\Phi_1})' = iD\widehat{\Phi_1} = -i\widehat{(x\Phi_1)},$$

and since $xy = -y'$ by (2.19), we find that

$$(\widehat{\Phi_1})' = -i\widehat{(-\Phi_1')} = \widehat{(-D\Phi_1)} = -\xi\widehat{\Phi_1},$$

where in the last identity we have again made use of (2.15). This shows that $\widehat{\Phi_1}$ solves the ODE (2.19) as well, and thus we conclude that $(\widehat{\Phi_1}/\Phi_1)' \equiv 0$, i.e., $\widehat{\Phi_1}/\Phi_1$ is constant (this would also have followed by the uniqueness theorem for solutions of ODEs).

But, $\Phi_1(0) = 1$, and by changing to polar coordinates, we see that

$$\begin{aligned}\widehat{\Phi_1}(0) &= \int_{\mathbb{R}} \mathrm{e}^{-x^2/2}\mathrm{d}x = \left(\iint \mathrm{e}^{-\frac{x^2+y^2}{2}}\mathrm{d}x\mathrm{d}y\right)^{1/2} \\ &= \left(\int_0^{2\pi}\int_0^{\infty} \mathrm{e}^{-r^2/2} r\,\mathrm{d}r\mathrm{d}\theta\right)^{1/2} = \sqrt{2\pi}.\end{aligned} \tag{2.20}$$

This shows that $\widehat{\Phi_1}/\Phi_1 = \frac{\sqrt{2\pi}}{1} = (2\pi)^{1/2}$, and thus

$$\widehat{\Phi_1} = (2\pi)^{1/2}\Phi_1.$$

By Fubini's theorem, this implies (2.17) for any $n \geq 1$, and then also (2.18), since

$$\Phi_n(0) = (2\pi)^{-n/2}\widehat{\Phi_n}(0) = (2\pi)^{-n/2}\int \Phi_n(x)\,\mathrm{d}x = (2\pi)^{-n}\int \widehat{\Phi_n}(x)\,\mathrm{d}x. \qquad \square$$

An alternative proof can be led by means of Cauchy's integral formula (see Exercise 2.3 (b)).

Proposition 2.25 *Let $f, g \in L^1(\mathbb{R}^n)$. Then*

$$\int f\hat{g}\,\mathrm{d}x = \int \hat{f}g\,\mathrm{d}x. \tag{2.21}$$

Proof. By Fubini's theorem, we have

$$\begin{aligned}\int f(x)\hat{g}(x)\,\mathrm{d}x &= \int \left(\int f(x)g(y)\mathrm{e}^{-ix\cdot y}\mathrm{d}y\right)\mathrm{d}x \\ &= \int \left(\int f(x)g(y)\mathrm{e}^{-ix\cdot y}\mathrm{d}x\right)\mathrm{d}y = \int \hat{f}(y)g(y)\mathrm{d}y. \qquad \square\end{aligned}$$

Theorem 2.26 (Fourier inversion formula) *Suppose $f \in \mathcal{L}^1(\mathbb{R}^n)$ is such that $\hat{f} \in \mathcal{L}^1(\mathbb{R}^n)$. Then*

$$f(x) = (2\pi)^{-n}\int_{\mathbb{R}^n} \hat{f}(\xi)\mathrm{e}^{ix\cdot\xi}\,\mathrm{d}\xi \tag{2.22}$$

for almost every $x \in \mathbb{R}^n$.

Proof. Let Φ_n be the Gaussian from Lemma 2.24, for which we already know an inversion formula, namely (2.17), and put

$$\varphi := (2\pi)^{-n}\widehat{\Phi_n} = (2\pi)^{-n/2}\Phi_n.$$

By (2.18), we have $\int \varphi\,\mathrm{d}x = 1$, and we may thus look at the approximate identity given by $\varphi_\varepsilon(x) := \varepsilon^{-n}\varphi\left(\frac{x}{\varepsilon}\right)$, $\varepsilon > 0$ (compare Example 2.12).

The key idea of the proof will consist in approximating the function f by "superpositions" of translates of scaled Gaussians, for which the Fourier inverse can be derived essentially from Lemma 2.26.

To this end, note that by Theorem 2.13 we have

$$\lim_{\varepsilon\to 0}\|f - f * \varphi_\varepsilon\|_1 = 0, \tag{2.23}$$

where $f * \varphi_\varepsilon = \int f(y)\lambda_y\varphi_\varepsilon\,\mathrm{d}y$.

Moreover, Proposition 2.1 shows that

$$\varphi_\varepsilon = (2\pi)^{-n}\widehat{\Phi_n(\varepsilon\cdot)}.$$

Since $\Phi_n(x) = \Phi_n(-x)$, we may thus invoke Proposition 2.25 to obtain

$$\begin{aligned} f * \varphi_\varepsilon(x) &= \int f(x-y)\,(2\pi)^{-n}\widehat{\Phi_n(\varepsilon\cdot)}(y)\,\mathrm{d}y \\ &= (2\pi)^{-n}\int f(x+y)\widehat{\Phi_n(\varepsilon\cdot)}(y)\,\mathrm{d}y \\ &= (2\pi)^{-n}\int (\lambda_{-x}f)^\wedge(\xi)\,\Phi_n(\varepsilon\xi)\,\mathrm{d}\xi \\ &= (2\pi)^{-n}\int \hat{f}(\xi)\mathrm{e}^{ix\cdot\xi}\,\Phi_n(\varepsilon\xi)\,\mathrm{d}\xi. \end{aligned}$$

Here, by assumption, $\hat{f}$ is integrable. Moreover, Φ_n is bounded and $\Phi_n(\varepsilon\xi)$ tends pointwise to $\Phi_n(0) = 1$ as $\varepsilon \to 0$. Thus, by Lebesgue's dominated convergence theorem,

$$\lim_{\varepsilon\to 0} f * \varphi_\varepsilon(x) = (2\pi)^{-n}\int \hat{f}(\xi)\mathrm{e}^{ix\cdot\xi}\,\mathrm{d}\xi$$

for every $x \in \mathbb{R}^n$.

On the other hand, (2.23) implies that there is a decreasing sequence $\varepsilon_j \searrow 0$ such that $f(x) = \lim_{j\to\infty} f * \varphi_{\varepsilon_j}(x)$ for almost every $x \in \mathbb{R}^n$. In combination, this yields (2.22). □

For instance, in dimension $n = 1$ the Fourier-inversion formula allows for the following *physical interpretation*: Assume that the function $f(t)$ describes the strength and sign of a time-dependent signal at time t, and assume that f satisfies the conditions in Theorem 2.26. Then

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\tau)\mathrm{e}^{i\tau t}\mathrm{d}t,$$

i.e., the signal f can be decomposed into "natural oscillations"

$$\mathrm{e}^{i\tau t} = \cos(\tau t) + i\sin(\tau t)$$

of "frequencies" $\tau \in \mathbb{R}$, where $\hat{f}(\tau)$ can be interpreted as the "amplitude" or signed "density distribution" of these frequencies.

Corollary 2.27 (Uniqueness theorem) *If $f, g \in L^1(\mathbb{R}^n)$ are functions such that $\hat{f} = \hat{g}$, then $f = g$.*

Proof. Suppose $f, g \in \mathcal{L}^1 = \mathcal{L}^1(\mathbb{R}^n)$ and $\hat{f} = \hat{g}$. Then $h := f - g \in \mathcal{L}^1$ and $\hat{h} = 0 \in \mathcal{L}^1$, and thus, by Theorem 2.26, $h(x) = 0$ for a.e. x. This implies that $f = g$ in $L^1(\mathbb{R}^n)$. □

Definition. The **Fourier cotransform** $\overline{\mathcal{F}}f$ of a function $f \in L^1(\mathbb{R}^n)$ is defined by

$$\overline{\mathcal{F}}f(\xi) := \int f(x)\mathrm{e}^{i\xi\cdot x}\,\mathrm{d}x = \hat{f}(-\xi).$$

Moreover, if f is any function on $\mathbb{R}^n$, we set

$$\check{f}(x) := f(-x), \qquad x \in \mathbb{R}^n.$$

If $f \in \mathcal{S}$, then $\check{f}$ is also in $\mathcal{S}$. Note also that for every $f \in L^1(\mathbb{R}^n)$,

$$(\overline{\mathcal{F}} f)^\vee = \mathcal{F} f, \quad \mathcal{F}(\check{f}) = \overline{\mathcal{F}} f, \tag{2.24}$$

$$(f * g)^\vee = \check{g} * \check{f} = \check{f} * \check{g}. \tag{2.25}$$

As a further consequence we obtain

Theorem 2.28 (a) *The Fourier transform $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is a linear automorphism of the vector space $\mathcal{S}(\mathbb{R}^n)$, whose inverse is given by $\mathcal{F}^{-1} = (2\pi)^{-n}\overline{\mathcal{F}}$.*

(b) *Suppose $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$ are Schwartz functions. Then their pointwise and their convolution product $\varphi\psi$ and $\varphi * \psi$ are also in $\mathcal{S}(\mathbb{R}^n)$, and*

$$(\varphi * \psi)^\wedge = \hat{\varphi}\,\hat{\psi}, \quad (\varphi\psi)^\wedge = (2\pi)^{-n}\hat{\varphi} * \hat{\psi}. \tag{2.26}$$

Proof. (a) If $\varphi \in \mathcal{S}$, then, by Corollary 2.20, $\mathcal{F}\varphi$ and $\overline{\mathcal{F}}\varphi$ are also in $\mathcal{S}$. Moreover, by the Fourier inversion formula (2.22), we have that

$$\varphi(x) = (2\pi)^{-n}\overline{\mathcal{F}}(\mathcal{F}\varphi)(x)$$

for a.e. $x \in \mathbb{R}^n$, and since φ is continuous, this identity even holds true for every $x \in \mathbb{R}^n$. By means of (2.24) we then also obtain that

$$\check{\varphi} = (2\pi)^{-n}\overline{\mathcal{F}}(\mathcal{F}\check{\varphi}) = (2\pi)^{-n}\overline{\mathcal{F}}(\overline{\mathcal{F}}\varphi) = (2\pi)^{-n}(\mathcal{F}\overline{\mathcal{F}}\varphi)^\vee,$$

hence

$$\varphi = (2\pi)^{-n}\overline{\mathcal{F}}(\mathcal{F}\varphi) = (2\pi)^{-n}\mathcal{F}(\overline{\mathcal{F}}\varphi),$$

which proves (a).

(b) Let $\varphi, \psi \in \mathcal{S}$. Then, by Leibniz' rule (cf. Exercise 2.12), for every $\alpha \in \mathbb{N}^n$,

$$D^\alpha(\varphi\psi) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} D^\beta \varphi\, D^{\alpha-\beta}\psi, \tag{2.27}$$

where, as usual, we have set

$$\binom{\alpha}{\beta} := \binom{\alpha_1}{\beta_1} \cdots \binom{\alpha_n}{\beta_n},$$

and where $\beta \le \alpha$ means that $\beta_j \le \alpha_j$ for $j = 1, \ldots, n$. It now follows immediately that $\varphi\psi \in \mathcal{S}$.

Next, by Proposition 2.3 we have $(\varphi * \psi)^\wedge = \hat{\varphi}\hat{\psi}$, so that (a) implies that $\varphi * \psi = \mathcal{F}^{-1}(\hat{\varphi}\hat{\psi}) \in \mathcal{S}$. If we replace in this identity φ by $\mathcal{F}^{-1}\varphi$ and ψ by $\mathcal{F}^{-1}\psi$, then we obtain that

$$\varphi\psi = \mathcal{F}(\mathcal{F}^{-1}\varphi * \mathcal{F}^{-1}\psi).$$

By means of the Fourier inversion formula, this yields

$$\begin{aligned}(\varphi\psi)^\wedge &= (2\pi)^{-2n}\mathcal{F}\mathcal{F}\big((\hat{\varphi})^\vee * (\hat{\psi})^\vee\big)\\ &= (2\pi)^{-2n}\mathcal{F}\mathcal{F}((\hat{\varphi} * \hat{\psi})^\vee)\\ &= (2\pi)^{-2n}\mathcal{F}\overline{\mathcal{F}}(\hat{\varphi} * \hat{\psi})\\ &= (2\pi)^{-n}\hat{\varphi} * \hat{\psi}.\end{aligned}$$

□

2.7 L^2-theory: Plancherel's Theorem

For $f, g \in L^2(\mathbb{R}^n)$, we shall denote by

$$(f, g) := \int_{\mathbb{R}^n} f(x)\overline{g(x)}\,\mathrm{d}x$$

the hermitian inner product on the Hilbert space $L^2(\mathbb{R}^n)$. The L^2-norm can then be written as $\|f\|_2 = (f, f)^{1/2}$.

Proposition 2.29 (Parseval's formula) *Let $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$. Then*

$$(\varphi, \psi) = (2\pi)^{-n}(\hat{\varphi}, \hat{\psi}).$$

Proof. Consider $f := \varphi * \psi^*$. By Theorem 2.28, $f \in \mathcal{S}(\mathbb{R}^n)$, so that we may apply the Fourier inversion formula:

$$\begin{aligned}(\varphi, \psi) &= \int \varphi(x)\psi^*(-x)\,\mathrm{d}x = \varphi * \psi^*(0) = f(0)\\ &= (2\pi)^{-n}\int \hat{f}(\xi)\,\mathrm{d}\xi.\end{aligned}$$

On the other hand, $\hat{f} = \hat{\varphi}\widehat{\psi^*} = \hat{\varphi}\overline{\hat{\psi}}$, and thus

$$(\varphi, \psi) = (2\pi)^{-n}(\hat{\varphi}, \hat{\psi}).$$

□

Theorem 2.30 (Plancherel's theorem) *The Fourier transform $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ allows for a unique extension to a bijective linear and isometric operator*

$$\tilde{\mathcal{F}} : L^2(\mathbb{R}^n, \mathrm{d}x) \to L^2(\mathbb{R}^n, \mathbf{d}\xi),$$

where $\mathbf{d}\xi$ denotes the scaled Lebesgue measure $\mathbf{d}\xi := (2\pi)^{-n}\mathrm{d}\xi$, also called the **Plancherel measure** *on $\mathbb{R}^n$.*

Proof. Let $\varphi \in \mathcal{S}(\mathbb{R}^n)$. Choosing $\psi = \varphi$ in Parseval's formula, we obtain

$$\begin{aligned} \|\mathcal{F}\varphi\|_{L^2(\mathbb{R}^n, \mathbf{d}\xi)} &= \left(\int |\mathcal{F}\varphi(\xi)|^2 (2\pi)^{-n} \mathrm{d}\xi \right)^{1/2} \\ &= \left(\int |\varphi(x)|^2 \mathrm{d}x \right)^{1/2} \\ &= \|\varphi\|_{L^2(\mathbb{R}^n, \mathrm{d}x)}. \end{aligned} \tag{2.28}$$

This shows that the Fourier transform $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is a linear isometry from the dense subspace $\mathcal{S}(\mathbb{R}^n)$ of $L^2(\mathbb{R}^n, \mathrm{d}x)$ onto the dense subspace $\mathcal{S}(\mathbb{R}^n)$ of $L^2(\mathbb{R}^n, \mathbf{d}\xi)$.

Our claim thus follows by standard arguments: if $f \in L^2(\mathbb{R}^n)$, then we can choose a sequence $\{\varphi_j\}_j$ in $\mathcal{S}(\mathbb{R}^n)$ which converges to f in $L^2(\mathbb{R}^n)$. The identity (2.28) then shows that the sequence of Fourier transforms $\{\widehat{\varphi_j}\}_j$ forms a Cauchy-sequence in $L^2(\mathbb{R}^n, \mathbf{d}\xi)$, and thus must have a limit function $g \in L^2(\mathbb{R}^n, \mathbf{d}\xi)$, by completeness of the space $L^2(\mathbb{R}^n, \mathbf{d}\xi)$.

One then easily checks that the limit function g does not depend on the choice of the approximating sequence $\{\varphi_j\}_j$. We may thus define the Fourier transform of f by $\tilde{\mathcal{F}} f := g$.

We leave it to the interested reader to check that the mapping $\tilde{\mathcal{F}} : L^2(\mathbb{R}^n, \mathrm{d}x) \to L^2(\mathbb{R}^n, \mathbf{d}x)$ defined in this way has all the desired properties. □

Remarks 2.31 (a) *For functions $f \in L^1 \cap L^2(\mathbb{R}^n)$, the* **Fourier–Plancherel transform** *$\tilde{\mathcal{F}} f$ of f as an L^2-function, and the pointwise defined Fourier transform $\hat{f}$ of f as an L^1-function, agree as elements of L^2.*

Indeed, an inspection of the proof of Corollary 2.14 reveals that for a given $f \in L^1 \cap L^2(\mathbb{R}^n)$, we can find a sequence $\{\varphi_j\}_j$ in $C_0^\infty(\mathbb{R}^n)$ such that $\lim_{j\to\infty} \|f - \varphi_j\|_1 = 0$ and $\lim_{j\to\infty} \|f - \varphi_j\|_2 = 0$. Then the sequence $\{\widehat{\varphi_j}\}_j$ will, on the one hand, converge uniformly towards $\hat{f}$, and on the other hand converge in L^2 towards $\tilde{\mathcal{F}} f$. But then $\hat{f}$ and $\tilde{\mathcal{F}} f$ must agree almost everywhere. □

In view of this remark, for $f \in L^2(\mathbb{R}^n)$ we shall in the sequel again call $\tilde{\mathcal{F}} f$ simply the "Fourier transform" of f, and usually denote it by $\hat{f}$, or $\mathcal{F} f$, in place of $\tilde{\mathcal{F}} f$, unless a more precise distinction of notation is really beneficial.

(b) *For arbitrary $f, g \in L^2(\mathbb{R}^n)$, Parseval's formula*

$$(f, g) = (2\pi)^{-n} (\hat{f}, \hat{g}). \tag{2.29}$$

will then remain valid.

2.8 Heisenberg's Commutation Relations and Spectral Decomposition of the Fourier Transform by Means of Hermite functions (*)

In Theorem 2.30 we have seen that the normalized Fourier transform

$$U_{\mathcal{F}} := (2\pi)^{-n/2}\tilde{\mathcal{F}}$$

is a unitary linear operator on the Hilbert space $L^2(\mathbb{R}^n)$. In functional analysis, it is shown that every normal linear operator on a Hilbert space can be "diagonalized," i.e., that it admits a unique spectral decomposition. This is, of course, also true for the unitary operator $U_{\mathcal{F}}$.

Our goal in this section is to explicitly determine the spectral decomposition of the Fourier transform $\tilde{\mathcal{F}}$, and thereby also of $U_{\mathcal{F}}$, without relying on the corresponding abstract results from functional analysis.

More precisely, we will show that the Hilbert space $L^2(\mathbb{R}^n)$ possesses an orthonormal basis $\{h_{\mathbf{k}}\}_{\mathbf{k}\in\mathbb{N}^n}$ of real-valued eigenfunctions of $\tilde{\mathcal{F}}$, where the functions $h_{\mathbf{k}}$ are tensor products of suitably normalized one-dimensional Hermite functions.

In Chapter 9, we will revisit this orthonormal basis and, with its help, provide a more elementary proof of the Schwartz kernel theorem than is usually found in the literature. This proof avoids the more advanced methods typically applied (which commonly require deeper knowledge of functional analysis and vector-valued integration theory, which I do not want to assume here).

Let us briefly recall some elementary facts from Hilbert space theory: Given the orthonormal basis $\{h_{\mathbf{k}}\}_{\mathbf{k}\in\mathbb{N}^n}$ of the Hilbert space $L^2(\mathbb{R}^n)$, we can associate to each function $f \in L^2(\mathbb{R}^n)$ its "sequence" of **Fourier–Hermite coefficients**

$$\check{f}(\mathbf{k}) := (f, h_{\mathbf{k}}) = \int_{\mathbb{R}^n} f(x)h_{\mathbf{k}}(x)\,\mathrm{d}x, \qquad \mathbf{k} \in \mathbb{N}^n.$$

Then the following **Plancherel–Hermite identity** holds:

$$\|f\|_{L^2}^2 = \sum_{\mathbf{k}\in\mathbb{N}^n} |\check{f}(\mathbf{k})|^2. \tag{2.30}$$

This identity holds in particular for any Schwartz function $f \in \mathcal{S}(\mathbb{R}^n)$. Later in Section 9.1, Proposition 9.3, we will see that Schwartz functions can even be characterized by the decay behavior of their Fourier–Hermite coefficients.

Let us now return to the spectral decomposition of the Fourier transform. We begin with the one-dimensional case $n = 1$. The key idea will be to reduce the spectral decomposition of the Fourier transform to that of a differential operator with polynomial coefficients, which is invariant under the Fourier transform in a certain sense. This is the classical **Hermite operator**

$$\mathcal{H}_1 := -\frac{\mathrm{d}^2}{\mathrm{d}x^2} + x^2 = D^2 + X^2,$$

where $D := \frac{1}{i}\frac{\mathrm{d}}{\mathrm{d}x}$ and X denotes the multiplication operator $f \mapsto xf$.

By Corollary 2.20, $\mathcal{H}_1$ has the following invariance property:

$$\mathcal{F}[\mathcal{H}_1\varphi] = \mathcal{H}_1(\mathcal{F}\varphi) \qquad \text{for all } \varphi \in \mathcal{S} = \mathcal{S}(\mathbb{R}). \tag{2.31}$$

The linear operators D and X are even continuous with respect to a natural topology on the space $\mathcal{S}$, which we will introduce in Chapter 4. However, in this section, we will not use this fact and will instead argue purely algebraically.

For any linear operators (i.e., linear mappings in the sense of linear algebra) $A, B : \mathcal{S} \to \mathcal{S}$, we denote their composition $A \circ B$ simply as AB, and define the **commutator** of A and B by

$$[A, B] := AB - BA,$$

i.e., $[A, B]\varphi := A(B\varphi) - B(A\varphi)$ for $\varphi \in \mathcal{S}$. Then (2.31) means precisely that the operators $\mathcal{F}$ and $\mathcal{H}_1$ commute on $\mathcal{S}$, i.e.,

$$[\mathcal{F}, \mathcal{H}_1] = 0.$$

Note also that

$$[A, B] = -[B, A].$$

The commutator $[\cdot, \cdot]$ defines a Lie algebra structure on the $\mathbb{C}$-algebra $\mathcal{L}(\mathcal{S})$ of all linear operators on $\mathcal{S}$ with values in $\mathcal{S}$ (see Exercise 2.20), though we will not delve further into this here.

The following computation rule for commutators, which is easy to verify, is often useful: For any linear operators $A, B, C : \mathcal{S} \to \mathcal{S}$, the *product rule* holds:

$$[A, BC] = [A, B]C + B[A, C]. \tag{2.32}$$

We call a linear operator A on $\mathcal{S}$ **formally self-adjoint** if

$$(A\varphi, \psi) = (\varphi, A\psi) \quad \text{for all } \varphi, \psi \in \mathcal{S}.$$

For example, the operators X and D are formally self-adjoint, as is $\mathcal{H}_1$.

However, the two fundamental operators D and X do not commute; instead, they satisfy **Heisenberg's commutation relation**

$$[D, X] = \frac{1}{i}I, \tag{2.33}$$

since $\frac{\mathrm{d}}{\mathrm{d}x}(x\varphi) - x(\frac{\mathrm{d}}{\mathrm{d}x}\varphi) = \varphi$.

Similarly, on $\mathbb{R}^n$ we have the following **Heisenberg commutation relations**:

$$[D_j, X_k] = \frac{1}{i}\delta_{jk}I, \quad [D_j, D_k] = 0 = [X_j, X_k], \qquad j, k = 1, \ldots, n, \tag{2.34}$$

where X_j denotes the multiplication operator $f \mapsto x_j f$, and where δ_{jk} again denotes Kronecker's symbol.

These commutation relations are of fundamental importance in many fields, particularly in the mathematical theory of partial differential equations (where they are implicitly present) and in physics, especially in the Schrödinger formulation of quantum mechanics. In this context, the operator X represents the *position operator*, and D (up to Planck's constant) represents the *momentum operator* for a one-dimensional particle. The operator $\mathcal{H}_1$ is the *Hamiltonian operator of the harmonic oscillator*.

Folland's monograph [12], which discusses numerous interesting questions about the mathematical and physical background of Heisenberg's commutation relations, contains an excellent introduction to these connections in Chapter 1 and is highly recommended for a deeper study of the topics of the present section.

For $n = 1$, equation (2.31) clearly implies the following fact: if $\varphi \in \mathcal{S} \setminus \{0\}$ is an eigenfunction of $\mathcal{H}_1$ with eigenvalue λ, i.e., if $\mathcal{H}_1\varphi = \lambda\varphi$, then it follows that

$$\mathcal{H}_1(\mathcal{F}\varphi) = \mathcal{F}(\mathcal{H}_1\varphi) = \lambda\mathcal{F}(\varphi),$$

i.e., along with φ, $\mathcal{F}(\varphi)$ is also an eigenfunction of $\mathcal{H}_1$ corresponding to the same eigenvalue λ. We will see that the eigenspaces of $\mathcal{H}_1$ are one-dimensional. Consequently, $\mathcal{F}(\varphi)$ must be a scalar multiple of φ, meaning that φ is also an eigenfunction of the Fourier transform $\mathcal{F}$.

We will therefore primarily focus on the spectral decomposition of $\mathcal{H}_1$. To this end, we introduce the two linear operators

$$\begin{aligned} Z &:= X + iD = x + \frac{\mathrm{d}}{\mathrm{d}x}, \\ Z^* &:= X - iD = x - \frac{\mathrm{d}}{\mathrm{d}x} \end{aligned}$$

on $\mathcal{S}$. Clearly, Z^* is the **formal adjoint operator** of Z, i.e., it holds that

$$(Z\varphi, \psi) = (\varphi, Z^*\psi)$$

for all $\varphi, \psi \in \mathcal{S}$. Furthermore, using (2.33), we see that the commutation relation

$$[Z, Z^*] = [X + iD, X - iD] = 2i[D, X] = 2I, \tag{2.35}$$

as well as

$$Z^*Z + I = (X - iD)(X + iD) + I = X^2 + D^2 - i[D, X] + I = \mathcal{H}_1 \tag{2.36}$$

hold. From this and the "product rule" (2.32), the following commutation relations are easily derived, the first one by induction on k (see Exercise 2.19):

$$\begin{aligned} &[Z, Z^{*k}] = k[Z, Z^*]Z^{*k-1} = 2kZ^{*k-1}, \\ &[Z, \mathcal{H}_1] = [Z, Z^*Z] = [Z, Z^*]Z = 2Z, \\ &[\mathcal{H}_1, Z^*] = [Z^*Z, Z^*] = Z^*[Z, Z^*] = 2Z^*. \end{aligned} \tag{2.37}$$

Using these identities, the following lemma can now be proved quickly:

Lemma 2.32 *If $\varphi \in \mathcal{S}$ is an eigenfunction of $\mathcal{H}_1$ with eigenvalue λ, i.e., if $\mathcal{H}_1\varphi = \lambda\varphi$, $\varphi \neq 0$, then*

$$\mathcal{H}_1(Z^*\varphi) = (\lambda + 2)(Z^*\varphi), \qquad \mathcal{H}_1(Z\varphi) = (\lambda - 2)(Z\varphi).$$

Proof. Using (2.37), we have

$$\mathcal{H}_1(Z^*\varphi) = [\mathcal{H}_1, Z^*]\varphi + Z^*(\mathcal{H}_1\varphi) = 2Z^*(\varphi) + \lambda Z^*(\varphi),$$

and the second identity is shown similarly. □

The operators Z^* and Z are therefore often referred to as the **raising operator** and **lowering operator**, respectively.

We begin by determining those functions φ for which $Z\varphi = 0$, i.e., the solutions to the ordinary differential equation

$$\frac{\mathrm{d}}{\mathrm{d}x}\varphi(x) + x\varphi(x) = 0.$$

It is evident that these solutions are of the form $\varphi(x) = c\mathrm{e}^{-x^2/2}$, where $c \in \mathbb{C}$.

This suggests to define the Gaussian

$$\tilde{h}_0(x) := \mathrm{e}^{-x^2/2} = \Phi_1(x)$$

as the "ground state." Note that (2.20) implies that

$$\|\tilde{h}_0\|_2 = (\tilde{h}_0, \tilde{h}_0) = \sqrt{\pi}.$$

Then, $Z\tilde{h}_0 = 0$, and hence, according to (2.36), it follows that $\mathcal{H}_1\tilde{h}_0 = \tilde{h}_0$.

By repeated application of the raising operator Z^*, we are thus led to defining, for any $k \in \mathbb{N}$,

$$\tilde{h}_k := Z^{*k}\tilde{h}_0.$$

Evidently, $\tilde{h}_k \in \mathcal{S}(\mathbb{R})$ for every $k \in \mathbb{N}$. As the last identity in (2.37) shows, $\mathcal{H}_1 Z^* = Z^*(\mathcal{H}_1 + 2I)$, and by induction on k, exploiting that $\tilde{h}_k = Z^*\tilde{h}_{k-1}$, we immediately deduce that

$$\mathcal{H}_1\tilde{h}_k = (2k + 1)\tilde{h}_k, \qquad k \in \mathbb{N}. \tag{2.38}$$

The real-valued functions $\{\tilde{h}_k\}_{k\in\mathbb{N}}$ even form an orthogonal system in $L^2(\mathbb{R})$. To see this, we consider the inner products $(\tilde{h}_j, \tilde{h}_k) = (\tilde{h}_k, \tilde{h}_j)$, assuming without loss of generality that $j \leq k$. For $j \geq 1$, using the first identity in (2.37), we have

$$\begin{aligned}(\tilde{h}_j, \tilde{h}_k) &= (Z^*\tilde{h}_{j-1}, \tilde{h}_k) = (\tilde{h}_{j-1}, ZZ^{*k}\tilde{h}_0) = (\tilde{h}_{j-1}, [Z, Z^{*k}]\tilde{h}_0)\\ &= (\tilde{h}_{j-1}, 2kZ^{*k-1}\tilde{h}_0) = 2k(\tilde{h}_{j-1}, \tilde{h}_{k-1}).\end{aligned}$$

By induction on j, this implies that for $j \le k$

$$(\tilde{h}_j, \tilde{h}_k) = 2^j k(k-1)\cdots(k-j+1)(\tilde{h}_0, \tilde{h}_{k-j}),$$

where $(\tilde{h}_0, \tilde{h}_{k-j}) = (Z^{k-j}\tilde{h}_0, \tilde{h}_0) = 0$ for $j < k$. Thus, for $j < k$, $(\tilde{h}_j, \tilde{h}_k) = 0$, and this also holds for $j = 0$. For $j = k$, we further find that

$$(\tilde{h}_k, \tilde{h}_k) = 2^k k!(\tilde{h}_0, \tilde{h}_0) = 2^k k!\sqrt{\pi}.$$

Consequently, the normalized functions

$$h_k := \frac{1}{\sqrt{2^k k!\sqrt{\pi}}}\,\tilde{h}_k, \qquad k \in \mathbb{N},$$

form an orthonormal system in the Hilbert space $L^2(\mathbb{R})$.

These functions are also eigenfunctions of the Fourier transform $\mathcal{F}$. Specifically,

$$\mathcal{F}h_k = (-i)^k\sqrt{2\pi}\,h_k, \qquad k \in \mathbb{N}. \tag{2.39}$$

Indeed, by Lemma 2.24, $\mathcal{F}\tilde{h}_0 = \sqrt{2\pi}\,\tilde{h}_0$. Furthermore, by Corollary 2.20, $\mathcal{F}D = X\mathcal{F}$ and $\mathcal{F}X = -D\mathcal{F}$, so that

$$\mathcal{F}Z^* = -iZ^*\mathcal{F}.$$

Using induction on k, the identity (2.39) is an immediate consequence.

The functions $\tilde{h}_k$ can be identified as the **classical Hermite functions**. To establish this, we use the identity

$$Z^* f(x) = (x - \frac{\mathrm{d}}{\mathrm{d}x})f(x) = -\mathrm{e}^{x^2/2}\frac{\mathrm{d}}{\mathrm{d}x}\big(\mathrm{e}^{-x^2/2}f(x)\big),$$

which by induction immediately generalizes to

$$Z^{*k} f(x) = (-1)^k \mathrm{e}^{x^2/2}\frac{\mathrm{d}^k}{\mathrm{d}x^k}\big(\mathrm{e}^{-x^2/2}f(x)\big), \qquad k \in \mathbb{N},$$

and thus allows us to write $\tilde{h}_k = Z^{*k}\tilde{h}_0$ as

$$\tilde{h}_k(x) = (-1)^k \mathrm{e}^{x^2/2}\frac{\mathrm{d}^k}{\mathrm{d}x^k}\mathrm{e}^{-x^2} = H_k(x)\mathrm{e}^{-x^2/2}, \tag{2.40}$$

where

$$H_k(x) := (-1)^k \mathrm{e}^{x^2}\frac{\mathrm{d}^k}{\mathrm{d}x^k}\mathrm{e}^{-x^2}$$

denotes the classical **Hermite polynomial** of degree $k \in \mathbb{N}$. Note that

$$H_k(x) = 2^k x^k + (\text{terms of degree} < k). \tag{2.41}$$

Before proving that the functions $h_k, k \in \mathbb{N}$, even form a Hilbert space basis of $L^2(\mathbb{R})$, we extend our discussion from $\mathbb{R}$ to $\mathbb{R}^n$.

For any multi-index $\mathbf{k} = (k_1, \dots, k_n) \in \mathbb{N}^n$ and $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, we define

$$h_{\mathbf{k}}(x) := h_{k_1}(x_1) \cdots h_{k_n}(x_n).$$

Using Fubini's theorem, we immediately see that the **normalized Hermite functions** $h_{\mathbf{k}}, \mathbf{k} \in \mathbb{N}^n$, form an orthonormal system in $L^2(\mathbb{R}^n)$. We define the **Hermite operator on** $\mathbb{R}^n$ by

$$\mathcal{H}_n := \sum_{j=1}^{n} \Big(-\frac{\partial^2}{\partial x_j^2} + x_j^2 \Big) = -\Delta + |x|^2.$$

Proposition 2.33 *The functions $h_{\mathbf{k}}, \mathbf{k} \in \mathbb{N}^n$, form a complete orthonormal system in $L^2(\mathbb{R}^n)$ and are simultaneous eigenfunctions of both the Hermite operator $\mathcal{H}_n$ and the Fourier transform $\mathcal{F}$. Specifically,*

$$\begin{aligned} \mathcal{H}_n h_{\mathbf{k}} &= (n + 2|\mathbf{k}|) h_{\mathbf{k}}, \\ \mathcal{F} h_{\mathbf{k}} &= (2\pi)^{n/2} (-i)^{|\mathbf{k}|} h_{\mathbf{k}}, \qquad \mathbf{k} \in \mathbb{N}^n. \end{aligned} \tag{2.42}$$

A significant application of this spectral decomposition of the Fourier transform can be found in Beckner [2].

Proof. The formulas in (2.42) follow immediately from the one-dimensional results (2.38) and (2.39), so it remains to show that the $h_{\mathbf{k}}, \mathbf{k} \in \mathbb{N}^n$, form a complete orthonormal system.

To this end, suppose that $f \in L^2(\mathbb{R}^n)$ is such that $(f, h_{\mathbf{k}}) = 0$ for every $\mathbf{k} \in \mathbb{N}^n$. As Hilbert space theory teaches us, it will then suffice to show that $f = 0$.

To this end, we define the ***n*-dimensional Hermite polynomials** $H_{\mathbf{k}}(x)$ on $\mathbb{R}^n$ by

$$H_{\mathbf{k}}(x) := H_{k_1}(x_1) \cdots H_{k_n}(x_n).$$

Then (2.41) shows that

$$H_{\mathbf{k}}(x) = 2^{|\mathbf{k}|} x^{\mathbf{k}} + (\text{terms of degree} < |\mathbf{k}|).$$

By induction on the degree d, we thus find that every polynomial of degree $\le d$ on $\mathbb{R}^n$ can be written as a linear combination of Hermite polynomials of degree $\le d$.

Since, by (2.40), $h_{\mathbf{k}}(x) = c_{\mathbf{k}} H_{\mathbf{k}}(x) \mathrm{e}^{-|x|^2/2}$ with constants $c_{\mathbf{k}} \neq 0$, it follows that

$$(f, P\mathrm{e}^{-|\cdot|^2/2}) = 0 \quad \text{for every polynomial } P \text{ on } \mathbb{R}^n.$$

Fix $\xi \in \mathbb{R}^n$. For any $M \in \mathbb{N}$, we have the estimate

$$\Big| \sum_{m=0}^{M} \frac{(-i\xi \cdot x)^m}{m!} \Big| \le \mathrm{e}^{|\xi \cdot x|},$$

and by the Cauchy–Schwarz inequality, the function $x \mapsto f(x)\mathrm{e}^{-|x|^2/2}\mathrm{e}^{|\xi \cdot x|}$ lies in $L^1(\mathbb{R}^n)$.

By the dominated convergence theorem, we may thus conclude that

$$\int f(x)\mathrm{e}^{-|x|^2/2}\mathrm{e}^{-i\xi\cdot x}\,\mathrm{d}x = \sum_{m=0}^{\infty}\int f(x)\mathrm{e}^{-|x|^2/2}\frac{(-i\xi\cdot x)^m}{m!}\,\mathrm{d}x = 0.$$

Using the uniqueness result in Corollary 2.27, we deduce that $f\mathrm{e}^{-|\cdot|^2/2} = 0$, and hence $f = 0$ in $L^2(\mathbb{R}^n)$. □

2.9 Exercises

2.1 For $a > 0$, let $f_a := \mathbf{1}_{[-a,a]}$ on $\mathbb{R}$. Compute:

a) the convolution $f_a * f_b$;
b) the Fourier transform of $g(x) := (1 - |x|)_+$.

2.2 Prove the formulas for the convolution product in Theorem 2.2 (ii).

2.3 Compute the Fourier transforms of the following functions on $\mathbb{R}$:

a) $f(x) := \mathrm{e}^{-|x|}$;
b) $g(x) := \mathrm{e}^{-x^2/2}$, by using Cauchy's integral formula.

2.4 Let $\alpha, \beta \in \mathbb{R}$, and let f be the function on $\mathbb{R}^n$ defined a.e. by

$$f(x) := |x - y|^{-\alpha}(1 + |x|)^{-\beta}.$$

Furthermore, let $1 \le p \le \infty$. For which values of α and β does f lie in $L^p(\mathbb{R}^n)$? When is f locally integrable?

2.5 Show that the space $\mathcal{T}$ of all step functions, i.e., all finite linear combinations of characteristic functions of cuboids, is dense in $\mathcal{L}^p(\mathbb{R}^n)$ for $1 \le p < \infty$ (i.e., for every function $f \in \mathcal{L}^p$, there exists a sequence $\{f_j\}_j$ in $\mathcal{T}$ such that $\lim_{j\to\infty} \|f - f_j\|_p = 0$). Does this also hold for $p = \infty$?

Hint: Using the inequality

$$|a - b|^p \le |a^p - b^p| \qquad \forall\, a, b \ge 0,$$

the case $p > 1$ can be reduced to the case $p = 1$, which is assumed to be known.

2.6 Let $\chi : \mathbb{R} \to S^1 := \{z \in \mathbb{C} : |z| = 1\}$ be a Lebesgue-measurable group homomorphism from the additive abelian group $(\mathbb{R}, +)$ into the multiplicative abelian group $(S^1, \cdot)$. Show that there exists a $\xi \in \mathbb{R}$ such that $\chi = e_\xi$.

Hint: Show that $\chi \in C^\infty(\mathbb{R})$ by considering the convolutions $\chi * \varphi$ for $\varphi \in C_0^\infty(\mathbb{R})$. Which differential equation does χ satisfy?

Supplement*: Show that there exist non-measurable homomorphisms $\chi : \mathbb{R} \to S^1$!

2.7 (Schur's Test) Let $X := \mathbb{R}^n, Y := \mathbb{R}^m$, and let $\mathcal{K} : X \times Y \to \mathbb{C}$ be a measurable "kernel function". Furthermore, assume there exists a constant $C \geq 0$ such that

$$\sup_{x\in X} \int_Y |\mathcal{K}(x,y)|\,\mathrm{d}y \leq C \quad \text{and} \quad \sup_{y\in Y} \int_X |\mathcal{K}(x,y)|\,\mathrm{d}x \leq C.$$

a) Show that for any $f \in L^p(Y)$ $(1 \leq p \leq \infty)$, the function

$$T_{\mathcal{K}} f(x) := \int_Y \mathcal{K}(x,y) f(y)\,\mathrm{d}y$$

is defined for almost every $x \in X$, and that the so defined mapping $T_{\mathcal{K}}$ is a continuous linear operator from $L^p(Y)$ to $L^p(X)$, with operator norm

$$\|T_{\mathcal{K}}\|_{L^p \to L^p} := \sup\{\|T_{\mathcal{K}} f\|_{L^p} : \|f\|_{L^p} \leq 1\} \leq C.$$

b) Use this to provide an alternative proof of Young's inequality.

Hint: Use Hölder's inequality to first show that for $1 \leq p < \infty$,

$$\left| \int \mathcal{K}(x,y) f(y)\,\mathrm{d}y \right| \leq C^{1/p'} \left(\int |\mathcal{K}(x,y)||f(y)|^p\,\mathrm{d}y \right)^{1/p}.$$

2.8 Let $f \in C^k(\mathbb{R}^n)$ be such that $D^\alpha f \in L^\infty$ for all derivatives D^α up to order k, and let $\{\varphi_\varepsilon\}_\varepsilon$ be an approximate identity on $\mathbb{R}^n$. Show that $\varphi_\varepsilon * f \to f$ uniformly on compact subsets as $\varepsilon \to 0$, and likewise for all derivatives up to order k.

2.9 Let $f \in \mathcal{L}^1(\mathbb{R}^n)$ be continuous at $x = 0$. Assuming that $\widehat{f}(\xi) \geq 0$ for all $\xi \in \mathbb{R}^n$, show that $\widehat{f} \in L^1(\mathbb{R}^n)$ and $f(0) = \frac{1}{(2\pi)^n} \int \widehat{f}(\xi)\,\mathrm{d}\xi$.

Hint: Choose a Dirac family $\psi_\varepsilon(x) = \varepsilon^{-n}\psi(\frac{x}{\varepsilon})$ with $\operatorname{supp}\psi \subset B_1(0)$ and $\psi(-x) = \psi(x)$. Set $\varphi_\varepsilon := \psi_\varepsilon * \psi_\varepsilon^*$. Consider $\frac{1}{(2\pi)^n} \int \widehat{f}(\xi)\widehat{\varphi_\varepsilon}(\xi)\,\mathrm{d}\xi$, and use Proposition 2.25 to derive an alternative expression for this integral.

2.10 Let $f \in L^1(\mathbb{R}^n)$. Show that $\widehat{f}$ is uniformly continuous.

2.11 Let $N \in \mathbb{N}$. Show that there exists a constant $C_N \geq 1$ such that

$$\frac{1}{C_N}(1+|x|^2)^{N/2} \leq \sum_{|\alpha|\leq N} |x^\alpha| \leq C_N(1+|x|^2)^{N/2} \quad \forall x \in \mathbb{R}^n.$$

2.12 Prove the higher-dimensional **Leibniz formula**:

$$D^\alpha(fg) = \sum_{\beta\leq\alpha} \binom{\alpha}{\beta} D^\beta f \cdot D^{\alpha-\beta} g, \quad f, g \in C^{|\alpha|}(\mathbb{R}^n). \tag{2.43}$$

Here, for $\alpha, \beta \in \mathbb{N}^n$, we write $\beta \leq \alpha$ if $\beta_j \leq \alpha_j$ for $j = 1, \ldots, n$, and

$$\alpha! := \alpha_1! \cdots \alpha_n!, \quad \binom{\alpha}{\beta} := \frac{\alpha!}{\beta!(\alpha-\beta)!} = \prod_{j=1}^n \binom{\alpha_j}{\beta_j}.$$

2.13 Let $N \in \mathbb{N}$, and consider the following norms on the space $\mathcal{S}(\mathbb{R}^n)$:

$$\|\varphi\|_{[N]} := \sum_{|\alpha|,|\beta|\le N} \sup_{x\in\mathbb{R}^n} |x^\alpha D^\beta \varphi(x)|,$$
$$\|\varphi\|_{\{N\}} := \sum_{|\alpha|,|\beta|\le N} \sup_{x\in\mathbb{R}^n} |D^\beta (x^\alpha \varphi)(x)|.$$

Prove that these norms are *equivalent*, i.e., there exists a constant $C = C_N \ge 1$ such that

$$\frac{1}{C}\|\varphi\|_{[N]} \le \|\varphi\|_{\{N\}} \le C\|\varphi\|_{[N]} \qquad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^n).$$

Hints: Show that there exist coefficients $c^{\alpha,\beta}_{\alpha',\beta'}, d^{\alpha,\beta}_{\alpha',\beta'} \in \mathbb{C}$ such that

$$D^\beta(x^\alpha\varphi) = \sum_{\alpha'\le\alpha,\,\beta'\le\beta} c^{\alpha,\beta}_{\alpha',\beta'}\, x^{\alpha'} D^{\beta'}\varphi, \tag{2.44}$$
$$x^\alpha(D^\beta\varphi) = \sum_{\alpha'\le\alpha,\,\beta'\le\beta} d^{\alpha,\beta}_{\alpha',\beta'}\, D^{\beta'}(x^{\alpha'}\varphi), \tag{2.45}$$

where $c^{\alpha,\beta}_{\alpha,\beta} = 1 = d^{\alpha,\beta}_{\alpha,\beta}$. For (2.44), use the Leibniz formula from Exercise 2.12, and derive (2.45) by induction on the value of $|\alpha| + |\beta|$.

Alternatively, both identities can be derived using Heisenberg's commutation relations (2.34).

2.14 Prove the following formulas:

a) $$\mathrm{e}^{-|\gamma|} = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\mathrm{e}^{i\gamma x}}{1+x^2}\,\mathrm{d}x, \qquad \gamma \in \mathbb{R}.$$

b) $$\mathrm{e}^{-\gamma} = \frac{1}{\sqrt{\pi}}\int_0^\infty \frac{\mathrm{e}^{-u}}{\sqrt{u}}\mathrm{e}^{-\frac{\gamma^2}{4u}}\,\mathrm{d}u, \qquad \gamma > 0.$$

Hint for b): Substitute $\frac{1}{1+x^2}$ in (a) with $\int_0^\infty \mathrm{e}^{-u(1+x^2)}\,\mathrm{d}u$.

2.15 a) Prove:

$$\int_{\mathbb{R}^n} \mathrm{e}^{-|x|-i\xi\cdot x}\mathrm{d}x = 2^n\pi^{\frac{n-1}{2}}\Gamma\left(\frac{n+1}{2}\right)\cdot\frac{1}{(1+|\xi|^2)^{\frac{n+1}{2}}}, \qquad \xi\in\mathbb{R}^n,$$

where $\Gamma(s) := \int_0^\infty \mathrm{e}^{-t}t^{s-1}\,\mathrm{d}t$, $s > 0$, is the Gamma function.

Hint: Use the "subordination formula" from Exercise 2.14 b) to represent $\mathrm{e}^{-|x|}$ as an average of scaled Gaussian functions $\mathrm{e}^{-|x|^2/4u}$.

b) For $y > 0$, the **Poisson kernel** P_y on $\mathbb{R}^n$ is defined by $\widehat{P_y}(\xi) := \mathrm{e}^{-y|\xi|}$, $\xi \in \mathbb{R}^n$. Show that

$$P_y(x) = \frac{c_n y}{(|x|^2+y^2)^{\frac{n+1}{2}}}, \quad \text{with } c_n := \frac{\Gamma\left(\frac{n+1}{2}\right)}{\pi^{\frac{n+1}{2}}}.$$

c) Show that the family of Poisson kernels $\{P_y\}_{y>0}$ is an approximate identity on $\mathbb{R}^n$.

2.16 Let $A(\mathbb{R}) := \mathcal{F}(L^1(\mathbb{R}))$ be the Fourier algebra of $\mathbb{R}$.

a) Prove that the space $C_0^2(\mathbb{R})$ of all twice continuously differentiable functions with compact support is contained in $A(\mathbb{R})$. Conclude that $A(\mathbb{R})$ is a dense subspace of $(C_\infty(\mathbb{R}), \|\cdot\|_\infty)$.
b) Show that $A(\mathbb{R}) \subsetneq C_\infty(\mathbb{R})$.

Hint to b): This requires familiarity with the open mapping theorem, to conclude that there is a constant $C > 0$ such that $\|f\|_{L^1} \le C\|\hat{f}\|_\infty$ for all $f \in L^1(\mathbb{R})$, if we assume that $A(\mathbb{R}) = C_\infty(\mathbb{R})$. Then make use of Exercise 2.1 a) to arrive at a contradiction.

2.17 Let $f \in L^2(\mathbb{R})$ be continuously differentiable such that $f' \in L^1(\mathbb{R})$. Further, assume f vanishes at infinity. Show that the limit

$$g(\xi) := \lim_{N\to\infty} \int_{-N}^{N} f(x)\, \mathrm{e}^{-i\xi x}\, \mathrm{d}x$$

exists for every $\xi \in \mathbb{R} \setminus \{0\}$, and that $g = \tilde{\mathcal{F}}(f)$ in $L^2(\mathbb{R})$.

2.18 (The Heisenberg Uncertainty Principle) Use the Heisenberg commutation relation (2.33), i.e.,

$$[D, X] = DX - XD = \frac{1}{i} I,$$

to prove the following inequality:

$$\|f\|_2^2 \le 2(2\pi)^{-1/2} \|xf\|_2 \|\widehat{\xi f}\|_2 \qquad \text{for all } f \in \mathcal{S}(\mathbb{R}).$$

Is the constant $2(2\pi)^{-1/2}$ in this inequality optimal?

(For an interpretation of this inequality in quantum physics, see, e.g., Stein–Shakarchi [40], Chapter 5.4.)

2.19 Prove the identities in (2.37).

2.20 (A bit of Algebra) Let A be an **algebra** over the field $\mathbb{K}$, i.e., a vector space over $\mathbb{K}$ (for simplicity, we assume $\mathbb{K}$ not to be of characteristic 2), which is additionally equipped with a bilinear product

$$A \times A \to A, \quad (a, b) \mapsto a \cdot b,$$

which satisfies the following for all $a, b, c \in A$ and all scalars $\lambda \in \mathbb{K}$:

$$\begin{aligned} (a + b) \cdot c &= a \cdot c + b \cdot c, \\ a \cdot (b + c) &= a \cdot b + a \cdot c, \\ \lambda(a \cdot b) &= (\lambda a) \cdot b = a \cdot (\lambda b). \end{aligned}$$

The algebra A is called **associative** if, additionally, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ holds for all $a, b, c \in A$ – this is not assumed here, however.

Let $\mathcal{L}(A)$ denote the $\mathbb{K}$-vector space of all linear endomorphisms $T : A \to A$ (which, with the addition of linear endomorphisms and the composition as the product, forms another, even associative, algebra). Let $D \in \mathcal{L}(A)$. D is called a **derivation** of A if the

following product rule holds for all $a, b \in A$:

$$D(a \cdot b) = (Da) \cdot b + a \cdot (Db).$$

For $a, b \in A$, the **commutator** of a with b is defined as $[a, b] := a \cdot b - b \cdot a$, and we set $D_a : A \to A,\ D_a(b) := [a, b]$. Show that:

i) D_a is a derivation of A.
ii) If D is a derivation of A, then for all $b_1, \ldots, b_n \in A$:

$$D(b_1 \cdots b_n) = \sum_{j=1}^{n} b_1 \cdots b_{j-1} \cdot D(b_j) \cdot b_{j+1} \cdots b_n.$$

iii) The mapping $(a, b) \mapsto [a, b]$ from $A \times A \to A$ is also bilinear, so it defines a new structure of an algebra on A, which we denote by $(A, +, [\cdot, \cdot])$. Show:
In this structure, $[a, b] = -[b, a]$ holds for all $a, b \in A$, as does $\tilde{D}_a(b) = 2[a, b]$, if $\tilde{D}_a$ denotes the derivation defined by a on $(A, +, [\cdot, \cdot])$.
Define for each element $a \in A$ the linear mapping $ad(a) : A \to A$ by

$$ad(a)(b) := [a, b], \qquad b \in A.$$

Since $ad(a) = \tilde{D}_a/2$, we see that $ad(a)$ is also a derivation on $(A, +, [\cdot, \cdot])$. Prove that this is equivalent to the statement that the following **Jacobi identity** holds in $(A, +, [\cdot, \cdot])$:

$$[a, [b, c]] + [c, [a, b]] + [b, [c, a]] = 0 \qquad \text{for all } a, b, c \in A.$$

This means that $(A, +, [\cdot, \cdot])$ is a so-called **Lie algebra**. The linear endomorphism $ad : a \mapsto ad(a)$ is then referred to as the **adjoint representation** of this Lie algebra on the $\mathbb{K}$-vector space A.



Chapter 3
Fourier Series and the Poisson Summation Formula

3.1 The Fourier Transform on $\mathbb{T}$

We now return to the theory of Fourier series, which we introduced in the first chapter. The goal will be to show that, with minor, mostly technical modifications, many ideas and methods of the previous chapter can be transferred to the theory of Fourier series. At the same time, we aim to place the theory in a more general context. Many advanced results on Fourier series and integrals, as well as their applications, can be found in the textbooks by Stein–Shakarchi [40], Deitmar [6], Katznelson [27], and the monograph by Stein and Weiss [43].

As preparation, we first want to show that the study of 2π-periodic functions on the real line can be mathematically simplified by considering them as functions on the one-dimensional torus, i.e., the unit circle.

Let Γ denote the discrete additive subgroup $\Gamma := 2\pi\mathbb{Z}$ of $\mathbb{R}$, and let $\mathbb{T} := \mathbb{R}/2\pi\mathbb{Z}$ be the (commutative) quotient group modulo Γ. For the neutral element $0 + 2\pi\mathbb{Z}$ of $\mathbb{T}$ we shall simply write 0. $\mathbb{T}$ is called the **one-dimensional torus**, since the mapping

$$\varphi : \mathbb{T} \to S^1, \quad x + 2\pi\mathbb{Z} \mapsto \mathrm{e}^{ix}$$

is a bijection between $\mathbb{T}$ and the unit circle S^1 in $\mathbb{C}$. The unit circle is endowed with the relative topology of its ambient space $\mathbb{C}$, and $\mathbb{T}$ with the topology of the circle S^1 transferred via the pull-back by φ.

Note next that a function $g : \mathbb{R} \to \mathbb{C}$ is 2π-periodic if and only if there exists a function $f : \mathbb{R}/2\pi\mathbb{Z} \to \mathbb{C}$ such that $g(x) = f(x + 2\pi\mathbb{Z})$.

Since, for example, the interval $A = [0, 2\pi[$ is a fundamental domain in $\mathbb{R}$ for the action of Γ, meaning that $\mathbb{R}$ is the disjoint union

$$\mathbb{R} = \dot{\bigcup_{\gamma\in\Gamma}} \gamma + A = \dot{\bigcup_{k\in\mathbb{Z}}} [2\pi k, 2\pi(k+1)[,$$

we also observe that any 2π-periodic function g on $\mathbb{R}$ is uniquely determined by its values on A.

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D. Müller, *An Invitation to Fourier Analysis and Distribution Theory*,
Universitext, https://doi.org/10.1007/978-3-032-31550-2_3

Remark 3.1 In place of the interval $[0, 2\pi[$, any other fundamental domain A could also be used here. A **fundamental domain** for the action of Γ on $\mathbb{R}$ is a subset $A \subset \mathbb{R}$ that contains exactly one representative from each orbit of Γ, i.e., from each coset $x + \Gamma$. Then $\mathbb{R}$ is the disjoint union of the translated sets $\gamma + A$, $\gamma \in \Gamma$, i.e.,

$$\mathbb{R} = \dot{\bigcup_{\gamma \in \Gamma}} \gamma + A.$$

This follows immediately from the fact that the mapping

$$\sigma : \Gamma \times A \to \mathbb{R}, \quad (\gamma, a) \mapsto \gamma + a$$

is bijective if and only if A is a fundamental domain.

To prove the latter, suppose that A is a fundamental domain. Let us then denote for any $x \in \mathbb{R}$ by $a_x \in A$ the unique element from A contained in $x + \Gamma$. Then $x \in a_x + \Gamma$, i.e., there is a unique $\gamma_x \in \Gamma$ such that $x = a_x + \gamma_x$. This proves that σ is surjective. And, if $x = \gamma + a$ for some $\gamma \in \Gamma$ and $a \in A$, then $a \in (x + \Gamma) \cap A$, hence $a = a_x$ and $\gamma = \gamma_x$, which establishes the injectivity of σ.

We leave it to the interested reader to verify that, conversely, if σ is bijective, then A is a fundamental domain.

In most applications, a fundamental domain is chosen as a half-open interval of length 2π, typically $[0, 2\pi[$ or $[-\pi, \pi[$, or at least a measurable fundamental domain, since we shall always work with measurable functions.

If $A \subset \mathbb{R}$ is a fundamental domain, then every 2π-periodic function f_{per} on $\mathbb{R}$ is uniquely determined by its restriction to A. A, in return, is mapped bijectively onto $\mathbb{T}$ under the quotient map

$$\pi : \mathbb{R} \to \mathbb{T}, \; x \mapsto x + \Gamma.$$

Thus, the mapping $f \mapsto f_{\text{per}} := f \circ \pi$, which assigns to each function f on $\mathbb{T}$ a 2π-periodic function f_{per} on $\mathbb{R}$, is bijective. Henceforth, we will generally identify functions on $\mathbb{T}$ with 2π-periodic functions on $\mathbb{R}$ and will occasionally denote f_{per} more concisely just by f when the meaning is clear from the context.

Definitions. (a) A function $f : \mathbb{T} \to \mathbb{C}$ is **Lebesgue measurable** (briefly: **measurable**) if the corresponding function $f_{\text{per}} : \mathbb{R} \to \mathbb{C}$ is Lebesgue measurable.

(b) A function f is **k-times (continuously) differentiable** if f_{per} is k-times continuously differentiable. $C^k(\mathbb{T})$ denotes the space of all k-times continuously differentiable functions on $\mathbb{T}$. If f is differentiable, then $f'_{\text{per}} : \mathbb{R} \to \mathbb{C}$ is also 2π-periodic, meaning there exists a unique function on $\mathbb{T}$, which we denote by f', such that $(f')_{\text{per}} = (f_{\text{per}})'$. We then write $\frac{\mathrm{d}}{\mathrm{d}t} f(t) := f'(t)$, $Df := \frac{1}{i} \frac{\mathrm{d}}{\mathrm{d}t} f$. Higher derivatives on $\mathbb{T}$ are defined analogously.

(c) Let $\mathcal{L}^p(\mathbb{T})$, $1 \le p \le \infty$, denote the space of measurable functions $f : \mathbb{T} \to \mathbb{C}$ such that the corresponding function f_{per} is p-integrable over the fundamental domain $[0, 2\pi[$, i.e., if $f_{\text{per}}|_{[0,2\pi[} \in \mathcal{L}^p([0, 2\pi[)$. $L^p(\mathbb{T})$ denotes the space of equivalence classes of $\mathcal{L}^p(\mathbb{T})$-functions modulo null functions (in Lebesgue's sense). As is common, we will often identify $\mathcal{L}^p$ with L^p.

The **integral** of an L^1-function $f \in L^1(\mathbb{T})$ is defined as

$$\int_{\mathbb{T}} f(t)\,\mathrm{d}t := \fint f_{\mathrm{per}}(x)\,\mathrm{d}x = \frac{1}{2\pi}\int_0^{2\pi} f_{\mathrm{per}}(x)\,\mathrm{d}x.$$

It can be shown (see Exercise 1.3) that for any measurable fundamental domain A, the following holds:

$$\int_{\mathbb{T}} f(t)\,\mathrm{d}t = \frac{1}{2\pi}\int_A f_{\mathrm{per}}(x)\,\mathrm{d}x = \frac{1}{2\pi}\int_{\mathbb{R}} f_{\mathrm{per}}|_A\,\mathrm{d}x. \tag{3.1}$$

Correspondingly, the L^p**-norm** of $f \in L^p(\mathbb{T})$ is defined by

$$\|f\|_p = \|f\|_{L^p(\mathbb{T})} := \begin{cases} \left(\int\limits_{\mathbb{T}} |f(t)|^p\,\mathrm{d}t\right)^{1/p}, & \text{if } 1 \le p < \infty, \\ \|f_{\mathrm{per}}|_A\|_\infty = \|f_{\mathrm{per}}\|_\infty, & \text{if } p = \infty. \end{cases} \tag{3.2}$$

For $f \in L^1(\mathbb{T})$, we have

$$\int\limits_{\mathbb{T}} f(t-s)\,\mathrm{d}t = \int\limits_{\mathbb{T}} f(t)\,\mathrm{d}t \quad \text{for all } s \in \mathbb{T}, \tag{3.3}$$

i.e., the integral over $\mathbb{T}$ is **translation-invariant.** Indeed, since the interval $[-s, -s+2\pi[$ forms a measurable fundamental domain, it follows from (3.1) that

$$\int\limits_{\mathbb{T}} f(t-s)\,\mathrm{d}t = \frac{1}{2\pi}\int\limits_{[0,2\pi[} f_{\mathrm{per}}(x-s)\,\mathrm{d}x = \frac{1}{2\pi}\int\limits_{[-s,-s+2\pi[} f_{\mathrm{per}}(x)\,\mathrm{d}x = \int\limits_{\mathbb{T}} f(t)\,\mathrm{d}t.$$

Thus, if $\mu = \mathrm{d}t$ denotes the corresponding measure on $\mathbb{T}$, defined for any measurable subset $M \subset \mathbb{T}$ by

$$|M| := \mu(M) := \int\limits_{\mathbb{T}} \mathbf{1}_M(t)\,\mathrm{d}t,$$

then this measure is also translation-invariant. Therefore, $\mathrm{d}t$ is referred to as the **Haar measure** of the (compact) group $\mathbb{T}$. Note that, unlike $\mathbb{R}$, the group $\mathbb{T}$ has finite measure, specifically satisfying

$$\mu(\mathbb{T}) = \int\limits_{\mathbb{T}} 1\,\mathrm{d}t = 1. \tag{3.4}$$

Thus, by Hölder's inequality, for every $p \in [1, \infty]$,

$$\begin{aligned} \int\limits_{\mathbb{T}} |f(t)|\,\mathrm{d}t &= \int\limits_{\mathbb{T}} 1 \cdot |f(t)|\,\mathrm{d}t \\ &\le \|\mathbf{1}\|_{L^{p'}(\mathbb{T})}\|f\|_{L^p(\mathbb{T})} = \|f\|_{L^p(\mathbb{T})}, \end{aligned}$$

i.e., in contrast to $\mathbb{R}^n$, we always have the inclusion

$$L^p(\mathbb{T}) \subset L^1(\mathbb{T}), \quad \text{for } 1 \le p \le \infty, \tag{3.5}$$

and

$$\|f\|_1 \le \|f\|_p \quad \text{for all } f \in L^p(\mathbb{T}). \tag{3.6}$$

Similarly to Theorem 2.2, it can be shown that for $f, g \in L^1(\mathbb{T})$, the **convolution**

$$f * g(t) := \int_{\mathbb{T}} f(s)g(t-s)\,\mathrm{d}s = \int_{\mathbb{T}} f(t-s)g(s)\,\mathrm{d}s$$

of f with g is well-defined for almost all $t \in \mathbb{T}$, that $f * g$ is integrable, and that

$$\|f * g\|_1 \le \|f\|_1 \|g\|_1.$$

Because of (3.5), the convolution of any $f \in L^p(\mathbb{T})$ with any $g \in L^r(\mathbb{T})$, where $1 \le p, r \le \infty$, is well-defined, in contrast to $\mathbb{R}^n$.

If we define the involution $f \mapsto f^*$ by $f^*(t) := \overline{f(-t)}$, it further follows, analogously to Theorem 2.2, that $(L^1(\mathbb{T}), +, *, ^*; \|\cdot\|_1)$ is a commutative involutive Banach algebra, the so-called **group algebra of the torus** $\mathbb{T}$.

Similar to Proposition 2.5, one can show that for $f \in L^1(\mathbb{T})$ and $g \in C^k(\mathbb{T})$, the convolution $f * g$ lies in $C^k(\mathbb{T})$, and that

$$D^j(f * g) = f * D^j g, \qquad 0 \le j \le k. \tag{3.7}$$

Furthermore, analogously to Corollary 2.14, for every p with $1 \le p < \infty$, the space $C^\infty(\mathbb{T})$ is dense in $L^p(\mathbb{T})$, as an analogue of Theorem 2.13 applies.

Definitions. (a) For $k \in \mathbb{Z}$, let $\tilde{e}_k : \mathbb{R} \to \mathbb{C}$ be the character on $\mathbb{R}$ defined by $\tilde{e}_k(x) := \mathrm{e}^{ikx}$, $x \in \mathbb{R}$. Since $\tilde{e}_k$ is 2π-periodic, it can be identified with a function $e_k : \mathbb{T} \to \mathbb{C}$, and we write suggestively

$$e_k(t) = \mathrm{e}^{ikt}, \ t \in \mathbb{T}.$$

Clearly, the function e_k is continuous, and it satisfies

$$e_k(s+t) = e_k(s)e_k(t) \quad \text{for all } s, t \in \mathbb{T},$$

i.e., e_k is a continuous homomorphism of the group $\mathbb{T}$ into the circle group S^1, i.e., a **character** of $\mathbb{T}$. It can be shown that every character of $\mathbb{T}$ has this form (this follows, e.g., easily from the result in Exercise 2.6).

(b) For $f \in L^1(\mathbb{T})$, we define, for $k \in \mathbb{Z}$,

$$\hat{f}(k) := \int_{\mathbb{T}} f(t)\overline{e_k(t)}\,\mathrm{d}t = \int_{\mathbb{T}} f(t)\mathrm{e}^{-ikt}\,\mathrm{d}t = \frac{1}{2\pi}\int_0^{2\pi} f_{\mathrm{per}}(x)\mathrm{e}^{-ikx}\,\mathrm{d}x.$$

$\hat{f}(k)$ is called the **k-th Fourier coefficient** of f, and the mapping

$$\hat{f} : \mathbb{Z} \to \mathbb{C},\ k \mapsto \hat{f}(k),$$

the **Fourier transform** of f.

If we denote by λ_s the **translation operator** $\lambda_s f(t) := f(t-s)$ for $s \in \mathbb{T}$, then for every $f \in L^1(\mathbb{T})$, essentially the same argument as in Proposition 2.1 shows that

$$\widehat{(\lambda_s f)}(k) = \mathrm{e}^{-iks}\hat{f}(k), \quad k \in \mathbb{Z}. \tag{3.8}$$

Moreover, the following analogue of Lemma 2.6 holds with a nearly identical proof:

Lemma 3.2 (Continuity of translation in $L^p(\mathbb{T})$) *Let $1 \le p \le \infty$, and let $f \in L^p(\mathbb{T})$. Then:*

$$\|\lambda_s f\|_p = \|f\|_p \qquad \textit{for all} \quad s \in \mathbb{T}. \tag{3.9}$$

Moreover, if $1 \le p < \infty$, then

$$\lim_{s\to 0} \|\lambda_s f - f\|_p = 0. \tag{3.10}$$

Definitions. An **approximate identity** on $\mathbb{T}$ is a family $\{\varphi_\varepsilon\}_{0<\varepsilon<\varepsilon_0}$ of integrable functions on $\mathbb{T}$ satisfying the following properties: there is a constant $C \ge 0$ such that

(i) $\int_{\mathbb{T}} \varphi_\varepsilon(t)\,\mathrm{d}t = 1$ for every $\varepsilon \in]0, \varepsilon_0[$;
(ii) $\int_{\mathbb{T}} |\varphi_\varepsilon(t)|\,\mathrm{d}t \le C$ for every $\varepsilon \in]0, \varepsilon_0[$;
(iii) for every neighborhood U of 0 in $\mathbb{T}$, we have $\lim_{\varepsilon\to 0} \int_{\mathbb{T}\setminus U} |\varphi_\varepsilon(t)|\,\mathrm{d}t = 0$.

If $\varphi_\varepsilon \in C^\infty(\mathbb{T})$, we call such an approximate identity a **Dirac family** on $\mathbb{T}$.

Similarly, a sequence $\{\phi_N\}_N$ in $C^\infty(\mathbb{T})$ is called a **Dirac sequence** if the obvious analogues of (i)–(ii) hold for all ϕ_N in place of the φ_ε, and (iii) holds for ϕ_N with limits taken as $N \to \infty$.

For example, for any decreasing null sequence $\varepsilon_N \searrow 0$, the sequence $\{\phi_N\}_N := \{\varphi_{\varepsilon_N}\}_N$ is a Dirac sequence on $\mathbb{T}$.

Since dilations are not available on the torus, in contrast to Euclidean space, one can no longer simply construct approximate identities by scaling a fixed function. Nevertheless, approximate identities and Dirac families or sequences can be constructed on the torus as well, however with a bit more of an effort.

Example: In the context of the so-called "Abel summation" of Fourier series (see, e.g., Stein–Shakarchi [40]), the **Poisson kernel** $P_r(t)$ on $\mathbb{T}$ is defined for $0 \le r < 1$ by

$$P_r(t) := \sum_{k=-\infty}^{\infty} r^{|k|}\mathrm{e}^{ikt}, \qquad t \in \mathbb{T}, \tag{3.11}$$

i.e., via its Fourier series expansion. By summing a geometric series, one easily obtains the following explicit expression for P_r:

$$P_r(t) = \frac{1 - r^2}{1 - 2r\cos t + r^2}, \qquad 0 \le r < 1. \tag{3.12}$$

Lemma 3.3 *Define* $\Phi_\varepsilon := P_{\mathrm{e}^{-\varepsilon}}$, $\varepsilon > 0$. *Then* $\{\Phi_\varepsilon\}_\varepsilon$ *is a Dirac family.*

Proof. By termwise integration of the series (3.11) (which is permissible since the series converges uniformly), we immediately obtain

$$\int_{\mathbb{T}} \Phi_\varepsilon(t)\,\mathrm{d}t = 1 \quad \text{for all } \varepsilon > 0.$$

Furthermore, due to the estimate

$$|2r\cos t| \le r^2 + (\cos t)^2 \le 1 + r^2, \tag{3.13}$$

we see that $P_r(t) \ge 0$ for every t, and so the second property (ii) of an approximate identity is also satisfied.

To prove the third property (iii), it suffices to show that for any $\delta \in]0, \pi[$, we have

$$\int_{\delta \le |t| \le \pi} P_r(t)\,\mathrm{d}t \to 0 \quad \text{as } r \to 1. \tag{3.14}$$

However, from the first inequality in (3.13) we have

$$1 - 2r\cos t + r^2 \ge 1 - (\cos t)^2 \ge 1 - (\cos\delta)^2 =: c_\delta > 0,$$

if $\delta \le |t| \le \pi/2$, and for $\pi/2 < |t| \le \pi$, we trivially have $1 - 2r\cos t + r^2 \ge c_\delta > 0$. Thus, for $\delta \le |t| \le \pi$, we obtain that $P_r(t) \le (1 - r^2)/c_\delta$, from which (3.14) immediately follows.

Since the series

$$\Phi_\varepsilon(t) = \sum_{k=-\infty}^{\infty} \mathrm{e}^{-|k|\varepsilon}\mathrm{e}^{ikt}$$

may be termwise differentiated arbitrarily many times with respect to t, we finally see that Φ_ε is a smooth function. □

Remark 3.4 By Proposition 1.3, the last part of the previous proof also shows that $\mathrm{e}^{-|k|\varepsilon} = \widehat{\Phi_\varepsilon}(k)$, so that the following Fourier inversion formula holds for Φ_ε:

$$\Phi_\varepsilon(t) = \sum_k \widehat{\Phi_\varepsilon}(k)\mathrm{e}^{ikt} \qquad \text{for every } t \in \mathbb{T}.$$

Remark 3.5 Another Dirac sequence, frequently used in Fourier analysis on $\mathbb{T}$, is the sequence $\{F_N\}_N$ of **Fejér kernels**, given by:

$$F_N(x) := \sum_{k=-N}^{N} \frac{N + 1 - |k|}{N + 1}\mathrm{e}^{ikx}, \qquad N \in \mathbb{N},$$

which arise naturally in the study of Cesàro means of Fourier series (see Exercise 3.2, as well as Exercises 3.6, 3.10, and 3.11 for interesting applications of these Fejér kernels).

With an almost identical proof, the following holds analogously to Theorem 2.13:

Theorem 3.6 *Let $\{\varphi_\varepsilon\}_{0<\varepsilon<\varepsilon_0}$ be an approximate identity on $\mathbb{T}$.*

1. *If $f \in L^p(\mathbb{T})$, $1 \le p < \infty$, then:*

$$\lim_{\varepsilon\to 0} \|f * \varphi_\varepsilon - f\|_p = 0.$$

2. *If $f \in \mathcal{L}^\infty(\mathbb{T})$ is uniformly continuous at the points of a set $V \subset \mathbb{T}$, then $f * \varphi_\varepsilon$ converges uniformly to f on V as $\varepsilon \to 0$.*

Analogous properties also hold if $\{\varphi_\varepsilon\}_{0<\varepsilon<\varepsilon_0}$ is replaced by a Dirac sequence $\{\phi_N\}_N$, and the limits are taken as $N \to \infty$ in place of $\varepsilon \to 0$.

Remark 3.7 The Poisson kernel $P_r(t)$ also plays a fundamental role in the theory of the Laplace equation on the unit disk

$$D := \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\}.$$

To see this, let us introduce polar coordinates $x_1 = r\cos\theta$, $x_2 = r\sin\theta$, $0 \le r < 1$, $0 \le \theta < 2\pi$. Then the function F defined on D by

$$F(r\cos\theta, r\sin\theta) := P_r(\theta)$$

is **harmonic**, i.e., $\Delta F = 0$ in D. Indeed, since in polar coordinates, the Laplacian Δ on $\mathbb{R}^2$ is given by

$$\Delta_{pc} = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2},$$

this follows directly from (3.11). Alternatively, this can also be derived from the fact that holomorphic and anti-holomorphic functions on $\mathbb{C}$ are harmonic, since with $z := x_1 + ix_2$:

$$r^{|k|}e^{ikt} = \begin{cases} z^k, & \text{if } k \ge 0, \\ \overline{z}^{|k|}, & \text{if } k < 0. \end{cases}$$

Now let $f \in L^1(\mathbb{T})$, and define the function

$$v(r,\theta) := f *_\theta P_r(\theta), \qquad 0 \le r < 1,\ 0 \le \theta < 2\pi,$$

(where $*_\theta$ denotes convolution with respect to the variable θ, i.e., convolution on $\mathbb{T}$). Then, by (3.7), it follows that

$$\Delta_{pc} v(r,\theta) = f *_\theta (\Delta_{pc} P)(r,\cdot))(\theta) = 0,$$

if we set $P(r,\theta) := P_r(\theta)$, since $\Delta_{pc} P \equiv 0$. The corresponding function

$$u(r\cos\theta, r\sin\theta) := v(r,\theta)$$

is thus harmonic in D.

Furthermore, by Theorem 3.6, we have:

$$\lim_{r\to 1} v(r,\cdot) = f \quad \text{in } L^1(\mathbb{T}).$$

Identifying again the torus $\mathbb{T}$ with the boundary $\partial D = S^1$ of the unit disk D, this shows that the restriction of u to the circle rS^1 with radius r converges as $r \to 1$ to the function f (at least in the L^1-sense). Thus, in this sense, u solves the **Dirichlet boundary value problem**:

$$\begin{aligned} \Delta u &= 0 \text{ in } D, \\ u|_{\partial D} &= f. \end{aligned}$$

With some additional effort, it can also be shown that

$$\lim_{r\to 1} v(r,\theta) = f(\theta) \text{ for almost every } \theta$$

(see, e.g., Stein–Weiss [43] for an analogous result in the upper half-plane).

Using Theorem 3.6, the proof of the Fourier inversion formula from Theorem 2.26 can now be easily transferred from $\mathbb{R}^n$.

We denote by $\ell^p(\mathbb{Z})$ the Banach space of all p-th power summable mappings $g : \mathbb{Z} \to \mathbb{C}$, equipped with the norm

$$\|g\|_{\ell^p} = \|g\|_p := \left(\sum_{k=-\infty}^{\infty} |g(k)|^p \right)^{1/p}, \qquad \text{for } 1 \le p < \infty,$$

and the supremum norm

$$\|g\|_{\ell^\infty} = \|g\|_\infty := \sup_{k\in\mathbb{Z}} |g(k)|, \qquad \text{for } p = \infty.$$

Theorem 3.8 (Fourier inversion formula on $\mathbb{T}$) *Suppose $f \in \mathcal{L}^1(\mathbb{T})$ is such that $\widehat{f} \in \ell^1(\mathbb{Z})$. Then*

$$f(t) = \sum_{k\in\mathbb{Z}} \widehat{f}(k) \mathrm{e}^{ikt} \tag{3.15}$$

for almost every $t \in \mathbb{T}$.

Proof. Let $\{\Phi_\varepsilon\}_{\varepsilon>0}$ be the Dirac family from Lemma 3.3. Then

$$\widehat{f * \Phi_\varepsilon}(k) = \widehat{f}(k)\widehat{\Phi_\varepsilon}(k) = \mathrm{e}^{-|k|\varepsilon}\widehat{f}(k).$$

Since $\widehat{f} \in \ell^1(\mathbb{Z})$, the series $\sum_k \widehat{f * \Phi_\varepsilon}(k)\mathrm{e}^{ikt} = \sum_k \mathrm{e}^{-|k|\varepsilon}\widehat{f}(k)\mathrm{e}^{ikt}$ converges for every $t \in \mathbb{T}$ to the Fourier series

$$Sf(t) = \sum_k \widehat{f}(k)\mathrm{e}^{ikt}$$

of f (which is absolutely convergent) as $\varepsilon \to 0$.

Furthermore, by Remark 3.4, we have

$$\begin{aligned}\sum_k \widehat{f}(k)\widehat{\Phi_\varepsilon}(k)\mathrm{e}^{ikt} &= \sum_k \int_{\mathbb{T}} f(s)\mathrm{e}^{-iks}\,\mathrm{d}s\,\mathrm{e}^{ikt}\,\widehat{\Phi_\varepsilon}(k)\\ &= \int_{\mathbb{T}} f(s)\sum_k \widehat{\Phi_\varepsilon}(k)\mathrm{e}^{ik(t-s)}\,\mathrm{d}s\\ &= \int_{\mathbb{T}} f(s)\Phi_\varepsilon(t-s)\,\mathrm{d}s\\ &= f * \Phi_\varepsilon(t),\end{aligned}$$

since the Fourier series of Φ_ε converges uniformly.

As $f * \Phi_\varepsilon$ converges to f in $L^1(\mathbb{T})$ by Theorem 3.6, there exists a subsequence $\{f*\Phi_{\varepsilon_j}\}_j$ that converges almost everywhere to f. This establishes the claimed inversion formula for f. □

Remarks 3.9 (On the Convergence of Fourier Series) The requirement $\hat{f} \in \ell^1(\mathbb{Z})$ in Theorem 3.8 is quite strong; for example, it is not satisfied by the indicator function f of an interval of length L with $0 < L < 2\pi$ (cf. Example 1.4, (1.20)). Nevertheless, as we shall see, according to Dirichlet's theorem (cf. Corollary 3.14), the Fourier series of such a function f converges to $f(t)$ at almost every point $t \in \mathbb{T}$.

In 1923, A. Kolmogorov proved that there exist L^1-functions on $\mathbb{T}$ whose Fourier series diverge pointwise everywhere (see, for instance, Katznelson [27], and Grafakos [16]).

For L^2-functions f on $\mathbb{T}$, however, N. Lusin had conjectured around 1915 that the Fourier series of such a function converges pointwise to the value $f(t)$ for almost every $t \in \mathbb{T}$. This famous conjecture of Lusin was finally proven in 1966 by L. Carleson [3] in a groundbreaking and highly profound work.

An extension of Carleson's theorem to the case of L^p-functions with any $p > 1$ was achieved about one year later by R. Hunt [26].

An alternative approach to this "Carleson–Hunt theorem" was developed in 1973 by C. Fefferman [10], whose ideas formed the foundation for a fundamental new methodology, which is now known as "time-frequency analysis". By further refinements of this methodology, M. Lacey and C. Thiele [29] devised a significantly simplified (but still quite demanding) proof in 2000, which is presented in a more detailed form in Grafakos' textbook [17].

For the Fourier transform on $\mathbb{T}$, the **Riemann–Lebesgue Lemma** also holds: $\hat{f}$ vanishes at infinity on $\mathbb{Z}$ for every $f \in L^1(\mathbb{T})$, i.e.,

$$\lim_{|k|\to\infty} |\hat{f}(k)| = 0 \quad \text{for every } f \in L^1(\mathbb{T}) \tag{3.16}$$

(cf. Exercise 3.2(d)). Using the formula

$$\widehat{D^j f}(k) = k^j \hat{f}(k), \quad k \in \mathbb{Z}, f \in C^j(\mathbb{T}), \tag{3.17}$$

the proof given in Theorem 2.22 can also be easily transferred. Indeed, (3.17) implies that

$$|\hat{f}(k)| \le \|D^j f\|_1 |k|^{-j}, \quad k \in \mathbb{Z} \setminus \{0\}, \tag{3.18}$$

if $f \in C^j(\mathbb{T})$.

Let now

$$c_\infty(\mathbb{Z}) = \{a : \mathbb{Z} \to \mathbb{C} : \lim_{|k|\to\infty} |a(k)| = 0\}$$

denote the Banach space of all "null" sequences on $\mathbb{Z}$, endowed with the supremum norm $\|\cdot\|_\infty$. It follows from (3.18) that $\hat{f} \in c_\infty(\mathbb{Z})$ if $f \in C^1(\mathbb{T})$. Furthermore, by Theorem 3.6 and (3.7), it is easily seen that the space $C^\infty(\mathbb{T}) \subset C^1(\mathbb{T})$ is dense in $L^1(\mathbb{T})$.

Analogous to Corollary 2.23, the following holds:

Theorem 3.10 *The Fourier transform $\mathcal{F} : L^1(\mathbb{T}) \to c_\infty(\mathbb{Z}), f \mapsto \hat{f}$, is a continuous, linear homomorphism from the involutive Banach algebra $(L^1(\mathbb{T}), +, *, ^*; \|\cdot\|_1)$ into the involutive Banach algebra $(c_\infty(\mathbb{Z}), +, \cdot, \overline{\ }; \|\cdot\|_\infty)$.*

In particular, for $f, g \in L^1(\mathbb{T})$, we have:

$$(f * g)^\wedge = \hat{f}\hat{g},$$
$$(f^*)^\wedge = \overline{\hat{f}}.$$

By the Fourier inversion formula, the **uniqueness theorem** also holds here, i.e., the Fourier transform $\mathcal{F} : L^1(\mathbb{T}) \to c_\infty(\mathbb{Z})$ is injective.

Finally, let us consider the Fourier transform on $L^2(\mathbb{T})$. Since $L^2(\mathbb{T}) \subset L^1(\mathbb{T})$, the Fourier transform $\hat{f}(k)$ here is even pointwise defined, and $\hat{f}$ lies in $c_\infty(\mathbb{Z})$ for $f \in L^2(\mathbb{T})$.

Theorem 3.11 (Plancherel's theorem for $\mathbb{T}$) *The Fourier transform $\mathcal{F}$ is a linear isometric operator from $L^2(\mathbb{T})$ onto $\ell^2(\mathbb{Z})$; in particular, for all $f, g \in L^2(\mathbb{T})$, Parseval's formula holds:*

$$\int_{\mathbb{T}} f(t)\overline{g(t)}\mathrm{d}t = \sum_{k\in\mathbb{Z}} \hat{f}(k)\overline{\hat{g}}(k). \tag{3.19}$$

Proof. Let $f \in L^2(\mathbb{T})$, and consider $\psi \in C^\infty(\mathbb{T})$. Then $f * \psi^* \in C^\infty(\mathbb{T})$, so $(f * \psi^*)^\wedge = \hat{f}\overline{\hat{\psi}} \in \ell^1(\mathbb{Z})$ (why?). By the Fourier inversion Theorem 3.8, it follows that

$$\begin{aligned}(f, \psi) &:= \int_{\mathbb{T}} f(t)\overline{\psi}(t)\mathrm{d}t = f * \psi^*(0) \\ &= \sum_{k\in\mathbb{Z}} \widehat{f * \psi^*}(k) = \sum_{k\in\mathbb{Z}} \hat{f}(k)\overline{\hat{\psi}}(k) \\ &=: (\hat{f}, \hat{\psi}).\end{aligned} \tag{3.20}$$

In particular, $\|f\|_2 = \|\hat{f}\|_2$ if f is smooth.

For general $f, g \in L^2(\mathbb{T})$, we choose a sequence $\{\psi_j\}_j \subset C^\infty(\mathbb{T})$ such that $g = \lim_{j\to\infty} \psi_j$ in $L^2(\mathbb{T})$. Using (3.20), we have:

$$(f,g) = \lim_{j\to\infty} (f,\psi_j) = \lim_{j\to\infty} (\hat{f}, \widehat{\psi_j}).$$

Since $\|\psi_j\|_2 = \|\widehat{\psi_j}\|_2$, the sequence $\{\widehat{\psi_j}\}$ forms a Cauchy sequence in $\ell^2(\mathbb{Z})$ and is thus convergent towards a limit $h \in \ell^2(\mathbb{Z})$. On the other hand (3.6) shows that the sequence $\{\psi_j\}_j$ also converges in $L^1(\mathbb{T})$ towards g, so that $\hat{g}(k) = \lim_{j\to\infty} \widehat{\psi_j}(k)$ for every $k \in \mathbb{Z}$. Thus, $h = \hat{g}$, i.e., $\hat{g} \in \ell^2(\mathbb{Z})$, and

$$(f,g) = \lim_{j\to\infty} (\hat{f}, \widehat{\psi_j}) = (\hat{f}, \hat{g}).$$

Consider finally the linear subspace

$$c_0(\mathbb{Z}) := \{a \in \ell^2(\mathbb{Z}) : a(k) = 0 \text{ for all but finitely many } k \in \mathbb{Z}\}$$

of $\ell^2(\mathbb{Z})$. Clearly, $c_0(\mathbb{Z}) \subset \mathcal{F}(L^2(\mathbb{T}))$ (cf. Proposition 1.3), and $c_0(\mathbb{Z})$ is dense in $\ell^2(\mathbb{Z})$. Moreover, since $\mathcal{F}$ is isometric and $L^2(\mathbb{T})$ is complete, $\mathcal{F}(L^2(\mathbb{T}))$ is complete, hence closed in $\ell^2(\mathbb{Z})$, and thus we see that $\mathcal{F}(L^2(\mathbb{T})) = \ell^2(\mathbb{Z})$. □

Corollary 3.12 *The characters $\{e_k\}_{k\in\mathbb{Z}}$ form a complete orthonormal system in $L^2(\mathbb{T})$.*

Proof. By the identities (1.13) in Proposition 1.1, $\{e_k\}_{k\in\mathbb{Z}}$ is an orthonormal system. Its completeness follows from the Plancherel identity $\|f\|_2 = \|\hat{f}\|_2$, since $(f, e_k) = \hat{f}(k)$, so that $\|f\|_2^2 = \sum_k |(f, e_k)|^2$. □

3.2 Remarks on the Concept of the Dual Group*

While the Fourier transform of an L^1-function on $\mathbb{R}^n$ is again a function on $\mathbb{R}^n$, the Fourier transform of an L^1-function on $\mathbb{T}$ is a function on the additive group $\mathbb{Z}$, not on $\mathbb{T}$! How can this difference be explained?

For every locally compact abelian group G, its dual group $\widehat{G}$ can be defined as follows:

$\widehat{G}$ consists of all **unitary characters** of G, i.e., all continuous group homomorphisms $\chi : G \to S^1$ from G into the multiplicative group $S^1 = \{z \in \mathbb{C} : |z| = 1\}$. Since the pointwise product $\chi_1\chi_2$ of any two characters $\chi_1, \chi_2 \in \widehat{G}$ is again a character of G, this defines an abelian product on $\widehat{G}$, making $\widehat{G}$ a group as well (the inverse of $\chi \in \hat{G}$ is simply the complex conjugate character $\overline{\chi}$). The topology on $\widehat{G} \subset C(G)$ is chosen to be the topology of locally uniform convergence on G. In this way, $\widehat{G}$ becomes a locally compact abelian group as well.

We also note that every locally compact abelian group G possesses a **Haar measure** dg, i.e., a translation-invariant, regular Borel measure that is strictly positive on all non-empty open sets. Furthermore, this measure is unique up to a strictly positive

factor. This generalizes the Lebesgue measure on $\mathbb{R}^n$. If G is compact (like $\mathbb{T}$), the Haar measure is typically normalized such that $\int_G 1 \, \mathrm{d}g = 1$. If G is discrete (like $\mathbb{Z}$), $\mathrm{d}g$ is chosen as the counting measure on G.

The **Fourier transform** $\mathcal{F}f$ of a function $f \in L^1(G)$ is then defined as the function on the dual group $\widehat{G}$ given by

$$\mathcal{F}f(\chi) = \hat{f}(\chi) := \int_G f(g)\overline{\chi(g)} \, \mathrm{d}g, \qquad \chi \in \hat{G}. \tag{3.21}$$

Using these concepts, many foundational aspects of Fourier analysis described earlier for $\mathbb{R}^n$ or $\mathbb{T}$ can indeed be generalized to arbitrary locally compact abelian groups (and, with the help of unitary representation theory, even to non-commutative groups). For further details and the subsequent considerations, I refer to the relevant literature, such as Katznelson [27], Rudin [32], Folland [13], or Deitmar [6].

For $G = \mathbb{R}^n$, the mapping $\xi \mapsto e_\xi$ defines an isomorphism between the locally compact abelian group $\mathbb{R}^n$ and its dual group $\widehat{\mathbb{R}^n}$ (see Exercise 2.6), i.e., $\widehat{\mathbb{R}^n} \simeq \mathbb{R}^n$. This is the deeper reason why the Fourier transform of a function on $\mathbb{R}^n$ can again be defined as a function on $\mathbb{R}^n$.

For the group $\mathbb{T}$, the unitary characters are precisely the characters $e_k(t) = \mathrm{e}^{ikt}$, $k \in \mathbb{Z}$, and the mapping $k \mapsto e_k$ defines an isomorphism between the locally compact abelian group $\mathbb{Z}$ and the dual group $\widehat{\mathbb{T}}$ of $\mathbb{T}$, i.e., $\widehat{\mathbb{T}} \simeq \mathbb{Z}$. This isomorphism allows us to regard the Fourier transform $\hat{f}$ of a function $f \in L^1(\mathbb{T})$, abstractly defined on $\widehat{\mathbb{T}}$ via (3.21), again as a function on $\mathbb{Z}$.

Finally, the unitary characters of the group $\mathbb{Z}$ are of the form

$$e_t(k) := e_k(t) = \mathrm{e}^{ikt}, \qquad k \in \mathbb{Z}, \tag{3.22}$$

where $t \in \mathbb{T}$, since every unitary character $\chi \in \widehat{\mathbb{Z}}$ is already determined by the value $\chi(1) \in S^1$. Thus, the mapping $t \mapsto e_t$ defines an isomorphism between the group $\mathbb{T}$ and the dual group $\widehat{\mathbb{Z}}$ of $\mathbb{Z}$, , i.e., $\widehat{\mathbb{Z}} \simeq \mathbb{T}$.

In particular, these considerations show that for the bi-dual groups, the following holds:

$$\widehat{\widehat{\mathbb{R}^n}} \simeq \mathbb{R}^n, \quad \widehat{\widehat{\mathbb{T}}} \simeq \mathbb{T}, \quad \widehat{\widehat{\mathbb{Z}}} \simeq \mathbb{T}.$$

More generally, according to **Pontryagin's duality theorem**, the bidual group $\widehat{\widehat{G}}$ of any locally compact abelian group G is always isomorphic to G, where the isomorphism from G to $\widehat{\widehat{G}}$ is given by

$$g \mapsto \chi_g, \qquad \text{with} \quad \chi_g(\eta) := \eta(g), \quad \eta \in \widehat{G}.$$

If we then define the **Fourier cotransform** $\overline{\mathcal{F}}h$ of a function $h \in L^1(\widehat{G})$ by

$$\overline{\mathcal{F}}h(g) := \int_{\hat{G}} h(\eta)\eta(g) \, \mathrm{d}\eta, \qquad g \in G,$$

then, with appropriate normalizations of the Haar measures on G and $\widehat{G}$, the Fourier inversion formula remains valid:

$$f(g) = \overline{\mathcal{F}}(\mathcal{F}f)(g) \qquad \text{for almost every } g \in G, \tag{3.23}$$

provided $f \in L^1(G)$ and $\hat{f} \in L^1(\widehat{G})$.

Finally, let us consider the case $G = \mathbb{T}$. Here, the Fourier cotransform $\overline{\mathcal{F}}a$ of a function $a \in \ell^1(\mathbb{Z})$ is given by the Fourier series

$$\overline{\mathcal{F}}a(t) := \sum_{k\in\mathbb{Z}} a(k)\mathrm{e}^{ikt}, \quad t \in \mathbb{T},$$

and the Fourier inversion formula (3.15) coincides with (3.23).

3.3 On Pointwise Convergence of Fourier Series

For $f \in \mathcal{L}^1(\mathbb{T})$ let us again denote by

$$S_N f(t) := \sum_{k=-N}^{N} \hat{f}(k)\mathrm{e}^{ikt}$$

the N-th partial sum of the Fourier series $\sum_{k\in\mathbb{Z}} \hat{f}(k)\mathrm{e}^{ikt}$. Then

$$\begin{aligned} S_N f(t) &= \sum_{k=-N}^{N} \frac{1}{2\pi}\int_0^{2\pi} f(s)\mathrm{e}^{ik(t-s)}\,\mathrm{d}s \\ &= \frac{1}{2\pi}\int_0^{2\pi} f(s)\left(\sum_{k=-N}^{N}\mathrm{e}^{ik(t-s)}\right)\mathrm{d}s, \end{aligned}$$

so that

$$S_N f(t) = f * D_N(t), \quad t \in \mathbb{T}, \tag{3.24}$$

where

$$D_N(x) := \sum_{k=-N}^{N} \mathrm{e}^{ikx}, \quad x \in \mathbb{R},$$

is the so-called **Dirichlet kernel** of order $N \in \mathbb{N}$. Summing a geometric series yields

$$D_N(x) = \begin{cases} \frac{\sin(N+\frac{1}{2})x}{\sin\frac{x}{2}}, & x \notin 2\pi\mathbb{Z}, \\ 2N+1, & x \in 2\pi\mathbb{Z}. \end{cases} \tag{3.25}$$

We want to provide sufficient conditions under which the Fourier series of f converges to $f(t)$ at a point $t \in \mathbb{T}$. For this purpose, we will use the following properties of D_N, which follow directly from its definition:

$$\int_{\mathbb{T}} D_N(t)\mathrm{d}t = 1, \tag{3.26}$$

$$\check{D}_N = D_N. \tag{3.27}$$

However, it can be shown (see Exercise 3.1) that

$$\int_{\mathbb{T}} |D_N(t)|\,\mathrm{d}t \sim \log N \quad \text{for } N \to \infty,$$

so the sequence D_N, in contrast to the sequence Φ_{ε_N} of Poisson kernels used in the proof of the Fourier inversion formula, unfortunately does not form an approximate identity. (This is why the question of pointwise convergence of Fourier series is generally very challenging.) Due to $\check{D}_N = D_N$, we will use the "symmetric" interval $[-\pi, \pi[$ as fundamental domain.

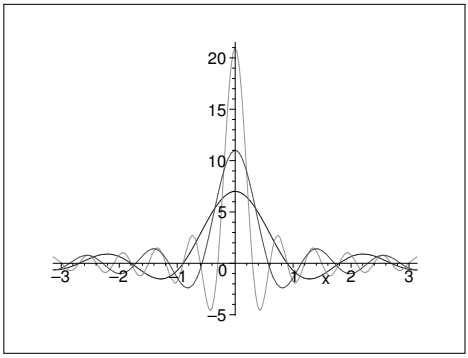


Fig. 3.1 Dirichlet kernels, $N = 3, 5, 10$

Theorem 3.13 (Dini) *Let $f \in \mathcal{L}^1(\mathbb{T})$, and let $t \in \mathbb{T}$. Suppose there exists an $A \in \mathbb{C}$ such that*

$$\int_0^\pi \left| \frac{f(t+s) + f(t-s)}{2} - A \right| \frac{\mathrm{d}s}{s} < \infty. \tag{3.28}$$

Then we conclude that

$$\lim_{N\to\infty} S_N f(t) = A.$$

Proof. From (3.24) and (3.27), we have

$$\begin{aligned} S_N f(t) &= f * D_N(t) = D_N * f(t) \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t-s) D_N(s)\, \mathrm{d}s \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t+s) D_N(s)\, \mathrm{d}s. \end{aligned}$$

By averaging over both plus and minus signs in s, we get

$$S_N f(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{f(t+s) + f(t-s)}{2} D_N(s)\, \mathrm{d}s.$$

Invoking also (3.26), we obtain that

$$S_N f(t) - A = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{f(t+s) + f(t-s)}{2} - A \right) D_N(s)\, \mathrm{d}s.$$

Let J_N denote the right-hand side of this equation. We aim to show that $\lim_{N\to\infty} J_N = 0$. To this end, for $s \in]-2\pi, 2\pi[$, set

$$h(s) := \begin{cases} \frac{1}{\sin(s/2)} - \frac{1}{s/2}, & s \neq 0, \\ 0, & s = 0. \end{cases}$$

It is straightforward to verify that h is analytic on $]-2\pi, 2\pi[$; in particular, $h \in C^\infty(]-2\pi, 2\pi[)$. Moreover, from (3.25), we get

$$\begin{aligned} 2\pi J_N = &\int_{-\pi}^{\pi} \left(\frac{f(t+s) - f(t-s)}{2} - A \right) \frac{\sin(N+\frac{1}{2})s}{s/2}\, \mathrm{d}s \\ &+ \int_{-\pi}^{\pi} \left(\frac{f(t+s) - f(t-s)}{2} - A \right) h(s) \sin(N+\frac{1}{2})s\, \mathrm{d}s. \end{aligned}$$

Define

$$\begin{aligned} f_1(s) &:= \left(\frac{f(t+s) + f(t-s)}{2} - A \right) \frac{1}{s/2}, \\ f_2(s) &:= \left(\frac{f(t+s) + f(t-s)}{2} - A \right) h(s). \end{aligned}$$

The theorem's assumptions imply that $f_1 \in \mathcal{L}^1(\mathbb{T})$. Since h is bounded on the compact interval $[-\pi, \pi]$, it follows that $f_2 \in \mathcal{L}^1(\mathbb{T})$ as well. Therefore:

$$\begin{aligned} 2\pi J_N &= \sum_{j=1}^{2} \int_{-\pi}^{\pi} f_j(s) \sin(N + \frac{1}{2})s \, \mathrm{d}s \\ &= \sum_{j=1}^{2} \frac{1}{2i} \left[\int_{-\pi}^{\pi} f_j(s) \mathrm{e}^{i(N+\frac{1}{2})s} \, \mathrm{d}s - \int_{-\pi}^{\pi} f_j(s) \mathrm{e}^{-i(N+\frac{1}{2})s} \, \mathrm{d}s \right]. \end{aligned}$$

Applying the Riemann–Lebesgue Lemma (on $\mathbb{R}$) to the above integrals in the bracket, we conclude that $J_N \to 0$ as $N \to \infty$. □

Corollary 3.14 (Dirichlet's Theorem) *Let $f : \mathbb{T} \to \mathbb{C}$ be piecewise continuously differentiable, i.e., there exists a partition $-\pi = \tau_0 < \tau_1 < \cdots < \tau_{m+1} = \pi$ of the interval $[-\pi, \pi]$ such that $f|_{]\tau_j, \tau_{j+1}[}$ can be extended to a continuously differentiable function on $[\tau_j, \tau_{j+1}]$ for each $j = 0, \ldots, m$. For $t \in [-\pi, \pi]$, denote by $f(t \pm 0)$ the one-sided limits*

$$f(t \pm 0) := \lim_{\substack{\tau \to 0 \\ \tau > 0}} f(t \pm \tau).$$

Then we have

$$\lim_{N \to \infty} S_N f(t) = \frac{f(t+0) + f(t-0)}{2}.$$

In particular, if f is continuous at t, it follows that

$$\lim_{N \to \infty} S_N f(t) = f(t).$$

Proof. Set $A := \frac{f(t+0)+f(t-0)}{2}$ in Dini's theorem. For $s > 0$, we then have

$$\frac{1}{s} \left(\frac{f(t+s) + f(t-s)}{2} - A \right) = \frac{1}{2} \left[\frac{f(t+s) - f(t+0)}{s} - \frac{f(t-0) - f(t-s)}{s} \right].$$

Since f is piecewise continuously differentiable, the limits

$$\frac{f(t+s) - f(t+0)}{s} \qquad \text{and} \qquad \frac{f(t-0) - f(t-s)}{s}$$

exist as $s \to 0$, coinciding with the right-sided derivative $f'(t+0)$ and the left-sided derivative $f'(t-0)$ of f at t, respectively. Thus, the integrand

$$g(s) := \left| \frac{f(t+s) + f(t-s)}{2} - A \right| \cdot \frac{1}{s}$$

is continuous at 0. Since f is piecewise continuous, this shows that g is piecewise continuous as well. In particular, $\int_0^{\pi} g(s) \, \mathrm{d}s < \infty$. Hence, Dirichlet's theorem follows from Dini's theorem (historically, however, Dini's theorem came after Dirichlet's). □

3.4 The Poisson Summation Formula

Let f be a "well-behaved" function on $\mathbb{R}$, for instance, a Schwartz function. Then we can associate f with a 2π-periodic function in the following two ways:

(a) Define
$$f_1(x) := 2\pi \sum_{\ell\in\mathbb{Z}} f(x + \ell 2\pi), \quad x \in \mathbb{R}.$$

(b) Consider the Fourier transform $\hat{f}$ restricted to integers and associate the coefficients $\hat{f}(k)$, $k \in \mathbb{Z}$, with the Fourier series
$$f_2(x) := \sum_{k\in\mathbb{Z}} \hat{f}(k)\mathrm{e}^{ikx}.$$

Then both f_1 and f_2 are evidently 2π-periodic, so we can view them as functions on $\mathbb{T}$. We will see that under suitable assumptions on f these functions f_1 and f_2 agree.

Theorem 3.15 *Let $f \in \mathcal{L}^1(\mathbb{R})$ be continuous and satisfy the following conditions:*

(i) *The series $\sum_{\ell\in\mathbb{Z}} f(x+\ell 2\pi)$ converges locally uniformly (and hence uniformly on any compact set) on $\mathbb{R}$.*
(ii) *The series $\sum_{k\in\mathbb{Z}} \hat{f}(k)$ converges absolutely.*

Then the **Poisson summation formula** *holds:*
$$2\pi \sum_{\ell\in\mathbb{Z}} f(2\pi\ell) = \sum_{k\in\mathbb{Z}} \hat{f}(k). \tag{3.29}$$

Remarks 3.16 (a) If f satisfies the assumptions of the theorem, then the same is true of any shifted function $\lambda_y f := f(\cdot - y)$. Applying the Poisson summation formula to $\lambda_{-x} f$ thus yields the following generalization of (3.29):
$$2\pi \sum_{\ell\in\mathbb{Z}} f(x + \ell 2\pi) = \sum_{k\in\mathbb{Z}} \hat{f}(k)\mathrm{e}^{ikx}. \tag{3.30}$$

(b) Let $a > 0$. Replacing f with $af(a\,\cdot)$, we obtain the following generalization of (3.29):
$$2\pi a \sum_{\ell\in\mathbb{Z}} f(\ell 2\pi a) = \sum_{k\in\mathbb{Z}} \hat{f}\left(\frac{k}{a}\right). \tag{3.31}$$

(c) Conditions (i) and (ii) of the theorem are satisfied if $f \in \mathcal{L}^1(\mathbb{R}) \cap C(\mathbb{R})$ satisfies estimates of the form
$$f(x) = O((1 + |x|)^{-1-\delta}) \quad \text{as } |x| \to \infty,$$
and
$$\hat{f}(\xi) = O((1 + |\xi|)^{-1-\delta}) \quad \text{as } |\xi| \to \infty,$$
for some $\delta > 0$. In particular, this is true for $f \in \mathcal{S}(\mathbb{R})$.

Proof. By (i), the function

$$f_1 : t \mapsto f_1(t) := 2\pi \sum_{\ell \in \mathbb{Z}} f(t + \ell 2\pi) \tag{3.32}$$

is continuous and 2π-periodic. We compute its Fourier coefficients: For $k \in \mathbb{Z}$, the local uniform convergence of the series (3.32) implies that

$$\hat{f}_1(k) = \frac{1}{2\pi} \int_0^{2\pi} f_1(t) \mathrm{e}^{-ikt} \mathrm{d}t = \sum_{\ell \in \mathbb{Z}} \int_{[0,2\pi[} f(t + \ell 2\pi) \mathrm{e}^{-ikt} \mathrm{d}t.$$

Translating the variable t in these integrals, this becomes

$$\hat{f}_1(k) = \sum_{\ell \in \mathbb{Z}} \int_{[\ell 2\pi, (\ell+1)2\pi[} f(x) \mathrm{e}^{-ikx} \mathrm{d}x = \int_{\mathbb{R}} f(x) \mathrm{e}^{-ikx} \mathrm{d}x = \hat{f}(k).$$

By (ii), the Fourier inversion formula (3.15) applies to f_1, and we conclude that

$$2\pi \sum_{\ell \in \mathbb{Z}} f(\ell 2\pi) = f_1(0) = \sum_{k \in \mathbb{Z}} \hat{f}_1(k) = \sum_{k \in \mathbb{Z}} \hat{f}(k).$$

Note that, due to the continuity of f_1 and condition (ii), the Fourier inversion formula (3.15) holds for every $t \in \mathbb{T}$. □

Examples 3.17 (a) Let $f = \mathbf{1}_{[-\frac{1}{2},\frac{1}{2}]} * \mathbf{1}_{[-\frac{1}{2},\frac{1}{2}]}$. Then $f(x) = (1 - |x|)_+$, and since

$$\widehat{\mathbf{1}_{[-\frac{1}{2},\frac{1}{2}]}}(\xi) = \frac{\sin(\xi/2)}{\xi/2},$$

it follows that

$$\hat{f}(\xi) = \frac{\sin^2(\xi/2)}{(\xi/2)^2}.$$

Choosing $a := 1/\pi$ in (3.31), we find that

$$2 \sum_{\ell \in \mathbb{Z}} f(2\ell) = \sum_{k \in \mathbb{Z}} \frac{\sin^2(k\pi/2)}{(k\pi/2)^2}.$$

Thus,

$$2 = 1 + 2 \sum_{j=0}^{\infty} \frac{1}{((2j+1)\pi/2)^2},$$

which implies

$$\sum_{j=0}^{\infty} \frac{1}{(2j+1)^2} = \frac{\pi^2}{8}.$$

(b) Choosing as f the Gaussian kernel $\Phi_1(x) = \mathrm{e}^{-x^2/2}$, using Lemma 2.24 and choosing $a := \sqrt{\frac{s}{2\pi}}$ in Remark 3.16(c), we find that

$$\sqrt{2\pi s} \sum_{\ell \in \mathbb{Z}} \mathrm{e}^{-\ell^2 \pi s} = \sqrt{2\pi} \sum_{k \in \mathbb{Z}} \mathrm{e}^{-\frac{k^2 \pi}{s}}, \quad \text{for every } s > 0.$$

This yields the well-known **functional equation**

$$s^{-1/2} \vartheta(1/s) = \vartheta(s), \quad s > 0, \tag{3.33}$$

for the **"theta function"**

$$\vartheta(s) := \sum_{n=-\infty}^{\infty} \mathrm{e}^{-\pi n^2 s}, \quad s > 0,$$

which is a special case of the more general Jacobi theta function Θ.

3.5 Multidimensional Fourier Series

The theory of Fourier series generalizes to higher dimensions without difficulty.

It is clear that any function f on the n-dimensional torus $\mathbb{T}^n$ can be identified with a multiple 2π-periodic function f_{per} on $\mathbb{R}^n$, i.e.,

$$f_{\mathrm{per}}(x_1 + 2\pi k_1, \ldots, x_n + 2\pi k_n) = f_{\mathrm{per}}(x_1, \ldots, x_n)$$

for all $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ and $k_1, \ldots, k_n \in \mathbb{Z}$.

The discrete subgroup $2\pi\mathbb{Z}$ of $\mathbb{R}$ is thus to be replaced here by the lattice group $\Gamma := (2\pi\mathbb{Z})^n$ in $\mathbb{R}^n$ (note that $\mathbb{T}^n \simeq \mathbb{R}^n/\Gamma$, under an obvious group isomorphism). As a fundamental domain for the action of Γ on $\mathbb{R}^n$, we can choose, for example, the cube $A := [0, 2\pi[^n$. The integral over the n-torus $\mathbb{T}^n$, with coordinates $t = (t_1, \ldots, t_n) \in \mathbb{T}^n$, is then defined by

$$\int_{\mathbb{T}^n} f(t)\, \mathrm{d}t := (2\pi)^{-n} \int_A f_{\mathrm{per}}(x)\, \mathrm{d}x.$$

The continuous characters of the group $\mathbb{T}^n$ are precisely the tensor products

$$e_{\mathbf{k}}(t) := \mathrm{e}^{i\mathbf{k}\cdot t} = \mathrm{e}^{ik_1 t_1} \cdots \mathrm{e}^{ik_n t_n}, \quad \mathbf{k} = (k_1, \ldots, k_n) \in \mathbb{Z}^n,$$

of characters of the one-dimensional torus. The Fourier transform of an integrable function $f : \mathbb{T}^n \to \mathbb{C}$ is defined by

$$\hat{f}(\mathbf{k}) := \int_{\mathbb{T}^n} f(t) \overline{e_{\mathbf{k}}(t)}\, \mathrm{d}t, \quad \mathbf{k} \in \mathbb{Z}^n,$$

where $\mathrm{d}t = \mathrm{d}t_1 \cdots \mathrm{d}t_n$ is the Haar measure on $\mathbb{T}^n$.

All results from Section 3.1 can be immediately and straightforwardly extended to the higher-dimensional case.

For example, in the proof of the Fourier inversion formula, we can use the n-fold tensor product $\Phi_\varepsilon^n(t) := \Phi_\varepsilon(t_1)\cdots\Phi_\varepsilon(t_n)$, $\varepsilon > 0$, of the "Poisson kernels" Φ_ε from the proof of Theorem 3.8 as an approximate identity. This yields, with a nearly identical proof, the **Fourier inversion formula on $\mathbb{T}^n$:**

$$f(t) = \sum_{\mathbf{k}\in\mathbb{Z}^n} \hat{f}(\mathbf{k})\mathrm{e}^{i\mathbf{k}\cdot t}, \quad \text{for a.e. } t \in \mathbb{T}^n, \tag{3.34}$$

if $f \in \mathcal{L}^1(\mathbb{T}^n)$ and $\sum_{\mathbf{k}\in\mathbb{Z}^n} |\hat{f}(\mathbf{k})| < \infty$.

Further details on the theory of higher-dimensional Fourier series can be found, for instance, in Stein–Weiss [43].

3.6 Exercises

3.1 Show that the L^1-norm of the Dirichlet kernel D_N behaves as follows for large N:

$$\int_{\mathbb{T}} |D_N(t)|\mathrm{d}t \sim \log N \quad \text{for } N \to \infty,$$

i.e., there exists a constant $C \geq 1$ such that $(1/C)\log N \leq \|D_N\|_1 \leq C \log N$ for all sufficiently large N.

3.2 For $N \in \mathbb{N}$, let $\sigma_N f(x)$ denote the **Cesàro mean**

$$\sigma_N f(x) := \frac{1}{N+1}\sum_{n=0}^{N} S_n f(x)$$

of the partial sums $S_n f(x)$ of the Fourier series of $f \in L^1(\mathbb{T})$. Show:

a) We can write $\sigma_N f = f * F_N$, where

$$F_N(x) := \frac{1}{N+1}\sum_{n=0}^{N} D_n(x) = \sum_{k=-N}^{N} \frac{N+1-|k|}{N+1}\mathrm{e}^{ikx}$$

denotes the **Fejér kernel** .

b) Let $y \in [-\pi, \pi[$. Then $F_N(0) = N+1$, and for $y \neq 0$:

$$F_N(y) = \frac{1}{N+1}\left(\frac{\sin\frac{(N+1)y}{2}}{\sin\frac{y}{2}}\right)^2.$$

c) The sequence $\{F_N\}_N$ forms an approximate identity on $\mathbb{T}$, i.e., it satisfies the conditions (i)–(iii) of an approximate identity as $N \to \infty$ in place of $\varepsilon \to 0$.

d) The space of trigonometric polynomials is dense in $L^1(\mathbb{T})$.

3.3 Let $\varphi(t) = \sum_{k=-n}^{n} a_k e^{ikt}$ be a trigonometric polynomial of degree $\leq n$ on $\mathbb{T}$. Show that φ has at most $2n$ zeros.

3.4 Let $f : \mathbb{T} \to \mathbb{R}$ be defined by

$$f(x) := \begin{cases} \pi/4, & x \in]-\pi/2, \pi/2[, \\ -\pi/4, & x \in [-\pi, -\pi/2[\cup]\pi/2, \pi[, \\ 0, & x = \pm\pi/2. \end{cases}$$

Let $S_N f(t) := \sum_{k=-N}^{N} \widehat{f}(k) e^{ikt}$ denote the N-th partial sum of the Fourier series of f.

i) Show that $f(t) = \lim_{N\to\infty} S_N f(t)$ for every $t \in \mathbb{T}$.

ii) Deduce that $\pi/4 = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} \pm \cdots$.

3.5 (Fejér's Lemma) Prove: If $f \in L^1(\mathbb{T})$ and $g \in L^\infty(\mathbb{T})$, then

$$\lim_{n\to\infty} \frac{1}{2\pi} \int_0^{2\pi} f(t) g(nt) \mathrm{d}t = \hat{f}(0)\hat{g}(0).$$

Hints. First, prove this for $f = e_k$, $k \in \mathbb{Z}$, using the decomposition $[0, 2\pi[= \bigcup_{k=0}^{n-1} [2\pi\frac{k}{n}, 2\pi\frac{k+1}{n}[$. For the general case, approximate f by trigonometric polynomials (cf. Exercise 3.2).

Thus, for $n \to \infty$, f and $g \circ (n\,\mathrm{Id})$ behave somewhat like independent random variables.

3.6 (*) Prove that for all $N \in \mathbb{N}$ and $t \in [0, 2\pi[$:

$$\left| \sum_{k=1}^{N} \frac{\sin kt}{k} \right| \leq \frac{\pi}{2} + 1.$$

Hints: Consider $f(t) := (\pi - t)/2$ on $[0, 2\pi[$ and use the properties of the Fejér kernel from Exercise 3.2. Compare $\sigma_N f(t)$ and $S_N f(t)$.

3.7 Let h be the function on $\mathbb{T}$ defined by $h(t) := t$ for $t \in]-\pi, \pi[$, and $h(\pi) := 0$, known as the "sawtooth function" when viewed as a 2π-periodic function on $\mathbb{R}$.

Note that $h(\pi - 0) = \pi$, $h(-\pi + 0) = -\pi$. Verify that in contrast the following **Gibbs phenomenon** holds:

$$\lim_{N\to\infty} S_N h(\pi - \pi/N) = A\pi, \quad \lim_{N\to\infty} S_N h(-\pi + \pi/N) = -A\pi,$$

with $A := \frac{2}{\pi} \int_0^\pi \frac{\sin t}{t} \mathrm{d}t$. Moreover, $A > 1.17$.

Hints: Express the partial sums of the Fourier series of h as Fourier-sine sums of the form $\sum_{j=1}^{N} b_j \sin jt$, and interpret $S_N h(\pi - \pi/N)$ as a Riemann sum. To estimate A from below, expand the sine in its Taylor series (at least the first 4 terms of this series must be computed!).

3.8 a) Let $f \in C^m(\mathbb{T})$. Show that for $j = 0, \ldots, m$:

$$(D^j f)^\wedge(k) = k^j \widehat{f}(k) \quad \text{for all } k \in \mathbb{Z},$$

and deduce that $\widehat{f}(k) = o(|k|^{-m})$ as $|k| \to \infty$.

b) Devise suitable analogs $\mathcal{S}(\mathbb{T})$ and $s(\mathbb{Z})$ for the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ on the groups $\mathbb{T}$ and $\mathbb{Z}$, respectively.
In particular, the Fourier transform on the torus should provide a linear isomorphism

$$\mathcal{F} : \mathcal{S}(\mathbb{T}) \to s(\mathbb{Z})$$

between these spaces.

3.9 (Reconstruction of Band-Limited Functions) The following result, due to Nyquist and Shannon, plays a significant role in information theory, as it shows how a signal can be reconstructed from measurements (i.e., "samples") taken at suitable (infinite) discrete time sequences.

Let f be a continuous and integrable function on $\mathbb{R}$, whose Fourier transform $\hat{f}$ is supported in the interval $I := [-\pi, \pi]$. Furthermore, assume there exists a $\delta > 0$ such that $f(x) = O((1 + |x|)^{-1-\delta})$. Then the function f is completely determined by its values on $\mathbb{Z}$.

a) More precisely, for every piecewise continuously differentiable function χ on $\mathbb{R}$, which is identically one on I and supported in the interval $I_a := [-a\pi, a\pi]$ ($a \geq 1$), the following **reconstruction formula** holds:

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{a} f\left(\frac{n}{a}\right) K\left(x - \frac{n}{a}\right), \quad x \in \mathbb{R}, \tag{I}$$

where $K := \mathcal{F}^{-1}\chi$.
For $\chi = \mathbf{1}_I$ and $a = 1$, we have in particular:

$$f(x) = \sum_{n=-\infty}^{\infty} f(n) \frac{\sin \pi(x-n)}{\pi(x-n)}, \quad x \in \mathbb{R}. \tag{II}$$

Note that in the case $a > 1$, where the "sampling density" is higher than in the case $a = 1$, we can even choose $\chi \in \mathcal{D}(\mathbb{R})$, so that $K \in \mathcal{S}(\mathbb{R})$, and the convergence of the series (I) is significantly faster than in the case $a = 1$.

b) *Hint:* First use the Poisson summation formula to show that

$$\hat{f}(\xi) = \sum_{n=-\infty}^{\infty} \frac{1}{a} f\left(\frac{n}{a}\right) \mathrm{e}^{-i\frac{n}{a}\xi} \chi(\xi).$$

c) Show that $\int_{-\infty}^{\infty} |f(x)|^2 \, \mathrm{d}x = \sum_{n=-\infty}^{\infty} |f(n)|^2$.

3.10 Let $f \in L^1(\mathbb{T})$ such that $\hat{f}(|n|) = -\hat{f}(-|n|) \geq 0$ for every $n \in \mathbb{Z}$. Show that

$$\sum_{n=1}^{\infty} \frac{1}{n} \hat{f}(n) < \infty.$$

Hints: Define $g(t) := \int_0^t f(s) \, \mathrm{d}s$, for $t \in [-\pi, \pi[$. Show that $\hat{g}(n) = \frac{1}{in} \hat{f}(n)$ for $n \neq 0$, and that $g * F_N(0) \to g(0)$ as $N \to \infty$ (cf. Exercise 3.2).

3.11 Let $\{a_n\}_{n \in \mathbb{N}}$ be a null sequence of non-negative real numbers, which is convex in the following sense:

$$a_{n-1} + a_{n+1} - 2a_n \geq 0 \quad \text{for all } n > 0.$$

Prove that there exists a non-negative function $f \in L^1(\mathbb{T})$ with $\hat{f}(n) = a_{|n|}$.

Hints: Show the following:

i) The sequence $a_n - a_{n+1}$ is monotonically decreasing, and $\sum_{n=0}^{\infty}(a_n - a_{n+1}) = a_0$.
ii) $\lim_{n \to \infty} n(a_n - a_{n+1}) = 0$.
iii) $\sum_{n=1}^{N} n(a_{n-1} + a_{n+1} - 2a_n) = a_0 - a_N - N(a_N - a_{N+1})$.
iv) The function

$$f(t) := \sum_{n=1}^{\infty} n(a_{n-1} + a_{n+1} - 2a_n) F_{n-1}(t)$$

has the desired properties, where F_N denotes the Fejér kernel from Exercise 3.2.

3.12 Using Exercises 3.10 and 3.11, prove that the series

$$\sum_{n=2}^{\infty} \frac{\cos nt}{\log n}$$

is the Fourier series of an L^1-function, while the "conjugate" series

$$\sum_{n=2}^{\infty} \frac{\sin nt}{\log n}$$

is not.

3.13 (*) Using Exercise 3.11, prove the following result:

For every sequence $\{c_n\}_{n\in\mathbb{Z}}$ *of non-negative real numbers with* $c_n \to 0$ *as* $|n| \to \infty$, *there exists a function* $f \in L^1(\mathbb{T})$ *such that* $|\hat{f}(n)| \geq c_n$ *for all* $n \in \mathbb{Z}$.

Thus, the Fourier coefficients of L^1-functions on $\mathbb{T}$ can decay arbitrarily slowly to 0.



Chapter 4
Tempered Distributions

4.1 Introduction

In many areas of analysis, and not least in physics and engineering, one encounters situations where one would like to differentiate functions that are not differentiable in the classical sense, or where it is not clear a priori whether they are differentiable. The theory of distributions, as developed by Laurent Schwartz (1915–2002) in the late 1940s, provides a mathematical framework that rigorously addresses this problem by extending the concept of differentiability to a broader class of objects than differentiable functions—these objects are called "distributions," or "generalized functions" (mostly in countries of the former Soviet Union). In this generalized sense, they are infinitely differentiable.

Physicists, such as Oliver Heaviside and Paul Dirac, had already worked with such "objects" in a formal way, but the theories of Schwartz and others placed their calculi on a solid foundation.

The core idea is to no longer view functions as pointwise-defined objects but instead in terms of their "action" on other, well-behaved functions, known as "test functions" (more precisely, as linear functionals on suitable spaces of test functions). Schwartz's distribution theory, say on $\mathbb{R}^n$, primarily employs the test function spaces $\mathcal{D} = C_0^\infty(\mathbb{R}^n)$ and $\mathcal{S} = \mathcal{S}(\mathbb{R}^n)$. Notably, both of these spaces consist of infinitely differentiable functions.

The underlying foundational approach, using in a suitable manner a duality theory to generalize concepts like pointwise derivatives to "weak derivatives," can also be traced back to work by Sergei Lwowitsch Sobolev (1908–1089).

Historical note. According to Kolmogorov and Fomin [28], the "method of introducing generalized functions ... originated in the work of S.L. Sobolev, published in 1935–36." In his autobiography [36], Schwartz writes in Chapter VI : "One of the precursors who came closest to distributions was the Soviet mathematician Sobolev." He explains in more detail Sobolev's achievements on this topic (which later also led to the theory of Sobolev spaces), but also the shortfalls, and comes to the conclusion that "Sobolev had

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D. Müller, *An Invitation to Fourier Analysis and Distribution Theory*,
Universitext, https://doi.org/10.1007/978-3-032-31550-2_4

taken a first few steps on the right road, and then had been forgotten...". Moreover, he writes "Naturally, I knew absolutely nothing about his work, since after I left the ENS I left for the military service and for the war. We became good friends after the war."

As for choosing the name "distribution", Schwartz explains that he was motivated somewhat by the notion of the distribution of electric charges in space.

Gårding [14] comments that "all these ideas (by Schwartz) were not entirely new, but when they appeared earlier, for instance in Sobolev [37], it was almost always in connection with specific problems". He also mentions earlier works by Weyl from 1940 and Friedrichs from 1944. Moreover, he writes that "Schwartz's broad attack, his radical use of infinitely differentiable functions and his conviction that distributions would be useful almost everywhere in analysis made the difference."

To describe the core idea of distribution theory in more detail, let us consider a "function" $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ (which is itself no longer pointwise defined!). We can associate with f a linear form u_f, for example, on $\mathcal{D}$, via

$$u_f(\varphi) := \int_{\mathbb{R}^n} f(x)\varphi(x)\,\mathrm{d}x, \qquad \varphi \in \mathcal{D}, \tag{4.1}$$

i.e., u_f is a function on $\mathcal{D} = C_0^\infty(\mathbb{R}^n)$. Importantly, f is already determined by the linear form u_f.

Proposition 4.1 *Let $f, g \in L^1_{loc}(\mathbb{R}^n)$ such that $u_f(\varphi) = u_g(\varphi)$ for all $\varphi \in \mathcal{D}$. Then $f = g$.*

Proof. It suffices to consider the case $f, g \in L^1(\mathbb{R}^n)$, since two locally integrable functions f and g are equal almost everywhere if and only if $f\chi = g\chi$ for every function $\chi \in \mathcal{D}$. Let $\{\varphi_j\}_j$ be a Dirac sequence in $\mathcal{D} = C_0^\infty(\mathbb{R}^n)$. For each $x \in \mathbb{R}^n$, we have

$$\begin{aligned} f * \varphi_j(x) &= \int f(y)\varphi_j(x-y)\,\mathrm{d}y = \int f(y)\check{\varphi}_j(y-x)\,\mathrm{d}y \\ &= \int f(y)(\lambda_x\check{\varphi}_j)(y)\,\mathrm{d}y = u_f(\lambda_x\check{\varphi}_j). \end{aligned}$$

Similarly, $g * \varphi_j(x) = u_g(\lambda_x\check{\varphi}_j)$. From $u_f = u_g$, it thus follows that $f * \varphi_j = g * \varphi_j$ for every $j \in \mathbb{N}$. Applying Theorem 2.13, we obtain in the limit as $j \to \infty$ that $f = g$ in $L^1(\mathbb{R}^n)$. □

According to Proposition 4.1, the mapping

$$L^1_{\mathrm{loc}}(\mathbb{R}^n) \ni f \mapsto u_f$$

is injective. In fact, using the more profound "Lebesgue Differentiation Theorem" (see, for example, Stein–Weiss [43, Ch. 1, Theorem 1.25]), it can be shown that for $f \in \mathcal{L}^1_{\mathrm{loc}}$, we even have

$$f(x) = \lim_{j\to\infty} f * \varphi_j(x) = \lim_{j\to\infty} u_f(\lambda_x\check{\varphi}_j) \qquad \text{for a.e. } x \in \mathbb{R}^n.$$

Thus, f can even be "reconstructed" from u_f.

The fundamental idea of distribution theory is, therefore, to no longer consider the function f itself, but rather the linear form u_f. One might then attempt to use all linear forms on $\mathcal{D}$ or $\mathcal{S}$ as "generalized functions." However, the algebraic dual spaces turn out to be far too large for practical analysis. The "correct" approach to the concept of distributions involves endowing the respective test function space with a suitable topology and then considering all **continuous** linear forms on this space as distributions.

Since this is technically simpler for the space $\mathcal{S}$ of Schwartz functions than for $\mathcal{D}$, and since it is even possible to define a Fourier transform for the continuous linear forms on $\mathcal{S}$ (i.e., the so-called "tempered distributions"), we will begin with these.

4.2 Locally Convex Vector Spaces and the Topology on $\mathcal{S}$

For $N \in \mathbb{N}$, the norm $\|\cdot\|_{(N)}$ on $\mathcal{S} = \mathcal{S}(\mathbb{R}^n)$ is defined by

$$\|\varphi\|_{(N)} := \max_{|\alpha| \le N} \sup_{x \in \mathbb{R}^n} (1 + |x|^2)^{N/2} |D^\alpha \varphi(x)|, \quad \varphi \in \mathcal{S}.$$

Note that for every $N \in \mathbb{N}$, there exists a constant $C_N \ge 1$ such that

$$\frac{1}{C_N}(1 + |x|^2)^{N/2} \le \sum_{|\alpha| \le N} |x^\alpha| \le C_N (1 + |x|^2)^{N/2} \quad \text{for all } x \in \mathbb{R}^n \tag{4.2}$$

(see Exercise 2.11).

We will often use the following shorthand:

$$\langle x \rangle := (1 + |x|^2)^{1/2}, \quad x \in \mathbb{R}^n.$$

The term $\langle x \rangle$ is sometimes referred to as the **japanese bracket** of x.

According to (4.2), a C^∞ function φ belongs to $\mathcal{S}$ if and only if $\|\varphi\|_{(N)} < \infty$ for all $N \in \mathbb{N}$.

It is therefore natural to introduce a topology on $\mathcal{S}$ such that a sequence $\{\varphi_k\}_k$ in $\mathcal{S}$ converges to $\varphi \in \mathcal{S}$ if and only if

$$\lim_{k \to \infty} \|\varphi - \varphi_k\|_{(N)} = 0 \quad \text{for every } N \in \mathbb{N}.$$

Since we will later consider even uncountable families of seminorms in the context of other spaces of test functions, we shall develop the construction of the topology in a more general framework.

Definitions. Let E be a vector space over the field $\mathbb{K}$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$.

a) A **seminorm** on E is a mapping

$$\|\cdot\| : E \to \mathbb{R}_{\ge 0}, \quad v \mapsto \|v\|,$$

with the following properties:

(i) $\|v + w\| \leq \|v\| + \|w\|$ for all $v, w \in E$;
(ii) $\|\lambda v\| = |\lambda|\|v\|$ for all $\lambda \in \mathbb{K}, v \in E$.

A seminorm $\|\cdot\|$ on E is a **norm** if, additionally, the following holds for every $v \in E$:

(iii) $\|v\| = 0 \iff v = 0$.

b) Let $\mathcal{N} = \{\|\cdot\|_i\}_{i\in I}$ be a family of seminorms on E. The family $\mathcal{N}$ is called **separating** if for every $v \in E$,

$$\|v\|_i = 0 \text{ for all } i \in I \iff v = 0.$$

In particular, this is true if one of the seminorms $\|\cdot\|_i$ is already a norm, as is the case for $E = \mathcal{S}$.

Example 4.2 The family $\mathcal{N} = \{\|\cdot\|_i\}_{i=1,\ldots,n}$ of seminorms on $E = \mathbb{R}^n$ given by $\|x\|_i := |x_i|$ is separating.

Next, let $\mathcal{N} = \{\|\cdot\|_i\}_{i\in I}$ be a given separating family of seminorms on E. By means of the seminorms in $\mathcal{N}$, we can define **generalized balls** in E as finite intersections of open balls with respect to finitely many of the seminorms $\|\cdot\|_i$. Specifically, for every finite subset $e = \{i_1, \ldots, i_m\} \subset I$ and every $r > 0$, let

$$B_{e,r}(v) := \{w \in E : \|w - v\|_{i_k} < r \text{ for } k = 1, \ldots, m\}.$$

Clearly, these generalized balls are convex.

A subset $U \subset E$ is called a **neighborhood** of $v \in E$ if there exists a generalized ball $B_{e,r}(v)$ contained in U. A subset $O \subset E$ is called **open** if it is a neighborhood of each of its points. Similar to normed spaces, it is easy to see using the triangle inequality for seminorms that, for example, the generalized balls $B_{e,r}(v)$ are open. Neighborhoods of 0 in E will also be called **zero neighborhoods**.

One can easily verify that the collection $\mathcal{T}_\mathcal{N}$ of all open subsets forms a **topology** on E. That is, both the empty set and the entire set E are open, any intersection of finitely many open sets is open, and the union of any family of open sets is open.

The sets $B_{e,r}(v)$ form a **neighborhood basis** at v. That is, for every neighborhood U of v, there exists a generalized ball $B_{e,r}(v) \subset U$.

It is worth noting that in place of the generalized balls $B_{e,r}(v)$, we could also have used the family of all finite intersections of open balls with arbitrary radii corresponding to finitely many of the seminorms $\|\cdot\|_i$. That is, generalized balls of the form

$$\{w \in E : \|w - v\|_i < r_i \quad \text{for } i \in e\},$$

where $e \subset I$ is finite, and $r_i > 0, i \in e$, are any positive numbers. These also evidently form a basis for the same topology.

Since we assume the family of seminorms $\mathcal{N}$ to be separating, the topology $\mathcal{T}_\mathcal{N}$ is **Hausdorff**.

Indeed, if $v \neq w$ in E, then there exists some $i \in I$ such that $r := \|v - w\|_i > 0$. Setting $e := \{i\}$, the disjoint open balls $B_{e,r/2}(v)$ and $B_{e,r/2}(w)$ then separate the points v and w.

For single-element sets $e = \{i\}$, we will occasionally write the ball $B_{e,r}(v)$ as $B_{i,r}(v)$.

A sequence $\{v_j\}_{j\in\mathbb{N}}$ in E is said to **converge** to $v \in E$ if, by definition, for every neighborhood U of v, there exists an index j_0 such that $v_j \in U$ for all $j \geq j_0$.

Lemma 4.3 *The sequence $\{v_j\}_{j\in\mathbb{N}}$ in E converges to $v \in E$ if and only if*

$$\lim_{j\to\infty} \|v_j - v\|_i = 0 \quad \textit{for all } i \in I. \tag{4.3}$$

Proof. If (4.3) holds and U is a neighborhood of v, then there exists a generalized ball $B_{e,r}(v) \subset U$. Assume that $e = \{i_1, \ldots, i_m\}$. By (4.3), for each $k = 1, \ldots, m$, there exists an index j_k such that $\|v_j - v\|_{i_k} < r$ for all $j \geq j_k$. For $j \geq \max\{j_1, \ldots, j_m\}$, it follows that $v_j \in B_{e,r}(v) \subset U$. Thus, v_j converges to v as $j \to \infty$.

Conversely, since every ball $B_{i,r}(v)$ is open, the reverse implication is immediate. □

Lemma 4.3 shows that the topology $\mathcal{T}_\mathcal{N}$ we have constructed delivers the desired notion of convergence *for sequences in E*. This holds in particular for the space $\mathcal{S}(\mathbb{R}^n)$, provided we choose the family of seminorms $\mathcal{N} = \{\|\cdot\|_{(N)}\}_{N\in\mathbb{N}}$.

In $\mathbb{R}^n$ and, more generally, e.g., in any metric spaces X, it is true that a subset $A \subset X$ is closed if and only if the limit of every sequence in A which is convergent in X again lies in A. Thus, the concept of convergence for sequences alone suffices to characterize closed sets – and hence the topology – since open sets are just the complements of closed sets.

Unfortunately, in the case of an uncountable family of seminorms $\mathcal{N}$, the topology $\mathcal{T}_\mathcal{N}$ is in general no longer determined by the notion of convergence for sequences, as one typically lacks countable neighborhood bases. In such cases, the concept of a sequence must be generalized to that of a **net**. We will only briefly touch on this point, as we will see later that in the context of distribution theory, one can ultimately still restrict oneself to the concept of convergence for sequences.

More comprehensive information on nets can be found in books on General Topology, e.g., in S. Willard [44].

Definitions. A set I is called **directed** if it is endowed with a relation $\leqslant$ satisfying the following properties:

(i) $i \leqslant i$ for all $i \in I$;
(ii) if $i_1 \leqslant i_2$ and $i_2 \leqslant i_3$, then $i_1 \leqslant i_3$;
(iii) for every $i_1, i_2 \in I$, there exists an $i_3 \in I$ such that $i_1 \leqslant i_3$ and $i_2 \leqslant i_3$.

The first two properties (i) and (ii) are familiar requirements for an order relation, whereas (iii) provides the positive orientation one is seeking for I.

Examples 4.4 (a) $\mathbb{N}$, endowed with the usual order $\leq$, is directed.

(b) One frequently regards an open interval of the form $]0, \varepsilon_0[$, $\varepsilon_0 > 0$, (such intervals often appear in the definition of Dirac families) as a directed set, by introducing for $\varepsilon_1, \varepsilon_2 \in]0, \varepsilon_0[$ the relation $\leqslant$ as follows:
$\varepsilon_1 \leqslant \varepsilon_2$ if and only if $\varepsilon_2 \leq \varepsilon_1$ with respect to the usual order $\leq$ on $]0, \varepsilon_0[$.

(c) Let X be a topological space, and let $x \in X$. Then the set $\mathcal{U}(x)$ of all neighborhoods of x forms a directed set under "reverse inclusion," i.e., if $U_1, U_2 \in \mathcal{U}(x)$, then $U_1 \leqslant U_2$ holds if and only if $U_2 \subset U_1$.

Definition. A **net** in a set X is a map $\nu : I \to X$ from a directed set I into the set X. We shall often refer to a net ν more suggestively by writing it as $\{x_i\}_{i\in I}$, with $x_i := \nu(i)$.

A net $\{x_i\}_{i\in I}$ in a topological space X **converges** to $x \in X$, written as $x_i \to x$ or $x = \lim x_i$, if for every neighborhood U of x, there exists an $i_0 \in I$ such that $x_i \in U$ for all $i \geqslant i_0$. Such an x is called a **limit** of the net.

Note that in general a convergent net does not have a unique limit, unless X is Hausdorff.

Examples 4.5 (a) Every sequence $\{x_j\}_{j\in\mathbb{N}}$ is a net with the index set $\mathbb{N}$. The usual notion of convergence for sequences coincides here with the one given by the net $\{x_j\}_{j\in\mathbb{N}}$.

(b) Let X be a topological space, and let $\mathcal{U}(x)$ denote the directed set of all neighborhoods of x. For each $U \in \mathcal{U}(x)$, let $x_U \in U$. Then the net $\{x_U\}_{U\in\mathcal{U}(x)}$ evidently converges to x.

Lemma 4.6 *Let X be a topological space, and let $A \subset X$. Then the set A is closed if and only if any limit of any net of points in A again lies in A (compare also Exercise 4.3).*

Proof. Let A be closed, i.e., let $\Omega := X \setminus A$ be open, and let $\{a_i\}_{i\in I}$ be a net of points in A with $\lim a_i = b \in X$. If $b \in \Omega$, then Ω is an open neighborhood of b. Since the net converges to b, there is some $i_0 \in I$ such that $a_{i_0} \in \Omega$, contradicting our assumption that $a_i \in A$ for every $i \in I$. Thus $b \in A$.

The reverse implication will be shown by contraposition. Suppose A is not closed. Then $\Omega := X \setminus A$ is not open, and thus there is some point $b \in \Omega$ for which Ω is not a neighborhood. But then every neighborhood $U \in \mathcal{U}(b)$ of b must contain a point $a_U \in A$. Thus, according to Example 4.5 (b), the net $\{a_U\}_{U\in\mathcal{U}(b)}$ of points in A converges to $b \notin A$. □

Definition. Let E be a vector space over the field $\mathbb{K}$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$, and let $\mathcal{N} = \{\|\cdot\|_i\}_{i\in I}$ and $\mathcal{N}^\sharp = \{\|\cdot\|_j^\sharp\}_{j\in J}$ be two separating families of seminorms on E. These families of seminorms are called **equivalent** if for each $i \in I$, there exists a finite subset $e \subset J$ and a constant $C > 0$ such that

$$\|v\|_i \leq C \sum_{j\in e} \|v\|_j^\sharp, \quad \text{for all } v \in E, \tag{4.4}$$

and if, conversely, every seminorm in $\mathcal{N}^\sharp$ can also be controlled by finitely many seminorms from $\mathcal{N}$ in this way.

This defines an equivalence relation on the set of all separating families of seminorms on E, as is easily seen.

Examples 4.7 (a) Given any separating family of seminorms $\mathcal{N} = \{\|\cdot\|_i\}_{i\in I}$ on E, one can define a new family $\widetilde{\mathcal{N}} := \{\|\cdot\|_e\}_{e\in\mathcal{F}}$, with index set $\mathcal{F} := \{e \subset I : e \text{ is finite}\}$, by setting

$$\|v\|_e := \max_{i\in e} \|v\|_i, \qquad e \in \mathcal{F}.$$

Since $\|v\|_e \le \sum_{i\in e} \|v\|_i$, this new family is obviously equivalent to $\mathcal{N}$. The family $\widetilde{\mathcal{N}}$ possesses the following additional **saturation property**:

For any finite collection of "indices" $e_1, \dots, e_m \in \mathcal{F}$, we have

$$\max_{j=1,\dots,m} \|\cdot\|_{e_j} \in \widetilde{\mathcal{N}}.$$

In particular, this implies that any generalized ball $B_{e,r}(v)$ with respect $\mathcal{N}$ can be written as

$$B_{e,r}(v) = \{w \in E : \|w - v\|_e < r\}, \quad e \in \mathcal{F}, r > 0,$$

i.e., it is a ball with respect to just one of the new seminorms from $\widetilde{\mathcal{N}}$, namely $\|\cdot\|_e$. These balls thus already form a neighborhood basis of $v \in E$.

(b) On $\mathbb{R}^n$, the family of seminorms $\mathcal{N}$ from Example 4.2 and the family $\mathcal{N}^\sharp := \{|\cdot|\}$ consisting of only the Euclidean norm are obviously equivalent. Thus, by Proposition 4.10, the family $\mathcal{N}$ again generates the Euclidean topology on $\mathbb{R}^n$ (compare also Exercise 4.1).

(c) On the space $\mathcal{S}(\mathbb{R}^n)$ (the Schwartz space), the following families of seminorms are, for example, equivalent (cf. Exercises 2.13, 4.2):

$$\|\varphi\|_{(N)} := \max_{|\alpha|\le N} \sup_{x\in\mathbb{R}^n} \langle x\rangle^N |D^\alpha \varphi(x)|, \qquad N \in \mathbb{N};$$

$$\|\varphi\|_{\alpha,\beta} := \sup_{x\in\mathbb{R}^n} |x^\alpha D^\beta \varphi(x)|, \qquad \alpha,\beta \in \mathbb{N}^n;$$

$$\|\varphi\|_{(\alpha,\beta)} := \sup_{x\in\mathbb{R}^n} |D^\beta (x^\alpha \varphi)(x)|, \qquad \alpha,\beta \in \mathbb{N}^n.$$

Let E and F be two vector spaces over $\mathbb{K}$. Assume that E is endowed with the separating family of seminorms $\mathcal{N}_E = \{\|\cdot\|_i\}_{i\in I}$, whose seminorms we will denote, for better distinction from those on F, by $\|\cdot\|_{E,i} := \|\cdot\|_i$.

Similarly, let $\mathcal{N}_F = \{\|\cdot\|_{F,j}\}_{j\in J}$ denote the according family of seminorms on F.

To better distinguish between the generalized balls in E and F, we will denote the generalized balls in E by $B_{E,e,r}(v)$ instead of $B_{e,r}(v)$ (where $v \in E$ and e is a finite subset of I), and the generalized balls in F by $B_{F,f,r}(w)$ (where $w \in F$ and f is a finite subset of J).

For linear maps between vector spaces E and F (endowed with separating families of seminorms $\mathcal{N}_E$ and $\mathcal{N}_F$, respectively), their continuity can be characterized analytically by the following proposition.

Proposition 4.8 (a) *Let $T : E \to F$ be a linear map. The following statements are equivalent:*

(i) *T is continuous.*
(ii) *T is continuous at $0 \in E$.*
(iii) *For every $j \in J$, there exists a finite index set $\{i_1, \dots, i_m\} \subset I$ and a constant $C \geq 0$ such that*

$$\|Tv\|_{F,j} \leq C \sum_{k=1,\dots,m} \|v\|_{E,i_k} \quad \text{for every } v \in E. \tag{4.5}$$

(b) *A linear form $u : E \to \mathbb{K}$ is continuous if and only if there exists a finite index set $\{i_1, \dots, i_m\} \subset I$ and a constant $C \geq 0$ such that*

$$|u(v)| \leq C \sum_{k=1,\dots,m} \|v\|_{i_k} \quad \text{for every } v \in E.$$

Proof. Since (b) represents the special case $F = \mathbb{K}$, it suffices to prove (a).

(i) $\Longrightarrow$ (ii) is clear.

(ii) $\Longrightarrow$ (iii): Suppose T is continuous at 0, and let $j \in J$. We aim to prove (4.5). By transitioning from $\mathcal{N}_E$ to $\widetilde{\mathcal{N}}_E$, and from $\mathcal{N}_F$ to $\widetilde{\mathcal{N}}_F$, as described in Example 4.7 (a), we may assume without loss of generality that the maximum of finitely many seminorms is again one of the seminorms in $\mathcal{N}_E$, or $\mathcal{N}_F$. It is then sufficient to show that for $j \in J$, there exists an index $i \in I$ and a constant $C \geq 0$ such that

$$\|Tv\|_{F,j} \leq C\|v\|_{E,i} \quad \text{for all } v \in E. \tag{4.6}$$

Indeed, the estimates of the form (4.5) for the families of seminorms $\mathcal{N}_E$ and $\mathcal{N}_F$ are equivalent to the estimates of the form (4.6) for the families $\widetilde{\mathcal{N}}_E$ and $\widetilde{\mathcal{N}}_F$, as can be easily verified.

The ball $B_{F,j,1}(0)$ is a zero neighborhood in F, so its preimage $T^{-1}(B_{F,j,1}(0))$ is a zero neighborhood in E, due to the continuity of T at 0. Thus, there exists a ball $B_{E,i,r}(0)$ with $r > 0$ that maps under T into $B_{F,j,1}(0)$, meaning $\|v\|_{E,i} < r$ implies $\|Tv\|_{F,j} < 1$.

By a scaling argument, inequality (4.6) follows, for example, with $C := 2/r$:

If $\|v\|_{E,i} \neq 0$, set $u := \frac{r}{2\|v\|_{E,i}} v$. Then $\|u\|_{E,i} < r$, and consequently $\|Tu\|_{F,j} < 1$. By the linearity of T, we deduce that

$$\|Tv\|_{F,j} = \frac{2}{r} \|Tu\|_{F,j} \|v\|_{E,i} < \frac{2}{r} \|v\|_{E,i}.$$

Finally, if $\|v\|_{E,i} = 0$, then $\|tv\|_{E,i} < r$ for every $t > 0$, which implies $t\|Tv\|_{F,j} = \|T(tv)\|_{F,j} < 1$ for every $t > 0$, hence $\|Tv\|_{F,j} = 0$. Thus, the inequality (4.6) holds in this case as well.

(iii) $\Longrightarrow$ (i): From (4.6), we immediately deduce by means of the linearity of T that

$$B_{E,i,r/C}(v) \subset T^{-1}(B_{F,j,r}(Tv)) \quad \text{for every } r > 0 \text{ and } v \in E,$$

which in return shows that T is continuous at every point $v \in E$. □

The direct product $E \times F$ of the vector spaces E and F is equipped with the family of seminorms

$$\|(v,w)\|_{i,j} := \|v\|_{E,i} + \|w\|_{F,j}, \quad (v,w) \in E \times F,\ (i,j) \in I \times J.$$

The topology induced by this family of seminorms is precisely the product topology on $E \times F$. For each $(v,w) \in E \times F$, the direct products of generalized balls $B_{E,e,r_1}(v) \times B_{F,f,r_2}(w)$ form a neighborhood basis of (v,w) in $E \times F$.

Thus, we have the following:

Corollary 4.9 *The vector space operations*

$$\begin{aligned} \mathrm{add} &: E \times E \to E,\ (v,w) \mapsto v + w, \\ \sigma &: \mathbb{K} \times E \to E,\ (\lambda, v) \mapsto \lambda v, \end{aligned}$$

are continuous.

Proof. For $v, w \in E$ and any $i \in I$, it is clear that

$$\|\mathrm{add}(v,w)\|_i = \|v+w\|_i \le \|v\|_i + \|w\|_i = \|(v,w)\|_{i,i}.$$

Since add is linear, its continuity follows directly from Proposition 4.8.

The mapping σ is bilinear and satisfies $\|\sigma(\lambda, v)\|_{E,i} = |\lambda|\|v\|_i$. We leave it to the reader to check that this implies the continuity of σ (or refer to Proposition A.7 in Appendix A.3). □

Using Proposition 4.8, it is also easy to characterize when two families of seminorms on a vector space E induce the same topology:

Proposition 4.10 *Let $\mathcal{N} = \{\|\cdot\|_i\}_{i\in I}$ and $\mathcal{N}^\sharp = \{\|\cdot\|_j^\sharp\}_{j\in J}$ be two separating families of seminorms on E. Then the topologies $\mathcal{T}_\mathcal{N}$ and $\mathcal{T}_{\mathcal{N}^\sharp}$ induced by them agree if and only if $\mathcal{N}$ and $\mathcal{N}^\sharp$ are equivalent.*

Proof. Let $T := \mathrm{Id}$ be the identity map on E. Denote the space E, when endowed with the family of seminorms $\mathcal{N}$, by E, and by $E^\sharp$, when endowed with the family $\mathcal{N}^\sharp$. Then $\mathcal{N}$ and $\mathcal{N}^\sharp$ generate the same topology if and only if $T : E \to E^\sharp$ and $T^{-1} : E^\sharp \to E$ are continuous. By Proposition 4.8, this is equivalent to $\mathcal{N}$ and $\mathcal{N}^\sharp$ being equivalent. □

A vector space endowed with a topology such that all singleton sets are closed and the vector space operations are continuous is called a **topological vector space**.

If there exists a neighborhood basis of 0 (or equivalently, of any point $v \in E$) consisting of convex neighborhoods, E is called a **locally convex** (topological) vector space.

If the topology in E is generated by a family of seminorms $\mathcal{N}$, then the generalized balls $B_{e,r}(v)$ form a neighborhood basis of v consisting of convex sets. Thus, the topological vector spaces we are considering here are locally convex.

Since every convex neighborhood of 0 in E can be associated with the corresponding Minkowski functional as a seminorm, the topology in any locally convex space E can, conversely, be generated by a separating family of seminorms.

For this and additional information on topological vector spaces, we refer, e.g., to Rudin [34].

We believe, however, that the approach chosen in this book—namely generating the topology on E by means of separating families of seminorms, which ties more easily to the theory of normed spaces—is also more advantageous for addressing analytical questions in the context of distribution theory.

4.2.1 The case of countably many seminorms

Let us now consider the special case where the separating family of seminorms $\mathcal{N} = \{\|\cdot\|_i\}_{i\in I}$ on E is at most countable, as is the case for $E = \mathcal{S}(\mathbb{R}^n)$. In this case, E can be endowed with a metric that induces the same topology $\mathcal{T}_\mathcal{N}$. Such a topological vector space is called **metrizable**.

To this end, let

$$d_i'(v,w) := \|v-w\|_i, \quad v,w \in E,$$

be the semimetric induced by $\|\cdot\|_i$. Then

$$d_i(v,w) := \frac{d_i'(v,w)}{1+d_i'(v,w)}$$

defines an equivalent semimetric which satisfies $d_i \le 1$.

Choose a countable family $\{\rho_i\}_{i\in I}$ of positive numbers $\rho_i > 0$ such that $\sum_i \rho_i \le 1$.

For $I = \mathbb{N}$, for example, one can choose $\rho_i := 2^{-i-1}$. Due to the separating property of $\mathcal{N}$, we then see that by

$$d(v,w) := \sum_{i\in I} \rho_i d_i(v,w) \tag{4.7}$$

a metric d is defined on E. This metric is clearly **invariant**, i.e.,

$$d(v_1+w, v_2+w) = d(v_1,v_2) \quad \text{for all } v_1,v_2,w \in E.$$

Lemma 4.11 (a) *Let $\{v_k\}_k$ be a sequence in E, and let $v \in E$. Then for the metric d in (4.7), we have $\lim_{k\to\infty} d(v_k,v) = 0$ if and only if*

$$\lim_{k\to\infty} \|v_k - v\|_i = 0 \quad \textit{for every } i \in I. \tag{4.8}$$

(b) *A sequence $\{v_k\}_k$ in E is a Cauchy sequence with respect to the metric d if and only if it is a Cauchy sequence with respect to each seminorm $\|\cdot\|_i$, $i \in I$.*

Proof. (a) Let $\varepsilon > 0$. Suppose (4.8) holds for every $i \in I$. We then first choose a finite subset $M \subset I$ such that $\sum_{i\in I\setminus M} \rho_i < \varepsilon/2$, and then some $k_0 \in \mathbb{N}$ such that $\|v - v_k\|_i < \varepsilon/2$ for all $k \ge k_0$ and $i \in M$. For $k \ge k_0$, we then have

$$d(v, v_k) \le \sum_{i \in M} \rho_i d_i(v, v_k) + \sum_{i \in I \setminus M} \rho_i \cdot 1$$
$$\le \sum_{i \in M} \rho_i \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \le \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus, $\lim_{k\to\infty} d(v, v_k) = 0$.

Conversely, assume that $\lim\limits_{k\to\infty} d(v, v_k) = 0$. Since

$$d_i(v, w) \le \frac{1}{\rho_i} d(v, w),$$

it follows for each $i \in I$ that $\lim\limits_{k\to\infty} d_i(v, v_k) = 0$, and hence,

$$\lim_{k\to\infty} \|v - v_k\|_i = \lim_{k\to\infty} d_i'(v, v_k) = 0.$$

(b) is shown analogously. □

Remark 4.12 If $\mathcal{N}$ is at most countable, then every point in E has a countable neighborhood basis. Therefore, a set in E is closed if and only if it is sequentially closed, and a mapping $\Phi : E \to X$ into a topological space X is continuous if and only if it is **sequentially continuous**, i.e., if for every convergent sequence $\{v_j\}_j$ in E, the following holds: $\Phi(\lim_{j\to\infty} v_j) = \lim_{j\to\infty} \Phi(v_j)$.

Definition. If $\mathcal{N}$ is at most countable and $d = d_{\mathcal{N}}$ is the metric on E constructed from $\mathcal{N}$ as in (4.7), then E is called a **Fréchet space** if (E, d) is a complete metric space.

If $\mathcal{N}$ consists of only one norm $\|\cdot\|$, then $(E, \mathcal{N})$ is a Fréchet space if and only if $(E, \|\cdot\|)$ is a Banach space.

4.2.2 On $\mathcal{S}$ as a topological vector space

Theorem 4.13 *$(\mathcal{S}(\mathbb{R}^n), \{\|\cdot\|_{(N)}\}_N)$ is a Fréchet space.*

Proof. Let $\{\varphi_k\}_k$ be a Cauchy sequence in $\mathcal{S}$. Then, by Lemma 4.11, it is a Cauchy sequence with respect to every norm $\|\cdot\|_{(N)}$, $N \in \mathbb{N}$. Using standard arguments, one concludes that for every $N \in \mathbb{N}$, there exists a uniquely determined function

$$\varphi_{(N)} \in \mathcal{S}^N := \{f \in C^N(\mathbb{R}^n) : \|f\|_{(N)} < \infty\}$$

such that

$$\lim_{k\to\infty} \|\varphi_{(N)} - \varphi_k\|_{(N)} = 0.$$

Since the sequence of norms $\|\cdot\|_{(N)}$ on $\mathcal{S}$ is monotonically increasing, i.e.,

$$\|\varphi\|_{(N)} \le \|\varphi\|_{(N+1)}, \tag{4.9}$$

it follows that $\varphi_{(N)} = \varphi_{(N+1)}$ for every $N \in \mathbb{N}$.

Setting $\varphi := \varphi_{(0)}$, it follows that $\varphi_{(N)} = \varphi$ for every $N \in \mathbb{N}$. Thus, $\varphi \in \mathcal{S}$, and

$$\lim_{k\to\infty} \|\varphi - \varphi_k\|_{(N)} = 0 \quad \text{for every } N \in \mathbb{N}.$$

Consequently, the sequence $\{\varphi_k\}_k$ converges in $\mathcal{S}$ to φ. □

The linear endomorphisms of $\mathcal{S}$ described in the following proposition are of particular importance in the theory of distributions.

Definitions. A function $f : \mathbb{R}^n \to \mathbb{C}$ has **polynomial growth** if there exist constants $C \geq 0$ and $N \in \mathbb{N}$ such that

$$|f(x)| \leq C(1 + |x|^2)^{N/2} = C\langle x\rangle^N \quad \text{for all } x \in \mathbb{R}^n.$$

A function $f \in C^\infty(\mathbb{R}^n)$ has **moderate growth** if every partial derivative $D^\alpha f$, $\alpha \in \mathbb{N}^n$, has polynomial growth. For example, all polynomial functions have moderate growth.

Proposition 4.14 *The following linear endomorphisms from $\mathcal{S}$ into $\mathcal{S}$ are continuous:*

(a) *Every partial derivative D^α, $\alpha \in \mathbb{N}^n$.*
(b) *Every multiplication operator*

$$M_f : \varphi \mapsto f\varphi, \quad \varphi \in \mathcal{S},$$

with a C^∞-function f of moderate growth.
(c) *The Fourier transform $\mathcal{F}$ and the Fourier cotransform $\overline{\mathcal{F}}$.*
(d) *The convolution*

$$C_\psi : \varphi \mapsto \varphi * \psi, \quad \varphi \in \mathcal{S},$$

with a Schwartz function $\psi \in \mathcal{S}$.

Proof. We employ Proposition 4.8 in the subsequent arguments.
(a) is immediately clear since

$$\|D^\alpha \varphi\|_{(N)} \leq \|\varphi\|_{(N+|\alpha|)}.$$

(b) Let $\varphi \in \mathcal{S}$, and let f have moderate growth. For $M \in \mathbb{N}$ and $|\alpha| \leq M$, we have

$$\langle x\rangle^M D^\alpha (f\varphi)(x) = \sum_{\beta\leq\alpha} \binom{\alpha}{\beta} \langle x\rangle^M D^\beta f(x) D^{\alpha-\beta}\varphi(x).$$

Since f has moderate growth, there exist constants $C_1 > 0$ and $\ell \in \mathbb{N}$ such that

$$|D^\beta f(x)| \leq C_1\langle x\rangle^\ell \quad \text{for all } \beta \text{ with } |\beta| \leq M.$$

Moreover, $|\alpha - \beta| \leq |\alpha| \leq M$ for $\beta \leq \alpha$, so

$$|D^{\alpha-\beta}\varphi(x)| \leq \|\varphi\|_{(M+N)}\langle x\rangle^{-N} \quad \text{for all } N \in \mathbb{N}.$$

Altogether, this gives

$$\langle x\rangle^M |D^\alpha (f\varphi)(x)| \le \sum_{\beta\le\alpha} \binom{\alpha}{\beta} C_1 \langle x\rangle^{M+\ell} \cdot \|\varphi\|_{(M+N)} \langle x\rangle^{-N} \le C\|\varphi\|_{(M+N)} \langle x\rangle^{M+\ell-N}.$$

Choosing $N \ge M + \ell$, we obtain

$$\|f\varphi\|_{(M)} \le C\|\varphi\|_{(M+N)}.$$

Thus, M_f is continuous.

(c) Since $\overline{\mathcal{F}}\varphi = (\mathcal{F}\varphi)^\vee$, it suffices to consider $\mathcal{F}$. Let $M \in \mathbb{N}$. We aim to show that for $\ell > n$,

$$\|\hat{\varphi}\|_{(M)} \le C\|\varphi\|_{(M+\ell)}.$$

According to (4.2), it is enough to prove that for $\ell > n$,

$$\sup_{\xi\in\mathbb{R}^n} |\xi^\beta D^\alpha \hat{\varphi}(\xi)| \le C\|\varphi\|_{(M+\ell)} \tag{4.10}$$

for all α, β with $|\alpha| \le M$, $|\beta| \le M$. By Corollary 2.20,

$$\sup_{\xi\in\mathbb{R}^n} |\xi^\beta D^\alpha \hat{\varphi}(\xi)| = \|\mathcal{F}(D^\beta(x^\alpha\varphi))\|_\infty \le \|D^\beta(x^\alpha\varphi)\|_1. \tag{4.11}$$

Furthermore,

$$\begin{aligned} |D^\beta(x^\alpha\varphi)(x)| &= \left|\sum_{\gamma\le\beta} \binom{\beta}{\gamma} (D^\gamma x^\alpha) D^{\beta-\gamma}\varphi(x)\right| \\ &\le C'(1+|x|^2)^{|\alpha|/2} \max_{\rho\le\beta} |D^\rho\varphi(x)| \\ &\le C''\langle x\rangle^{M+\ell} \max_{|\rho|\le M} |D^\rho\varphi(x)| \cdot \langle x\rangle^{-\ell} \\ &\le C''\|\varphi\|_{(M+\ell)} \langle x\rangle^{-\ell}. \end{aligned}$$

From this, it follows that $\|D^\beta(x^\alpha\varphi)\|_1 \le C\|\varphi\|_{(M+\ell)}$, and thus, combined with (4.11), also (4.10), provided that $\ell > n$.

(d) For $\varphi, \psi \in \mathcal{S}$, it follows from the Fourier inversion formula (see Theorem 2.28) that

$$\varphi * \psi = (2\pi)^{-n}\overline{\mathcal{F}}(\widehat{\varphi * \psi}) = (2\pi)^{-n}\overline{\mathcal{F}}(\hat{\varphi}\hat{\psi}),$$

which implies

$$C_\psi(\varphi) = (2\pi)^{-n}\overline{\mathcal{F}} \circ M_{\hat{\psi}} \circ \mathcal{F}(\varphi).$$

Since $\mathcal{F}$, $\overline{\mathcal{F}}$, and $M_{\hat{\psi}}$ are continuous on $\mathcal{S}$ by parts (b) and (c), it follows that C_ψ is also continuous. □

Shorter proofs, which, however, provide less information, can be carried out using the closed graph theorem (see, for example, Rudin [34]).

Definition. A linear functional (i.e., a linear form) on $\mathcal{S}(\mathbb{R}^n)$ which is continuous with respect to the topology of $\mathcal{S}(\mathbb{R}^n)$ is called a **tempered distribution**. By $\mathcal{S}'(\mathbb{R}^n)$ we denote the complex vector space of all tempered distributions, i.e., the topological dual space of $\mathcal{S}(\mathbb{R}^n)$.

A linear functional u on $\mathcal{S}(\mathbb{R}^n)$ is thus a tempered distribution if and only if there exist constants $C \geq 0$ and $N \in \mathbb{N}$ such that

$$|u(\varphi)| \leq C\|\varphi\|_{(N)} \quad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{4.12}$$

The space $\mathcal{S}' = \mathcal{S}'(\mathbb{R}^n)$ is endowed with the weak*-topology. This topology is more generally defined as follows:

Definition. Suppose E is a topological vector space over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, and let E' denote its **topological dual space**, i.e., the space of all continuous linear functionals $v' : E \to \mathbb{K}$. To any $v \in E$, we associated the seminorm $\|\cdot\|_v$ on E' defined by

$$\|v'\|_v := |v'(v)|, \quad v' \in E'.$$

The **weak*-topology** on E' is the locally convex topology induced by the family of seminorms $\{\|\cdot\|_v\}_{v\in E}$.

This family is evidently separating. Moreover, by Proposition 4.8, the point evaluation

$$E' \ni v' \mapsto v'(v) \in \mathbb{K}$$

is continuous on E' for each $v \in E$. Indeed, one easily verifies that the weak*-topology is the coarsest topology on E' for which all these point evaluations are continuous.

In particular, a sequence or a net $\{v'_j\}_j$ in E' converges to $v' \in E'$ in the weak*-topology if and only if, for every $v \in E$,

$$\lim_{j\to\infty} v'_j(v) = v'(v).$$

Thus, the weak*-topology is the "**topology of pointwise convergence**".

For example, a sequence or net $\{u_j\}_j$ in $\mathcal{S}'$ of tempered distributions converges to a tempered distribution u if and only if, for every Schwartz function φ,

$$\lim_{j\to\infty} u_j(\varphi) = u(\varphi).$$

As a direct consequence of the Banach–Steinhaus theorem, which also holds in Fréchet spaces (see Theorem A.6 in the Appendix), we obtain the following:

Proposition 4.15 *Let $\{u_j\}_j$ be a sequence of tempered distributions, and suppose that for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$, the limit*

$$u(\varphi) := \lim_{j\to\infty} u_j(\varphi)$$

exists. Then u is a tempered distribution too.

Examples 4.16 (a) A function $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ is called **tempered**, if there exist constants $C \geq 0$, $M \in \mathbb{N}$ such that

$$\int_{|x|\leq r} |f(x)|\mathrm{d}x \leq Cr^M \quad \text{for all } r \geq 1. \tag{4.13}$$

For example, every function of moderate growth is tempered, as is every L^p-function (why?).
If f is tempered, let u_f denote the linear form

$$u_f(\varphi) := \int f(x)\varphi(x)\,\mathrm{d}x, \quad \varphi \in \mathcal{S}, \tag{4.14}$$

on $\mathcal{S}$. Then u_f is continuous.
Indeed, let $N > M$. Clearly,

$$|\varphi(x)| \leq \|\varphi\|_{(N)}\langle x\rangle^{-N}, \qquad x \in \mathbb{R}^n.$$

Using (4.13), it follows that

$$\begin{aligned}
|u_f(\varphi)| \leq \int |f(x)\varphi(x)|\,\mathrm{d}x &= \int_{|x|\leq 1} |f(x)\varphi(x)|\,\mathrm{d}x \\
&\quad + \sum_{j=0}^{\infty} \int_{2^j\leq|x|\leq 2^{j+1}} |f(x)\varphi(x)|\,\mathrm{d}x \\
&\leq \|\varphi\|_{(N)} \left\{ \int_{|x|\leq 1} |f(x)|\,\mathrm{d}x + \sum_{j=0}^{\infty} \int_{2^j\leq|x|\leq 2^{j+1}} |f(x)2^{-jN}|\,\mathrm{d}x \right\} \\
&\leq \|\varphi\|_{(N)} \left\{ C_1 + C\sum_{j=0}^{\infty} 2^{(j+1)M}2^{-jN} \right\} \\
&= C_2\|\varphi\|_{(N)},
\end{aligned}$$

since $2^{(j+1)M}2^{-jN} = 2^M 2^{-(N-M)j}$, where $N - M > 0$. Thus, u_f is a tempered distribution.
We will henceforth view the tempered function f also as the tempered distribution u_f, and often write $f(\varphi)$ in place of $u_f(\varphi)$.

(b) Let $x \in \mathbb{R}^n$. We denote by δ_x the linear form

$$\delta_x(\varphi) := \varphi(x), \quad \varphi \in \mathcal{S},$$

on $\mathcal{S}$. Since $|\varphi(x)| \leq \|\varphi\|_{(0)}$, δ_x is continuous.
The tempered distribution δ_x is called the **point measure** at x, often also, with an abuse of notation, the "**Dirac delta function**" at x.
For $x = 0$, one often writes simply $\delta_0 = \delta$.

More generally, if μ is a **bounded Radon measure** on $\mathbb{R}^n$, i.e., a continuous linear form on the Banach space $(C_\infty(\mathbb{R}^n), \|\cdot\|_\infty)$ of all continuous functions on $\mathbb{R}^n$ vanishing at infinity, then for $\varphi \in \mathcal{S} \subset C_\infty(\mathbb{R}^n)$,

$$|\mu(\varphi)| \le \|\mu\|_1 \|\varphi\|_\infty = \|\mu\|_1 \|\varphi\|_{(0)},$$

where $\|\mu\|_1$ denotes the operator norm of the linear form $\mu : C_\infty(\mathbb{R}^n) \to \mathbb{C}$. Thus, $\mu|_{\mathcal{S}}$ is a tempered distribution, which is usually still denoted simply by μ.

Remark 4.17 By the Riesz representation theorem, the bounded Radon measures on $\mathbb{R}^n$ can be identified with the finite, regular complex Borel measures on $\mathbb{R}^n$; $\|\mu\|_1$ then just agrees with the total variation of the complex measure μ. See Rudin [33] for details.

(c) Let $f \in L^1(\mathbb{R}^n)$, and set again $f_\varepsilon(x) := \varepsilon^{-n} f(\frac{x}{\varepsilon})$, $\varepsilon > 0$. For each $\varphi \in \mathcal{S}$, we then have

$$f_\varepsilon(\varphi) = \int \varepsilon^{-n} f\big(\tfrac{x}{\varepsilon}\big)\varphi(x)\,\mathrm{d}x = \int f(x)\varphi(\varepsilon x)\,\mathrm{d}x,$$

so by Lebesgue's dominated convergence theorem,

$$\lim_{\varepsilon\to 0} f_\varepsilon(\varphi) = \int f(x)\mathrm{d}x\,\varphi(0).$$

Thus, we obtain

$$\lim_{\varepsilon\to 0} f_\varepsilon = \Big(\int f(x)\,\mathrm{d}x\Big)\,\delta_0 \qquad \text{in } \mathcal{S}'(\mathbb{R}^n).$$

In particular, $\lim_{\varepsilon\to 0}\varphi_\varepsilon = \delta_0$ (in $\mathcal{S}'$) for any Dirac family $\{\varphi_\varepsilon\}_\varepsilon$.

(d) Let $M, N \in \mathbb{N}$, and let u_λ be the function defined on $\mathbb{R}$ by $u_\lambda(x) := \lambda^N x^M \mathrm{e}^{i\lambda x}$. For each $\varphi \in \mathcal{S}$, we have

$$u_\lambda(\varphi) = \lambda^N \widehat{(x^M\varphi)}(-\lambda),$$

and since $\widehat{(x^M\varphi)}$ clearly lies in $\mathcal{S}$, it follows that

$$\lim_{|\lambda|\to\infty} u_\lambda = 0 \quad \text{in } \mathcal{S}'(\mathbb{R}).$$

Although each function u_λ grows polynomially, the family of tempered distributions u_λ nevertheless converges to zero due to the oscillatory factor $\mathrm{e}^{i\lambda x}$.

To emphasize the "duality" between $\mathcal{S}'$ and $\mathcal{S}$, we often write

$$\langle u, \varphi\rangle := u(\varphi), \quad \text{for } u \in \mathcal{S}',\ \varphi \in \mathcal{S};$$

similarly, we write $\langle f, \varphi\rangle := u_f(\varphi)$ when f is a tempered function. Clearly, then

$$\langle f, \varphi\rangle = \int f(x)\varphi(x)\,\mathrm{d}x, \qquad \varphi \in \mathcal{S}. \tag{4.15}$$

4.3 Operations on Tempered Distributions

Many operations that we know for functions, such as differentiation, multiplication, convolution with suitable functions, or Fourier transform, can be extended to tempered distributions, using the scheme described below. It should also be noted that, henceforth, we will refer to linear homomorphisms of vector spaces as **linear operators**.

Definition. Let $T : \mathcal{S} \to \mathcal{S}$ be a linear operator, for which there exists a continuous linear operator ${}^tT : \mathcal{S} \to \mathcal{S}$ such that

$$\langle T\psi, \varphi\rangle = \langle \psi, {}^tT\varphi\rangle \quad \text{for all } \varphi, \psi \in \mathcal{S}. \tag{4.16}$$

Then, for $u \in \mathcal{S}'$, the linear functional Tu on $\mathcal{S}$ is defined by

$$(Tu)(\varphi) = \langle Tu, \varphi\rangle := \langle u, {}^tT\varphi\rangle, \quad \varphi \in \mathcal{S}, \tag{4.17}$$

i.e., $Tu = u \circ {}^tT$. This shows that Tu is continuous on $\mathcal{S}$, i.e., $Tu \in \mathcal{S}'$.

Because of (4.16), ${}^tT$ is referred to as the **formally transposed operator**.

For $u = u_f$, with $f \in \mathcal{S}$, it then follows immediately from (4.16) that

$$\langle Tu_f, \varphi\rangle = \langle u_f, {}^tT\varphi\rangle = \langle f, {}^tT\varphi\rangle = \langle Tf, \varphi\rangle = \langle u_{Tf}, \varphi\rangle,$$

i.e.,

$$Tu_f = u_{Tf} \quad \text{for all } f \in \mathcal{S}. \tag{4.18}$$

Thus, our definition of Tu extends T from $\mathcal{S}$ to $\mathcal{S}'$ (strictly speaking, we should have denoted this extension with a different symbol!).

The operator $T : \mathcal{S}' \to \mathcal{S}'$ defined by (4.17) is continuous with respect to the weak*-topology, because:

If $u = \lim u_j$ in $\mathcal{S}'$, then for $\varphi \in \mathcal{S}$, we have

$$(Tu)(\varphi) = u({}^tT\varphi) = \lim u_j({}^tT\varphi) = \lim(Tu_j)(\varphi),$$

i.e., $Tu = \lim Tu_j$ in $\mathcal{S}'$.

Note here that a mapping f between two arbitrary topological spaces X and Y is continuous if and only if it is **net-continuous***, i.e., if for every net $\{x_j\}_j$ in X converging to a point $x \in X$, the net $\{f(x_j)\}_j$ in Y converges to $f(x)$* (compare Exercise 4.3; the easy proof can also be found in any textbook on set topology, such as [44]).

4.3.1 Differentiation of Tempered Distributions

Let $\alpha \in \mathbb{N}^n$. By Proposition 4.14, the differential operator D^α is a continuous linear operator on $\mathcal{S}(\mathbb{R}^n)$. Moreover, for $\varphi, \psi \in \mathcal{S}$, integration by parts yields

$$\langle D^\alpha\psi, \varphi\rangle = \int D^\alpha\psi(x)\varphi(x)\,\mathrm{d}x = (-1)^{|\alpha|}\int \psi(x)D^\alpha\varphi(x)\,\mathrm{d}x = \langle \psi, (-1)^{|\alpha|}D^\alpha\varphi\rangle,$$

i.e., ${}^tD^\alpha = (-1)^{|\alpha|}D^\alpha$. Hence, for $u \in \mathcal{S}'$, the **partial derivative** $D^\alpha u \in \mathcal{S}'$ is defined by

$$\langle D^\alpha u, \varphi\rangle := (-1)^{|\alpha|}\langle u, D^\alpha\varphi\rangle, \quad \varphi \in \mathcal{S}.$$

Examples 4.18 (i) Let H denote the **Heaviside function** on $\mathbb{R}$, i.e., $H = \mathbf{1}_{[0,\infty[}$. For $\varphi \in \mathcal{S}(\mathbb{R})$, we have

$$\begin{aligned}\left\langle \frac{\mathrm{d}H}{\mathrm{d}x}, \varphi \right\rangle &= -\langle H, \frac{\mathrm{d}\varphi}{\mathrm{d}x}\rangle = -\int_{-\infty}^{\infty} H(x)\varphi'(x)\,\mathrm{d}x \\ &= -\int_0^\infty \varphi'(x)\,\mathrm{d}x = \varphi(0) = \langle \delta_0, \varphi\rangle.\end{aligned}$$

Thus, the "generalized derivative" or "derivative of H as a tempered distribution" is given by

$$H' := \frac{\mathrm{d}H}{\mathrm{d}x} = \delta_0. \tag{4.19}$$

(ii) For $x \in \mathbb{R}^n$, $D^\alpha\delta_x$ is given by

$$\langle D^\alpha\delta_x, \varphi\rangle = (-1)^{|\alpha|}D^\alpha\varphi(x), \quad \varphi \in \mathcal{S}.$$

(iii) The function f on $\mathbb{R}$ defined by

$$f(x) := \begin{cases}\log|x|, & x \neq 0,\\ 0, & x = 0,\end{cases}$$

is tempered. For $\varphi \in \mathcal{S}$ and $\varepsilon > 0$, we have

$$\langle \frac{\mathrm{d}f}{\mathrm{d}x}, \varphi\rangle = -\int_{-\infty}^{\infty} \log|x|\varphi'(x)\,\mathrm{d}x = -\lim_{\varepsilon\to 0}\int_{|x|\geq\varepsilon} \log|x|\varphi'(x)\,\mathrm{d}x.$$

Integration by parts gives

$$\langle \frac{\mathrm{d}f}{\mathrm{d}x}, \varphi\rangle = \lim_{\varepsilon\to 0}\left\{(\log\varepsilon)\,(\varphi(\varepsilon) - \varphi(-\varepsilon)) + \int_{|x|\geq\varepsilon}\frac{1}{x}\varphi(x)\,\mathrm{d}x.\right\}$$

Since $\varphi(\varepsilon) - \varphi(-\varepsilon) = O(\varepsilon)$ and $\lim_{\varepsilon\to 0}\varepsilon\log\varepsilon = 0$, it follows that

$$\langle \frac{\mathrm{d}f}{\mathrm{d}x}, \varphi\rangle = \lim_{\varepsilon\to 0}\int_{|x|\geq\varepsilon}\frac{1}{x}\varphi(x)\,\mathrm{d}x.$$

The tempered distribution defined by

$$\langle p.v.\frac{1}{x}, \varphi\rangle := p.v.\int_{-\infty}^{\infty}\frac{1}{x}\,\varphi(x)\,\mathrm{d}x := \lim_{\varepsilon\to 0}\int_{|x|\geq\varepsilon}\frac{1}{x}\,\varphi(x)\,\mathrm{d}x$$

is called the **Cauchy principal value** of the function $x \mapsto \frac{1}{x}$ (note that the latter not locally integrable near 0!).

Accordingly,

$$p.v.\int_{-\infty}^{\infty}\frac{1}{x}\,\varphi(x)\,\mathrm{d}x := \lim_{\varepsilon\to 0}\int_{|x|\geq\varepsilon}\frac{1}{x}\,\varphi(x)\,\mathrm{d}x$$

is referred to as the Cauchy principal value of the corresponding integral. In the sense of distributions, we thus have

$$\frac{\mathrm{d}}{\mathrm{d}x}\log|x| = p.v.\frac{1}{x}. \tag{4.20}$$

4.3.2 Multiplying a tempered distribution by a function of moderate growth

Let $f \in C^\infty(\mathbb{R}^n)$ be a function of moderate growth. According to Proposition 4.14, the multiplication operator M_f is continuous on $\mathcal{S}$. Since

$$\langle M_f\psi,\varphi\rangle = \int (f\psi)\varphi\,\mathrm{d}x = \int \psi(f\varphi)\,\mathrm{d}x = \langle\psi, M_f\varphi\rangle,$$

we have ${}^tM_f = M_f$. Therefore, the **product** of the tempered distribution $u \in \mathcal{S}'$ with f is defined by $fu := M_f u$, i.e.,

$$\langle fu,\varphi\rangle = \langle u, f\varphi\rangle, \qquad \varphi\in\mathcal{S}.$$

Caution. The product of two arbitrary tempered distributions unfortunately cannot be defined!

For example, the product of the Dirac delta function δ_0 with itself does not make sense, at least not in the sense of distributions. We shall not give a strict proof of this, but add an observation which supports this assertion:

If $\{\varphi_\varepsilon\}_\varepsilon$ is a Dirac family, we have seen that $\lim_{\varepsilon\to 0}\varphi_\varepsilon = \delta_0$ in $\mathcal{S}'$. However, since $\varphi_\varepsilon\varphi_\varepsilon = \varepsilon^{-n}\psi_\varepsilon$, where $\psi := \varphi_1\varphi_1$, the product $\varphi_\varepsilon\varphi_\varepsilon$ does not converge to any limit in $\mathcal{S}'$ as $\varepsilon\to 0$.

4.3.3 The Fourier transform of a tempered distribution

According to Proposition 2.25, we have

$$\langle\mathcal{F}\psi,\varphi\rangle = \langle\psi,\mathcal{F}\varphi\rangle \quad \text{for all } \varphi\in\mathcal{S},$$

i.e., ${}^t\mathcal{F} = \mathcal{F}$. Furthermore, the Fourier transform is continuous as an operator on $\mathcal{S}$ by Proposition 4.14. Thus, the **Fourier transform** $\hat{u} = \mathcal{F}u \in \mathcal{S}'$ of the tempered distribution u is defined by

$$\langle\hat{u},\varphi\rangle := \langle u,\hat{\varphi}\rangle, \quad \varphi\in\mathcal{S}.$$

Analogously, due to

$$\begin{aligned}\langle \overline{\mathcal{F}}\psi, \varphi\rangle &= \int \hat{\psi}(-\xi)\varphi(\xi)\,\mathrm{d}\xi = \int \hat{\psi}(\xi)\check{\varphi}(\xi)\,\mathrm{d}\xi \\ &= \int \psi(\xi)\widehat{\check{\varphi}}(\xi)\,\mathrm{d}\xi = \int \psi(\xi)\hat{\varphi}(-\xi)\,\mathrm{d}\xi \\ &= \langle \psi, \overline{\mathcal{F}}\varphi\rangle,\end{aligned}$$

the **Fourier cotransform** $\overline{\mathcal{F}}u \in \mathcal{S}'$ of $u \in \mathcal{S}'$ is defined by

$$\langle \overline{\mathcal{F}}u, \varphi\rangle := \langle u, \overline{\mathcal{F}}\varphi\rangle, \quad \varphi \in \mathcal{S}.$$

For the Fourier transform of tempered distributions, the Fourier inversion formula holds universally:

Theorem 4.19 *For every $u \in \mathcal{S}'(\mathbb{R}^n)$, we have*

$$u = (2\pi)^{-n}\overline{\mathcal{F}}(\mathcal{F}u) = (2\pi)^{-n}\mathcal{F}(\overline{\mathcal{F}}u).$$

Proof. For every $\varphi \in \mathcal{S}$, Theorem 2.28 gives $\varphi = (2\pi)^{-n}\overline{\mathcal{F}}\mathcal{F}\varphi = (2\pi)^{-n}\mathcal{F}\overline{\mathcal{F}}\varphi$. It follows that

$$\langle (2\pi)^{-n}\mathcal{F}(\overline{\mathcal{F}}u), \varphi\rangle = (2\pi)^{-n}\langle \overline{\mathcal{F}}u, \mathcal{F}\varphi\rangle = \langle u, (2\pi)^{-n}\overline{\mathcal{F}}\mathcal{F}\varphi\rangle = \langle u, \varphi\rangle,$$

so $(2\pi)^{-n}\overline{\mathcal{F}}\mathcal{F}u = u$. Similarly, one shows $(2\pi)^{-n}\mathcal{F}\overline{\mathcal{F}}u = u$. □

Examples 4.20 (i) The Fourier transform of the point measure δ_x is given by

$$\langle \widehat{\delta_x}, \varphi\rangle = \langle \delta_x, \hat{\varphi}\rangle = \hat{\varphi}(x) = \langle e_{-x}, \varphi\rangle,$$

i.e.,

$$\widehat{\delta_x} = e_{-x}, \qquad x \in \mathbb{R}^n. \tag{4.21}$$

In particular, $\widehat{\delta_0} = 1$.

Similarly to (4.21), one shows that $\overline{\mathcal{F}}\delta_x = e_x$. Using Theorem 4.19, we thus have

$$\mathcal{F}e_x = \mathcal{F}(\overline{\mathcal{F}}\delta_x) = (2\pi)^n\delta_x,$$

i.e.,

$$\widehat{e_x} = (2\pi)^n\delta_x, \qquad x \in \mathbb{R}^n. \tag{4.22}$$

(ii) As one can quickly verify, a tempered distribution v on $\mathbb{R}$ is defined by setting

$$v := \sum_{k\in\mathbb{Z}} \delta_k,$$

i.e., more specifically, by

$$\langle v, \varphi\rangle := \sum_{k=-\infty}^{\infty} \varphi(k), \qquad \varphi \in \mathcal{S}.$$

By Poisson's summation formula (3.29), we have

$$\langle \hat{v}, \varphi \rangle = \langle v, \hat{\varphi} \rangle = \sum_{k \in \mathbb{Z}} \hat{\varphi}(k) = 2\pi \sum_{\ell \in \mathbb{Z}} \varphi(2\pi\ell).$$

It follows that

$$\mathcal{F}\left(\sum_{k=-\infty}^{\infty} \delta_k \right) = 2\pi \sum_{\ell=-\infty}^{\infty} \delta_{2\pi\ell}. \tag{4.23}$$

4.3.4 The convolution of a tempered distribution with a Schwartz function

Let $\varphi, \psi, \eta \in \mathcal{S}$. As before, let C_ψ denote the convolution operator $C_\psi \eta := \eta * \psi$. Then, according to Fubini's theorem, we have

$$\begin{aligned} \langle C_\psi \eta, \varphi \rangle &= \langle \eta * \psi, \varphi \rangle = \iint \eta(y)\psi(x-y)\,\mathrm{d}y\,\varphi(x)\,\mathrm{d}x \\ &= \iint \eta(y)\check{\psi}(y-x)\varphi(x)\,\mathrm{d}x\,\mathrm{d}y \\ &= \int \eta(y) \int \varphi(x)\check{\psi}(y-x)\,\mathrm{d}x\,\mathrm{d}y = \langle \eta, \varphi * \check{\psi} \rangle, \end{aligned}$$

i.e.,

$${}^t C_\psi = C_{\check{\psi}}.$$

According to Proposition 4.14, the operator ${}^t C_\psi$ is therefore continuous on $\mathcal{S}$.

Thus, the **convolution** $u * \psi$ of the distribution $u \in \mathcal{S}'$ with the Schwartz function ψ is defined by

$$\langle u * \psi, \varphi \rangle := \langle u, \varphi * \check{\psi} \rangle. \tag{4.24}$$

Similarly, $\psi * u \in \mathcal{S}'$ is defined by

$$\langle \psi * u, \varphi \rangle := \langle u, \check{\psi} * \varphi \rangle, \qquad \varphi \in \mathcal{S}.$$

Since $\check{\psi} * \varphi = \varphi * \check{\psi}$, it follows that

$$u * \psi = \psi * u. \tag{4.25}$$

There is also a second canonical way of defining the convolution of u with ψ. To this end, note that for $f, \psi \in \mathcal{S}$, we have

$$f * \psi(x) = \int f(y)\check{\psi}(y-x)\,\mathrm{d}y = \langle f, \lambda_x \check{\psi} \rangle,$$

where for each $x \in \mathbb{R}^n$, the function $\lambda_x \check{\psi}$ belongs to $\mathcal{S}$. Thus, we define *pointwise* the convolution product for $u \in \mathcal{S}'$ and $\psi \in \mathcal{S}$ as

$$u \circledast \psi(x) := \langle u, \lambda_x \check{\psi} \rangle, \qquad x \in \mathbb{R}^n. \tag{4.26}$$

We will show that both definitions are equivalent, i.e., that

$$u * \psi = u_{u \circledast \psi}. \tag{4.27}$$

We need a few lemmata.

Lemma 4.21 *Let $e_j := (0, \dots, 1, 0, \dots, 0)$ denote the j-th unit vector of the canonical basis of $\mathbb{R}^n$. For $\varphi \in \mathcal{S}$ and $h \in \mathbb{R} \setminus \{0\}$, define*

$$\varphi_h(x) := \frac{\varphi(x + he_j) - \varphi(x)}{h}, \quad x \in \mathbb{R}^n.$$

Then

$$\frac{\partial \varphi}{\partial x_j} = \lim_{h \to 0} \varphi_h$$

in the topology of $\mathcal{S}$.

Proof. Since the Fourier transform $\mathcal{F} : \mathcal{S} \to \mathcal{S}$ is an isomorphism of topological vector spaces (Proposition 4.14 and Theorem 2.28), it suffices to show that

$$\widehat{\frac{\partial \varphi}{\partial x_j}} = \lim_{h \to 0} \widehat{\varphi_h} \quad \text{in } \mathcal{S},$$

i.e., that

$$\lim_{h \to 0} \rho_h \hat{\varphi} = 0 \quad \text{in } \mathcal{S}, \tag{4.28}$$

where we define

$$\rho_h(\xi) := \frac{\mathrm{e}^{ih\xi_j} - 1}{h} - i\xi_j.$$

Now, for $N \in \mathbb{N}$, $\alpha \in \mathbb{N}^n$, we have

$$\langle \xi \rangle^N D^\alpha (\rho_h \hat{\varphi}) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} \langle \xi \rangle^N (D^\beta \rho_h)(D^{\alpha - \beta} \hat{\varphi}). \tag{4.29}$$

Furthermore,

$$\begin{aligned} \frac{\mathrm{e}^{ih\xi_j} - 1}{h} - i\xi_j &= i\xi_j \int_0^1 (\mathrm{e}^{ih\xi_j t} - 1)\, \mathrm{d}t \\ &= -h\xi_j^2 \int_0^1 \int_0^t \mathrm{e}^{ih\xi_j s}\, \mathrm{d}s \mathrm{d}t. \end{aligned}$$

This implies that $|\rho_h(\xi)| \le |h|\xi_j^2$. Furthermore, $|D_j\rho_h(\xi)| = |e^{ih\xi_j} - 1| \le |h||\xi_j|$, and for $k \ge 2$, $|D_j^k\rho_h(\xi)| = |h^{k-1}e^{ih\xi_j}| = |h|^{k-1}$. Hence,

$$|D^\beta\rho_h(\xi)| \le \begin{cases} |h|\xi_j^2, & \text{if } |\beta| = 0, \\ |h||\xi_j|, & \text{if } |\beta| = 1, \\ |h|^{|\beta|-1}, & \text{if } |\beta| \ge 2. \end{cases}$$

From this, it is straightforward to see that the expression in (4.29) converges uniformly to 0 as $h \to 0$. Consequently,

$$\lim_{h\to 0} \|\rho_h\hat{\varphi}\|_{(N)} = 0.$$

This proves (4.28). □

Lemma 4.22 *Let $\varphi, \psi \in \mathcal{S}$. For $\varepsilon > 0$, denote by η_ε the "Riemann sum"*

$$\eta_\varepsilon := \sum_{k\in\mathbb{Z}^n} \varphi(\varepsilon k)\varepsilon^n \lambda_{\varepsilon k}\check{\psi}. \tag{4.30}$$

Then for every $N \in \mathbb{N}$,

$$\sum_{k\in\mathbb{Z}^n} |\varphi(\varepsilon k)|\varepsilon^n \|\lambda_{\varepsilon k}\check{\psi}\|_{(N)} < \infty, \tag{4.31}$$

i.e., the series (4.30) converges in $\mathcal{S}$ to a Schwartz function η_ε, which is given pointwise by

$$\eta_\varepsilon(x) = \sum_{k\in\mathbb{Z}^n} \varphi(\varepsilon k)\varepsilon^n \psi(\varepsilon k - x), \quad x \in \mathbb{R}^n.$$

Furthermore,

$$\lim_{\varepsilon\to 0} \eta_\varepsilon = \varphi * \check{\psi} \quad \text{in } \mathcal{S}. \tag{4.32}$$

Proof. It is easy to see (see Exercise 4.4) that for every $N \in \mathbb{N}$, there exists a constant $C = C_N \ge 0$ such that

$$\|\lambda_x\check{\psi}\|_{(N)} \le C\langle x\rangle^N \quad \text{for all } x \in \mathbb{R}^n. \tag{4.33}$$

Moreover, for every $M \in \mathbb{N}$,

$$|\varphi(x)| \le \|\varphi\|_{(M)}\langle x\rangle^{-M}.$$

It follows that

$$|\varphi(\varepsilon k)|\varepsilon^n \|\lambda_{\varepsilon k}\check{\psi}\|_{(N)} \le C\|\varphi\|_{(M)}\varepsilon^n\langle \varepsilon k\rangle^{-M+N}.$$

By comparison with a corresponding integral, one can easily conclude that

$$\sup_{0<\varepsilon<1} \sum_{k\in\mathbb{Z}^n} \varepsilon^n\langle \varepsilon k\rangle^{-M+N} < \infty, \tag{4.34}$$

provided $M > N + n$. Hence, (4.31) follows.

Now, let $\eta_\varepsilon = \sum_{k\in\mathbb{Z}^n} \varphi(\varepsilon k)\varepsilon^n \lambda_{\varepsilon k}\check{\psi} \in \mathcal{S}(\mathbb{R}^n)$. Define

$$F(x, y) := \varphi(y)\check{\psi}(x - y), \quad (x, y) \in \mathbb{R}^n \times \mathbb{R}^n.$$

It is easy to verify (e.g., using the partial Fourier transform in x, or the linear change of coordinates $x' := x - y, y' := y$) that $F \in \mathcal{S}(\mathbb{R}^n \times \mathbb{R}^n)$. Moreover,

$$\eta_\varepsilon(x) = \sum_{k\in\mathbb{Z}^n} F(x, \varepsilon k)\varepsilon^n$$

and

$$\varphi * \check{\psi}(x) = \int F(x, y)\,\mathrm{d}y,$$

so that

$$\eta_\varepsilon(x) - \varphi * \check{\psi}(x) = \sum_{k\in\mathbb{Z}^n} \int_{\varepsilon(k+Q)} [F(x, \varepsilon k) - F(x, y)]\,\mathrm{d}y, \tag{4.35}$$

where $Q := [0, 1[^n$ denotes the unit cube.

Let $\alpha \in \mathbb{N}^n, |\alpha| \le N$. By the mean value theorem, for every $y \in \varepsilon(k + Q)$,

$$|D_x^\alpha F(x, \varepsilon k) - D_x^\alpha F(x, y)| \le \sqrt{n}\varepsilon \sup_{z\in\varepsilon(k+Q)} |D_x^\alpha \nabla_y F(x, z)|.$$

Moreover, we have

$$\langle (x, y)\rangle^{2N} |D_x^\alpha \nabla_y F(x, y)| \lesssim \|F\|_{(2N+1)},$$

and

$$\langle x\rangle\langle y\rangle \le \langle (x, y)\rangle^2,$$

thus

$$\langle x\rangle^N |D_x^\alpha \nabla_y F(x, z)| \lesssim \|F\|_{(2N+1)}\langle z\rangle^{-N} \le C\|F\|_{(2N+1)}\langle \varepsilon k\rangle^{-N},$$

if $z \in \varepsilon(k + Q)$. Consequently, it follows that

$$\sup_{y\in\varepsilon(k+Q)} \langle x\rangle^N |D_x^\alpha F(x, \varepsilon k) - D_x^\alpha F(x, y)| \le \varepsilon C\|F\|_{(2N+1)}\langle \varepsilon k\rangle^{-N},$$

i.e., using (4.35), for $|\alpha| \le N$,

$$\begin{aligned}
&\sup_{x\in\mathbb{R}^n} \langle x\rangle^N |D^\alpha [\eta_\varepsilon - \varphi * \check{\psi}](x)| \\
&\le \varepsilon C\|F\|_{(2N+1)} \sum_{k\in\mathbb{Z}^n} \int_{\varepsilon(k+Q)} \langle \varepsilon k\rangle^{-N}\mathrm{d}y \\
&= \varepsilon C\|F\|_{(2N+1)} \sum_{k\in\mathbb{Z}^n} \varepsilon^n \langle \varepsilon k\rangle^{-N}.
\end{aligned}$$

For $N > n$, it follows (cf. (4.34)) that

$$\|\eta_\varepsilon - \varphi * \check{\psi}\|_{(N)} \le \varepsilon C \|F\|_{(2N+1)}.$$

This implies (4.32). □

Theorem 4.23 *Let $u \in \mathcal{S}'(\mathbb{R}^n), \psi \in \mathcal{S}(\mathbb{R}^n)$. Then the following hold:*

(a) *$u \circledast \psi$ belongs to $C^\infty(\mathbb{R}^n)$ and has moderate growth. Furthermore,*

$$D^\alpha(u \circledast \psi) = (D^\alpha u) \circledast \psi = u \circledast D^\alpha \psi$$

for every $\alpha \in \mathbb{N}^n$, where in the middle term D^α refers to differentiation in the sense of distributions, while in the other terms it refers to classical differentiation.

(b) *$u * \psi = u_f$, with $f := u \circledast \psi$.*

(c) *$(u * \psi)^\wedge = \hat{\psi}\hat{u}$.*

(d) *$(u * \psi) * \varphi = u * (\psi * \varphi)$ for all $\varphi \in \mathcal{S}(\mathbb{R}^n)$.*

(e) *$\hat{u} * \hat{\psi} = (2\pi)^n (\psi u)^\wedge$.*

Proof. (a) Let $j \in \{1, \dots, n\}, h \in \mathbb{R} \setminus \{0\}$. Then for fixed x,

$$\frac{u \circledast \psi(x + he_j) - u \circledast \psi(x)}{h} = \langle u, \varphi_h \rangle,$$

where we set $\varphi(y) := (\lambda_x \check{\psi})(y) = \psi(x - y)$, and

$$\varphi_h(y) := \frac{\varphi(y - he_j) - \varphi(y)}{h}.$$

By Lemma 4.21, $\frac{\partial \varphi}{\partial y_j} = -\lim_{h \to 0} \varphi_h$ in $\mathcal{S}$, and hence

$$\lim_{h \to 0} \langle u, \varphi_h \rangle = -\left\langle u, \frac{\partial \varphi}{\partial y_j} \right\rangle.$$

It follows that $u \circledast \psi$ is partially differentiable in x_j, and

$$\frac{\partial}{\partial x_j}(u \circledast \psi)(x) = -\left\langle u, \frac{\partial \varphi}{\partial y_j} \right\rangle = \left\langle u, \lambda_x \left(\frac{\partial \psi}{\partial y_j} \right)^\vee \right\rangle,$$

i.e.,

$$D_j(u \circledast \psi)(x) = u \circledast D_j \psi(x), \quad \text{for every } x \in \mathbb{R}^n.$$

Thus, $u \circledast \psi$ is partially differentiable. Moreover, by (4.33), $\nabla(u \circledast \psi)$ is locally bounded, and hence $u \circledast \psi$ is continuous (use the mean value theorem from one-dimensional analysis iteratively!). By induction on the order of differentiation, we obtain from the above special case that $u \circledast \psi \in C^\infty$, and that

$$D^\alpha(u \circledast \psi) = u \circledast D^\alpha \psi, \quad \text{for all } \alpha \in \mathbb{N}^n. \tag{4.36}$$

Furthermore, (4.33) shows that $u \circledast \psi$ has polynomial growth, and the same holds for $D^\alpha(u \circledast \psi)$ by (4.36), i.e., $u \circledast \psi$ has moderate growth. Finally,

$$(D^\alpha u) \circledast \psi(x) = (-1)^{|\alpha|}\langle u, D^\alpha(\lambda_x \check{\psi})\rangle = \langle u, \lambda_x (D^\alpha \psi)^\vee\rangle = u \circledast D^\alpha \psi(x).$$

(b) Let $\varphi \in \mathcal{S}$, and set again

$$\eta_\varepsilon := \sum_{k\in\mathbb{Z}^n} \varphi(\varepsilon k)\varepsilon^n \lambda_{\varepsilon k}\check{\psi}, \qquad \varepsilon > 0.$$

By Lemma 4.22, $\varphi * \check{\psi} = \lim_{\varepsilon\to 0} \eta_\varepsilon$ in $\mathcal{S}$. By the continuity of u, it follows that

$$\langle u * \psi, \varphi\rangle = \langle u, \varphi * \check{\psi}\rangle = \lim_{\varepsilon\to 0}\langle u, \eta_\varepsilon\rangle.$$

Moreover, the series (4.30) for η_ε converges in $\mathcal{S}$, so that

$$\begin{aligned}\langle u, \eta_\varepsilon\rangle &= \sum_{k\in\mathbb{Z}^n} \varphi(\varepsilon k)\varepsilon^n \langle u, \lambda_{\varepsilon k}\check{\psi}\rangle \\ &= \sum_{k\in\mathbb{Z}^n} \varphi(\varepsilon k) u \circledast \psi(\varepsilon k)\varepsilon^n.\end{aligned}$$

Clearly,

$$\sum_{k\in\mathbb{Z}^n} \varphi(\varepsilon k) u \circledast \psi(\varepsilon k)\varepsilon^n$$

is a Riemann sum approximating the integral

$$\int_{\mathbb{R}^n} \varphi(y) u \circledast \psi(y)\,\mathrm{d}y,$$

i.e.,

$$\lim_{\varepsilon\to 0}\langle u, \eta_\varepsilon\rangle = \int \varphi(y) u \circledast \psi(y)\,\mathrm{d}y = \langle u \circledast \psi, \varphi\rangle.$$

It follows that $\langle u * \psi, \varphi\rangle = \langle u \circledast \psi, \varphi\rangle$ for all $\varphi \in \mathcal{S}$.

(c) For $\varphi \in \mathcal{S}$, we have by Theorem 2.28,

$$\begin{aligned}\langle (u * \psi)^\wedge, \varphi\rangle &= \langle u * \psi, \hat{\varphi}\rangle = \langle u, \hat{\varphi} * \check{\psi}\rangle \\ &= \langle u, \hat{\varphi} * (\mathcal{F}^{-1}\check{\psi})^\wedge\rangle = (2\pi)^n \langle u, \mathcal{F}\big(\varphi(\mathcal{F}^{-1}\check{\psi})\big)\rangle \\ &= \langle \hat{u}, \varphi\,(2\pi)^n \mathcal{F}^{-1}\check{\psi}\rangle = \langle \hat{u}, \varphi\hat{\psi}\rangle \\ &= \langle \hat{\psi}\hat{u}, \varphi\rangle.\end{aligned}$$

The identities in (d) and (e) are left as an exercise (see Exercise 4.11). □

Remarks 4.24 (a) In view of the identity (4.27) proved in Theorem 4.23, we will henceforth identify the distribution $u * \psi$ with the C^∞-function $u \circledast \psi$.

(b) The proof of Theorem 4.23 heavily relies on the continuity of tempered distributions on the space $\mathcal{S}$, highlighting the importance of this continuity requirement for distributions.

4.4 Regularization: Friedrichs Mollification

Let $\varphi \in \mathcal{S}(\mathbb{R}^n)$ with $\int \varphi(x)\,\mathrm{d}x = 1$, and consider the approximate identity

$$\varphi_\varepsilon(x) := \varepsilon^{-n}\varphi\left(\frac{x}{\varepsilon}\right), \qquad \varepsilon > 0.$$

Proposition 4.25 *For every $\psi \in \mathcal{S}$, we have*

$$\lim_{\varepsilon\to 0} \psi * \varphi_\varepsilon = \psi \quad \text{in } \mathcal{S}. \tag{4.37}$$

If $u \in \mathcal{S}'(\mathbb{R}^n)$, and if we set $u_\varepsilon := u * \varphi_\varepsilon \in C^\infty(\mathbb{R}^n)$, then:

$$u = \lim_{\varepsilon\to 0} u_\varepsilon \quad \text{in } \mathcal{S}'.$$

Proof. Let $\psi \in \mathcal{S}(\mathbb{R}^n)$. Then we have

$$\hat{\psi} - \widehat{\psi * \varphi_\varepsilon} = (1 - \widehat{\varphi_\varepsilon})\,\hat{\psi},$$

and since the Fourier transform is a topological isomorphism from $\mathcal{S}$ to $\mathcal{S}$, it thus suffices to show that for every $\eta \in \mathcal{S}$,

$$(1 - \widehat{\varphi_\varepsilon})\eta \to 0 \text{ in } \mathcal{S}, \text{ as } \varepsilon \to 0. \tag{4.38}$$

Now, we have

$$h_\varepsilon(x) := 1 - \widehat{\varphi_\varepsilon}(x) = 1 - \hat{\varphi}(\varepsilon x) = \hat{\varphi}(0) - \hat{\varphi}(\varepsilon x),$$

so that, using the mean value inequality, we obtain:

$$|D^\beta h_\varepsilon(x)| \le C_\beta \begin{cases} \varepsilon|x|, & \text{if } |\beta| = 0; \\ \varepsilon^{|\beta|}, & \text{if } |\beta| > 0. \end{cases}$$

From this, together with the Leibniz' rule, (4.38) follows, and thus (4.37) as well.

Finally, since $\varphi \in \mathcal{S}$, the function $\check{\varphi}$ also belongs to $\mathcal{S}$ and likewise satisfies $\int \check{\varphi}(x)\,\mathrm{d}x = 1$. Thus, if $u \in \mathcal{S}'$ and $\psi \in \mathcal{S}$, then (4.37) implies that

$$\langle u, \psi\rangle = \lim_{\varepsilon\to 0}\langle u, \psi * \check{\varphi}_\varepsilon\rangle = \lim_{\varepsilon\to 0}\langle u * \varphi_\varepsilon, \psi\rangle = \lim_{\varepsilon\to 0}\langle u_\varepsilon, \psi\rangle. \qquad \square$$

The smooth functions $u_\varepsilon := u * \varphi_\varepsilon$ are often called **Friedrichs mollifications** of the tempered distribution u.

Conclusion. Every tempered distribution can be approximated "arbitrarily closely" in the sense of distributions by smooth functions.

If, in the proof of Theorem 4.25, we choose φ such that $\hat{\varphi}$ has compact support, then (4.38) implies in particular

Proposition 4.26 *$C_0^\infty(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$.*

4.5 Exercises

4.1 Let E be a locally convex vector space whose topology is generated by the separating family of seminorms $\mathcal{N}$. If $p : E \to \mathbb{R}_{\geq 0}$ is any continuous seminorm on E, set $N_p := \{x \in E : p(x) = 0\}$. Prove:

a) N_p is a closed linear subspace of E, and

$$p(x + y) = p(x) \qquad \text{for all } x \in E, y \in N_p.$$

b) If E is finite-dimensional, and if $\|\cdot\|$ is any norm on E, then the families of seminorms $\mathcal{N}$ and $\mathcal{N}^\sharp := \{\|\cdot\|\}$ are equivalent.
Hints. Assume without loss of generality that the family of seminorms $\mathcal{N}$ is saturated. Then:

(i) If $p \in \mathcal{N}$ is such that $\dim N_p > 0$, then there exists some $q \in \mathcal{N}$ such that $\dim N_{p+q} < \dim N_p$. Conclude that at least one of the seminorms in $\mathcal{N}$ is a norm.
(ii) Use the well-known fact that any two norms on a finite-dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$ are equivalent.

4.2 Show that the following families of seminorms on $\mathcal{S}(\mathbb{R}^n)$ are pairwise equivalent:

$$\|\varphi\|_{(N)} := \max_{|\alpha| \leq N} \sup_{x \in \mathbb{R}^n} \langle x \rangle^N |D^\alpha \varphi(x)|, \qquad N \in \mathbb{N};$$

$$\|\varphi\|_{\alpha,\beta} := \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta \varphi(x)|, \qquad \alpha, \beta \in \mathbb{N}^n;$$

$$\|\varphi\|_{(\alpha,\beta)} := \sup_{x \in \mathbb{R}^n} |D^\beta (x^\alpha \varphi)(x)|, \qquad \alpha, \beta \in \mathbb{N}^n.$$

Hint: Use, e.g., Exercise 2.13.

4.3 (On the Concept of a Net) Let X be a topological space, and let A be a subset of X. A point $x \in X$ is called a **contact point** of A if every neighborhood U of x contains a point of A. We denote the **closure** of the set A by $\overline{A}$, i.e., the set of all contact points of A. Prove:

a) Let $x \in X$. Then $x \in \overline{A}$ if and only if there exists a net in A that converges to x.
b) $A \subset X$ is closed (i.e., $X \setminus A$ is open) if and only if $\overline{A} = A$.
c) If $B \subset X$ is closed, and if $A \subset B$, then $\overline{A} \subset B$.

d) Let Y also be a topological space. A mapping $f : X \to Y$ is continuous if and only if it is "net-continuous," i.e., if for any net $\{x_i\}_{i\in I}$ in X converging to some point $x \in X$, the net $\{f(x_i)\}_{i\in I}$ converges to $f(x)$ in Y.

4.4 Prove:

a) For each $N \in \mathbb{N}$, there exists a constant $C = C_N \geq 0$ such that

$$\|\lambda_x \psi\|_{(N)} \leq C\|\psi\|_{(N)}\langle x\rangle^N \qquad \text{for all } \psi \in \mathcal{S}(\mathbb{R}^n),\ x \in \mathbb{R}^n.$$

b) Let $\varepsilon > 0$, and $N > n$. Then

$$\sum_{k\in\mathbb{Z}^n} \langle \varepsilon k\rangle^{-N} < \infty.$$

4.5 a) Let $u \in \mathcal{S}'(\mathbb{R}^n)$, and let $f \in C^\infty(\mathbb{R}^n)$ be a function of moderate growth. Show that for the product $fu \in \mathcal{S}'$, the Leibniz formula

$$D^\alpha(fu) = \sum_{\beta\leq\alpha} \binom{\alpha}{\beta}(D^\beta f)(D^{\alpha-\beta}u)$$

holds.

b) Show that for $\alpha, \beta \in \mathbb{N}^n$:

$$x^\alpha(D^\beta\delta_0) = \begin{cases} i^{|\alpha|}\frac{\beta!}{(\beta-\alpha)!}D^{\beta-\alpha}\delta_0, & \text{if } \alpha \leq \beta, \\ 0, & \text{otherwise.} \end{cases}$$

c) Compute the distribution $f''' \in \mathcal{S}'(\mathbb{R})$, where $f(x) := |x|^3$, $x \in \mathbb{R}$.

d) For $t > 0$, let v_t be the function defined on $\mathbb{R}$ by

$$v_t(x) := \begin{cases} te^{itx}, & \text{if } x > 0, \\ 0, & \text{if } x \leq 0. \end{cases}$$

Determine $\lim\limits_{t\to\infty} v_t$ in $\mathcal{S}'(\mathbb{R})$.

4.6 Show that for $\alpha \in \mathbb{N}^n$, the Fourier transform of $D^\alpha\delta_0$ is the monomial ξ^α, i.e.,

$$\widehat{D^\alpha\delta_0}(\xi) = \xi^\alpha, \qquad \xi \in \mathbb{R}^n.$$

Note that this immediately implies that the distributions $D^\alpha\delta_0$, $\alpha \in \mathbb{N}^n$, are linearly independent (a fact that can also be proved by applying these distributions to suitable polynomials).

4.7 Let $u \in \mathcal{S}'(\mathbb{R})$. Show that $u' = 0$ if and only if u is a constant function.

Hint. Choose $\chi \in \mathcal{D} = \mathcal{D}(\mathbb{R}) := C_0^\infty(\mathbb{R})$ with $\int \chi(x)\mathrm{d}x = 1$, and write any $\varphi \in \mathcal{D}(\mathbb{R})$ in the form

$$\varphi = \psi' + \alpha\chi, \text{ with } \psi \in \mathcal{D}(\mathbb{R}),\ \alpha \in \mathbb{C}.$$

Also use the fact that $\mathcal{D}$ is dense in $\mathcal{S}$. Alternatively, make use of Friedrichs mollification.

4.8 Let $u \in \mathcal{S}'(\mathbb{R}^n)$, and let P be a complex polynomial. Prove:

$$\widehat{P(D)u} = P\widehat{u} \quad \text{and} \quad \widehat{Pu} = P(-D)\widehat{u}.$$

4.9 a) Let $f : \mathbb{R}^n \to \mathbb{C}$ be a function such that for all $\varphi \in \mathcal{S} = \mathcal{S}(\mathbb{R}^n)$, the product $f\varphi$ is integrable (which ensures that the linear form $u_f(\varphi) := \int f(x)\varphi(x)\,\mathrm{d}x$ is well-defined on $\mathcal{S}$).
Show that f is a tempered function.

b*) Let $f : \mathbb{R}^n \to \mathbb{C}$ be a function such that for all $\varphi \in \mathcal{S}$, the product $M_f\varphi := f\varphi$ also lies in $\mathcal{S}$. Show that f is smooth and of moderate growth.
Hint. It can be assumed that $M_f : \mathcal{S} \to \mathcal{S}$ is continuous (this follows, e.g., using the closed graph theorem from functional analysis – why?).

4.10 a) Let $\varphi \in \mathcal{S}(\mathbb{R}^n)$. Show that the mapping $x \mapsto \lambda_x\varphi$ is continuous from $\mathbb{R}^n$ to $\mathcal{S}(\mathbb{R}^n)$.

b) Define the translation operator λ_x on $\mathcal{S}'(\mathbb{R}^n)$ and prove that for $u \in \mathcal{S}'(\mathbb{R}^n)$:

$$\widehat{\lambda_x u} = e_{-x}\widehat{u} \quad \text{and} \quad \widehat{e_x u} = \lambda_x\widehat{u} \quad \text{for all } x \in \mathbb{R}^n.$$

4.11 Let $u \in \mathcal{S}'(\mathbb{R}^n)$, $\varphi, \psi \in \mathcal{S}(\mathbb{R}^n)$. Show:

a) $(u * \psi) * \varphi = u * (\psi * \varphi)$.
b) $\hat{u} * \hat{\psi} = (2\pi)^n(\psi u)^\wedge$.

4.12 For $1 \le p < \infty$ and $N \in \mathbb{N}$, let

$$\|\varphi\|_{(p,N)} := \max_{|\alpha|,|\beta| \le N} \|x^\beta D^\alpha \varphi\|_p, \quad \varphi \in \mathcal{S}(\mathbb{R}^n).$$

Show:

a) There exists a constant C_p and an $N \in \mathbb{N}$ such that

$$\|\varphi\|_\infty \le C_p \|\varphi\|_{(p,N)} \quad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^n).$$

Hint: Apply integrations by parts in all variables $x_1, \dots, x_n$.

b) The families of seminorms $\{\|\cdot\|_{(p,N)}\}_N$ and $\{\|\cdot\|_{(N)}\}_N$ are equivalent on $\mathcal{S}(\mathbb{R}^n)$.

4.13 Let $f \in L^1(\mathbb{T})$, and let $f_{\mathrm{per}} : \mathbb{R} \to \mathbb{C}$ denote the corresponding 2π-periodic function on $\mathbb{R}$. Show that f_{per} is tempered, and that

$$\widehat{f_{\mathrm{per}}} = 2\pi \sum_{k \in \mathbb{Z}} \hat{f}(k)\delta_k$$

in the sense of tempered distributions on $\mathbb{R}$.

4.14 (The Hilbert Transform – Basic Properties) Let $H \in \mathcal{S}'(\mathbb{R})$ be the Cauchy principal value of $1/(\pi x)$ (cf. Example 4.18 (iii)), i.e.,

$$\langle H, \varphi \rangle := \frac{1}{\pi} p.v. \int_{-\infty}^{\infty} \frac{\varphi(x)}{x} \mathrm{d}x, \quad \varphi \in \mathcal{S}(\mathbb{R}),$$

and let

$$H_\varepsilon(x) := \frac{x}{\pi(\varepsilon^2 + x^2)}, \qquad \varepsilon > 0.$$

a) Show that

$$H_\varepsilon \to H \quad \text{in } \mathcal{S}'(\mathbb{R}) \text{ as } \varepsilon \to 0.$$

b) Let $S_\varepsilon(y) := \mathrm{e}^{-\varepsilon|y|}\operatorname{sign} y$, $y \in \mathbb{R}$, $\varepsilon > 0$. Determine $\widehat{S_\varepsilon}$, and deduce from a) that

$$\widehat{H} = -i \operatorname{sign} .$$

c) The **Hilbert transform** of the function $\varphi \in \mathcal{S}(\mathbb{R})$ is defined by

$$\mathcal{H}\varphi := \varphi * H.$$

Show that $\|\mathcal{H}\varphi\|_2 = \|\varphi\|_2$, so that $\mathcal{H}$ uniquely extends to an isometric linear operator on $L^2(\mathbb{R})$.

d) Let $\varphi \in \mathcal{S}(\mathbb{R})$. Show that $\mathcal{H}\varphi$ is smooth. Is it even true that $\mathcal{H}\varphi \in \mathcal{S}(\mathbb{R})$ for every $\varphi \in \mathcal{S}(\mathbb{R})$?

4.15 (The Hilbert Transform – Relations to Complex Analysis) Let

$$\mathcal{H} : L^2(\mathbb{R}) \to L^2(\mathbb{R})$$

be the Hilbert transform defined in Exercise 4.14. Recall that for $\varphi \in \mathcal{S}(\mathbb{R})$, we may write $\mathcal{H}\varphi = \varphi * H$, where H is the Cauchy-principal value $p.v.(\frac{1}{\pi x})$, and that for arbitrary $f \in L^2(\mathbb{R})$, we have

$$\widehat{\mathcal{H}f} = -i \operatorname{sign} \hat{f}.$$

Let g be a real-valued function in $L^2(\mathbb{R})$, and put $h := \mathcal{H}g$ and

$$f := g + ih = g + i\mathcal{H}g \in L^2(\mathbb{R}).$$

Moreover, for $z = x + iy \in \mathbb{C}$ with $y > 0$, we set (at least formally)

$$F(z) := \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\xi) \mathrm{e}^{iz\xi} \, \mathrm{d}\xi.$$

a) Show that the integral in the definition of F is absolutely convergent, and that F is holomorphic in the upper half-plane $D := \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$.
Hints: Show that $\hat{f}(\xi) = 0$ for $\xi < 0$, and use Morera's theorem.

b) Show that

$$F(x + iy) = g * P_y(x) + ih * P_y(x),$$

where $P_y(x) := \frac{1}{\pi} \frac{y}{x^2+y^2}$ denotes the Poisson kernel on $\mathbb{R}$ (see Exercise 2.15 b)).

Setting $F_y(x) := F(x + iy),\ y > 0$, we have in particular

$$\operatorname{Im} F_y = \mathcal{H}(\operatorname{Re} F_y) \quad \text{for every } y > 0. \tag{4.39}$$

c) Use Exercise 2.15 c) to show that

$$\lim_{y \to 0} F_y = f = g + i\mathcal{H}g \quad \text{in } L^2(\mathbb{R}).$$

Thus, $f = g + i\mathcal{H}g$ is, in this sense, the boundary value of a holomorphic function in the complex upper half-plane. Furthermore, as in (4.39), the Hilbert transform establishes the passage from the real to the imaginary part even for these boundary values.

Chapter 5
Distributions in Open Subsets of $\mathbb{R}^n$

5.1 Basic Definitions

We now aim to "localize" the concept of tempered distributions in order to introduce distributions in arbitrary open subsets of $\mathbb{R}^n$.

Let $\Omega \subset \mathbb{R}^n$ be a given, non-empty open subset of $\mathbb{R}^n$. The space of **test functions** on Ω is chosen as

$$\mathcal{D}(\Omega) := C_0^\infty(\Omega) \hookrightarrow C_0^\infty(\mathbb{R}^n),$$

i.e., the space of all C^∞-functions on Ω whose support is compact in the topological space Ω. The arrow $\hookrightarrow$ indicates that the corresponding map is an embedding, which is given here by the map $\mathcal{D}(\Omega) \ni \varphi \mapsto \tilde{\varphi}$, where $\tilde{\varphi}$ denotes again the trivial extension of φ to $\mathbb{R}^n$, i.e.,

$$\tilde{\varphi}(x) := \begin{cases} \varphi(x), & \text{if } x \in \Omega, \\ 0, & \text{if } x \in \mathbb{R}^n \setminus \Omega. \end{cases}$$

To motivate our subsequent definition of distributions in Ω, let us first consider a measurable function $f : \Omega \to \mathbb{C}$.

Recall that f is called **locally integrable** if for every point $x \in \Omega$, there exists a neighborhood of x in Ω over which f is integrable.

As is well known and easily seen, the function f is locally integrable, i.e., $f \in L^1_{\mathrm{loc}}(\Omega)$ (where, as usual, we identify f with its equivalence class modulo null functions in $L^1_{\mathrm{loc}}(\Omega)$), if and only if f is integrable over every compact subset $K \subset \Omega$. This is evidently equivalent to the fact that for every function $\chi \in \mathcal{D}(\Omega)$, the trivial extension $\widetilde{f\chi}$ of $f\chi$ to $\mathbb{R}^n$ belongs to $L^1(\mathbb{R}^n)$.

We also observe that the operations of trivial extension from Ω to $\mathbb{R}^n$ and restriction to Ω are *dual* in the following sense:

For all $g \in L^1(\Omega)$ and $h \in L^\infty(\mathbb{R}^n)$, we have

$$\langle \tilde{g}, h \rangle := \int_{\mathbb{R}^n} \tilde{g}(x) h(x)\, \mathrm{d}x = \int_\Omega g(x) h|_\Omega(x)\, \mathrm{d}x =: \langle g, h|_\Omega \rangle.$$

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D. Müller, *An Invitation to Fourier Analysis and Distribution Theory*,
Universitext, https://doi.org/10.1007/978-3-032-31550-2_5

In particular, for every $f \in L^1_{\mathrm{loc}}(\Omega)$ and $\phi \in \mathcal{S}(\mathbb{R}^n)$,

$$\begin{aligned}\langle \widetilde{f\chi}, \phi\rangle &= \int_{\mathbb{R}^n} \widetilde{f\chi}(x)\phi(x)\,\mathrm{d}x = \int_\Omega f(x)(\chi\,\phi|_\Omega)(x)\,\mathrm{d}x \\ &= \langle f, \chi\,\phi|_\Omega\rangle,\end{aligned} \tag{5.1}$$

where $\phi \in \mathcal{S}(\mathbb{R}^n)$ and $\chi \in \mathcal{D}(\Omega)$ imply $\varphi := \chi\,\phi|_\Omega \in \mathcal{D}(\Omega)$. This observation motivates the following definitions:

Definitions. (a) Let $u : \mathcal{D}(\Omega) \to \mathbb{C}$ be a linear functional, i.e., a linear form on the $\mathbb{C}$-vector space $\mathcal{D}(\Omega)$ in the sense of linear algebra. If $\chi \in \mathcal{D}(\Omega)$, let $\widetilde{u\chi}$ denote the linear form on $\mathcal{S}(\mathbb{R}^n)$ defined as follows:

$$\widetilde{u\chi}(\phi) := u(\chi\,\phi|_\Omega), \qquad \phi \in \mathcal{S}(\mathbb{R}^n).$$

(b) The linear functional $u : \mathcal{D}(\Omega) \to \mathbb{C}$ is called a **distribution** in Ω if $\widetilde{u\chi}$ is a tempered distribution on $\mathbb{R}^n$ for every $\chi \in \mathcal{D}(\Omega)$.

Thus distributions in Ω "look locally" like tempered distributions.

The distributions in Ω clearly form a linear subspace $\mathcal{D}'(\Omega)$ of the space of all linear forms on $\mathcal{D}(\Omega)$.

(c) If $f \in L^1_{\mathrm{loc}}(\Omega)$, let u_f denote the linear functional

$$u_f(\varphi) := \int_\Omega f(x)\varphi(x)\,\mathrm{d}x = \langle f, \varphi\rangle, \qquad \varphi \in \mathcal{D}(\Omega),$$

on $\mathcal{D}(\Omega)$. From (5.1), it follows that for $\phi \in \mathcal{S}(\mathbb{R}^n)$, $\chi \in \mathcal{D}(\Omega)$:

$$\widetilde{u_f\chi}(\phi) = u_f(\chi\,\phi|_\Omega) = \langle f, \chi\,\phi|_\Omega\rangle = \langle \widetilde{f\chi}, \phi\rangle,$$

i.e.,

$$\widetilde{u_f\chi} = u_{\widetilde{f\chi}} \quad \text{on } \mathcal{S}(\mathbb{R}^n). \tag{5.2}$$

Note that since χ has compact support, the function $\widetilde{f\chi}$ is tempered. Thus, the identity (5.2) holds in the sense of tempered distribution, i.e., in $\mathcal{S}'(\mathbb{R}^n)$.

Thus, for locally integrable functions f, the operation of forming the tempered distribution $\widetilde{u_f\chi}$ matches with forming the tempered distribution obtained from the trivial extension of the function $f\chi$ to $\mathbb{R}^n$. In particular, we see that $u_f \in \mathcal{D}'(\Omega)$ is a distribution in Ω.

Identifying, as usual, the function $f \in L^1_{\mathrm{loc}}(\Omega)$ with the distribution $u_f \in \mathcal{D}'(\Omega)$, we thus have an embedding

$$L^1_{\mathrm{loc}}(\Omega) \hookrightarrow \mathcal{D}'(\Omega). \tag{5.3}$$

Since not every locally integrable function on $\mathbb{R}^n$ is a tempered distribution (but only the tempered functions, see Exercise 4.9), this shows that the space $\mathcal{D}'(\mathbb{R}^n)$ is significantly larger than $\mathcal{S}'(\mathbb{R}^n)$.

For example, the function $f(x) = \mathrm{e}^x$, $x \in \mathbb{R}$, lies in $L^1_{\mathrm{loc}}(\mathbb{R}) \subset \mathcal{D}'(\mathbb{R})$, but not in $\mathcal{S}'(\mathbb{R})$.

(d) For $u \in \mathcal{D}'(\Omega)$ and $\varphi \in \mathcal{D}(\Omega)$, we will often write

$$\langle u, \varphi \rangle := u(\varphi),$$

to emphasize the duality between $\mathcal{D}'(\Omega)$ and $\mathcal{D}(\Omega)$.

Next, we aim to directly characterize the "continuity property" of a distribution without referring to the topology of $\mathcal{S}(\mathbb{R}^n)$. To this end, we need further definitions.

Definitions. Let Ω be a non-empty open subset of $\mathbb{R}^n$.

(a) For $N \in \mathbb{N}$, let $\|\cdot\|_N$ denote the following C^N-norm on $\mathcal{D}(\Omega)$:

$$\|\varphi\|_N := \max_{|\alpha| \le N} \sup_{x \in \Omega} |D^\alpha \varphi(x)|, \quad \varphi \in \mathcal{D}(\Omega).$$

Evidently, $\{\|\cdot\|_N\}$ is an ascending sequence of norms on $\mathcal{D}(\Omega)$.

(b) If K is a compact subset of $\mathbb{R}^n$, let

$$\mathcal{D}_K := \{\varphi \in C_0^\infty(\mathbb{R}^n) : \operatorname{supp} \varphi \subset K\}.$$

Note that if $K \subset \Omega$, then the linear mapping $\iota : \varphi \mapsto \varphi|_\Omega$ allows us to identify the space $\mathcal{D}_K$ with the subspace

$$\mathcal{D}_K(\Omega) := \{\varphi \in C_0^\infty(\Omega) : \operatorname{supp} \varphi \subset K\}$$

of $\mathcal{D}(\Omega)$ (the inverse mapping to ι is given by the trivial extension $\varphi \mapsto \tilde{\varphi}$).

In this way we can also regard

$$\mathcal{D}_K \hookrightarrow \mathcal{D}(\Omega)$$

as an embedded subspace of $\mathcal{D}(\Omega)$, and we shall often do so without further comment in what follows.

Evidently, the norms $\|\cdot\|_N$ and $\|\cdot\|_{(N)}$ on $\mathcal{D}_K$ are equivalent: there exists a constant $C_K \ge 1$ such that

$$\|\varphi\|_N \le \|\varphi\|_{(N)} \le C_K \|\varphi\|_N, \quad \text{for all } \varphi \in \mathcal{D}_K. \tag{5.4}$$

From now on, we endow $\mathcal{D}_K$ with the locally convex topology generated by the sequence of C^N-norms $\{\|\cdot\|_N\}_{N \in \mathbb{N}}$.

Proposition 5.1 *$\mathcal{D}_K$ is a Fréchet space whose topology coincides with the relative topology of $\mathcal{S}(\mathbb{R}^n)$ on $\mathcal{D}_K$, i.e., a subset $U \subset \mathcal{D}_K$ is open if and only if there exists an open subset $V \subset \mathcal{S}(\mathbb{R}^n)$ such that $U = V \cap \mathcal{D}_K$.*

Proof. Clearly, $\mathcal{D}_K$ forms a closed subspace of $\mathcal{S}(\mathbb{R}^n)$ and is therefore complete with respect to the topology of $\mathcal{S}(\mathbb{R}^n)$. This topology, however, coincides on $\mathcal{D}_K$ with that defined by the norms $\|\cdot\|_N$, $N \in \mathbb{N}$, due to (5.4). □

Remark 5.2 With (5.4), one can easily show the following (see also Theorem 5.3): A linear functional u on $\mathcal{D}(\Omega)$ is a distribution in Ω if and only if, for every compact subset $K \subset \Omega$, there exist constants $C_K \geq 0$ and $N_K \in \mathbb{N}$ such that

$$|u(\varphi)| \leq C_K \sum_{|\alpha| \leq N_K} \sup_{x \in \Omega} |D^\alpha \varphi(x)| \quad \text{for all } \varphi \in \mathcal{D}_K.$$

This is the definition originally used by Schwartz in [35]. The advantage of our definition lies in the fact that it allows us to quickly reduce most statements about distributions in Ω to the corresponding statements already proven for tempered distributions.

(c) Let $\{K_j\}_j$ be a sequence of compact subsets of Ω. We shall call it a **compact exhaustion** of Ω, if the following holds:

(i) $K_j \subset K_{j+1}^0$ for all j;
(ii) $\Omega = \bigcup_j K_j$.

Clearly, these conditions imply

(iii) For every compact set $K \subset \Omega$ there exists a j such that $K \subset K_j^0$.

Compact exhaustions exist for every non-empty open subset Ω of $\mathbb{R}^n$.

For example, one can construct such an exhaustion as follows:

Let $\mathcal{B}$ denote the family of all compact Euclidean balls $\overline{B}_r(x)$ that lie entirely within Ω and whose radius r and coordinates x_j of the center x are all rational. This family is countable, so we can choose a sequence $\{B_j\}_j$ of such balls which runs through all balls in $\mathcal{B}$. The increasing sequence of compact sets

$$K'_j := B_0 \cup \cdots \cup B_j, \quad j \in \mathbb{N},$$

then satisfies (ii), as one quickly verifies (even the sequence of the open interiors $(K'_j)^0$ of these sets covers Ω). Since each of the sets K'_j is covered by the open interior of $K'_{k(j)}$ for some sufficiently large $k(j) \in \mathbb{N}$, we can finally extract a subsequence $\{K_j\}_j = \{K'_{k_j}\}_j$ that also satisfies (i).

(d) We say that a sequence $\{A_j\}_j$ of subsets of Ω **tends to infinity** in Ω or is **locally finite** if for every compact set $K \subset \Omega$ there exists an index j_0 such that $A_j \cap K = \emptyset$ for all $j \geq j_0$, i.e., if K is intersected by at most finitely many of the sets A_j.

Note that we also allow $A_j = \emptyset$.

For example, the sets $A_j := K_{j+1} \setminus K_j$ tend to infinity in Ω if the K_j form a compact exhaustion of Ω.

In a certain sense, this definition suggests imagining the boundary points of Ω, if $\Omega \neq \mathbb{R}^n$, also as infinitely distant points.

(e) For any subset $A \subset \Omega$ and any $N \in \mathbb{N}$, we define the seminorm

$$\|f\|_{A,N} := \max_{|\alpha| \leq N} \sup_{x \in A} |D^\alpha f(x)|,$$

if $A \neq \emptyset$, and $\|f\|_{A,N} := 0$, if $A = \emptyset$, where $f : \Omega \to \mathbb{C}$ can be any C^N-function.

If $\{A\} := \{A_j\}_j$ is a sequence of subsets of Ω tending to infinity, $\{N\} := \{N_j\}_j$ is a sequence of natural numbers, and $\{\varepsilon\} := \{\varepsilon_j\}_j$ is a sequence of positive numbers $\varepsilon_j > 0$, then the following seminorm is defined on $\mathcal{D}(\Omega)$:

$$\|\varphi\|_{\{A\},\{N\},\{\varepsilon\}} := \sum_j \frac{1}{\varepsilon_j} \|\varphi\|_{A_j,N_j}.$$

For example, if $\|\varphi\|_{\{A\},\{N\},\{\varepsilon\}} \le 1$, then in particular $\|\varphi\|_{A_j,N_j} \le \varepsilon_j$ for all j, i.e., these seminorms allow to measure how fast the function φ and its derivatives vanish at infinity.

We denote by $\mathcal{N}_{\mathcal{D}(\Omega)}$ the separating family of all such seminorms $\|\varphi\|_{\{A\},\{N\},\{\varepsilon\}}$ on $\mathcal{D}(\Omega)$. Clearly, the sum of any finite number of seminorms from $\mathcal{N}_{\mathcal{D}(\Omega)}$ also belongs to $\mathcal{N}_{\mathcal{D}(\Omega)}$.

Theorem 5.3 *Let $\Omega \neq \emptyset$ be open in $\mathbb{R}^n$, and let u be a linear form on $\mathcal{D}(\Omega)$. The following statements are equivalent:*

(a) *u is a distribution, i.e., $u \in \mathcal{D}'(\Omega)$.*

(b) *For every compact set $K \subset \Omega$, the restriction $u|_{\mathcal{D}_K}$ of u to $\mathcal{D}_K \subset \mathcal{D}(\Omega)$ is continuous on the Fréchet space $(\mathcal{D}_K, \{\|\cdot\|_N\}_N)$, i.e., there exist constants $C_K \ge 0$ and $N_K \in \mathbb{N}$ such that*

$$|u(\varphi)| \le C_K \|\varphi\|_{N_K} \qquad \text{for all } \varphi \in \mathcal{D}_K. \tag{5.5}$$

(c) *There exists a seminorm $\|\cdot\|_{\{A\},\{N\},\{\varepsilon\}}$ such that*

$$|u(\varphi)| \le \|\varphi\|_{\{A\},\{N\},\{\varepsilon\}} \quad \text{for all } \varphi \in \mathcal{D}(\Omega). \tag{5.6}$$

(d) *It holds that $\lim_{j\to\infty} u(\varphi_j) = 0$ for every sequence $\{\varphi_j\}_j$ in $\mathcal{D}(\Omega)$ tending to 0 in the following sense:*
For every $N \in \mathbb{N}$, we have $\lim_{j\to\infty} \|\varphi_j\|_N = 0$, and additionally, all functions φ_j are supported in a common compact set $K \subset \Omega$.

Proof. (a) $\Longrightarrow$ (b): Let $u \in \mathcal{D}'(\Omega)$, and let $K \subset \Omega$ be compact. Choose $\chi \in \mathcal{D}(\Omega)$ such that $\chi \equiv 1$ on a neighborhood of K. Then for $\varphi \in \mathcal{D}_K$, we have $\varphi = \chi\varphi$, so that

$$u(\varphi) = u(\chi\,\varphi|_\Omega) = \widetilde{u\chi}(\varphi), \quad \varphi \in \mathcal{D}_K.$$

Denoting by $\iota : \mathcal{D}_K \hookrightarrow \mathcal{S}$ the embedding of $\mathcal{D}_K$ into $\mathcal{S}$, we get $u|_{\mathcal{D}_K} = \widetilde{u\chi} \circ \iota$. Since both $\iota : \mathcal{D}_K \to \mathcal{S}$ and $\widetilde{u\chi} : \mathcal{S} \to \mathbb{C}$ are continuous, it follows that $u|_{\mathcal{D}_K}$ is continuous on $\mathcal{D}_K$. The inequality (5.5) now follows immediately from Proposition 4.8 (b).

(b) $\Longrightarrow$ (c): Let $\{K_j\}_{j\in\mathbb{N}}$ be a compact exhaustion of Ω, and choose $\chi_j \in \mathcal{D}(\Omega)$ such that $\chi_j \equiv 1$ on a neighborhood of K_j and $\operatorname{supp}\chi_j \subset K_{j+1}$. Setting $\eta_0 := \chi_0$ and $\eta_j := \chi_j - \chi_{j-1}$ for $j \ge 1$, then for every $\varphi \in \mathcal{D}(\Omega)$

$$\varphi = \sum_j \eta_j \varphi,$$

where only finitely many terms in this series are nonzero. Since for $j \geq 1$ the support of η_j is contained in $K_{j+1} \setminus K_{j-1}$, it follows from (5.5) and the Leibniz formula that

$$\begin{aligned} |u(\varphi)| &= \left| \sum_j u(\eta_j \varphi) \right| \leq \sum_j |u(\eta_j \varphi)| \\ &\leq \sum_j C_{K_{j+1}} \|\eta_j \varphi\|_{N_{K_{j+1}}} \\ &\leq \sum_j C_j \|\varphi\|_{A_j, N_j}, \end{aligned}$$

if we set $A_0 := K_1$, $N_0 := N_{K_1}$, $A_j := K_{j+1} \setminus K_{j-1}$, $N_j := N_{K_{j+1}}$ for $j \geq 1$, and choose the constants $C_j > 0$ sufficiently large. Putting $\varepsilon_j := 1/C_j$, (5.6) follows.

(c) $\Longrightarrow$ (d): This is immediately clear, since only finitely many of the sets A_j intersect the compact set K, which is assumed to contain the supports of all φ_j.

(d) $\Longrightarrow$ (a): Assume that the linear functional u on $\mathcal{D}(\Omega)$ is continuous in the sense of (d), and let $\chi \in \mathcal{D}(\Omega)$. We must show that the linear form $\widetilde{u\chi}$ on $\mathcal{S} = \mathcal{S}(\mathbb{R}^n)$ is continuous. Due to the linearity of $\widetilde{u\chi}$, it suffices to prove continuity at 0.

Let $\{\phi_j\}_j$ be a null sequence in $\mathcal{S}$. Setting $\varphi_j := \chi \phi_j|_\Omega \in \mathcal{D}(\Omega)$, we have

$$\widetilde{u\chi}(\phi_j) = u(\varphi_j).$$

Let $K := \operatorname{supp} \chi$. Then $\operatorname{supp} \varphi_j \subset K$ for every $j \in \mathbb{N}$. Denote by $\tilde{\chi} \in \mathcal{D}(\mathbb{R}^n)$ the trivial extension of χ to $\mathbb{R}^n$. By Proposition 4.14 (b), since $\{\phi_j\}_j$ is a null sequence in $\mathcal{S}$, the sequence $\tilde{\chi}\,\phi_j$ also converges to 0 in $\mathcal{S}$, where $\tilde{\chi}\,\phi_j \in \mathcal{D}_K$ for every $j \in \mathbb{N}$. By Proposition 5.1, the sequence $\tilde{\chi}\,\phi_j$ thus converges to 0 in $\mathcal{D}_K$, and we have $\varphi_j = (\tilde{\chi}\,\phi_j)|_\Omega$. By (d), it follows that

$$\lim_{j\to\infty} \widetilde{u\chi}(\phi_j) = \lim_{j\to\infty} u(\varphi_j) = 0. \qquad \square$$

Definition. We say that a sequence $\{\varphi_j\}_j$ in $\mathcal{D}(\Omega)$ is a **$\mathcal{D}$-null sequence** if it converges to 0 in the sense of Theorem 5.3 (d), i.e., if there exists a compact set $K \subset \Omega$ such that $\operatorname{supp} \varphi_j \subset K$ for every j, and $\lim\limits_{j\to\infty} \|\varphi_j\|_N = 0$ for every $N \in \mathbb{N}$.

Analogous to tempered distributions, distributions can also be understood as continuous linear functionals on the space $\mathcal{D}(\Omega)$, provided that a suitable topology is introduced on it.

Definitions. (a) The space $\mathcal{D}(\Omega)$ is endowed with the topology generated by the family of seminorms $\|\varphi\|_{\{A\},\{N\},\{\varepsilon\}}$ from $\mathcal{N}_{\mathcal{D}(\Omega)}$.

With this topology, $\mathcal{D}(\Omega)$ becomes a locally convex topological vector space, and from Theorem 5.3 we immediately obtain

Proposition 5.4 *A linear functional on $\mathcal{D}(\Omega)$ is a distribution if and only if it is continuous on $\mathcal{D}(\Omega)$, i.e., $\mathcal{D}'(\Omega)$ is the topological dual space $\mathcal{D}(\Omega)'$ of $\mathcal{D}(\Omega)$.*

(b) We endow $\mathcal{D}'(\Omega)$ with the weak*-topology with respect to $\mathcal{D}(\Omega)$.

Proposition 5.5 *Let $\{u_j\}_j$ be a sequence of distributions in $\mathcal{D}'(\Omega)$, and suppose that for every $\varphi \in \mathcal{D}(\Omega)$ the limit*

$$u(\varphi) := \lim_{j\to\infty} u_j(\varphi)$$

exists. Then $u \in \mathcal{D}'(\Omega)$.

Proof. Let $\chi \in \mathcal{D}(\Omega)$. Then, clearly,

$$\widetilde{u\chi}(\phi) = \lim_{j\to\infty} \widetilde{u_j\chi}(\phi) \quad \text{for all } \phi \in \mathcal{S}(\mathbb{R}^n),$$

and thus, by Proposition 4.15, $\widetilde{u\chi}$ is a tempered distribution. Therefore, $u \in \mathcal{D}'(\Omega)$. □

(c) Let $u \in \mathcal{D}'(\Omega)$. If the natural number $N = N_K \in \mathbb{N}$ in inequality (5.5) can be chosen independently of the compact set $K \subset \Omega$, then the distribution u is said to be of **finite order**. The smallest natural number N with this property is called the **order** of the distribution u.

Examples 5.6 (i) Let $f \in L^1_{\mathrm{loc}}(\Omega)$. Then we have

$$|u_f(\varphi)| = |\langle f, \varphi\rangle| = \left|\int_K f(x)\varphi(x)\,\mathrm{d}x\right|$$
$$\leq \int_K |f(x)|\,\mathrm{d}x\, \|\varphi\|_\infty = C_K\|\varphi\|_0 \qquad \text{for all } \varphi \in \mathcal{D}_K.$$

Thus, f, or more precisely u_f, has order 0.

(ii) Analogous to Example 4.18 (ii), for $x \in \Omega$, the distribution $D^\alpha\delta_x \in \mathcal{D}'(\Omega)$ is given by

$$\langle D^\alpha\delta_x, \varphi\rangle := (-1)^{|\alpha|}D^\alpha\varphi(x), \quad \varphi \in \mathcal{D}(\Omega).$$

Then we obtain

$$|\langle D^\alpha\delta_x, \varphi\rangle| = |D^\alpha\varphi(x)| \leq \|\varphi\|_{|\alpha|}.$$

Thus, the order of $D^\alpha\delta_x$ is bounded by $|\alpha|$, but indeed it is exactly $N = |\alpha|$ (cf. Remark 6.3).

(iii) The distribution $\sum_{j=0}^\infty D^j\delta_j$ on $\mathbb{R}$ clearly has no finite order.

(d) Let $\Omega_1 \subset \Omega$ be non-empty open subsets of $\mathbb{R}^n$. If $\tilde\varphi \in \mathcal{D}(\Omega)$ denotes the trivial extension of $\varphi \in \mathcal{D}(\Omega_1)$ to Ω, i.e.,

$$\tilde\varphi(x) := \begin{cases} \varphi(x), & \text{if } x \in \Omega_1, \\ 0, & \text{if } x \in \Omega \setminus \Omega_1, \end{cases} \tag{5.7}$$

then the **restriction** $u|_{\Omega_1}$ of the distribution $u \in \mathcal{D}'(\Omega)$ to Ω_1 is defined by

$$u|_{\Omega_1}(\varphi) := u(\tilde\varphi),\ \varphi \in \mathcal{D}(\Omega_1). \tag{5.8}$$

Theorem 5.3 (b) immediately shows that $u|_{\Omega_1}$ is a distribution in Ω_1.

We say that u **vanishes** on Ω_1 if $u|_{\Omega_1} = 0$. This is clearly equivalent to

$$u(\varphi) = 0 \qquad \text{for all } \varphi \in \mathcal{D}(\Omega) \text{ with } \operatorname{supp}\varphi \subset \Omega_1. \tag{5.9}$$

More generally, we say that the distributions u and v in $\mathcal{D}'(\Omega)$ **agree on** Ω_1 if their restrictions to Ω_1 are equal.

Lemma 5.7 *A distribution $u \in \mathcal{D}'(\Omega)$ vanishes on Ω if and only if for every point $x \in \Omega$ there exists an open neighborhood U_x of x in Ω on which u vanishes.*

Proof. One direction is clear. Conversely, suppose that for every $x \in \Omega$ there exists an open neighborhood U_x in Ω on which u vanishes. Let $\varphi \in \mathcal{D}(\Omega)$. We must show that $u(\varphi) = 0$.

Let K be the compact support of φ, and choose finitely many points $x_1, \dots, x_k \in K$ such that

$$K \subseteq \bigcup_{j=1}^{k} U_{x_j}.$$

By Lemma 2.17, we can then choose a smooth partition of unity subordinate to the U_{x_j}, i.e., functions $\chi_1, \dots, \chi_k \in \mathcal{D}(\Omega)$ with $\operatorname{supp}\chi_j \subset U_{x_j}$, such that $\sum_{j=1}^{k} \chi_j \equiv 1$ on a neighborhood of K. Then $\varphi = \sum_{j=1}^{k} \varphi\chi_j$, where $\operatorname{supp}\varphi\chi_j \subset U_{x_j}$. By (5.9), since u vanishes on U_{x_j}, it follows that:

$$u(\varphi) = \sum_{j=1}^{k} u(\varphi\chi_j) = 0. \qquad \square$$

Corollary 5.8 *Let $u \in \mathcal{D}'(\Omega)$. If $\{U_\alpha\}_{\alpha\in A}$ is a family of open subsets $U_\alpha \subset \Omega$ on which u vanishes, then u also vanishes on the union $\bigcup_{\alpha\in A} U_\alpha$.*

Definition. Let $u \in \mathcal{D}'(\Omega)$. By Corollary 5.8, there exists a largest open set $U \subset \Omega$ on which u vanishes, namely the union of all open sets on which u vanishes. The complement $\Omega \setminus U$ is called the **support** of the distribution u and is denoted by $\operatorname{supp} u$. Clearly, $\operatorname{supp} u$ is closed in Ω, and we have

$$u(\varphi) = 0 \qquad \text{for all } \varphi \in \mathcal{D}(\Omega) \text{ with } \operatorname{supp}\varphi \cap \operatorname{supp} u = \emptyset. \tag{5.10}$$

5.2 Restriction of Tempered Distributions to Open Subsets of $\mathbb{R}^n$, Embedding of $\mathcal{S}'$ into the Space $\mathcal{D}'(\mathbb{R}^n)$.

Let Ω be a non-empty open subset of $\mathbb{R}^n$, and let $u \in \mathcal{S}'(\mathbb{R}^n)$ be a tempered distribution. Then, according to Theorem 5.3, the restriction $u|_{\mathcal{D}(\Omega)}$ of the linear functional u to the subspace $\mathcal{D}(\Omega)$ of $\mathcal{S}(\mathbb{R}^n)$ (where we again identify functions in $\mathcal{D}(\Omega)$ with their trivial extensions to $\mathbb{R}^n$) is clearly a distribution in Ω.

Furthermore, if $\Omega = \mathbb{R}^n$, then according to Proposition 4.26 the space $\mathcal{D}(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$, so that u is already determined by its restriction $u|_{\mathcal{D}(\mathbb{R}^n)}$. Thus, the linear mapping

$$\iota : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{D}'(\mathbb{R}^n),\ u \mapsto u|_{\mathcal{D}(\mathbb{R}^n)}$$

is injective. Moreover, it is clearly continuous with respect to the weak*-topologies on $\mathcal{S}'$ and $\mathcal{D}'$, respectively, i.e., ι is a topological embedding.

Accordingly, from now on, we will also consider tempered distributions as distributions on $\mathbb{R}^n$ and simply write u in place of $u|_{\mathcal{D}(\mathbb{R}^n)}$.

5.3 Operations on Distributions

Similar to tempered distributions, we also want to define common operations such as partial derivatives and multiplication by suitable functions for distributions in open subsets of $\mathbb{R}^n$. In most cases, we can easily reduce this to the operations already introduced for tempered distributions using criterion (d) from Theorem 5.3.

Let $\Omega \subset \mathbb{R}^n$, $\Omega_1 \subset \mathbb{R}^{n_1}$ and $\Omega_2 \subset \mathbb{R}^{n_2}$ be non-empty open subsets of Euclidean spaces.

Definition. Let $T : \mathcal{D}(\Omega_1) \to \mathcal{D}(\Omega_2)$ be a linear operator for which there exists a continuous linear operator

$${}^tT : \mathcal{D}(\Omega_2) \to \mathcal{D}(\Omega_1)$$

(the **formally transposed operator**) such that

$$\langle T\psi, \varphi\rangle = \langle \psi, {}^tT\varphi\rangle \qquad \text{for all } \psi \in \mathcal{D}(\Omega_1), \varphi \in \mathcal{D}(\Omega_2). \tag{5.11}$$

Then, for $u \in \mathcal{D}'(\Omega_1)$, the distribution $Tu \in \mathcal{D}'(\Omega_2)$ is defined by $Tu := u \circ {}^tT$, i.e.,

$$\langle Tu, \varphi\rangle = \langle u, {}^tT\varphi\rangle, \quad \varphi \in \mathcal{D}(\Omega_2). \tag{5.12}$$

The continuity of ${}^tT$ can easily be reduced to a null sequence continuity condition.

Definition. Let $A : \mathcal{D}(\Omega) \to X$ be a linear map into a locally convex topological vector space X. The operator A is said to be **$\mathcal{D}$-null sequentially continuous** if

$$\lim_{j\to 0} A\varphi_j = 0 \text{ in } X \text{ for every } \mathcal{D}\text{-null sequence } \{\varphi_j\}_j \text{ in } \mathcal{D}(\Omega).$$

Specifically for the operator $A := {}^tT : \mathcal{D}(\Omega_2) \to \mathcal{D}(\Omega_1)$ in (5.12) we thus immediately obtain

Lemma 5.9 *If the linear operator ${}^tT : \mathcal{D}(\Omega_2) \to \mathcal{D}(\Omega_1)$ is $\mathcal{D}$-null sequentially continuous, then for every distribution $u \in \mathcal{D}'(\Omega_1)$, the linear functional on $\mathcal{D}(\Omega_2)$ defined by $Tu := u \circ {}^tT$ is a distribution in Ω_2.*

Proof. If $\{\varphi_j\}_j$ is a $\mathcal{D}$-null sequence in $\mathcal{D}(\Omega_2)$ and $u \in \mathcal{D}'(\Omega_1)$, then (d) in Theorem 5.3 shows that the sequence $Tu(\varphi_j) = u({}^tT(\varphi_j))$ converges in $\mathbb{C}$ to 0. Hence, by criterion (d) in Theorem 5.3, Tu is a distribution in Ω_2. □

We next consider somewhat more generally linear operators

$$A : \mathcal{D}(\Omega_2) \to \mathcal{D}(\Omega_1). \tag{5.13}$$

The verification of $\mathcal{D}$-null sequential continuity for such operators A can, in turn, be easily reduced via localization to the continuity of operators between spaces of Schwartz functions.

Definition. Let A be a linear operator as in (5.13). Then, for each $\chi \in \mathcal{D}(\Omega_2)$, we define the linear mapping

$$\widetilde{A\chi} : \mathcal{S}(\mathbb{R}^{n_2}) \to \mathcal{S}(\mathbb{R}^{n_1})$$

by

$$\widetilde{A\chi}(\phi) := A(\chi\, \phi|_{\Omega_2}), \quad \phi \in \mathcal{S}(\mathbb{R}^{n_2}).$$

Note here that $\chi\, \phi|_{\Omega_2} \in \mathcal{D}(\Omega_2)$, so that $A(\chi\, \phi|_{\Omega_2}) \in \mathcal{D}(\Omega_1) \subset \mathcal{S}(\mathbb{R}^{n_1})$, if we once again interpret the space $\mathcal{D}(\Omega_1)$ as a subspace of $\mathcal{S}(\mathbb{R}^{n_1})$ as in Section 5.2.

Lemma 5.10 (Continuity on $\mathcal{D}$ – Continuity on $\mathcal{S}$) *Assume that for every function $\chi \in \mathcal{D}(\Omega_2)$, there exists a compact set $K_1 \subset \Omega_1$ such that the following hold:*

(i) $\widetilde{A\chi}(\mathcal{S}(\mathbb{R}^{n_2})) \subset \mathcal{D}_{K_1}$.
(ii) *The linear operator $\widetilde{A\chi} : \mathcal{S}(\mathbb{R}^{n_2}) \to \mathcal{S}(\mathbb{R}^{n_1})$ is continuous.*

Then A is a $\mathcal{D}$-null sequentially continuous operator.

Proof. Let $\{\varphi_j\}_j$ be a $\mathcal{D}$-null sequence in $\mathcal{D}(\Omega_2)$. Then, in particular, the supports of the φ_j are contained in a common compact set $K_2 \subset \Omega_2$, i.e., $\varphi_j \in \mathcal{D}_{K_2}$ for all j. Choose $\chi \in \mathcal{D}(\Omega_2)$ such that $\chi \equiv 1$ on a neighborhood of K_2, and then select a compact set $K_1 \subset \Omega_1$ such that $\widetilde{A\chi}(\mathcal{S}(\mathbb{R}^{n_2})) \subset \mathcal{D}_{K_1}$. For $\varphi \in \mathcal{D}_{K_2}$, we then clearly have

$$A\varphi = \widetilde{A\chi}(\varphi).$$

Furthermore, by Proposition 5.1, the topology on $\mathcal{D}_{K_i}$ coincides for $i = 1, 2$ with the relative topology of $\mathcal{S}(\mathbb{R}^{n_i})$ on $\mathcal{D}_{K_i}$. Hence, due to the continuity of $\widetilde{A\chi} : \mathcal{S}(\mathbb{R}^{n_2}) \to \mathcal{S}(\mathbb{R}^{n_1})$, the sequence $A\varphi_j$ converges to 0 in $\mathcal{D}_{K_1} \subset \mathcal{D}(\Omega_1)$, and is thus a $\mathcal{D}$-null sequence in $\mathcal{D}(\Omega_1)$. □

Remark 5.11 The $\mathcal{D}$-null sequential continuity of A, as well as the condition in Lemma 5.10, are in fact even equivalent to the continuity of A—see Theorem 5.22 for details.

Lemmas 5.9 and 5.10 immediately imply

Corollary 5.12 *If the linear operator $A := {}^tT$ in (5.12) satisfies the conditions (i) and (ii) of Lemma 5.10, then for every distribution $u \in \mathcal{D}'(\Omega_1)$, the linear functional $Tu := u \circ {}^tT$ defines a distribution $Tu \in \mathcal{D}'(\Omega_2)$.*

The scheme above of defining the distribution Tu applies in general dimensions $n_1, n_2 \geq 1$. In the subsequent discussions, however, we will only make use of it when $n_1 = n_2$.

For example, in Section 5.1, we essentially introduced the **restriction** $u|_{\Omega_1}$ of a distribution $u \in \mathcal{D}'(\Omega)$ to an open subset Ω_1 of Ω according to this scheme. Indeed:

Here,

$$T : \mathcal{D}(\Omega) \to \mathcal{D}(\Omega_1), \quad \varphi \mapsto \varphi|_{\Omega_1},$$

is the operator of restriction to Ω_1, and

$$^tT : \mathcal{D}(\Omega_1) \to \mathcal{D}(\Omega), \quad \psi \mapsto \tilde{\psi},$$

is the operator of trivial extension to Ω (cf. (5.7)), since for $\varphi \in \mathcal{D}(\Omega)$ and $\psi \in \mathcal{D}(\Omega_1)$, we clearly have

$$\int \varphi|_{\Omega_1}(x)\psi(x)\,\mathrm{d}x = \int \varphi(x)\tilde{\psi}(x)\,\mathrm{d}x.$$

The operator ^tT is evidently $\mathcal{D}$-null sequence continuous. For $u|_{\Omega_1}$, we then have precisely

$$Tu(\psi) = u(^tT\psi) = u(\tilde{\psi}) = u|_{\Omega_1}(\psi)$$

(cf. (5.8)).

5.3.1 Differentiation of distributions

The operator $D^\alpha : \mathcal{D}(\Omega) \to \mathcal{D}(\Omega)$ is evidently continuous, and we have $^tD^\alpha = (-1)^{|\alpha|}D^\alpha$. Therefore, for $u \in \mathcal{D}'(\Omega)$, the partial derivative $D^\alpha u \in \mathcal{D}'(\Omega)$ is defined by

$$\langle D^\alpha u, \varphi\rangle := (-1)^{|\alpha|}\langle u, D^\alpha\varphi\rangle, \quad \varphi \in \mathcal{D}(\Omega).$$

5.3.2 The multiplication of a distribution by a smooth function

Let $f \in C^\infty(\Omega)$, and let $M_f : \mathcal{D}(\Omega) \to \mathcal{D}(\Omega)$ be the multiplication operator defined by

$$M_f(\varphi) := f\varphi.$$

For each $\chi \in \mathcal{D}(\Omega)$, it is evident that

$$\widetilde{M_f\chi}(\phi) = M_{\widetilde{f\chi}}(\phi), \quad \phi \in \mathcal{S} = \mathcal{S}(\mathbb{R}^n).$$

Since $\widetilde{f\chi} \in C_0^\infty(\mathbb{R}^n)$, it follows from Proposition 4.14 that $\widetilde{M_f\chi} : \mathcal{S} \to \mathcal{S}$ is continuous. Furthermore, $\widetilde{M_f\chi}(\mathcal{S}) \subset \mathcal{D}_K$ if $K := \operatorname{supp}\chi$. Since moreover $^tM_f = M_f$, we thus see that the operator $^tM_f = M_f$ satisfies the assumptions of Corollary 5.12, so that $M_f u := u \circ {}^tM_f = u \circ M_f$ defines a distribution in $\mathcal{D}'(\Omega)$.

We therefore define the **product** of the function $f \in C^\infty(\Omega)$ with the distribution $u \in \mathcal{D}'(\Omega)$ by $fu := M_f u$, i.e., by

$$\langle fu, \varphi \rangle := \langle u, f\varphi \rangle, \quad \varphi \in \mathcal{D}(\Omega).$$

It is easy to show (see Exercise 5.2) that

$$\operatorname{supp}(fu) \subset \operatorname{supp} f \cap \operatorname{supp} u. \tag{5.14}$$

5.3.3 The convolution of a distribution with a test function

Let $\Omega = \mathbb{R}^n$ and let $\psi \in \mathcal{D} := \mathcal{D}(\mathbb{R}^n)$. Define

$$C_\psi : \mathcal{D} \to \mathcal{D}, \quad \varphi \mapsto \varphi * \psi,$$

as the convolution operator with ψ. For each $\chi \in \mathcal{D}$, we have

$$\widetilde{(C_\psi \chi)}\phi = (\chi\phi) * \psi = \tilde{C}_\psi \circ M_\chi(\phi), \quad \phi \in \mathcal{S},$$

where $\tilde{C}_\psi$ denotes the convolution operator with ψ on $\mathcal{S}$. According to Proposition 4.14 (b), (d), it follows that the operator

$$\widetilde{C_\psi \chi} = \tilde{C}_\psi \circ M_\chi : \mathcal{S} \to \mathcal{S}$$

is continuous. Furthermore, let K be the compact set

$$K := \operatorname{supp} \chi + \operatorname{supp} \psi.$$

For $\phi \in \mathcal{S}$, we then have

$$\operatorname{supp}(\tilde{C}_\psi \chi)\phi = \operatorname{supp}(\chi\phi) * \psi \subset \operatorname{supp}(\chi\phi) + \operatorname{supp} \psi \subset K,$$

i.e., $\widetilde{C_\psi \chi}(\mathcal{S}) \subset \mathcal{D}_K$. Thus, by Lemma 5.10, C_ψ is $\mathcal{D}$-null sequentially continuous.

Furthermore, similarly to convolution on $\mathcal{S}$, it holds that ${}^t C_\psi = C_{\check{\psi}}$. Therefore, the **convolution** $u * \psi \in \mathcal{D}'(\mathbb{R}^n)$ of the distribution $u \in \mathcal{D}'$ with the test function $\psi \in \mathcal{D}$ is defined by

$$\langle u * \psi, \varphi \rangle := \langle u, \varphi * \check{\psi} \rangle, \quad \varphi \in \mathcal{D}.$$

Similarly, $\psi * u$ is defined by

$$\langle \psi * u, \varphi \rangle := \langle u, \check{\psi} * \varphi \rangle,$$

and then clearly

$$u * \psi = \psi * u. \tag{5.15}$$

Moreover, one easily shows that

$$\operatorname{supp}(u * \psi) \subset \operatorname{supp} u + \operatorname{supp} \psi. \tag{5.16}$$

Similar to the convolution of a tempered distribution with a Schwartz function, one can also define *pointwise* a convolution product as follows:

$$u \circledast \psi(x) := \langle u, \lambda_x \check{\psi} \rangle, \quad x \in \mathbb{R}^n. \tag{5.17}$$

Analogous to Theorem 4.23, we then have:

Theorem 5.13 *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ and $\psi \in \mathcal{D}(\mathbb{R}^n)$. Then:*

(a) *$u \circledast \psi \in C^\infty(\mathbb{R}^n) \subset L^1_{\mathrm{loc}}(\mathbb{R}^n)$, and*

$$D^\alpha(u \circledast \psi) = (D^\alpha u) \circledast \psi = u \circledast D^\alpha \psi \quad \text{for all } \alpha \in \mathbb{N}^n,$$

where in the middle term, D^α refers to the derivative in the sense of distributions, while in the other terms, it denotes the classical derivative.

(b) *$u * \psi = u_f$, where $f := u \circledast \psi$.*

(c) *$(u * \psi) * \varphi = u * (\psi * \varphi)$ for all $\varphi \in \mathcal{D}(\mathbb{R}^n)$.*

Proof. For $\chi \in \mathcal{D}(\mathbb{R}^n)$, $\widetilde{u\chi} \in \mathcal{S}'$ is a tempered distribution. Choose $r > 0$ such that $\operatorname{supp} \psi \subset \overline{B}_r(0)$, and χ such that $\chi \equiv 1$ on $\overline{B}_{R+r}(0)$, $R > 0$. For $\varphi \in \mathcal{D}$ with $\operatorname{supp} \varphi \subset \overline{B}_R(0)$, it follows that

$$\langle u * \psi, \varphi \rangle = \langle u, \varphi * \check{\psi} \rangle = \langle u, \chi(\varphi * \check{\psi}) \rangle = \langle \widetilde{u\chi}, \varphi * \check{\psi} \rangle = \langle \widetilde{u\chi} * \psi, \varphi \rangle,$$

since $\operatorname{supp} \varphi * \check{\psi} \subset \overline{B}_R(0) + \overline{B}_r(0) = \overline{B}_{R+r}(0)$, i.e.,

$$u * \psi = \widetilde{u\chi} * \psi \quad \text{on } \mathcal{D}_{\overline{B}_R(0)}.$$

And, by Theorem 4.23, we have

$$\widetilde{u\chi} * \psi = \widetilde{u\chi} \circledast \psi.$$

Furthermore, for $x \in \overline{B}_R(0)$,

$$\widetilde{u\chi} \circledast \psi(x) = \langle \widetilde{u\chi}, \lambda_x \check{\psi} \rangle = \langle u, \chi(\lambda_x \check{\psi}) \rangle,$$

where $\operatorname{supp} \lambda_x \check{\psi} = x - \operatorname{supp} \psi \subset \overline{B}_{R+r}(0)$, so that $\chi(\lambda_x \check{\psi}) = \lambda_x \check{\psi}$. It follows that

$$\widetilde{u\chi} \circledast \psi(x) = u \circledast \psi(x) \quad \text{for all } x \in \overline{B}_R(0). \tag{5.18}$$

Setting $f := u \circledast \psi$, we obtain overall that

$$u * \psi|_{B_R(0)} = u_f|_{B_R(0)}.$$

Since $R > 0$ was arbitrary, (b) follows.

Similarly, (a) and (c) follow from (5.18) and the analogous statements in Theorem 4.23. □

Due to this theorem, we will henceforth, similarly to the case of tempered distributions, always identify the distribution $u * \psi$ with the C^∞-function $u \circledast \psi$.

5.3.4 Coordinate change

Let $\Phi : \Omega_1 \to \Omega_2$ be a C^∞-diffeomorphism from the non-empty open subset Ω_1 of $\mathbb{R}^n$ onto the open subset Ω_2 of $\mathbb{R}^n$. Since, for all functions $\psi \in \mathcal{D}(\Omega_2)$ and $\varphi \in \mathcal{D}(\Omega_1)$, the change of variables formula

$$\langle \psi \circ \Phi, \varphi \rangle = \int_{\Omega_1} \psi \circ \Phi(x)\, \varphi(x)\, \mathrm{d}x = \int_{\Omega_2} \psi(y)\, \varphi \circ \Phi^{-1}(y) |\det D\Phi^{-1}(y)|\, \mathrm{d}y$$

holds, we define for $u \in \mathcal{D}'(\Omega_2)$ the distribution $u \circ \Phi \in \mathcal{D}'(\Omega_1)$ by

$$\langle u \circ \Phi, \varphi \rangle := \langle u, \varphi \circ \Phi^{-1} |\det D\Phi^{-1}| \rangle, \qquad \varphi \in \mathcal{D}(\Omega_1). \tag{5.19}$$

Remarks. (a) The distribution $u \circ \Phi$ can actually be defined more generally for any **submersion** $\Phi : \Omega_1 \to \Omega_2$, i.e., any C^∞ mapping from an open subset Ω_1 of $\mathbb{R}^n$ to an open subset Ω_2 of $\mathbb{R}^m$, for which the total differential $D\Phi(x)$ is surjective at every point $x \in \Omega_1$ (see, e.g., Hörmander [23]).

(b) One can introduce distributions on arbitrary C^∞ manifolds M by assigning to each coordinate chart $(U_\alpha, \kappa_\alpha)$ of a given atlas of the manifold a distribution u_α in the corresponding coordinate domain $\Omega_\alpha := \kappa_\alpha(U_\alpha) \subset \mathbb{R}^n$, requiring that these distributions transform into each other under the transition maps $\Phi_{\beta,\alpha}$ between any two of these charts (U_β, κ_β) and $(U_\alpha, \kappa_\alpha)$ in the sense of (5.19).

Alternatively, one can attempt again to define distributions as the continuous linear functionals on $\mathcal{D}(M) := C_0^\infty(M)$. This leads to the concept of "distribution density."

However, these two concepts lead to different notions, and only when a "smooth reference measure" is given on M, as provided, for example, by the Riemannian volume element on a Riemannian manifold, can both concepts be converted into one another (see Hörmander [23]). A measure $\mathrm{d}\mu$ on M is here called **smooth** if, under each coordinate map $\kappa_\alpha : U_\alpha \to \Omega_\alpha \subset \mathbb{R}^n$ of an open chart domain $U_\alpha \subset M$, its image measure $(\kappa_\alpha)_*(\mathrm{d}\mu)$ on Ω_α has a smooth density ρ_α with respect to Lebesgue measure, i.e., if $(\kappa_\alpha)_*(\mathrm{d}\mu) = \rho_\alpha\, \mathrm{d}x$, with $\rho_\alpha \in C^\infty(\Omega_\alpha)$.

Such a smooth reference measure allows one, on the one hand, to identify suitable functions f on M again with measures $f\mathrm{d}\mu$, and thus with the distributional density u_f, which is defined by

$$u_f(\varphi) := \int_M f\varphi\, \mathrm{d}\mu, \qquad \varphi \in C_0^\infty(M).$$

Furthermore, it allows one, similarly to $\mathbb{R}^n$, to introduce formally transposed operators via $\langle T\varphi, \psi\rangle = \langle \varphi, {}^tT\psi\rangle$, if we here set

$$\langle \varphi, \psi\rangle := \int_M \varphi\psi \, \mathrm{d}\mu.$$

Now, if $\{u_\alpha\}_\alpha$ is a distribution on M, where $u_\alpha \in \mathcal{D}'(\Omega_\alpha)$ for every α, then one can associate to it a linear functional v on $\mathcal{D}(M)$ as follows: If, for example, $\varphi \in \mathcal{D}(M)$ is supported in the chart domain U_α, one sets

$$v(\varphi) := \langle u_\alpha, (\varphi \circ \kappa_\alpha^{-1})\, \rho_\alpha\rangle.$$

If $\varphi \in \mathcal{D}(M)$ is also supported in the chart domain U_β, and $\Phi_{\beta,\alpha} := \kappa_\beta \circ \kappa_\alpha^{-1}$ denotes the transition map between Ω_α and Ω_β (more precisely, the respective subsets corresponding to the intersection $U_\alpha \cap U_\beta$), then, due to our assumptions on the family of distributions u_α, namely $u_\alpha = u_\beta \circ \Phi_{\beta,\alpha}$, it follows from (5.19) that

$$\langle u_\alpha, (\varphi \circ \kappa_\alpha^{-1})\, \rho_\alpha\rangle = \langle u_\beta, (\varphi \circ \kappa_\beta^{-1})\rho_\alpha \circ \Phi_{\alpha,\beta}^{-1} \,|\det D\Phi_{\alpha,\beta}^{-1}|\rangle.$$

Moreover, it is clear that $\rho_\beta \, \mathrm{d}x$ is the image measure $(\Phi_{\alpha,\beta})_*(\rho_\alpha \, \mathrm{d}x)$ of $\rho_\alpha \mathrm{d}x$ under the transition map $\Phi_{\alpha,\beta}$, i.e.,

$$(\Phi_{\alpha,\beta})_*(\rho_\alpha \, \mathrm{d}x) = \rho_\beta \, \mathrm{d}x.$$

From this, using the change of variables formula for the Lebesgue measure $\mathrm{d}x$, one easily obtains

$$\rho_\alpha \circ \Phi_{\alpha,\beta}^{-1} \,|\det D\Phi_{\alpha,\beta}^{-1}| = \rho_\beta.$$

Altogether, we thus obtain

$$\langle u_\alpha, (\varphi \circ \kappa_\alpha^{-1})\, \rho_\alpha\rangle = \langle u_\beta, (\varphi \circ \kappa_\beta^{-1})\, \rho_\beta\rangle.$$

This shows that the definition of v does not depend on the choice of the coordinate chart, and using partitions of unity, one finally shows that v is well-defined, linear, and continuous on all of $\mathcal{D}(M)$. Thus, we have assigned a distributional density v on $\mathcal{D}(M)$ to the distribution $\{u_\alpha\}_\alpha$, and conversely, one can show that to each distributional density v one can assign a distribution $\{u_\alpha\}_\alpha$ in such a way that the described processes are mutually inverse. For further information on this, see Hörmander [23].

Example 5.14 Let $\Phi := \exp : \mathbb{R} \to]0, \infty[$, $x \mapsto \mathrm{e}^x$ be the exponential mapping, and let $y_0 \in]0, \infty[$ be a fixed point in the image domain. For $\varphi \in \mathcal{D}(\mathbb{R})$, we have

$$\begin{aligned}\langle \delta'_{y_0} \circ \exp, \varphi\rangle &= \langle \delta'_{y_0}, \varphi \circ \log \,|\log'|\rangle = -\frac{\mathrm{d}}{\mathrm{d}y}\left(\frac{\varphi \circ \log(y)}{y}\right)(y_0) \\ &= -\frac{\varphi'(\log y_0)}{y_0^2} + \frac{\varphi(\log y_0)}{y_0^2}.\end{aligned}$$

Thus, we obtain

$$\delta'_{y_0} \circ \exp = \mathrm{e}^{-2x_0} \left(\delta'_{x_0} + \delta_{x_0} \right), \quad \text{with} \quad x_0 := \log y_0.$$

5.4 $\mathcal{D}(\Omega)$ as a Topological Vector Space*

The results we will prove in this and the following section are of more theoretical significance—those primarily interested in the practical handling of distributions can skip these sections without any issue or return to them later.

In this section, we aim to examine the topology of $\mathcal{D}(\Omega)$ in more detail, where Ω is again assumed to be a non-empty open subset of $\mathbb{R}^n$.

As can be easily seen, for each compact set $K \subset \Omega$, the restrictions of the seminorms from $\mathcal{N}_{\mathcal{D}(\Omega)}$ to $\mathcal{D}_K$ form a system of seminorms on $\mathcal{D}_K$ that is equivalent to the family $\{\|\cdot\|_N\}_{N\in\mathbb{N}}$ of C^N-norms. From this, we immediately obtain

Proposition 5.15 *For every compact set $K \subset \Omega$, the topology of $\mathcal{D}_K$ coincides with the relative topology of $\mathcal{D}(\Omega)$, i.e., a subset $U \subset \mathcal{D}_K$ is open if and only if there exists an open subset $V \subset \mathcal{D}(\Omega)$ such that $U = V \cap \mathcal{D}_K$.*

As we will see, the space $\mathcal{D}(\Omega)$ is unfortunately not metrizable. However, the notion of convergence for sequences in $\mathcal{D}(\Omega)$ can still be easily described (see also Theorem 5.3 (d)).

Proposition 5.16 *Let $\{\varphi_j\}_j$ be a sequence in $\mathcal{D}(\Omega)$ and $\varphi \in \mathcal{D}(\Omega)$. The following statements are equivalent:*

(i) $\varphi = \lim\limits_{j\to\infty} \varphi_j$ *in* $\mathcal{D}(\Omega)$.

(ii) *All φ_j and φ are supported in a common compact set $K \subset \Omega$, and φ_j converges to φ in $\mathcal{D}_K$, i.e., $\lim\limits_{j\to\infty} \|\varphi - \varphi_j\|_N = 0$ for every $N \in \mathbb{N}$.*

Proof. (ii) $\Longrightarrow$ (i) follows immediately from Proposition 5.15.

Conversely, suppose (i) holds. First, we observe that every C^N-norm $\|\cdot\|_N$ is contained in $\mathcal{N}_{\mathcal{D}(\Omega)}$ (choose for $\{A\}$ the sequence $\Omega, \emptyset, \emptyset, \ldots$, and set $N_j := N$, $\varepsilon_j := 1$). Hence, in particular, $\lim\limits_{j\to\infty} \|\varphi - \varphi_j\|_N = 0$ for every $N \in \mathbb{N}$. It remains to show that all φ_j are supported in a common compact subset of Ω.

If this were not the case, there would exist a sequence of points $\{x_j\}_j$ in Ω, which tends to infinity in Ω (in the sense described in Section 5.1), as well as a subsequence $\{\varphi_{k_j}\}_j$ of $\{\varphi_j\}_j$ such that $(\varphi - \varphi_{k_j})(x_j) \neq 0$ for every j. Choosing $A_j := \{x_j\}$, $N_j := 0$, and $\varepsilon_j := |(\varphi - \varphi_{k_j})(x_j)|$, for the associated seminorm, we would then have

$$\|\varphi - \varphi_{k_j}\|_{\{A\},\{N\},\{\varepsilon\}} \geq \frac{1}{\varepsilon_j} \|\varphi - \varphi_{k_j}\|_{0,\{x_j\}} = \frac{|(\varphi - \varphi_{k_j})(x_j)|}{\varepsilon_j} = 1$$

for every j, contradicting (i). □

Definition. A sequence or, more generally, a net $\{v_j\}_{j\in J}$ in a topological vector space E is called a **Cauchy sequence**, or a **Cauchy net** if, for every neighborhood U of 0 in E, there exists an index $j_0 \in J$ such that

$$v_j - v_k \in U \quad \text{for all } j, k \geqslant j_0.$$

If the topology on E is induced by the seminorms $\{\|\cdot\|_i\}_{i\in I}$, then this is equivalent to the condition that for every $\varepsilon > 0$ and every $i \in I$, there exists a $j_0 \in J$ such that

$$\|v_j - v_k\|_i < \varepsilon \quad \text{for all } j, k \geqslant j_0. \tag{5.20}$$

Theorem 5.17 *$\mathcal{D}(\Omega)$ is* **complete***, i.e., every Cauchy sequence, and more generally every Cauchy net in $\mathcal{D}(\Omega)$, converges in $\mathcal{D}(\Omega)$.*

Proof. Let $\{\varphi_\nu\}_\nu$ be a Cauchy net in $\mathcal{D}(\Omega)$. Then, in particular, $\{\varphi_\nu\}_\nu$ is a Cauchy net with respect to each C^N-norm $\|\cdot\|_N$, and hence, by well-known theorems from real analysis, $\{\varphi_\nu\}_\nu$ converges uniformly on Ω to a C^∞ function φ on Ω, and each derivative $D^\alpha\varphi_\nu$ converges uniformly to $D^\alpha\varphi$. (The usual proof for Cauchy sequences easily carries over to Cauchy nets.)

Similar to the proof of Theorem 5.16, we show that φ has compact support, i.e., φ lies in $\mathcal{D}(\Omega)$.

Otherwise, there would exist a sequence of points $\{x_j\}_j$ in Ω tending to infinity in Ω, such that $\varphi(x_j) \neq 0$ for all j. Let us then choose $A_j := \{x_j\}$, $N_j := 0$, and $\varepsilon_j := |\varphi(x_j)|$, and set

$$\begin{aligned}\|\psi\|' &:= \|\psi\|_{\{A\},\{N\},\{\varepsilon\}} \\ &= \sum_j \frac{|\psi(x_j)|}{|\varphi(x_j)|}.\end{aligned}$$

Since $\{\varphi_\nu\}_\nu$ is a Cauchy net with respect to $\|\cdot\|'$, there would exist some ν_0 and a constant $C > 0$ such that

$$\|\varphi_\nu\|' \leq C \quad \text{for all } \nu \geqslant \nu_0,$$

(why?), hence

$$\sum_j \frac{|\varphi_\nu(x_j)|}{|\varphi(x_j)|} \leq C \quad \text{for all } \nu \geqslant \nu_0. \tag{5.21}$$

However, for every j we have $\lim_\nu \varphi_\nu(x_j) = \varphi(x_j)$, which leads to a contradiction to (5.21).

It remains to show that φ_ν converges to φ in $\mathcal{D}(\Omega)$. To this end, we consider an arbitrary seminorm $\|\cdot\| := \|\cdot\|_{\{A\},\{N\},\{\varepsilon\}}$ from $\mathcal{N}_{\mathcal{D}(\Omega)}$ and fix some $\varepsilon > 0$.

Then there exists an index ν_0 such that

$$\|\varphi_\nu - \varphi_\mu\| < \varepsilon/3 \quad \text{for all } \nu, \mu \geqslant \nu_0.$$

Furthermore, there exists a j_0 such that $\operatorname{supp}\varphi \cap A_j = \emptyset$ for all $j \geq j_0$. Choose $j_1 \geq j_0$ so that

$$\sum_{j>j_1} \frac{1}{\varepsilon_j} \|\varphi_{\nu_0}\|_{A_j,N_j} < \varepsilon/3.$$

For $\nu \geqslant \nu_0$, we then have

$$\sum_{j>j_1} \frac{1}{\varepsilon_j} \|\varphi_{\nu}\|_{A_j,N_j} \leq \|\varphi_\nu - \varphi_{\nu_0}\| + \varepsilon/3 < 2\varepsilon/3.$$

It follows that

$$\|\varphi - \varphi_\nu\| = \sum_{j=0}^{j_1} \frac{1}{\varepsilon_j} \|\varphi - \varphi_\nu\|_{A_j,N_j} + \sum_{j>j_1} \frac{1}{\varepsilon_j} \|\varphi_\nu\|_{A_j,N_j} < \sum_{j=0}^{j_1} \frac{1}{\varepsilon_j} \|\varphi - \varphi_\nu\|_{N_j} + 2\varepsilon/3.$$

Since for each j we have that $\lim_\nu \|\varphi - \varphi_\nu\|_{N_j} = 0$, there finally exists a $\nu_1 \geqslant \nu_0$ such that

$$\sum_{j=0}^{j_1} \frac{1}{\varepsilon_j} \|\varphi - \varphi_\nu\|_{N_j} < \varepsilon/3$$

for every $\nu \geqslant \nu_1$. For $\nu \geqslant \nu_1$, we thus obtain $\|\varphi - \varphi_\nu\| < \varepsilon$. □

If $\{K_j\}_j$ is any compact exhaustion of Ω, then clearly

$$\mathcal{D}(\Omega) = \bigcup_j \mathcal{D}_{K_j}. \tag{5.22}$$

This shows that the space $\mathcal{D}(\Omega)$ is, in a certain sense, a "limit" of an increasing sequence of subspaces $\mathcal{D}_{K_j}$. The following theorem shows that this is even true in a topological sense.

Theorem 5.18 *The locally convex topological vector space $\mathcal{D}(\Omega)$ is the "inductive limit" of the topological vector spaces $\mathcal{D}_K$, $K \subset \Omega$ compact, i.e., a convex subset of $\mathcal{D}(\Omega)$ is a neighborhood of zero in $\mathcal{D}(\Omega)$ if and only if its intersection with each $\mathcal{D}_K$ is a neighborhood of zero in $\mathcal{D}_K$.*

Proof. One implication is clear by Proposition 5.15.

Conversely, let $U \subset \mathcal{D}(\Omega)$ be a convex subset whose intersection with each $\mathcal{D}_K$ is a zero neighborhood in $\mathcal{D}_K$. It remains to show that U is a zero neighborhood in $\mathcal{D}(\Omega)$.

To this end, assume that $\{K_j\}_j$ is a compact exhaustion of Ω. For each j, there exist $N_j \in \mathbb{N}$ and $\delta_j > 0$ such that the ball

$$\{\psi \in \mathcal{D}_{K_{j+1}} : \|\psi\|_{N_j} < \delta_j\}$$

is contained in $U \cap \mathcal{D}_{K_{j+1}}$.

As in the proof of Theorem 5.3, we choose a sequence $\{\eta_j\}_j$ in $\mathcal{D}(\Omega)$ such that $\sum_j \eta_j \equiv 1$ in Ω and $\operatorname{supp}\eta_j \subset A_j$, where $A_0 := K_1$ and $A_j := K_{j+1} \setminus K_{j-1}$ for $j \geq 1$. Any $\varphi \in \mathcal{D}(\Omega)$ can then be decomposed as a finite sum

$$\varphi = \sum_j 2^{-j-1}(2^{j+1}\eta_j\varphi).$$

By the convexity of U, it follows that φ already lies in U if each function $2^{j+1}\eta_j\varphi$ lies in U.

Using Leibniz's rule, one easily sees that for each j, there exists a constant $k_j \geq 0$ such that for all $\varepsilon_j > 0$, the following holds:

$$\|\varphi\|_{A_j,N_j} \leq \varepsilon_j \implies \|2^{j+1}\eta_j\varphi\|_{N_j} \leq k_j\varepsilon_j. \tag{5.23}$$

We now choose a sequence $\{\varepsilon_j\}_j$, $\varepsilon_j > 0$, such that $k_j\varepsilon_j < \delta_j$ for all j, and define

$$\|\cdot\| := \|\varphi\|_{\{A\},\{N\},\{\varepsilon\}},$$

with the above chosen A_j, N_j, and ε_j.

Let now $\varphi \in \mathcal{D}(\Omega)$ with $\|\varphi\| < 1$. Then $\frac{1}{\varepsilon_j}\|\varphi\|_{A_j,N_j} < 1$, hence, by (5.23),

$$\|2^{j+1}\eta_j\varphi\|_{N_j} < \delta_j.$$

That is, $2^{j+1}\eta_j\varphi \in U$, since $\operatorname{supp}(2^{j+1}\eta_j\varphi) \subset K_{j+1}$. Thus, φ in U.

This shows that the ball $\{\varphi \in \mathcal{D}(\Omega) : \|\varphi\| < 1\}$ lies in U, hence U is a neighborhood of zero in $\mathcal{D}(\Omega)$. □

Remark 5.19 Since the space $\mathcal{D}_K$ is metrizable, a mapping on $\mathcal{D}_K$ is continuous if and only if it is sequentially continuous.

This immediately leads to

Theorem 5.20 *Let $T : \mathcal{D}(\Omega) \to X$ be a linear mapping into a locally convex topological vector space X. Then the following properties are equivalent:*

(i) *T is continuous.*
(ii) *$T|_{\mathcal{D}_K} : \mathcal{D}_K \to X$ is continuous for every compact set $K \subset \Omega$.*
(iii) *$\lim_{j\to 0} T\varphi_j = 0$ for every $\mathcal{D}$-null sequence $\{\varphi_j\}_j$ in $\mathcal{D}(\Omega)$, i.e., T is $\mathcal{D}$-null sequentially continuous.*

Proof. By the definition of the topology on $\mathcal{D}(\Omega)$, the topology of $\mathcal{D}_K$ coincides with the relative topology of $\mathcal{D}(\Omega)$ whenever $K \subset \Omega$ is compact. This immediately implies (i) $\implies$ (ii), and (ii) $\implies$ (iii) is clear.

Conversely, from Remark 5.19, we see that (iii) $\implies$ (ii).

Finally, assume (ii) holds, and let V be a zero neighborhood in X. We show that

$$U := T^{-1}(V) \subset \mathcal{D}(\Omega)$$

is a zero neighborhood in $\mathcal{D}(\Omega)$. Without loss of generality, we may assume that V is convex. Then the same holds for U, since T is linear, and by assumption, $U \cap \mathcal{D}_K = (T|_{\mathcal{D}_K})^{-1}(V)$ is a zero neighborhood in $\mathcal{D}_K$ for every compact set $K \subset \Omega$.

Thus, by Theorem 5.18, U is a zero neighborhood in $\mathcal{D}(\Omega)$, so T is continuous at 0. By the linearity of T, T is then continuous at every point $\varphi \in \mathcal{D}(\Omega)$, proving (i). □

Remarks 5.21 (a) The topology of $\mathcal{D}(\Omega)$ is not metrizable.

(b) The space $\mathcal{D}(\Omega)$, endowed with the topology generated by the sequence of C^N-norms $\|\cdot\|_N$, is not complete (see Exercise 5.3).

Regarding (a), choosing a compact exhaustion $\{K_j\}_j$ of Ω, we have

$$\mathcal{D}(\Omega) = \bigcup_{j=0}^{\infty} \mathcal{D}_{K_j}.$$

However, each subspace $\mathcal{D}_{K_j}$ clearly has empty interior relative to $\mathcal{D}(\Omega)$ (otherwise, $\mathcal{D}_{K_j}$ would contain a zero neighborhood of $\mathcal{D}(\Omega)$, implying $\mathcal{D}_{K_j} = \mathcal{D}(\Omega)$).

If $\mathcal{D}(\Omega)$ were metrizable, then by Theorem 5.17, it would be a Fréchet space. Consequently, by the Baire category theorem (specifically Corollary A.3 in Appendix A), the set $\bigcup_{j=0}^{\infty} \mathcal{D}_{K_j}$ would have no interior points in $\mathcal{D}(\Omega)$, leading to a contradiction.

5.5 Continuous Linear Mappings Between Spaces of Test Functions*

Here we give a characterization of the continuous linear mappings between spaces of test functions. Again, let $\Omega_1 \subset \mathbb{R}^{n_1}$ and $\Omega_2 \subset \mathbb{R}^{n_2}$ be non-empty open sets.

Theorem 5.22 *Let $T : \mathcal{D}(\Omega_1) \to \mathcal{D}(\Omega_2)$ be a linear mapping. Then the following are equivalent:*

(i) *T is continuous.*
(ii) *For every sequence $\{\varphi_j\}_j$ in $\mathcal{D}(\Omega_1)$, the following holds:*
If $\varphi_j \to \varphi$ in $\mathcal{D}(\Omega_1)$, then $T\varphi_j \to T\varphi$ in $\mathcal{D}(\Omega_2)$.
(iii) *For every compact set $K_1 \subset \Omega_1$, there exists a compact set $K_2 \subset \Omega_2$ such that $T|_{\mathcal{D}_{K_1}}$ maps $\mathcal{D}_{K_1}$ continuously into $\mathcal{D}_{K_2}$.*
(iv) *For every $\chi \in \mathcal{D}(\Omega_1)$, there exists a compact set $K \subset \Omega_2$ such that:*
$\widetilde{T\chi}(\mathcal{S}(\mathbb{R}^n)) \subset \mathcal{D}_K$, and the mapping $\widetilde{T\chi} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is continuous.

Proof. The equivalence of (i) and (ii) follows immediately from Theorem 5.20.

(i) $\Longrightarrow$ (iii): Let T be continuous, and let $K_1 \subset \Omega_1$ be compact. It suffices to show that there exists a compact set $K_2 \subset \Omega_2$ such that $T(\mathcal{D}_{K_1}) \subset \mathcal{D}_{K_2}$; the continuity of $T|_{\mathcal{D}_{K_1}} : \mathcal{D}_{K_1} \to \mathcal{D}_{K_2}$ is then clear, since $\mathcal{D}_{K_i}$ carries the relative topology of $\mathcal{D}(\Omega_i)$, $i = 1, 2$.

Suppose that no such compact set $K_2 \subset \Omega_2$ exists. Then choose a compact exhaustion $\{K^j\}_{j\in\mathbb{N}}$ of Ω_2. For each j, there would then exist a point $y_j \in \Omega_2 \setminus K^j$ and a function $\psi_j \in \mathcal{D}_{K_1}$ such that $T\psi_j(y_j) \neq 0$. Set $\varphi_j := \frac{\psi_j}{\|\psi_j\|_j}$. Then we have

$$\|\varphi_j\|_N \leq \|\varphi_j\|_j = 1 \quad \text{for all } j \geq N,$$

and $\varepsilon_j := \frac{|T\varphi_j(y_j)|}{j+1} > 0$. Then the seminorm $\|\cdot\|'$ defined by

$$\|\psi\|' := \sum_j \frac{1}{\varepsilon_j}|\psi(y_j)|, \quad \psi \in \mathcal{D}(\Omega_2),$$

lies in $\mathcal{N}_{\mathcal{D}(\Omega_2)}$. Due to the continuity of T, there exists an $N \in \mathbb{N}$ and a constant $C \geq 0$ such that

$$\|T\varphi\|' \leq C\|\varphi\|_N \quad \text{for all } \varphi \in \mathcal{D}_{K_1}.$$

In particular, for every $j \geq N$ we obtain

$$j+1 = \frac{|T\varphi_j(y_j)|}{\varepsilon_j} \leq \|T\varphi_j\|' \leq C\|\varphi_j\|_N \leq C.$$

This yields a contradiction.

(iii) $\Longrightarrow$ (iv): Let $\chi \in \mathcal{D}(\Omega_1)$, and set $K_1 := \operatorname{supp}\chi$. For $\phi \in \mathcal{S}$, we then have

$$\widetilde{T\chi}(\phi) = T(\phi^0), \text{ where } \phi^0 := \chi\ \phi|_{\Omega_1} \in \mathcal{D}_{K_1}.$$

Since multiplication by χ is a continuous mapping on $\mathcal{S}$, the continuity of $\widetilde{T\chi} : \mathcal{S} \to \mathcal{S}$ follows immediately from (iii) and Proposition 5.1.

(iv) $\Longrightarrow$ (i): By Theorem 5.20, $T : \mathcal{D}(\Omega_1) \to \mathcal{D}(\Omega_2)$ is continuous if and only if $T|_{\mathcal{D}_K} : \mathcal{D}_K \to \mathcal{D}(\Omega_2)$ is continuous for every compact set $K \subset \Omega_1$. Let K be such a compact set, and choose $\chi \in \mathcal{D}(\Omega_1)$ such that $\chi \equiv 1$ on a neighborhood of K. For $\varphi \in \mathcal{D}_K$, we then have $T\varphi = \widetilde{T\chi}(\varphi)$. Thus, by (iv) and Proposition 5.1, the restriction $T|_{\mathcal{D}_K} : \mathcal{D}_K \to \mathcal{S}$ is continuous, and the image lies in some space $\mathcal{D}_{K_2}$. Using Proposition 5.1 again, we conclude that

$$T|_{\mathcal{D}_K} : \mathcal{D}_K \to \mathcal{D}_{K_2} \hookrightarrow \mathcal{D}(\Omega_2)$$

is continuous. □

5.6 Exercises

5.1 a) Let $\Omega_1 \subset \Omega$ be non-empty open subsets of $\mathbb{R}^n$, and let $f \in L^1_{\mathrm{loc}}(\Omega)$. Show: $u_f|_{\Omega_1} = u_{f|_{\Omega_1}}$.

b) Determine $\operatorname{supp} D^\alpha\delta_x$ for every $x \in \mathbb{R}^n$ and every $\alpha \in \mathbb{N}^n$.

5.2 Let $f \in C^\infty(\Omega)$ and $u \in \mathcal{D}'(\Omega)$.

a) Show that the product rule

$$D_j(fu) = (D_j f)u + f(D_j u)$$

also holds under these assumptions.

b) Show that $\operatorname{supp} fu \subset \operatorname{supp} f \cap \operatorname{supp} u$.

c) Suppose $fu = 0$. Conclude that $\operatorname{supp} u \subset f^{-1}(\{0\})$. Does the converse hold too?

5.3 Show that the space $\mathcal{D}(\Omega)$, endowed with the topology induced by the sequence of norms $\|\cdot\|_N$, is not complete.

Chapter 6
Distributions with Compact Support

In this chapter, we will show that every compactly supported distribution $u \in \mathcal{D}'(\Omega)$ can be extended to a linear functional on the space

$$\mathcal{E}(\Omega) := C^\infty(\Omega),$$

even further on the space $\mathcal{E}(\mathbb{R}^n) := C^\infty(\mathbb{R}^n)$. This will in particular allow us to regard any distribution with compact support as a tempered distribution, so that its Fourier transform is defined. It is even possible to define its Fourier–Laplace transform as an entire holomorphic function.

Definition. Let $\Omega \neq \emptyset$ be open in $\mathbb{R}^n$. The set of all distributions $u \in \mathcal{D}'(\Omega)$ with compact support clearly forms a linear subspace of $\mathcal{D}'(\Omega)$, which we denote by $\mathcal{E}'(\Omega)$. I.e.,

$$\mathcal{E}'(\Omega) := \{u \in \mathcal{D}'(\Omega) : \operatorname{supp} u \text{ is compact}\}.$$

6.1 Extension of $u \in \mathcal{E}'(\Omega)$ to $\mathcal{E}(\Omega)$ and $\mathcal{E}(\mathbb{R}^n)$

Proposition 6.1 *Let $u \in \mathcal{E}'(\Omega)$, and let $\chi \in \mathcal{D}(\Omega)$ be such that $\chi \equiv 1$ in a neighborhood of the support of u.*

(a) *Then $\chi u = u$ on $\mathcal{D}(\Omega)$.*
(b) *By setting*

$$u_{\mathcal{E}}(\varphi) := u(\chi\varphi), \quad \varphi \in C^\infty(\Omega),$$

we can extend u from $\mathcal{D}(\Omega)$ to a linear form $u_{\mathcal{E}}$ on the space $C^\infty(\Omega)$. This extension does not depend on the choice of χ.
(c) *There exists an $N \in \mathbb{N}$ and, for each compact neighborhood K of $\operatorname{supp} u$ in Ω, a constant $C_K \geq 0$ such that*

$$|u_{\mathcal{E}}(\varphi)| \leq C_K \max_{|\alpha| \leq N} \sup_{x \in K} |D^\alpha \varphi(x)| = C_K \|\varphi\|_{K,N} \tag{6.1}$$

for all $\varphi \in C^\infty(\Omega)$. In particular, u has finite order, and we may choose N in (6.1) as the order of u.

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D. Müller, *An Invitation to Fourier Analysis and Distribution Theory*,
Universitext, https://doi.org/10.1007/978-3-032-31550-2_6

Proof. (a) Let $\varphi \in \mathcal{D}(\Omega)$. Then

$$\langle u, \varphi \rangle = \langle u, \chi\varphi \rangle + \langle u, (1-\chi)\varphi \rangle = \langle \chi u, \varphi \rangle + \langle u, (1-\chi)\varphi \rangle.$$

Since $(1-\chi)\varphi \equiv 0$ in a neighborhood of $\operatorname{supp} u$, i.e., $\operatorname{supp}((1-\chi)\varphi) \cap \operatorname{supp} u = \emptyset$, it follows from (5.10) that $\langle u, (1-\chi)\varphi \rangle = 0$, and consequently $\langle u, \varphi \rangle = \langle \chi u, \varphi \rangle$.

(b) Let χ_1 be another test function in $\mathcal{D}(\Omega)$ that is identically 1 in a neighborhood of $\operatorname{supp} u$. For $\varphi \in C^\infty(\Omega)$, we then have

$$\langle u, \chi\varphi \rangle - \langle u, \chi_1\varphi \rangle = \langle u, (\chi - \chi_1)\varphi \rangle.$$

Since $(\chi - \chi_1)\varphi$ vanishes in a neighborhood of $\operatorname{supp} u$, it follows that $\langle u, (\chi - \chi_1)\varphi \rangle = 0$. This shows that $u_{\mathcal{E}}$ is well-defined. Moreover, (a) implies that

$$u_{\mathcal{E}}|_{\mathcal{D}(\Omega)} = u,$$

which completes the proof of (b).

(c) Since K is a neighborhood of the support of u, there exists, by Lemma 2.15, a function $\chi \in \mathcal{D}(\Omega)$ such that $\chi \equiv 1$ in a neighborhood of $\operatorname{supp} u$ and $\operatorname{supp} \chi \subset K$. By (a), we have $u = \chi u$, and by Theorem 5.3(b), there exist constants $N = N_K \in \mathbb{N}$ and $C(K) \geq 0$ such that for all $\varphi \in C^\infty(\Omega)$,

$$|u_{\mathcal{E}}(\varphi)| = |u(\chi\varphi)| \leq C(K)\|\chi\varphi\|_N,$$

since $\operatorname{supp}(\chi\varphi) \subset K$. By the Leibniz rule and since $\operatorname{supp} \chi \subset K$,

$$\|\chi\varphi\|_N \leq C_K \max_{|\alpha| \leq N} \sup_{x \in K} |D^\alpha \varphi(x)| = C_K \|\varphi\|_{K,N},$$

which establishes (6.1).

A priori, N may depend on K. However, we may assume without loss of generality that each of our compact sets K is contained in a fixed compact set $L \subset \Omega$, for which we chose N according to Theorem 5.3. Thus, N can be chosen independently of K, and we may even take N as the order of u. □

Definitions. (a) We endow the space $\mathcal{E}(\Omega)$ with the locally convex topology induced by the family of seminorms

$$\mathcal{N}_{\mathcal{E}(\Omega)} := \{\|\cdot\|_{K,N}\}_{K \subset \Omega \text{ compact}, N \in \mathbb{N}},$$

which is clearly the topology of uniform convergence of all derivatives on compact subsets.

If $\{K_j\}_{j \in \mathbb{N}}$ is a compact exhaustion of Ω, and if we choose in place of $\mathcal{N}_{\mathcal{E}(\Omega)}$ the corresponding family of seminorms in which only the compact sets $K = K_j$ are used, then we obtain an equivalent countable family of seminorms. This shows that the space $\mathcal{E}(\Omega)$ is metrizable.

Moreover, it is easily seen that $\mathcal{E}(\Omega)$ is complete and thus a Fréchet space.

By Proposition 6.1(c), every distribution with compact support in Ω extends naturally to a continuous linear functional on $\mathcal{E}(\Omega)$.

Conversely, every continuous linear functional $u_\mathcal{E}$ on $\mathcal{E}(\Omega)$ satisfies an estimate (6.1) for a suitable compact subset $K \subset \Omega$ and suitable constants C_K and N. Thus its restriction $u := u_\mathcal{E}|_{\mathcal{D}(\Omega)}$ to $\mathcal{D}(\Omega) \subset \mathcal{E}(\Omega)$ is a distribution in Ω with compact support in K. Its natural extension coincides again with $u_\mathcal{E}$ (see also Exercise 6.1).

Thus, the space $\mathcal{E}'(\Omega)$ of distributions with compact support in Ω can indeed be identified with the topological dual space $\mathcal{E}(\Omega)'$ of $\mathcal{E}(\Omega)$ by means of the linear isomorphism $\mathcal{E}'(\Omega) \to \mathcal{E}(\Omega)', u \mapsto u_\mathcal{E}$.

(b) The proof of Proposition 6.1 shows in a similar way that by putting

$$\widetilde{u_\mathcal{E}}(\varphi) := u_\mathcal{E}(\varphi|_\Omega) = u(\chi\,\varphi|_\Omega), \quad \varphi \in C^\infty(\mathbb{R}^n), \tag{6.2}$$

we may even extend the distribution $u \in \mathcal{E}'(\Omega)$ in a natural way from the space $\mathcal{D}(\Omega)$ "trivially" to a linear functional $\widetilde{u_\mathcal{E}}$ on the space $\mathcal{E}(\mathbb{R}^n)$. This extension again does not depend on the choice of χ, and an inequality analogous to (6.1) holds:

$$|\widetilde{u_\mathcal{E}}(\varphi)| \le C \max_{|\alpha|\le N} \sup_{x\in K} |D^\alpha \varphi(x)| \quad \text{for all } \varphi \in \mathcal{E}(\mathbb{R}^n), \tag{6.3}$$

for some suitable compact set $K \subset \Omega$ and certain constants C and N. We denote the restriction of $\widetilde{u_\mathcal{E}}$ to the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ by

$$\tilde{u} := \widetilde{u_\mathcal{E}}|_{\mathcal{S}(\mathbb{R}^n)}, \qquad u \in \mathcal{E}'(\Omega).$$

Note that for any $\chi \in \mathcal{D}(\Omega)$ such that $\chi \equiv 1$ in a neighborhood of $\operatorname{supp} u$, we have

$$\tilde{u} = \widetilde{u\chi}. \tag{6.4}$$

Moreover, if $\varphi \in \mathcal{D}(\Omega)$, and if $\tilde{\varphi} \in \mathcal{D}(\mathbb{R}^n)$ denotes again the trivial extension of φ to $\mathbb{R}^n$, then clearly

$$\langle \tilde{u}, \tilde{\varphi} \rangle = \langle u, \varphi \rangle. \tag{6.5}$$

Thus, any distribution $u \in \mathcal{E}'(\Omega)$ with compact support in Ω can naturally be identified by means of (6.2) with the continuous linear functional $\widetilde{u_\mathcal{E}} \in \mathcal{E}(\mathbb{R}^n)'$ on $\mathcal{E}(\mathbb{R}^n)$.

As (6.4) shows, its restriction $\tilde{u}$ to the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ is a tempered distribution (this also follows immediately from (6.3)).

Simplification of notation. In the following, with a slight abuse of notation, we will in general denote all these extensions of $u \in \mathcal{E}'(\Omega)$—$u_\mathcal{E}$ on $\mathcal{E}(\Omega)$, $\widetilde{u_\mathcal{E}}$ on $\mathcal{E}(\mathbb{R}^n)$, and $\tilde{u}$ on $\mathcal{S}(\mathbb{R}^n)$—simply as u, whenever it is clear from the context which space they are acting on.

In particular, for $u \in \mathcal{E}'(\Omega)$, the Fourier transform $\hat{u} \in \mathcal{S}'$ is well-defined (specifically by $\widehat{u\chi}$, if χ is as in Proposition 6.1).

(c) More generally, the Fourier–Laplace transform of $u \in \mathcal{E}'(\Omega)$ can be defined as follows: For $\zeta = (\zeta_1, \dots, \zeta_n) \in \mathbb{C}^n$, let $e_\zeta : \mathbb{R}^n \to \mathbb{C}$ denote the function

$$e_\zeta(x) := \mathrm{e}^{i\zeta \cdot x}, \quad x \in \mathbb{R}^n,$$

where

$$\zeta \cdot x := \zeta_1 x_1 + \cdots + \zeta_n x_n.$$

Since $e_\zeta \in \mathcal{E}(\mathbb{R}^n)$, we can thus define for $u \in \mathcal{E}'(\Omega)$ the function

$$\mathcal{FL}u : \mathbb{C}^n \to \mathbb{C}$$

by

$$\mathcal{FL}u(\zeta) := u(e_{-\zeta}), \qquad \zeta \in \mathbb{C}^n.$$

Note. More specifically, for any function $\chi \in \mathcal{D}(\Omega)$ that is identically 1 on a neighborhood of the support of u, we have by definition:

$$\mathcal{FL}u(\zeta) := u(\chi\, e_{-\zeta}|_\Omega), \qquad \zeta \in \mathbb{C}^n. \tag{6.6}$$

$\mathcal{FL}u$ is called the **Fourier–Laplace transform** of u. We will study this in more detail in Section 6.5.

(d) For $u \in \mathcal{E}'(\mathbb{R}^n)$ and $\psi \in C^\infty(\mathbb{R}^n)$, the **convolution** $u * \psi$ can be defined in the usual way, both by

$$\langle u * \psi, \varphi \rangle := \langle u, \varphi * \check{\psi} \rangle, \quad \varphi \in \mathcal{D}(\mathbb{R}^n),$$

and pointwise by

$$u \circledast \psi(x) := \langle u, \lambda_x \check{\psi} \rangle, \quad x \in \mathbb{R}^n.$$

6.2 Distributions with a Single-Point Support

We now investigate the question of what distributions supported at a single point look like.

Theorem 6.2 *Let $y \in \mathbb{R}^n$, and let $u \in \mathcal{E}'(\mathbb{R}^n)$ with $\operatorname{supp} u = \{y\}$. Furthermore, let N be the order of u. Then there exist constants $a_\alpha \in \mathbb{C}$ for $|\alpha| \le N$ such that*

$$u = \sum_{|\alpha| \le N} a_\alpha D^\alpha \delta_y. \tag{6.7}$$

Proof. Without loss of generality, we may assume $y = 0$ by considering the shifted distribution $\lambda_y u$ defined by

$$\langle \lambda_{-y} u, \varphi \rangle := \langle u, \lambda_y \varphi \rangle, \qquad \varphi \in \mathcal{D} := \mathcal{D}(\mathbb{R}^n),$$

whose support is given by $\{0\}$.

Choose then $\chi \in \mathcal{D}$ with $\chi(x) = 1$ for $|x| \le 1/2$ and $\chi(x) = 0$ for $|x| \ge 1$, and define

$$\chi_{(\varepsilon)}(x) := \chi\left(\frac{x}{\varepsilon}\right), \quad 0 < \varepsilon < 1.$$

Then $\operatorname{supp} \chi_{(\varepsilon)} \subset \overline{B}_\varepsilon(0)$. Furthermore, by Proposition 6.1, we have

$$u = \chi_{(\varepsilon)} u \quad \text{for every } \varepsilon > 0.$$

Choosing, for example, $K := \overline{B}_1(0)$ in (6.1), it follows that

$$|u(\varphi)| \le C \max_{|\alpha| \le N} \|D^\alpha(\chi_{(\varepsilon)}\varphi)\|_\infty \quad \text{for all } \varphi \in \mathcal{E}(\mathbb{R}^n). \tag{6.8}$$

Here, we already consider u as a linear functional on $\mathcal{E} = \mathcal{E}(\mathbb{R}^n)$.
Furthermore,

$$D^\alpha(\chi_{(\varepsilon)}\varphi)(x) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} \varepsilon^{-|\beta|} (D^\beta \chi)\left(\frac{x}{\varepsilon}\right) D^{\alpha-\beta}\varphi(x),$$

so that

$$\|D^\alpha(\chi_{(\varepsilon)}\varphi)\|_\infty \le C_1 \sum_{\beta \le \alpha} \varepsilon^{-|\beta|} \sup_{|x| \le \varepsilon} |D^{\alpha-\beta}\varphi(x)|. \tag{6.9}$$

First, we assume that $D^\gamma\varphi(0) = 0$ for all $\gamma \in \mathbb{N}^n$ with $|\gamma| \le N$, meaning that φ vanishes to order $N+1$ at 0. Then, the Taylor expansion of φ around 0 shows that $\varphi(x) = O(|x|^{N+1})$. Accordingly,

$$D^\varrho\varphi(x) = O(|x|^{N+1-|\varrho|}), \quad \text{for } |\varrho| \le N.$$

Thus, for $|\alpha - \beta| \le N$,

$$\sup_{|x| \le \varepsilon} |D^{\alpha-\beta}\varphi(x)| \le C_{\alpha-\beta}\, \varepsilon^{N+1-|\alpha-\beta|} = C_{\alpha-\beta}\varepsilon^{N+1-|\alpha|+|\beta|},$$

so that, using (6.9),

$$\|D^\alpha(\chi_{(\varepsilon)}\varphi)\|_\infty \le C_2 \sum_{\beta \le \alpha} \varepsilon^{-|\beta|}\varepsilon^{N+1-|\alpha|+|\beta|} \le C_3\varepsilon^{N+1-|\alpha|},$$

if $|\alpha| \le N$. Thus, by (6.8), in the limit as $\varepsilon \to 0$ we obtain:

If φ vanishes to order $N+1$ at 0, then $u(\varphi) = 0$.

Next, if $\varphi \in \mathcal{E}$ is arbitrary, then by Taylor's formula,

$$\varphi(x) = \sum_{|\alpha| \le N} \frac{\varphi^{(\alpha)}(0)}{\alpha!} x^\alpha + r(x),$$

where $r \in C^\infty(\mathbb{R}^n)$ vanishes to order $N+1$ at 0. From the previous result, we thus have $u(r) = 0$. Defining

$$P_\alpha(x) := \frac{(-i)^{|\alpha|}}{\alpha!} x^\alpha,$$

then $P_\alpha \in \mathcal{E}$, and it follows that

$$u(\varphi) = u\left(\sum_{|\alpha| \le N} (-1)^{|\alpha|} D^\alpha \varphi(0)\, P_\alpha \right) = \sum_{|\alpha| \le N} u(P_\alpha)(-1)^{|\alpha|} D^\alpha \varphi(0),$$

i.e.,

$$u = \sum_{|\alpha| \le N} a_\alpha D^\alpha \delta_0,$$

with $a_\alpha := u(P_\alpha) \in \mathbb{C}$. □

Remark 6.3 Since the distributions $D^\alpha \delta_0, \alpha \in \mathbb{N}^n$, are linearly independent (see, e.g., Exercise 4.6), in combination with Example 5.6(ii) the identity (6.7) shows in particular that $D^\alpha \delta_y$ has order $|\alpha|$.

This in return implies that at least one coefficient a_α with $|\alpha| = N$ in (6.7) must be nonzero, as otherwise the order of u would be strictly less than N.

6.3 Convolution with a Compactly Supported Distribution

Under suitable conditions on distributions $u, v \in \mathcal{D}'(\mathbb{R}^n)$, their convolution product $u * v \in \mathcal{D}'$ is well-defined. We now show that this is, for instance, the case if one of the two factors has compact support.

Assume for instance that $u \in \mathcal{E}' \subset \mathcal{S}'$ is a distribution with compact support, and consider the linear operator C_u of convolution with u on $\mathcal{S}$, i.e.,

$$C_u \varphi := u * \varphi, \quad \varphi \in \mathcal{S}.$$

By Theorem 4.23, for $\varphi \in \mathcal{S}$, we have $C_u \varphi \in C^\infty(\mathbb{R}^n)$, and

$$D^\alpha (C_u \varphi)(x) = u * D^\alpha \varphi(x) = u(\lambda_x (D^\alpha \varphi)^\vee).$$

Furthermore, if N is the order of u and K a compact neighborhood of the support of u, then by (6.1),

$$|u(\psi)| \le C \max_{|\beta| \le N} \sup_{y \in K} |D^\beta \psi(y)| \quad \text{for all } \psi \in \mathcal{E}.$$

Setting $D^\alpha \varphi =: \psi_\alpha$, we thus obtain:

$$|D^\alpha (C_u(\varphi)(x)| \le C_1 \max_{|\beta| \le N} \sup_{y \in K} |D^\beta \psi_\alpha(x - y)|. \tag{6.10}$$

From this, it follows quickly that for every $M \in \mathbb{N}$

$$\|C_u\varphi\|_{(M)} \leq C_M \|\varphi\|_{(M+N)}. \tag{6.11}$$

In particular, C_u maps the space $\mathcal{S}$ continuously into itself.
Next, if we define the distribution $\check{u} \in \mathcal{D}'$ for $u \in \mathcal{D}'$ by

$$\langle \check{u}, \varphi \rangle := \langle u, \check{\varphi} \rangle, \quad \varphi \in \mathcal{D},$$

then for all $\varphi, \psi \in \mathcal{S}$:

$$\begin{aligned} \langle C_u\varphi, \psi \rangle &= \langle u * \varphi, \psi \rangle = \langle u, \psi * \check{\varphi} \rangle \\ &= \langle u, (\varphi * \check{\psi})^{\vee} \rangle = \langle \check{u}, \varphi * \check{\psi} \rangle \\ &= \langle \check{u} * \psi, \varphi \rangle = \langle \varphi, \check{u} * \psi \rangle, \end{aligned}$$

which shows that

$${}^t C_u = C_{\check{u}}.$$

Definition. If $u \in \mathcal{E}'(\mathbb{R}^n)$ and $v \in \mathcal{S}'(\mathbb{R}^n)$, then we define the distribution $u*v \in \mathcal{S}'(\mathbb{R}^n)$ by

$$\langle u * v, \varphi \rangle := \langle v, \check{u} * \varphi \rangle, \quad \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{6.12}$$

Analogously, we define $v * u \in \mathcal{S}'(\mathbb{R}^n)$ by

$$\langle v * u, \varphi \rangle := \langle v, \varphi * \check{u} \rangle, \quad \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{6.13}$$

Proposition 6.4 *Let $u, v \in \mathcal{S}'(\mathbb{R}^n)$, and assume that at least one of these distributions has compact support. Then the following hold:*

(a) $u * v = v * u$.
(b) $(u * v) * \varphi = u * (v * \varphi)$ *for all* $\varphi \in \mathcal{S}(\mathbb{R}^n)$.
(c) $\operatorname{supp}(u * v) \subset \operatorname{supp} u + \operatorname{supp} v$.
(d) $D^\alpha(u * v) = (D^\alpha u) * v = u * (D^\alpha v)$ *for all* $\alpha \in \mathbb{N}^n$.
(e) $\langle u * v, \varphi \rangle = \langle u, \varphi * \check{v} \rangle$ *for all* $\varphi \in \mathcal{S}(\mathbb{R}^n)$.

Proof. The proof is left as an exercise (see Exercise 6.2). □

Analogously, by choosing φ from $\mathcal{D}(\mathbb{R}^n)$ in (6.12) and (6.13), one can also define the convolutions $u*v$ and $v*u$ of a distribution $u \in \mathcal{E}'(\mathbb{R}^n)$ with a distribution $v \in \mathcal{D}'(\mathbb{R}^n)$. Indeed, using standard arguments, we again see that for every $\chi \in \mathcal{D}$, the operator

$$\widetilde{C_u\chi} : \mathcal{S} \to \mathcal{S}, \quad \phi \mapsto C_u(\chi\phi) = u * (\chi\phi)$$

is continuous and that $\operatorname{supp} \widetilde{C_u\chi}(\phi) \subset \operatorname{supp} u + \operatorname{supp} \chi$ is contained in a compact set depending on χ (and u). Thus, by Lemma 5.10, the linear operator $C_u : \mathcal{D} \to \mathcal{D}$, $\varphi \mapsto u * \varphi$ is $\mathcal{D}$-null sequentially continuous. Furthermore, we then have

Proposition 6.5 *The statements of Proposition 6.4 remain valid when replacing $\mathcal{S}'(\mathbb{R}^n)$ by $\mathcal{D}'(\mathbb{R}^n)$ and $\mathcal{S}(\mathbb{R}^n)$ by $\mathcal{D}(\mathbb{R}^n)$.*

Corollary 6.6 *Let $\delta = \delta_0$, and let $u \in \mathcal{D}'(\mathbb{R}^n)$. Then we have*

$$u = \delta * u = u * \delta,$$

and more generally,

$$D^\alpha u = (D^\alpha \delta) * u = u * (D^\alpha \delta) \quad \text{for every } \alpha \in \mathbb{N}^n. \tag{6.14}$$

According to (6.14), the differential operator D^α thus agrees with the operator of convolution with the distribution $D^\alpha \delta$.

Proof. For $\varphi \in \mathcal{D}$, we have

$$\langle \check{\delta}, \varphi \rangle = \langle \delta, \check{\varphi} \rangle = \check{\varphi}(0) = \varphi(0) = \langle \delta, \varphi \rangle,$$

i.e., $\check{\delta} = \delta$. Furthermore,

$$\delta * \varphi(x) = \langle \delta, \lambda_x \check{\varphi} \rangle = \check{\varphi}(0 - x) = \varphi(x),$$

i.e., $\delta * \varphi = \varphi$. It follows that

$$\langle \delta * u, \varphi \rangle = \langle u, \check{\delta} * \varphi \rangle = \langle u, \delta * \varphi \rangle = \langle u, \varphi \rangle,$$

hence $\delta * u = u$.

The remaining statements follow from Proposition 6.5. □

6.4 Classical Derivative Versus Generalized Derivative

The following theorem, which is a variant of a classical lemma by du Bois Raymond from the calculus of variations, demonstrates that the generalized derivative in the sense of distributions indeed extends the classical concept of differentiation for functions.

Let again $\Omega \subset \mathbb{R}^n$ be a non-empty open subset.

Theorem 6.7 *Let $u, f \in C(\Omega)$ and $j \in \{1, \ldots, n\}$.*

(a) *If $D_j u = f$ in the classical sense, then $D_j u = f$ also holds in the distributional sense.*

(b) *If $D_j u = f$ in the distributional sense, then u is partially differentiable with respect to x_j in the classical sense, and it also holds that $D_j u = f$ in the classical sense.*

Proof. (a) If $D_j u = f$ in the classical sense, then u is continuously partially differentiable with respect to x_j. For $\varphi \in \mathcal{D}(\Omega)$, it follows by integration by parts that

$$-\langle u, D_j \varphi \rangle = -\int u(x) D_j \varphi(x)\, \mathrm{d}x = \int D_j u(x) \varphi(x)\, \mathrm{d}x = \langle f, \varphi \rangle.$$

Thus, $D_j u = f$ in the distributional sense.

To prove (b), we assume that $D_j u = f$ in the distributional sense in $\mathcal{D}'(\Omega)$. If $\chi \in \mathcal{D}(\Omega)$, then the product rule also holds for χu, i.e.,

$$D_j(\chi u) = (D_j \chi)u + \chi(D_j u) \tag{6.15}$$

(see Exercise 5.2). By assumption, the right-hand side is a continuous function, and χu has compact support in Ω. This shows that it suffices to prove the statement for $u \in \mathcal{E}'(\Omega)$. Furthermore, we may assume without loss of generality that $\Omega = \mathbb{R}^n$.

Thus, let $u \in C_0(\mathbb{R}^n) \subset \mathcal{E}'(\mathbb{R}^n)$. We choose a Dirac sequence $\{\varphi_k\}_k$ in $\mathcal{D}(\mathbb{R}^n)$ and define $u_k := u * \varphi_k$, $f_k := f * \varphi_k$ (Friedrichs mollification of u and f). Then, according to Theorem 5.13,

$$D_j u_k = (D_j u) * \varphi_k = f * \varphi_k = f_k,$$

where u_k and f_k belong to $C_0^\infty(\mathbb{R}^n)$. Furthermore, by Theorem 2.13 (b), as $k \to \infty$, the functions u_k converge uniformly to u, and the functions $D_j u_k = f_k$ converge uniformly to f. By a standard argument, this implies that u is partially differentiable with respect to x_j and that $D_j u = f$ in the classical sense. □

6.5 Paley–Wiener Theorems

In this section, we examine the properties of the Fourier–Laplace transform of functions and distributions with compact support. These results are significant, for example, in the theory of partial differential equations.

Definitions. (a) A *continuous* function f on an open subset U of $\mathbb{C}^n$ is called **holomorphic** in U if it is separately holomorphic in each variable, i.e., if for every $j = 1, \dots, n$ and every $(\zeta_1, \dots, \zeta_n) \in U$, the function

$$f_j : w \mapsto f(\zeta_1, \dots, \zeta_{j-1}, w, \zeta_{j+1}, \dots, \zeta_n)$$

is holomorphic in a neighborhood of ζ_j in $\mathbb{C}$.

A function that is holomorphic in all of $\mathbb{C}^n$ is called an **entire** function.

(b) Let $\zeta = (\zeta_1, \dots, \zeta_n) \in \mathbb{C}^n$, and write $\zeta_j = \xi_j + i\eta_j$ with $\xi_j, \eta_j \in \mathbb{R}$. We then set $\xi := (\xi_1, \dots, \xi_n), \eta := (\eta_1, \dots, \eta_n) \in \mathbb{R}^n$, and write $\zeta = \xi + i\eta$. The vectors

$$\xi =: \operatorname{Re} \zeta \quad \text{and} \quad \eta =: \operatorname{Im} \zeta$$

are called the **real** and **imaginary part** of ζ. We consider $\mathbb{R}^n$ as the set of all $\zeta \in \mathbb{C}^n$ with $\operatorname{Im} \zeta = 0$. Furthermore, we define

$$|\zeta| := (|\zeta_1|^2 + \cdots + |\zeta_n|^2)^{1/2}, \quad \zeta^\alpha := \zeta_1^{\alpha_1} \cdots \zeta_n^{\alpha_n}, \quad \alpha \in \mathbb{N}^n.$$

Recall that for $\zeta \in \mathbb{C}^n$, the function e_ζ is defined by

$$e_\zeta(x) := \mathrm{e}^{i\zeta \cdot x}, \qquad x \in \mathbb{R}^n.$$

Proposition 6.8 *Let $u \in \mathcal{E}'(\mathbb{R}^n)$ be a distribution with compact support. Then its Fourier–Laplace transform*

$$\mathcal{FL}u(\zeta) := u(e_{-\zeta}), \qquad \zeta \in \mathbb{C}^n,$$

is an entire function on $\mathbb{C}^n$. Furthermore, $\mathcal{FL}u|_{\mathbb{R}^n} = \hat{u}$.

Proof. Let $\{\varphi_\varepsilon\}_{\varepsilon>0}$ be a Dirac family in $\mathcal{D}$, and set $u_\varepsilon := u * \varphi_\varepsilon \in \mathcal{D}$. For the function u_ε, the Fourier–Laplace transform is given by the integral

$$\mathcal{FL}u_\varepsilon(\zeta) = \int_{\mathbb{R}^n} u_\varepsilon(x) \mathrm{e}^{-i\zeta\cdot x} \mathrm{d}x.$$

Since for fixed x, the integrand is clearly holomorphic in ζ, Morera's theorem applied to each variable $\zeta_1, \ldots, \zeta_n$ shows that $\mathcal{FL}u_\varepsilon$ is an entire function.

On the other hand, since $\varphi_\varepsilon(x) = \varepsilon^{-n}\varphi\left(\frac{x}{\varepsilon}\right)$, we have

$$\mathcal{FL}u_\varepsilon(\zeta) = (u * \varphi_\varepsilon)(e_{-\zeta}) = u(e_{-\zeta} * \check{\varphi}_\varepsilon),$$

where

$$e_{-\zeta} * \check{\varphi}_\varepsilon(x) = \check{\varphi}_\varepsilon * e_{-\zeta}(x) = \int \varphi_\varepsilon(-y)\mathrm{e}^{-i\zeta\cdot(x-y)}\,\mathrm{d}y = \int \varphi_\varepsilon(y)\mathrm{e}^{-i\zeta\cdot y}\,\mathrm{d}y\,\mathrm{e}^{-i\zeta\cdot x}$$
$$= \int \varphi(y)\mathrm{e}^{-i\varepsilon\zeta\cdot y}\,\mathrm{d}y\,\mathrm{e}^{-i\zeta\cdot x}.$$

It follows that

$$\mathcal{FL}u_\varepsilon(\zeta) = \mathcal{FL}\varphi(\varepsilon\zeta)\,\mathcal{FL}u(\zeta). \tag{6.16}$$

Since $\mathcal{FL}\varphi$ is continuous,

$$\lim_{\varepsilon\to 0} \mathcal{FL}\varphi(\varepsilon\zeta) = \hat{\varphi}(0) = 1,$$

uniformly on every compact subset of $\mathbb{C}^n$. From (6.16), we conclude that for sufficiently small $\varepsilon > 0$,

$$\mathcal{FL}u(\zeta) = \frac{\mathcal{FL}u_\varepsilon(\zeta)}{\mathcal{FL}\varphi(\varepsilon\zeta)},$$

showing that $\mathcal{FL}u$ is holomorphic.

For $\zeta = \xi \in \mathbb{R}^n$, we have $\mathcal{FL}u_\varepsilon(\xi) = \widehat{u_\varepsilon}(\xi)$, so by (6.16),

$$\widehat{u_\varepsilon}(\xi) = \hat{\varphi}(\varepsilon\xi)\,\mathcal{FL}u(\xi).$$

Thus, for $\psi \in \mathcal{D}$:

$$\langle \hat{u}_\varepsilon, \psi\rangle = \int \hat{\varphi}(\varepsilon\xi)\mathcal{FL}u(\xi)\psi(\xi)\,\mathrm{d}\xi. \tag{6.17}$$

Moreover, by Proposition 4.25

$$u = \lim_{\varepsilon\to 0} u_\varepsilon \quad \text{in } \mathcal{S}',$$

hence

$$\hat{u} = \lim_{\varepsilon \to 0} \widehat{u_\varepsilon} \quad \text{in } \mathcal{S}'.$$

Taking limits as $\varepsilon \to 0$,

$$\langle \hat{u}, \psi \rangle = \int \mathcal{F}\mathcal{L}u(\xi)\psi(\xi)\,\mathrm{d}\xi = \langle \mathcal{F}\mathcal{L}u|_{\mathbb{R}^n}, \psi \rangle.$$

Thus, indeed, $\mathcal{F}\mathcal{L}u|_{\mathbb{R}^n} = \hat{u}$. □

Example 6.9 We have $\mathcal{F}\mathcal{L}(D^\alpha \delta_y)(\zeta) = \zeta^\alpha \mathrm{e}^{-i\zeta\cdot y}$.

Indeed, if $\chi \in \mathcal{D}(\Omega)$ is identically 1 on a neighborhood of y, then by (6.6)

$$\begin{aligned}\mathcal{F}\mathcal{L}(D^\alpha \delta_y)(\zeta) &= \langle D^\alpha \delta_y, \chi e_{-\zeta} \rangle = (-1)^{|\alpha|} D^\alpha(\chi e_{-\zeta})(y) \\ &= (-1)^{|\alpha|} D^\alpha e_{-\zeta}(y) = \zeta^\alpha \mathrm{e}^{-i\zeta\cdot y},\end{aligned}$$

where we have used Leibniz' formula in the second-last identity.

For $r \geq 0$ let $\overline{B}_r := \overline{B}_r(0)$.

Theorem 6.10 (a) *For every $N \in \mathbb{N}$, there exists a constant $C_N \geq 0$ such that for every $r \geq 0$, the following holds:*
If $\varphi \in \mathcal{D}$ is supported in the closed ball $\overline{B}_r$, then

$$|\mathcal{F}\mathcal{L}\varphi(\zeta)| \leq C_N r^n \|\varphi\|_N \,(1 + |\zeta|)^{-N} \mathrm{e}^{r|\operatorname{Im}\zeta|}, \qquad \zeta \in \mathbb{C}^n. \tag{6.18}$$

(b) *Conversely, if f is an entire function on $\mathbb{C}^n$, and if for every $N \in \mathbb{N}$ there is a constant $C_N \geq 0$ such that*

$$|f(\zeta)| \leq C_N (1 + |\zeta|)^{-N} \mathrm{e}^{r|\operatorname{Im}\zeta|} \quad \text{for every } \zeta \in \mathbb{C}^n, \tag{6.19}$$

then there is a unique function $\varphi \in \mathcal{D}$ with support in $\overline{B}_r$ such that $f = \mathcal{F}\mathcal{L}\varphi$.

Proof. (a) For $x \in \overline{B}_r$, we have

$$|\mathrm{e}^{-i\zeta\cdot x}| = \mathrm{e}^{\operatorname{Im}\zeta\cdot x} \leq \mathrm{e}^{|\operatorname{Im}\zeta|\,|x|} \leq \mathrm{e}^{r|\operatorname{Im}\zeta|}.$$

Let $\varphi \in \mathcal{D}_{\overline{B}_r}$, and set $f := \mathcal{F}\mathcal{L}\varphi$. Then

$$f(\zeta) = \int_{\mathbb{R}^n} \varphi(x)\mathrm{e}^{-i\zeta\cdot x}\,\mathrm{d}x,$$

and by integrations by parts, we obtain

$$\zeta^\alpha f(\zeta) = \int_{\mathbb{R}^n} (D^\alpha\varphi)(x)\mathrm{e}^{-i\zeta\cdot x}\,\mathrm{d}x, \quad \alpha \in \mathbb{N}^n.$$

It follows for $|\alpha| \leq N$ that

$$|\zeta^\alpha f(\zeta)| \leq \int_{\overline{B}_r} \|\varphi\|_N \,\mathrm{e}^{r|\operatorname{Im}\zeta|}\,\mathrm{d}x = Cr^n \|\varphi\|_N \mathrm{e}^{r|\operatorname{Im}\zeta|},$$

and thus also

$$(1+|\zeta|)^N |f(\zeta)| \le C_N r^n \|\varphi\|_N \, \mathrm{e}^{r|\operatorname{Im}\zeta|},$$

which proves (6.18).

(b) Conversely, let the entire function f satisfy (6.19). We define φ by

$$\varphi(x) := (2\pi)^{-n} \int_{\mathbb{R}^n} f(\xi) \mathrm{e}^{ix\cdot\xi} \, \mathrm{d}\xi, \qquad x \in \mathbb{R}^n.$$

Then (6.19), together with Proposition 2.18, shows that $\varphi \in C^\infty(\mathbb{R}^n)$. We now claim that for all $x \in \mathbb{R}^n$ and all $\zeta_2, \dots, \zeta_n \in \mathbb{C}$, the integral

$$F(\eta_1) := \int_{-\infty}^{\infty} f(\xi_1 + i\eta_1, \zeta_2, \dots, \zeta_n) \mathrm{e}^{i(x_1(\xi_1+i\eta_1)+x_2\zeta_2+\cdots+x_n\zeta_n)} \, \mathrm{d}\xi_1$$

is independent of $\eta_1 \in \mathbb{R}$.

To see this, we consider for $R > 0$ the closed contour Γ_R in the $(\xi_1 + i\eta_1)$-plane, which connects the points $-R$, R, $R + i\eta_1$, $-R + i\eta_1$, and back to $-R$.

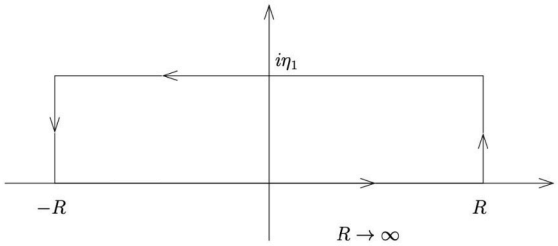


Fig. 6.1 Contour Γ_R

Setting

$$h(\zeta_1) := f(\zeta_1, \zeta_2, \dots, \zeta_n) \mathrm{e}^{i(x_1\zeta_1+x_2\zeta_2+\cdots+x_n\zeta_n)},$$

the Cauchy integral theorem implies that

$$\int_{\Gamma_R} h(\zeta_1) \, \mathrm{d}\zeta_1 = 0.$$

Furthermore, for fixed $\eta_1, \zeta_2, \dots, \zeta_n$ and $x \in \mathbb{R}^n$, (6.19) gives

$$|h(\pm R + is)| \le C(1+R)^{-N}, \quad \text{for } |s| \le |\eta_1|.$$

Thus, as $R \to \infty$, the integrals along the vertical edges of Γ_R tend to 0. Taking the limit as $R \to \infty$, we obtain

$$\int_{-\infty}^{\infty} h(\xi_1) \, \mathrm{d}\xi_1 = \int_{-\infty}^{\infty} h(\xi_1 + i\eta_1) \, \mathrm{d}\xi_1,$$

i.e., $F(0) = F(\eta_1)$ for every $\eta_1 \in \mathbb{R}$.

Applying the same argument to the remaining coordinates, we conclude that

$$\varphi(x) = (2\pi)^{-n} \int_{\mathbb{R}^n} f(\xi + i\eta) \mathrm{e}^{ix\cdot(\xi+i\eta)}\, \mathrm{d}\xi$$

for every $\eta \in \mathbb{R}^n$.

Now let $x \in \mathbb{R}^n$, $x \neq 0$, and choose $\eta := \lambda \frac{x}{|x|}$ with $\lambda > 0$. Then $x \cdot \eta = \lambda|x|$, $|\eta| = \lambda$, and thus, using (6.19), we obtain

$$|f(\xi + i\eta)\mathrm{e}^{ix\cdot(\xi+i\eta)}| \leq C_N(1 + |\xi|)^{-N}\mathrm{e}^{\lambda(r-|x|)}.$$

For $N > n$, it follows for all $\lambda > 0$ that

$$|\varphi(x)| \leq C\, \mathrm{e}^{\lambda(r-|x|)}.$$

Letting $\lambda \longrightarrow \infty$, we conclude that $\varphi(x) = 0$ whenever $|x| > r$, and thus $\operatorname{supp}\varphi \subset \overline{B}_r$, i.e., $\varphi \in \mathcal{D}_{\overline{B}_r} \subset L^1$.

Finally, by the Fourier inversion formula, we have $\hat{\varphi} = f|_{\mathbb{R}^n}$, i.e., $g := \mathcal{F}\mathcal{L}\varphi$ and f are entire functions that agree on $\mathbb{R}^n$.

By the identity theorem, it follows that

$$g(\zeta_1, \xi_2, \ldots, \xi_n) = f(\zeta_1, \xi_2, \ldots, \xi_n) \quad \text{for all } \zeta_1 \in \mathbb{C}, \xi_2, \ldots, \xi_n \in \mathbb{R},$$

and repeated application of the identity theorem to the remaining coordinates yields $g = f$. Similarly, one shows using the identity theorem that φ is unique. □

Theorem 6.11 *An entire function f on $\mathbb{C}^n$ is the Fourier–Laplace transform of a distribution with support in the closed ball $\overline{B}_r$ if and only if there exist constants $C \geq 0$ and $N \in \mathbb{N}$ such that*

$$|f(\zeta)| \leq C(1 + |\zeta|)^N \mathrm{e}^{r|\operatorname{Im}\zeta|} \quad \textit{for all } \zeta \in \mathbb{C}^n. \tag{6.20}$$

Proof. We first prove the necessity of condition (6.20). Let $u \in \mathcal{E}'(\mathbb{R}^n)$ with $\operatorname{supp} u \subset \overline{B}_r$, and let N be the order of u. Further, set $f(\zeta) := \mathcal{F}\mathcal{L}u(\zeta)$.

By Proposition 6.1, there exists a constant $A \geq 0$ such that

$$|u(\varphi)| \leq A \max_{|\alpha|\leq N} \sup_{|x|\leq r+1} |D^\alpha \varphi(x)|, \qquad \varphi \in \mathcal{D}.$$

Now, let $h \in C^\infty(\mathbb{R})$ be such that $h(t) = 1$ for $t \leq 1/2$ and $h(t) = 0$ for $t \geq 1$.

Then, for $\zeta \in \mathbb{C}^n$, $\zeta \neq 0$, the function

$$\varphi_\zeta(x) := \mathrm{e}^{-i\zeta\cdot x} h(|\zeta|(|x| - r)), \quad x \in \mathbb{R}^n,$$

belongs to $C_0^\infty(\mathbb{R}^n)$, and satisfies

$$\varphi_\zeta(x) = \begin{cases} \mathrm{e}^{-i\zeta\cdot x}, & \text{if } |x| < r + \frac{1}{2|\zeta|}, \\ 0, & \text{if } |x| > r + \frac{1}{|\zeta|}. \end{cases} \tag{6.21}$$

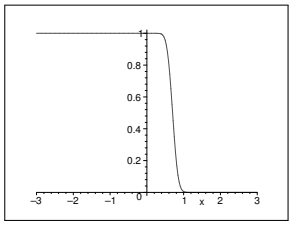


Fig. 6.2 Graph of h

In particular, φ_ζ coincides with $e_{-\zeta}$ on a neighborhood of $\overline{B}_r$, so that

$$f(\zeta) = u(e_{-\zeta}) = u(\varphi_\zeta).$$

This yields

$$|f(\zeta)| \le A \max_{|\alpha|\le N} \sup_x |D^\alpha \varphi_\zeta(x)|.$$

Now, the function $x \mapsto q(x) := |x|$ is smooth on a neighborhood of the set where $r + \frac{1}{2|\zeta|} \le |x| \le r + \frac{1}{|\zeta|}$, and we obtain

$$D_j \varphi_\zeta(x) = -\zeta_j \varphi_\zeta(x) + |\zeta| D_j q(x) \mathrm{e}^{-i\zeta\cdot x} h'(|\zeta|(|x| - r)).$$

Note that the second summand is supported in the region where $r + \frac{1}{2|\zeta|} \le |x| \le r + \frac{1}{|\zeta|}$, on which $D_j q$ is bounded by 1. Using (6.21), we obtain

$$|D_j \varphi_\zeta(x)| \le C(1 + |\zeta|) \sup_{|x'|\le r+\frac{1}{|\zeta|}} |\mathrm{e}^{-i\zeta\cdot x'}|.$$

Arguing in a similar way for higher order derivatives of φ_ζ, we obtain more generally

$$|D^\alpha \varphi_\zeta(x)| \le C_\alpha (1 + |\zeta|)^{|\alpha|} \sup_{|x'|\le r+\frac{1}{|\zeta|}} |\mathrm{e}^{-i\zeta\cdot x'}|.$$

But, if $|x'| \le r + \frac{1}{|\zeta|}$, then

$$|\mathrm{e}^{-i\zeta\cdot x'}| = \mathrm{e}^{(\mathrm{Im}\,\zeta)\cdot x'} \le \mathrm{e}^{|\mathrm{Im}\,\zeta|\,|x'|} \le \mathrm{e}^{r|\mathrm{Im}\,\zeta|+1},$$

so that

$$\sup_x |D^\alpha \varphi_\zeta(x)| \le C'_\alpha (1 + |\zeta|)^{|\alpha|} \mathrm{e}^{r|\mathrm{Im}\,\zeta|}.$$

Thus, we obtain (6.20).

Conversely, suppose f is an entire function on $\mathbb{C}^n$ satisfying the inequality (6.20).

Since $g := f|_{\mathbb{R}^n}$ has polynomial growth and is therefore a tempered function, we have $u := (2\pi)^{-n}\overline{\mathcal{F}} g \in \mathcal{S}'$, and by Theorem 4.19 we get $\hat{u} = g$.

We choose a Dirac family $\{\varphi_\varepsilon\}_{\varepsilon>0}$ such that $\operatorname{supp}\varphi_\varepsilon \subset \overline{B}_\varepsilon$, and set

$$f_\varepsilon(\zeta) := f(\zeta)\,\mathcal{F}\mathcal{L}\varphi_\varepsilon(\zeta), \quad \zeta \in \mathbb{C}^n.$$

By Theorem 6.10 (a) and (6.20), f_ε satisfies the estimate (6.19) with $r+\varepsilon$ in place of r. Thus, by Theorem 6.10 (b), there exist unique functions $u_\varepsilon \in \mathcal{D}$ with support in $\overline{B}_{r+\varepsilon}$ such that $f_\varepsilon = \mathcal{F}\mathcal{L}u_\varepsilon$.

Let $\psi \in \mathcal{D}$ with $\operatorname{supp}\psi \cap \overline{B}_r = \emptyset$. Set $\phi := \mathcal{F}^{-1}\psi \in \mathcal{S}$. Then $\psi = \hat{\phi}$, and for sufficiently small $\varepsilon > 0$, we have $\psi u_\varepsilon = 0$. Since $\phi g \in L^1(\mathbb{R}^n)$ and $\lim_{\varepsilon\to 0} \widehat{\varphi_\varepsilon}(\xi) = \lim_{\varepsilon\to 0} \hat{\varphi}(\varepsilon\xi) = 1$, we obtain by the dominated convergence theorem

$$\begin{aligned}\langle u,\psi\rangle = \langle u,\hat{\phi}\rangle = \langle \hat{u},\phi\rangle &= \int_{\mathbb{R}^n} f(\xi)\phi(\xi)\,\mathrm{d}\xi \\ &= \lim_{\varepsilon\to 0}\int_{\mathbb{R}^n} f_\varepsilon(\xi)\phi(\xi)\,\mathrm{d}\xi = \lim_{\varepsilon\to 0}\int_{\mathbb{R}^n} \hat{u}_\varepsilon(\xi)\phi(\xi)\,\mathrm{d}\xi \\ &= \lim_{\varepsilon\to 0}\int_{\mathbb{R}^n} u_\varepsilon(\xi)\hat{\phi}(\xi)\,\mathrm{d}\xi = \lim_{\varepsilon\to 0}\int_{\mathbb{R}^n} u_\varepsilon(\xi)\psi(\xi)\,\mathrm{d}\xi = 0.\end{aligned}$$

Thus, we conclude that $\operatorname{supp} u \subset \overline{B}_r$.

Finally, since $\mathcal{F}\mathcal{L}u$ and f are entire functions that agree on $\mathbb{R}^n$, the identity theorem implies $\mathcal{F}\mathcal{L}u = f$. □

In the following two chapters, we will discuss a few interesting applications of Fourier analysis and distribution theory to the theory of partial differential equations, before developing the theory of distributions a bit further in the last Chapter 9.

For further results on distributions and their applications we refer the interested reader for instance to the classical standard works of Schwartz [35] and Gel'fand–Shilov [15], as well as Rudin [34], Hörmander [23], and Duistermaat [8], among others.

6.6 Exercises

6.1 Let Ω be a non-empty open subset of $\mathbb{R}^n$. Show that:

$\mathcal{D}(\Omega)$ is dense in $\mathcal{E}(\Omega)$. Furthermore, for every continuous linear form v on $\mathcal{E}(\Omega)$, the linear form $v|_{\mathcal{D}(\Omega)}$ belongs to $\mathcal{E}'(\Omega)$, and

$$v = (v|_{\mathcal{D}(\Omega)})_{\mathcal{E}}.$$

6.2 Prove Proposition 6.4.

6.3 Let $K = K_r := \{x \in \mathbb{R}^n : |x| \le r\}$, $r \ge 0$, and let $u \in \mathcal{E}'(\mathbb{R}^n)$ such that $\operatorname{supp} u \subset K$.

Show that if $f \in C^\infty(\mathbb{R}^n)$ vanishes on K and $r > 0$, then $u(f) = 0$.

Does this also hold for $r = 0$?

6.4 Let $\Delta := \sum_{j=1}^{n} \frac{\partial^2}{\partial x_j^2}$ denote the **Laplace operator** on $\mathbb{R}^n$. Show that:

a) If $u \in \mathcal{S}'(\mathbb{R}^n)$ is harmonic, i.e., if $\Delta u = 0$, then u is a polynomial function. Why does this not contradict the harmonicity of the function $e^{x_1} \cos x_2$?
b) Use a) to give a new proof of Liouville's theorem from complex analysis, according to which every bounded entire function on $\mathbb{C}$ is constant.

Hint. Use the Fourier transform and Exercise 5.2.

6.5 (Homogeneous Distributions) For $r > 0$, let $\mathfrak{D}_r : \mathbb{R}^n \to \mathbb{R}^n$ denote the **dilation** $\mathfrak{D}_r x := rx,\ x \in \mathbb{R}^n$. If $u \in \mathcal{D}'(\mathbb{R}^n)$, then let $u \circ \mathfrak{D}_r$ be the distribution defined by

$$\langle u \circ \mathfrak{D}_r, \varphi \rangle := r^{-n} \langle u, \varphi \circ \mathfrak{D}_{r^{-1}} \rangle, \qquad \varphi \in \mathcal{D}(\mathbb{R}^n).$$

u is called **homogeneous of degree** $\alpha \in \mathbb{C}$ if $u \circ \mathfrak{D}_r = r^\alpha u$ for all $r > 0$. Show that:

i) If $u \in \mathcal{S}'(\mathbb{R}^n)$ is homogeneous of degree α, then $\hat{u}$ is homogeneous of degree $-\alpha - n$.
ii) If $u \in \mathcal{D}'(\mathbb{R}^n)$ is homogeneous of degree $\alpha \in \mathbb{C}$ and $\beta \in \mathbb{N}^n$, then $D^\beta u$ is homogeneous of degree $\alpha - |\beta|$.
iii) $D^\beta \delta_0$ is homogeneous of degree $-|\beta| - n$.
iv) If $\operatorname{supp} u = \{0\}$ and u is homogeneous of degree $-n$, then $u = c\delta_0,\ c \in \mathbb{C}$.
v) Give an example of a homogeneous distribution of degree -1 on $\mathbb{R}$ that does not have compact support.
vi)* Determine all homogeneous distributions of degree -1 on $\mathbb{R}$.

Chapter 7
Fundamental Solutions

7.1 On the Significance of Fundamental Solutions for Linear Partial Differential Equations with Constant Coefficients

Let $P(D) = \sum_{|\alpha|\le k} a_\alpha D^\alpha$ be a linear differential operator with constant coefficients $a_\alpha \in \mathbb{C}$. In the following we shall write simply δ for the Dirac δ_0.

Definition. A distribution $K \in \mathcal{D}'(\mathbb{R}^n)$ is called a **fundamental solution** for $P(D)$ if

$$P(D)K = \delta.$$

Theorem 7.1 *If $K \in \mathcal{D}'(\mathbb{R}^n)$ is a fundamental solution for $P(D)$, and if $\varphi \in \mathcal{D}(\mathbb{R}^n)$ (respectively $\varphi \in \mathcal{S}(\mathbb{R}^n)$, if $K \in \mathcal{S}'(\mathbb{R}^n)$), then $u := K * \varphi \in C^\infty(\mathbb{R}^n)$ solves the differential equation*

$$P(D)u = \varphi \tag{7.1}$$

in the classical sense.

Proof. We prove the case $K \in \mathcal{D}'(\mathbb{R}^n)$; for $K \in \mathcal{S}'(\mathbb{R}^n)$ and $\varphi \in \mathcal{S}(\mathbb{R}^n)$, the proof proceeds analogously. For $\varphi \in \mathcal{D}(\mathbb{R}^n)$, we have by Theorem 5.13

$$P(D)u = P(D)(K * \varphi) = (P(D)K) * \varphi = \delta * \varphi = \varphi,$$

where $u \in C^\infty(\mathbb{R}^n)$. □

Remarks 7.2 (a) If the distribution $K * v$ is well-defined, then $u := K * v$ also satisfies the differential equation $P(D)u = v$ (in the sense of distributions) for more general distributions v. For example, by Proposition 6.5, this holds for $v \in \mathcal{E}'(\mathbb{R}^n)$, i.e., when v has compact support.

(b) The differential equation $P(D)u = \varphi$ generally has infinitely many solutions u.

Example 7.3 $P(D) = \frac{\mathrm{d}}{\mathrm{d}x} - a$ on $\mathbb{R}$, with $a \in \mathbb{C}$.

(i) If $a = 0$, then Heaviside's function $H := \mathbf{1}_{[0,\infty[}$ is clearly a fundamental solution (cf. Example 4.18 (i)), and thus so is $c + H$ for any constant $c \in \mathbb{C}$.

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D. Müller, *An Invitation to Fourier Analysis and Distribution Theory*,
Universitext, https://doi.org/10.1007/978-3-032-31550-2_7

Conversely, if K is any fundamental solution for $\frac{\mathrm{d}}{\mathrm{d}x}$, then $u := K - H$ satisfies the differential equation

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{\mathrm{d}K}{\mathrm{d}x} - \frac{\mathrm{d}H}{\mathrm{d}x} = \delta - \delta = 0.$$

This implies that u is a constant function, because:

If $\{\varphi_\varepsilon\}_\varepsilon$ is a Dirac family, and if we set $u_\varepsilon := \varphi_\varepsilon * u$, then $\frac{\mathrm{d}u_\varepsilon}{\mathrm{d}x} = \varphi_\varepsilon * \frac{\mathrm{d}u}{\mathrm{d}x} = 0$, and thus, by Theorem 6.7, we see that for every $\varepsilon > 0$, the smooth function u_ε is constant: $u_\varepsilon = c_\varepsilon$, $c_\varepsilon \in \mathbb{C}$. As $\varepsilon \to 0$, however, u_ε converges to u in $\mathcal{D}(\mathbb{R})$, and thus the constants c_ε converge to a constant $c \in \mathbb{C}$, i.e., $u = c$.

Consequently, the "general" fundamental solution for $\frac{\mathrm{d}}{\mathrm{d}x}$ is

$$K = c + \mathbf{1}_{[0,\infty[}, \qquad \text{with } c \in \mathbb{C}.$$

(ii) If $a \in \mathbb{C}$ is arbitrary, set $\gamma_a(x) := \mathrm{e}^{ax}$, $x \in \mathbb{R}$. By the product rule, we then have for $u \in \mathcal{D}'(\mathbb{R})$

$$\left(\frac{\mathrm{d}}{\mathrm{d}x} - a\right)(\gamma_a u) = \gamma_a' u + \gamma_a u' - a\gamma_a u = \gamma_a \frac{\mathrm{d}u}{\mathrm{d}x}.$$

Since moreover $\gamma_a \delta_0 = \delta_0$, we see that

$$H_a(x) := (\gamma_a H)(x) = \mathrm{e}^{ax}\, \mathbf{1}_{[0,\infty[}(x)$$

is a fundamental solution for $\frac{\mathrm{d}}{\mathrm{d}x} - a$.

Questions:

a) What does the general fundamental solution for $\frac{\mathrm{d}}{\mathrm{d}x} - a$ look like?
b) Note that H_a belongs to $\mathcal{S}'$ if and only if $\operatorname{Re} a \le 0$. Does there nevertheless exist a tempered fundamental solution for every $a \in \mathbb{C}$ (cf. Exercise 7.1)?

Finally, note that if $\varphi \in \mathcal{D}(\mathbb{R})$, then by Theorem 7.1,

$$u(x) := H_a * \varphi(x) = \langle H_a, \lambda_x \check{\varphi} \rangle = \int_0^\infty \mathrm{e}^{ay} \varphi(x - y)\, \mathrm{d}y = \int_{-\infty}^x \mathrm{e}^{ax} \mathrm{e}^{-ay} \varphi(y)\, \mathrm{d}y$$

is a solution of the inhomogeneous differential equation

$$\frac{\mathrm{d}u}{\mathrm{d}x} - au = \varphi.$$

Of course, this can also be verified directly.

This should be compared with *Duhamel's formula* from the theory of ordinary differential equations!

We now explicitly determine fundamental solutions for two important partial differential operators, namely the heat operator and the Laplace operator.

7.2 The Heat Equation in $\mathbb{R}^n$

The **heat operator**

$$\frac{\partial}{\partial t} - \sum_{j=1}^{n} \frac{\partial^2}{\partial {x_j}^2} = \frac{\partial}{\partial t} - \Delta_x$$

on $\mathbb{R} \times \mathbb{R}^n$, with coordinates (t, x), $x = (x_1, \ldots, x_n)$, has the following physical interpretation:

The temperature $u(t, x)$ at the location $x \in \mathbb{R}^n$ at time $t \in \mathbb{R}$ in a "homogeneous isotropic medium with thermal conductivity = 1, without external heat supply" satisfies the homogeneous heat equation

$$\left(\frac{\partial}{\partial t} - \Delta_x\right) u = 0.$$

We consider the initial value problem

$$\begin{aligned} \frac{\partial u}{\partial t} - \Delta_x u &= 0 \quad \text{on } \mathbb{R}^{n+1}_+, \\ u(0, x) &= f(x), \end{aligned} \tag{7.2}$$

where $\mathbb{R}^{n+1}_+$ denotes the upper half-space $\mathbb{R}^{n+1}_+ := \mathbb{R}_{\geq 0} \times \mathbb{R}^n$. This problem is physically meaningful:

Given the temperature distribution f at time $t = 0$, determine the temperature distribution at a later time $t > 0$.

First, let $f \in \mathcal{S}(\mathbb{R}^n)$. We solve (7.2) by using the partial Fourier transform in x, initially arguing formally. We use the following suggestive notation for the partial Fourier transform of u with respect to x:

$$u(t, \widehat{\xi}) := \int_{\mathbb{R}^n} u(t, x) \mathrm{e}^{-i\xi \cdot x} \, \mathrm{d}x.$$

Then (7.2) transforms into

$$\begin{aligned} \frac{\partial}{\partial t} u(t, \widehat{\xi}) + |\xi|^2 u(t, \widehat{\xi}) &= 0, \quad t \geq 0, \\ u(0, \widehat{\xi}) &= \hat{f}(\xi). \end{aligned}$$

Solving this initial value problem for fixed $\xi \in \mathbb{R}^n$ in t, we immediately obtain

$$u(t, \widehat{\xi}) = \hat{f}(\xi) \mathrm{e}^{-t|\xi|^2}, \qquad t \geq 0.$$

Using Lemma 2.24 and Proposition 2.1, we quickly see that for $t > 0$

$$\mathrm{e}^{-t|\xi|^2} = \widehat{W_t}(\xi), \qquad \xi \in \mathbb{R}^n, \tag{7.3}$$

where

$$W_t(x) := (4\pi t)^{-n/2} \mathrm{e}^{-\frac{|x|^2}{4t}}, \quad x \in \mathbb{R}^n,\ t > 0.$$

W_t is called the **heat kernel** (or also the Gauss–Weierstrass kernel). By inverse Fourier transform in ξ, we finally obtain

$$u(t,x) = f * W_t(x), \qquad t > 0. \tag{7.4}$$

Conversely, one can now easily verify that, for instance for $f \in \mathcal{S}$, the function u defined by (7.4) indeed satisfies the heat equation:

Since $W_t \in \mathcal{S}$ for $t > 0$, it follows that $u(t,\cdot) \in \mathcal{S}$ for every $t > 0$, so that all the formal calculations we have carried out are retrospectively justified. Furthermore, this solution u satisfies the initial condition in the following sense:

$$\lim_{t \searrow 0} \|f - u(t,\cdot)\|_\infty = 0, \qquad f \in \mathcal{S}, \tag{7.5}$$

i.e., for $t \to 0$, the functions $u(t,\cdot)$ converge uniformly to the initial value f.

To see this, we observe that W_t is positive, that

$$\|W_t\|_1 = \int W_t(x)\,\mathrm{d}x = \widehat{W}_t(0) = 1,$$

and that

$$W_t(x) = t^{-n/2} W_1(t^{-1/2}x), \qquad t > 0. \tag{7.6}$$

Thus, $\{W_t\}_{t>0}$ is a Dirac family in $\mathcal{S}(\mathbb{R}^n)$, so that (7.5) follows from Theorem 2.13.

Remark 7.4 By (7.3), we have $\widehat{W_s}\,\widehat{W_t} = \widehat{W_{s+t}}$ for any $s,t > 0$, hence

$$W_s * W_t = W_{s+t}, \qquad s,t > 0. \tag{7.7}$$

Furthermore, we have

$$\|f * W_t\|_p \le \|f\|_p \|W_t\|_1 = \|f\|_p$$

for $1 \le p \le \infty$. Thus, the operators $\{T_t\}_{t>0}$ with $T_t f := f * W_t$ form a so-called "contraction semigroup" on $L^p(\mathbb{R}^n)$ for $1 \le p \le \infty$, known as the **heat semigroup**.

We can now easily construct a fundamental solution for the heat operator. To this end, we set for $(t,x) \in \mathbb{R}^{n+1}$

$$W(t,x) := \begin{cases} W_t(x), & \text{if } t > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Then $W \ge 0$, and for $N > 0$,

$$\int_{|t|\le N} \int_{|x|\le N} |W(t,x)|\,\mathrm{d}x\,\mathrm{d}t \le \int_0^N \int_{\mathbb{R}^n} W_t(x)\,\mathrm{d}x\,\mathrm{d}t = N,$$

so that W is a tempered function on $\mathbb{R}^{n+1}$. Since $W(t,x)$ vanishes to infinite order on the hyperplane $\{t=0\}$ outside the origin $(0,0)$, we further see that

$$W|_{\mathbb{R}^{n+1}\setminus\{0\}} \in C^\infty(\mathbb{R}^{n+1}\setminus\{0\}). \tag{7.8}$$

Theorem 7.5 *W is a fundamental solution for $\frac{\partial}{\partial t}-\Delta_x$ on $\mathbb{R}^{n+1}$.*

Proof. We set for $\varepsilon>0$

$$W^\varepsilon(t,x) := \begin{cases} W(t,x), & \text{if } t>\varepsilon,\\ 0, & \text{otherwise.}\end{cases}$$

By the dominated convergence theorem, we then have

$$W = \lim_{\varepsilon\to 0} W^\varepsilon \quad \text{in } \mathcal{D}'.$$

Thus, it suffices to show that

$$\lim_{\varepsilon\to 0}\left(\frac{\partial}{\partial t}-\Delta_x\right)W^\varepsilon = \delta_0 \quad \text{in } \mathcal{D}'.$$

Let $\varphi\in\mathcal{D}(\mathbb{R}^{n+1})$. Integrations by parts yield

$$\begin{aligned}\left\langle\left(\frac{\partial}{\partial t}-\Delta_x\right)W^\varepsilon,\varphi\right\rangle &= \left\langle W^\varepsilon, -\left(\frac{\partial}{\partial t}+\Delta_x\right)\varphi\right\rangle\\ &= -\int_{\mathbb{R}^n}\int_\varepsilon^\infty W(t,x)\left(\frac{\partial}{\partial t}+\Delta_x\right)\varphi(t,x)\,\mathrm{d}t\,\mathrm{d}x\\ &= \int_{\mathbb{R}^n} W(\varepsilon,x)\varphi(\varepsilon,x)\,\mathrm{d}x\\ &\quad+\int_\varepsilon^\infty\int_{\mathbb{R}^n}\left(\frac{\partial}{\partial t}-\Delta_x\right)W(t,x)\varphi(t,x)\,\mathrm{d}x\,\mathrm{d}t.\end{aligned}$$

Now, for $t>0$, $W(t,x)$ clearly satisfies the heat equation, i.e.,

$$\left(\frac{\partial}{\partial t}-\Delta_x\right)W(t,x)=0.$$

It follows that

$$\left\langle\left(\frac{\partial}{\partial t}-\Delta_x\right)W^\varepsilon,\varphi\right\rangle = \int_{\mathbb{R}^n} W_\varepsilon(x)\varphi(\varepsilon,x)\,\mathrm{d}x,$$

and since φ is uniformly continuous and $\{W_\varepsilon\}_{\varepsilon>0}$ forms an approximate identity, it is easy to conclude that $\lim_{\varepsilon\searrow 0}\int W_\varepsilon(x)\varphi(\varepsilon,x)\,\mathrm{d}x=\varphi(0,0)$. Thus, we obtain

$$\lim_{\varepsilon\to 0}\left\langle\left(\frac{\partial}{\partial t}-\Delta_x\right)W^\varepsilon,\varphi\right\rangle = \langle\delta_{(0,0)},\varphi\rangle. \qquad \square$$

7.3 The Laplace Operator Δ and Powers Δ^m

Let again

$$\Delta := \sum_{j=1}^{n} \frac{\partial^2}{\partial x_j^2}$$

be the Laplace operator on $\mathbb{R}^n$. A distribution $u \in \mathcal{D}'(\mathbb{R}^n)$ is called **harmonic** if

$$\Delta u = 0. \tag{7.9}$$

For example, any "stationary", i.e., time-independent solution of the heat equation is harmonic as a function of x. However, harmonic functions also arise in many other contexts, e.g., as electric potentials or as gravitational potentials in mechanics.

We aim to compute a fundamental solution for Δ. There are various methods available for this.

Our approach will be based on the concept of "analytic families of distributions", which is also useful in many other contexts, and it will further utilize some previous results on heat kernels.

This approach will in fact allow us to determine fundamental solutions in $\mathcal{S}'$ for all powers Δ^m, $m \in \mathbb{N}_{>0}$, of Δ.

We start with the following simple idea: If $\Delta^m K = \delta$, then for the Fourier transform $\hat{K}$ of K, we have

$$(-|\xi|^2)^m \hat{K}(\xi) = 1.$$

Thus, at least formally, we may solve for $\hat{K}$ by

$$\hat{K}(\xi) = (-1)^m |\xi|^{-2m},$$

provided that $|\cdot|^{-2m}$ makes sense at all as a tempered distribution. By Fourier inversion, we should then obtain a fundamental solution K for Δ^m.

Clearly, the function $|\cdot|^{-2m}$ is locally integrable (and thus also tempered) if and only if $2m < n$, i.e., for $m = 1$ if $n \geq 3$.

This suggests to consider more general powers $|\xi|^{-\alpha}$. Note that $|\cdot|^{-\alpha}$ is a tempered function on $\mathbb{R}^n$ (and hence also a tempered distribution) if and only if $\operatorname{Re}\alpha < n$.

7.3.1 Riesz potentials

We therefore define, for $\operatorname{Re}\alpha < n$, the tempered distribution $R_\alpha \in \mathcal{S}'(\mathbb{R}^n)$ via its Fourier transform:

$$\widehat{R_\alpha}(\xi) := |\xi|^{-\alpha}, \quad \alpha \in \mathbb{C},\ \operatorname{Re}\alpha < n. \tag{7.10}$$

To compute R_α, we consider, for $\operatorname{Re}\alpha > 0$, the "subordination integral"

$$\int_0^\infty \mathrm{e}^{-t|\xi|^2} t^{\alpha/2-1}\,\mathrm{d}t = |\xi|^{-\alpha}\int_0^\infty \mathrm{e}^{-t} t^{\alpha/2-1}\,\mathrm{d}t = |\xi|^{-\alpha}\Gamma(\alpha/2). \tag{7.11}$$

Here, Γ is Euler's **Gamma function**, given for $\operatorname{Re} z > 0$ by

$$\Gamma(z) := \int_0^\infty \mathrm{e}^{-t} t^{z-1}\,\mathrm{d}t.$$

As is well-known, the Gamma function satisfies the functional equation

$$\Gamma(z+1) = z\Gamma(z), \qquad \operatorname{Re} z > 0, \tag{7.12}$$

i.e., $\Gamma(z) = \frac{\Gamma(z+1)}{z}$. By iteration, it implies for $\operatorname{Re} z > 0$:

$$\Gamma(z) = \frac{\Gamma(z+m)}{z(z+1)\cdots(z+m-1)}, \tag{7.13}$$

for every $m \in \mathbb{N}_{>0}$. The right-hand side of (7.13) clearly has a meromorphic extension to the half-plane $\{\operatorname{Re} z > -m\}$. Since $m \in \mathbb{N}$ is arbitrary, this shows that Γ has a meromorphic extension to all of $\mathbb{C}$, with simple poles at the points $z = -m$, $m \in \mathbb{N}$. Thus, Γ is defined on all of $\mathbb{C}$ as a meromorphic function.

Moreover, the identity (7.13) remains valid for this meromorphic extension at all points $z \neq 0, -1, -2, \ldots$, that is, whenever z is not a pole of Γ. This follows immediately from the identity theorem of complex analysis (see, e.g., Stein–Shakarchi [41], Chapter 6).

Furthermore, (see Stein–Shakarchi [41], Chapter 6, Theorem 1.4)

$$\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z, \tag{7.14}$$

so that Γ has no zeros in $\mathbb{C}$, and hence $1/\Gamma$ is an entire holomorphic function.

By (7.11), we have for $\xi \neq 0$:

$$|\xi|^{-\alpha} = \frac{1}{\Gamma(\frac{\alpha}{2})}\int_0^\infty \mathrm{e}^{-t|\xi|^2} t^{\alpha/2-1}\mathrm{d}t, \qquad \operatorname{Re}\alpha > 0. \tag{7.15}$$

Lemma 7.6 *For $0 < \operatorname{Re}\alpha < n$, the distribution R_α defined by (7.10) agrees with the tempered function given by*

$$R_\alpha(x) = \pi^{-n/2}2^{-\alpha}\frac{\Gamma\left(\frac{n-\alpha}{2}\right)}{\Gamma\left(\frac{\alpha}{2}\right)}|x|^{\alpha-n}. \tag{7.16}$$

Proof. Let $\varphi \in \mathcal{S}(\mathbb{R}^n)$. Then, for $\operatorname{Re}\alpha < n$, we have by Fourier inversion in $\mathcal{S}'$

$$\langle R_\alpha, \varphi\rangle = \langle \widehat{R_\alpha}, (2\pi)^{-n}\overline{\mathcal{F}\varphi}\rangle.$$

For $0 < \operatorname{Re}\alpha < n$, however, the integral in (7.15) is absolutely convergent, and thus for these values of α, we obtain

$$\begin{aligned}\langle R_\alpha, \varphi\rangle &= \frac{1}{(2\pi)^n \Gamma(\frac{\alpha}{2})} \int_{\mathbb{R}^n} \int_0^\infty \mathrm{e}^{-t|\xi|^2} t^{\alpha/2-1}\, \mathrm{d}t\, \overline{\mathcal{F}\varphi}(\xi)\, \mathrm{d}\xi \\ &= \frac{1}{(2\pi)^n \Gamma(\frac{\alpha}{2})} \int_0^\infty \int_{\mathbb{R}^n} \mathrm{e}^{-t|\xi|^2} \overline{\mathcal{F}\varphi}(\xi)\, \mathrm{d}\xi\, t^{\alpha/2-1}\, \mathrm{d}t.\end{aligned}$$

By (7.3), Proposition 2.25 and the Fourier inversion formula, the inner integral is given by

$$\int_{\mathbb{R}^n} \widehat{W_t}(\xi)\overline{\mathcal{F}\varphi}(\xi)\, \mathrm{d}\xi = \int_{\mathbb{R}^n} W_t(x)\, \mathcal{F}\overline{\mathcal{F}\varphi}(x)\, \mathrm{d}x = (2\pi)^n \int_{\mathbb{R}^n} W_t(x)\varphi(x)\, \mathrm{d}x.$$

Thus, we obtain

$$\langle R_\alpha, \varphi\rangle = \frac{1}{\Gamma(\frac{\alpha}{2})} \int_0^\infty \int_{\mathbb{R}^n} (4\pi t)^{-n/2} \mathrm{e}^{-\frac{|x|^2}{4t}} t^{\alpha/2-1} \varphi(x)\, \mathrm{d}x\, \mathrm{d}t. \tag{7.17}$$

Now,

$$\begin{aligned}\int_0^\infty \mathrm{e}^{-\frac{|x|^2}{4t}} t^{\frac{\alpha-n}{2}-1}\, \mathrm{d}t &= \int_0^\infty \mathrm{e}^{-t\frac{|x|^2}{4}} t^{\frac{n-\alpha}{2}-1}\, \mathrm{d}t \\ &= \left(\frac{4}{|x|^2}\right)^{\frac{n-\alpha}{2}} \int_0^\infty \mathrm{e}^{-t} t^{\frac{n-\alpha}{2}-1}\, \mathrm{d}t = 2^{n-\alpha}\Gamma(\tfrac{n-\alpha}{2})|x|^{\alpha-n},\end{aligned}$$

where the integral is absolutely convergent. Since $|\cdot|^{\alpha-n}$ is also locally integrable, we can apply Fubini's theorem in (7.17), obtaining

$$\langle R_\alpha, \varphi\rangle = \pi^{-n/2} 2^{-\alpha} \frac{\Gamma\left(\frac{n-\alpha}{2}\right)}{\Gamma(\frac{\alpha}{2})} \int_{\mathbb{R}^n} |x|^{\alpha-n} \varphi(x)\, \mathrm{d}x. \qquad \square$$

The functions $R_\alpha, 0 < \operatorname{Re}\alpha < n$, are also called **Riesz potentials**.

7.3.2 An analytic family of distributions

The Riesz potentials depend analytically on the parameter α in the following sense:

Definition. Let $V \neq \emptyset$ be an open subset of $\mathbb{C}$, and let $\{u_\alpha\}_{\alpha\in V}$ be a family of distributions $u_\alpha \in \mathcal{D}'(\Omega)$ in an open subset $\Omega \subset \mathbb{R}^n$ (respectively $u_\alpha \in \mathcal{S}'(\mathbb{R}^n)$). If for every test function $\varphi \in \mathcal{D}(\Omega)$ (respectively $\varphi \in \mathcal{S}(\mathbb{R}^n)$), the function

$$\alpha \mapsto \langle u_\alpha, \varphi\rangle$$

is holomorphic on V, then the family $\{u_\alpha\}_{\alpha\in G}$ is called an **analytic family of distributions (respectively tempered distributions)**.

We claim that the family of Riesz potentials $\{R_\alpha\}_{0<\operatorname{Re}\alpha<n}$ is such an analytic family of tempered distributions.

To this end, we first observe that the coefficient function

$$\gamma_n(\alpha) := \pi^{-n/2} 2^{-\alpha} \frac{\Gamma\left(\frac{n-\alpha}{2}\right)}{\Gamma\left(\frac{\alpha}{2}\right)}$$

is analytic for $0 < \operatorname{Re}\alpha < n$.

Furthermore, if we fix $0 < a < b < n$, then for α in the strip where $a < \operatorname{Re}\alpha < b$, we have

$$|\,|x|^{\alpha-n}| \le F_{a,b}(x),$$

where $F_{a,b}$ denotes the tempered function

$$F_{a,b}(x) := \begin{cases} |x|^{a-n}, & \text{if } |x| \le 1, \\ |x|^{b-n}, & \text{if } |x| > 1. \end{cases}$$

Thus, for every $\varphi \in \mathcal{S}(\mathbb{R}^n)$, the function

$$f_\varphi : \alpha \mapsto \int_{\mathbb{R}^n} |x|^{\alpha-n} \varphi(x)\,\mathrm{d}x$$

is well-defined and continuous on the strip $a < \operatorname{Re}\alpha < b$. By Morera's theorem, it is even holomorphic on this strip, since for any closed contour σ in it, by Fubini's theorem we have

$$\oint_\sigma \int_{\mathbb{R}^n} |x|^{\alpha-n} \varphi(x)\,\mathrm{d}x\,\mathrm{d}\alpha = \int_{\mathbb{R}^n} \oint_\sigma \mathrm{e}^{(\alpha-n)\log|x|}\,\mathrm{d}\alpha\,\varphi(x)\,\mathrm{d}x = 0$$

(note that the inner integral vanishes for $x \neq 0$ by Cauchy's theorem).

Since this argument holds for any $0 < a < b < n$, we see that the family of distributions $\{R_\alpha\}_{0<\operatorname{Re}\alpha<n}$ is indeed analytic.

In a next step, we will extend this family analytically. First, we observe that the coefficient function $\gamma_n(\alpha)$ is meromorphic on $\mathbb{C}$, with simple poles at the points of the "**exceptional set**"

$$E := \{n + 2k : \ k \in \mathbb{N}\}.$$

(Note that $\Gamma(x) > 0$ for $x > 0$, so that $1/\Gamma(\frac{\alpha}{2}) \neq 0$ for $\alpha \in E$.)

Lemma 7.7 *The family of Riesz potentials $\{R_\alpha\}_{0<\operatorname{Re}\alpha<n}$ can be uniquely extended to an analytic family of tempered distributions $\{R_\alpha\}_{\alpha\in\mathbb{C}\setminus E}$ on $\mathbb{C} \setminus E$.*

For $\operatorname{Re}\alpha > 0$, R_α is then given by the tempered function

$$R_\alpha(x) = \gamma_n(\alpha)|x|^{\alpha-n},$$

and $R_0 = \delta$. Moreover, the family of distributions $\{|\cdot|^{\alpha-n}\}_{\alpha\in H}$ is analytic on the half plane $H := \{\alpha \in \mathbb{C} : \operatorname{Re}\alpha > 0\}$, and thus specifically at all points of the exceptional set E.

Proof. For $\operatorname{Re}\alpha > 0$, we define R_α by

$$R_\alpha(x) = \gamma_n(\alpha)|x|^{\alpha-n}.$$

Our previous arguments for the range $0 < \operatorname{Re}\alpha < n$ remain valid for $\operatorname{Re}\alpha > 0$ and show that the family $\{|\cdot|^{\alpha-n}\}_{\alpha\in H}$ is analytic on H. Thus, the family $\{R_\alpha\}$ is analytic on $H \setminus E$ and has simple poles at the points in E. Furthermore, for every Schwartz function φ, it follows from Lemma 7.6 and (7.10) that

$$\langle R_\alpha, \varphi\rangle = \langle |\cdot|^{-\alpha}, (2\pi)^{-n}\overline{\mathcal{F}}\varphi\rangle, \qquad 0 < \operatorname{Re}\alpha < n, \tag{7.18}$$

and, from what has already been explained, the family of tempered distributions $\{|\cdot|^{-\alpha}\}_{\operatorname{Re}\alpha<n}$ is analytic. We have previously defined the distributions R_α for $\operatorname{Re}\alpha < n$ by $\widehat{R_\alpha} = |\cdot|^{-\alpha}$, i.e., by

$$\langle R_\alpha, \varphi\rangle := \langle |\cdot|^{-\alpha}, (2\pi)^{-n}\overline{\mathcal{F}}\varphi\rangle, \qquad \operatorname{Re}\alpha < n.$$

Thus, an analytic continuation of the family R_α is evidently given even for the range $\operatorname{Re}\alpha < n$. Overall, we have thereby defined $\{R_\alpha\}_\alpha$ as an analytic family of tempered distributions on all of $\mathbb{C} \setminus E$.

Our continuation is unique since if $\{\tilde{R}_\alpha\}_{\alpha\in\mathbb{C}\setminus E}$ is any analytic continuation of it, then for every $\varphi \in \mathcal{D}$ we have that

$$\langle R_\alpha, \varphi\rangle = \langle \tilde{R}_\alpha, \varphi\rangle \quad \text{for } 0 < \operatorname{Re}\alpha < n,$$

so that, by the identity theorem of complex analysis, $\langle R_\alpha, \varphi\rangle = \langle \tilde{R}_\alpha, \varphi\rangle$ for every $\alpha \in \mathbb{C} \setminus E$.

Finally, since $\widehat{R_0} = |\cdot|^0 = 1$, it follows that $R_0 = \delta$. □

7.3.3 A fundamental solution for Δ^m

We are now in a position to prove the following theorem.

Theorem 7.8 *Let $m \in \mathbb{N}_{>0}$, and define*

$$K_m(x) := \begin{cases} (-1)^m \pi^{-n/2} 2^{-2m} \frac{\Gamma\left(\frac{n}{2}-m\right)}{\Gamma(m)} |x|^{2m-n}, & \text{if } 2m-n \notin 2\mathbb{N}, \\ \frac{(-1)^{1-\frac{n}{2}} \pi^{-n/2} 2^{1-2m}}{\Gamma(m+1-\frac{n}{2})\Gamma(m)} |x|^{2m-n} \log|x|, & \text{if } 2m-n \in 2\mathbb{N}. \end{cases}$$

Then $K_m \in \mathcal{S}'(\mathbb{R}^n)$ is a fundamental solution for Δ^m, i.e., $\Delta^m K_m = \delta$.

(Note that for $2m - n = 2k \in 2\mathbb{N}$, the function $x \mapsto |x|^{2m-n}$ is a polynomial of degree $2k < 2m$, so that $\Delta^m(|\cdot|^{2m-n}) \equiv 0$.)

Proof. Clearly, K_m is always locally integrable and tempered, i.e., $K_m \in \mathcal{S}'(\mathbb{R}^n)$.

(a) The case $2m - n \notin 2\mathbb{N}$, i.e., $2m \notin E$.
By Lemma 7.7, we have

$$K_m = (-1)^m R_{2m},$$

and by Lemma 7.6 and (7.10), for $0 < \operatorname{Re}\alpha < n$,

$$(\Delta^m R_\alpha)^\wedge(\xi) = (-|\xi|^2)^m \widehat{R_\alpha}(\xi) = (-1)^m |\xi|^{2m-\alpha},$$

which implies

$$(\Delta^m R_\alpha)^\wedge = (-1)^m |\cdot|^{-(\alpha-2m)}.$$

This implies the following identity:

$$(-\Delta)^m R_\alpha = R_{\alpha-2m}, \tag{7.19}$$

which is a special case of a more general analogous identity for powers of linear differential operators with constant coefficients, proven by Atiyah [1], as well as Bernstein and Gel'fand [4].

The left-hand side of (7.19) is clearly holomorphic as a family of distributions on all of $\mathbb{C}\backslash E$, and the right-hand side is well-defined and holomorphic on $\mathbb{C}\backslash(2m+E) \supset \mathbb{C}\backslash E$. By the identity theorem, (7.19) thus remains valid for all $\alpha \in \mathbb{C} \setminus E$, in particular for $\alpha = 2m$. It follows that

$$\Delta^m K_m = (-1)^m \Delta^m R_{2m} = R_0 = \delta.$$

(b) The case $2m - n \in 2\mathbb{N}$, i.e., $2m = n + 2k \in E$, with $k \in \mathbb{N}$.
Note that here n is even. We set $k := m - n/2 \in \mathbb{N}$.

This case is more complicated since the function γ_n has a simple pole at $\alpha = 2m$, which we first want to remove. Defining β by

$$\alpha = 2m + \beta = n + 2k + \beta,$$

it follows from (7.13) that for small $\beta \neq 0$,

$$\gamma_n(\alpha) = \pi^{-n/2} 2^{-n-2k-\beta} \frac{\Gamma\left(-k - \frac{\beta}{2}\right)}{\Gamma\left(m + \frac{\beta}{2}\right)} = \frac{\pi^{-n/2} 2^{-n-2k-\beta} \Gamma\left(1 - \frac{\beta}{2}\right)}{-\frac{\beta}{2}\left(-\frac{\beta}{2} - 1\right) \cdots \left(-\frac{\beta}{2} - k\right) \Gamma\left(m + \frac{\beta}{2}\right)}.$$

This implies that the function

$$\varrho(\alpha) := (\alpha - 2m)\gamma_n(\alpha) = \beta\gamma_n(\alpha)$$

is holomorphic at $\alpha = 2m$, and for $\beta \to 0$ we obtain

$$\varrho(2m) = \frac{\pi^{-n/2} 2^{1-n-2k}}{(-1)^{k+1} k! \Gamma(m)} = \frac{\pi^{-n/2} 2^{1-2m} (-1)^{1+m-\frac{n}{2}}}{\Gamma(m+1-\frac{n}{2})\Gamma(m)}. \tag{7.20}$$

(Note that $m \geq 1$ and that $\Gamma(k+1) = k!$.)

Now, setting

$$Q_\alpha(x) := (\alpha - 2m)R_\alpha(x) = \varrho(\alpha)|x|^{\alpha-n}, \qquad \alpha \in (H \setminus E) \cup \{2m\},$$

it follows from Lemma 7.7 that the family $\{Q_\alpha\}_\alpha$ is analytic on $(H \setminus E) \cup \{2m\}$, and from (7.19) we have

$$\Delta^m Q_\alpha = (-1)^m(\alpha - 2m)R_{\alpha-2m} \tag{7.21}$$

on $(H \setminus E) \cup \{2m\}$, in particular on the disk $U := \{\alpha \in \mathbb{C} : |\alpha - 2m| < 1\}$.

Since the function $\alpha \mapsto \langle R_{\alpha-2m}, \varphi\rangle$ is analytic on U, we may differentiate (7.21) with respect to α at $\alpha = 2m$ and obtain

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\alpha}\Big|_{\alpha=2m}\langle\Delta^m Q_\alpha, \varphi\rangle &= \frac{\mathrm{d}}{\mathrm{d}\alpha}\Big|_{\alpha=2m}(-1)^m(\alpha-2m)\langle R_{\alpha-2m}, \varphi\rangle \\ &= (-1)^m\langle R_0, \varphi\rangle + (-1)^m\,(2m-2m)\frac{\mathrm{d}}{\mathrm{d}\alpha}\Big|_{\alpha=2m}\langle R_{\alpha-2m}, \varphi\rangle \\ &= (-1)^m\langle\delta, \varphi\rangle. \end{aligned} \tag{7.22}$$

On the other hand,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\alpha}\Big|_{\alpha=2m}\langle\Delta^m Q_\alpha, \varphi\rangle &= \frac{\mathrm{d}}{\mathrm{d}\alpha}\Big|_{\alpha=2m}\left\{\rho(\alpha)\int |x|^{\alpha-n}\Delta^m\varphi(x)\,\mathrm{d}x\right\} \\ &= \varrho'(2m)\int |x|^{2k}\Delta^m\varphi(x)\,\mathrm{d}x \\ &\quad +\varrho(2m)\int |x|^{2m-n}\log|x|\,\Delta^m\varphi(x)\,\mathrm{d}x.\end{aligned}$$

Since $2k < 2m$, we have $\Delta^m(|\cdot|^{2k}) = 0$, and thus, by integration by parts,

$$\int |x|^{2k}\Delta^m\varphi(x)\,\mathrm{d}x = \int \Delta^m(|\cdot|^{2k})\varphi\,\mathrm{d}x = 0 \quad \text{for all } \varphi \in \mathcal{S}.$$

Thus, together with (7.20), we obtain

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\Big|_{\alpha=2m}\langle\Delta^m Q_\alpha, \varphi\rangle = (-1)^m\int K_m\Delta^m\varphi\,\mathrm{d}x = (-1)^m\langle\Delta^m K_m, \varphi\rangle.$$

Together with (7.22), we conclude that $\Delta^m K_m = \delta$. □

In particular, defining for $m = 1$ the **Newtonian potential** $N = N_n$ on $\mathbb{R}^n$ by

$$N(x) := \begin{cases} -\frac{\Gamma\left(\frac{n}{2}-1\right)}{4\pi^{n/2}}|x|^{2-n}, & \text{if } n \neq 2, \\ \frac{1}{2\pi}\log|x|, & \text{if } n = 2, \end{cases}$$

we see that N is a fundamental solution for Δ, i.e.,

$$\Delta N = \delta. \tag{7.23}$$

Since

$$\Gamma(1/2) = \int_0^\infty \mathrm{e}^{-t} t^{-1/2} \mathrm{d}t = 2 \int_0^\infty \mathrm{e}^{-u^2} \, \mathrm{d}u = \sqrt{\pi}, \tag{7.24}$$

it follows that

$$-\frac{\Gamma(-1/2)}{4\pi^{1/2}} = \frac{2\Gamma(1/2)}{4\pi^{1/2}} = \frac{1}{2},$$

and thus, for dimension $n = 1$,

$$N(x) = N_1(x) = \frac{1}{2}|x|.$$

Of course, it is also easy to see directly that

$$\Delta N_1 = \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\frac{1}{2}|x|\right) = \delta.$$

7.4 Distributions as Derivatives of Functions

Using Theorem 7.8, we now show that every distribution can be locally represented as a sum of derivatives of continuous functions (cf. the statement of Theorem 7.9, also in Rudin [34, Theorem 6.26]).

Thus, $\mathcal{D}'(\Omega)$ is, in a certain sense, the smallest space that contains all continuous functions on Ω and in which one can differentiate arbitrarily often (in a generalized sense).

Theorem 7.9 *Let $u \in \mathcal{E}'(\mathbb{R}^n)$. Then there exists a continuous function $f \in C(\mathbb{R}^n)$ and an $m \in \mathbb{N}$ such that*

$$u = \Delta^m f.$$

Proof. Let N be the order of $u \in \mathcal{E}'(\mathbb{R}^n)$, and choose $m \in \mathbb{N}$ such that $2m > N + n + 1$. We additionally assume that $2m - n$ is odd – the other case can be handled similarly with minimal additional effort. Let $K = K_m \in \mathcal{S}'(\mathbb{R}^n)$ be the fundamental solution for Δ^m from Theorem 7.8, i.e.,

$$\Delta^m K = \delta.$$

K then has the form

$$K(x) = c|x|^{2m-n}.$$

This function is homogeneous of degree $2m - n$ (i.e., $K(rx) = r^{2m-n}K(x)$ for every $r > 0$), smooth outside the origin, and continuous on all of $\mathbb{R}^n$, with $K(0) = 0$.

Every partial derivative $D_j K$ is therefore also smooth outside the origin and homogeneous of degree $2m - n - 1$. From this, one easily concludes that there exists a constant C_j such that

$$|D_j K(x)| \le C_j |x|^{2m-n-1} \quad \text{for all } x \neq 0.$$

Furthermore, it is immediately clear that $D_j K(0) = 0$. Altogether, we see that the function $D_j K$ is continuous on all of $\mathbb{R}^n$, vanishes at the origin, is smooth outside the origin, and is homogeneous of degree $2m - n - 1$.

Iterating this argument, we conclude by induction in an analogous manner that the function K is even N-times continuously differentiable, where for every $\alpha \in \mathbb{N}^n$ with $|\alpha| \leq N$, we have that $D^\alpha K(0) = 0$, $D^\alpha K$ is smooth outside the origin, and homogeneous of degree $2m - n - |\alpha|$.

From now on, we only use that $K \in \mathcal{S}'(\mathbb{R}^n)$ lies $C^N(\mathbb{R}^n)$.

Let us put $f := u * K \in \mathcal{S}'(\mathbb{R}^n)$. By Proposition 6.4, we then have

$$\Delta^m f = u * (\Delta^m K) = u * \delta = u,$$

so it remains only to show that f is a continuous function.

To this end, let $\{\varphi_j\}_j$ be a Dirac sequence in $\mathcal{D}(\mathbb{R}^n)$, and define $f_j := f * \varphi_j = u * K_j$, with

$$K_j := K * \varphi_j \in C^\infty(\mathbb{R}^n).$$

Since $f_j \in C^\infty(\mathbb{R}^n) \subset C(\mathbb{R}^n)$, it suffices to show that the sequence $\{f_j\}_j$ is locally uniformly convergent:

Indeed, if $\tilde{f} := \lim f_j$ is the continuous limit function, then naturally $\tilde{f} = \lim f_j$ in $\mathcal{D}'(\mathbb{R}^n)$. On the other hand, by Proposition 4.25, $f = \lim f_j$ in $\mathcal{S}'(\mathbb{R}^n)$, so that $\tilde{f} = f$.

Altogether, we have thus reduced the proof to showing that the sequence $\{f_j\}$ is a Cauchy sequence on each compact subset $A \subset \mathbb{R}^n$ with respect to the sup-norm on A.

Now, if B is a compact neighborhood of the support of u, then by (6.1), there exists a constant $C \geq 0$ such that

$$|u(\psi)| \leq C \max_{|\alpha| \leq N} \sup_{y \in B} |D^\alpha \psi(y)| \quad \text{for all } \psi \in C^\infty(\mathbb{R}^n).$$

Let $A \subset \mathbb{R}^n$ be compact. Then, for $x \in A$ and $j, k \in \mathbb{N}$, we obtain

$$|f_j(x) - f_k(x)| = |u(\lambda_x(\check{K}_j - \check{K}_k))| \leq C \max_{|\alpha| \leq N} \sup_{y \in B} |D^\alpha (K_j - K_k)(x - y)|,$$

i.e.,

$$\sup_{x \in A} |f_j(x) - f_k(x)| \leq C \max_{|\alpha| \leq N} \sup_{z \in A - B} |D^\alpha (K_j - K_k)(z)|. \tag{7.25}$$

For $|\alpha| \leq N$, however,

$$D^\alpha K_j = (D^\alpha K) * \varphi_j,$$

and, since K is N-times continuously differentiable, the distributional derivative $D^\alpha K$ agrees according to Theorem 6.7 with the corresponding classical derivative of order α, so in particular $D^\alpha K \in C(\mathbb{R}^n)$.

Thus, by Theorem 2.13 (b), the sequence $(D^\alpha K) * \varphi_j$ converges uniformly on the compact set $A - B$ to $D^\alpha K$ as $j \to \infty$, and thus, by (7.25), the sequence $\{f_j|_A\}_j$ forms a Cauchy sequence on A with respect to the sup-norm. □

Corollary 7.10 *Let $u \in \mathcal{D}'(\Omega)$, and let $K \subset \Omega$ be compact. Then there exist an open neighborhood V of K in Ω, an $m = m(K) \in \mathbb{N}$, and a continuous function $f \in C(V)$ such that*

$$u|_V = \Delta^m f \text{ on } V.$$

Proof. Apply Theorem 7.9 to χu, where $\chi \in \mathcal{D}(\Omega)$ is identically 1 on a neighborhood of K. □

7.5 The Theorem of Malgrange and Ehrenpreis

Around 1954–1955, Malgrange [31] and Ehrenpreis [9] independently proved that every nontrivial linear differential operator

$$P(D) = \sum_{|\alpha| \le m} a_\alpha D^\alpha \tag{7.26}$$

with constant coefficients $a_\alpha \in \mathbb{C}$ has a fundamental solution $K \in \mathcal{D}'(\mathbb{R}^n)$.

A natural idea to define such a fundamental solution is again the following:

If $P(D)K = \delta$, by taking the Fourier transforms we obtain $P(\xi)\hat{K}(\xi) = 1$ (assuming $K \in \mathcal{S}'$), so that, at least on a formal level, K should be given by the Fourier inversion formula as

$$K(x) = (2\pi)^{-n} \int_{\mathbb{R}^n} \frac{1}{P(\xi)} e^{ix\cdot\xi}\, d\xi. \tag{7.27}$$

Unfortunately, however, as we have already seen in the example of the powers of the Laplace operator, the zeros of $P(\xi)$ generally prevent $1/P$ from being locally integrable at all, so it is not clear how to define $1/P$ as a distribution.

Thus, (7.27) will generally not make sense, even if one interprets the inverse Fourier transform in (7.27) in the sense of distributions, i.e., as

$$\langle K, \varphi \rangle = (2\pi)^{-n} \int_{\mathbb{R}^n} \frac{1}{P(\xi)} \hat{\varphi}(-\xi)\, d\xi, \qquad \varphi \in \mathcal{D}. \tag{7.28}$$

The idea to resolve this problem is, for instance in dimension $n = 1$, to move the real "contour" $\mathbb{R}$ of integration in (7.28) into the complex domain and choose a suitable contour that avoids the zeros of P.

I will present here a simplified approach (compared to the ones used in the original works by Malgrange and Ehrenpreis), based on an idea by L. Nirenberg and described in a very similar manner by Folland in [11].

Let m be the order of $P(D)$, i.e., the degree of the polynomial $P(\xi) = \sum_{|\alpha| \le m} a_\alpha \xi^\alpha$. Then there exists a $\beta \in \mathbb{N}^n$ with $|\beta| = m$ and $a_\beta \neq 0$. Let

$$P_m(\xi) := \sum_{|\alpha| = m} a_\alpha\, \xi^\alpha$$

be the so-called **principal symbol** of the differential operator $P(D)$, i.e., the principal part of its symbol $P(\xi)$.

Since $P_m \neq 0$, there exists a unit vector $\eta \in \mathbb{R}^n$ with $P_m(\eta) \neq 0$. By means of a rotation of the coordinate system, we may assume that $\eta = (0, \ldots, 0, 1)$. Since

$$P_m(\xi) = \sum_{|\alpha|=m} a_\alpha \, \xi_1^{\alpha_1} \cdots \xi_{n-1}^{\alpha_{n-1}} \xi_n^{\alpha_n},$$

we then have $P_m(\eta) = a_{(0,\ldots,0,m)}$, and by replacing P with $\frac{1}{a_{(0,\ldots,0,m)}} P$, we may assume without loss of generality that $a_{(0,\ldots,0,m)} = 1$. Thus, P can be written as

$$P(\xi) = P(\xi', \xi_n) = \xi_n^m + \sum_{k=0}^{m-1} Q_k(\xi') \xi_n^k, \tag{7.29}$$

with $\xi' := (\xi_1, \ldots, \xi_{n-1}) \in \mathbb{R}^{n-1}$, and polynomials $Q_k(\xi')$ of degree $\leq m - k$. For fixed ξ', P is thus a polynomial of degree m in the single variable ξ_n, and we denote its complex roots, listed with their multiplicities, by

$$\lambda_1(\xi'), \ldots, \lambda_m(\xi') \in \mathbb{C},$$

i.e.,

$$P(\xi', \lambda_j(\xi')) = 0, \quad j = 1, \ldots, m.$$

We further assume that these are always lexicographically ordered.

The **lexicographical order** on $\mathbb{C}$ is defined such that $z < w$ if $\operatorname{Im} z < \operatorname{Im} w$, or if $\operatorname{Im} z = \operatorname{Im} w$ and $\operatorname{Re} z < \operatorname{Re} w$.

Lemma 7.11 *For each $j = 1, \ldots, m$, the function $\xi' \mapsto \operatorname{Im} \lambda_j(\xi')$ is continuous on $\mathbb{R}^{n-1}$.*

Proof. Let $\xi_0' \in \mathbb{R}^{n-1}$ and $\varepsilon > 0$. Consider the complex disks $B_j := B_\varepsilon(\lambda_j(\xi_0')) \subset \mathbb{C}$ of radius ε centered at $\lambda_j(\xi_0') \in \mathbb{C}$ for $j = 1, \ldots, m$. By possibly decreasing ε, we may assume that the closed disks $\overline{B}_i$ and $\overline{B}_j$ are disjoint whenever $\lambda_i(\xi_0') \neq \lambda_j(\xi_0')$.

Now let $\lambda_j(\xi') = \alpha_j(\xi') + i\beta_j(\xi')$, where $\alpha_j(\xi'),\ \beta_j(\xi') \in \mathbb{R}$. Since $\beta_j \leq \beta_{j+1}$, we can choose

$$1 = j_1 < j_2 < \cdots < j_\ell = m + 1$$

such that

$$\beta_{j_1}(\xi_0') < \beta_{j_2}(\xi_0') < \cdots < \beta_{j_{\ell-1}}(\xi_0')$$

and

$$\beta_j(\xi_0') = \beta_{j_s}(\xi_0') \quad \text{for } j = j_s, \ldots, j_{s+1} - 1 \quad (s = 1, \ldots, \ell - 1).$$

By further decreasing ε, we may assume that the intervals $[\beta_{j_s}(\xi_0') - \varepsilon,\ \beta_{j_s}(\xi_0') + \varepsilon]$, $s = 1, \ldots, \ell - 1$, are pairwise disjoint.

We now consider the holomorphic function

$$f : z \mapsto P(\xi_0', z)$$

on $\mathbb{C}$. Clearly, $f(z) \neq 0$ for every $z \in \bigcup_{j=1}^{m} \partial B_j$ (where ∂B_j denotes the boundary of B_j), and since P is continuous on $\mathbb{C}^n$, there exists $\delta > 0$ such that

$$|P(\xi', z) - f(z)| < |f(z)| \quad \text{for every } z \in \bigcup_{j=1}^{m} \partial B_j, \text{ if } |\xi' - \xi'_0| < \delta.$$

By Rouché's theorem, for each $\xi' \in B_\delta(\xi'_0)$, the function $z \mapsto P(\xi', z)$ has the same number of zeros (counting multiplicities) in each disk B_j as $f(z) = P(\xi'_0, z)$.

Due to the choice of ε, there must thus be $j_2 - j_1$ roots $\lambda_j(\xi')$ with $\beta_j(\xi') \in [\beta_{j_1}(\xi'_0) - \varepsilon, \beta_{j_1}(\xi'_0) + \varepsilon]$, $j_3 - j_2$ roots with $\beta_j(\xi') \in [\beta_{j_2}(\xi'_0) - \varepsilon, \beta_{j_2}(\xi'_0) + \varepsilon]$, and so on and so forth. That is, for $j = j_s, \ldots, j_{s+1} - 1$, we always have $\beta_j(\xi') \in [\beta_{j_s}(\xi'_0) - \varepsilon, \beta_{j_s}(\xi'_0) + \varepsilon]$. In particular,

$$|\beta_j(\xi') - \beta_j(\xi'_0)| < \varepsilon, \quad \text{if } j = j_s, \ldots, j_{s+1} - 1,$$

and thus

$$|\operatorname{Im} \lambda_j(\xi') - \operatorname{Im} \lambda_j(\xi'_0)| < \varepsilon, \quad j = 1, \ldots, m,$$

whenever $|\xi' - \xi'_0| < \delta$, i.e., $\operatorname{Im} \lambda_j$ is continuous at the point ξ'_0 for every j. □

Caution: The functions λ_j themselves may be discontinuous (see Figure 7.1)!

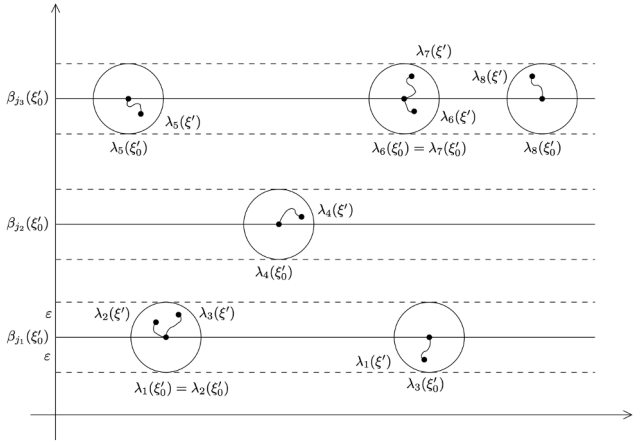


Fig. 7.1 Roots of $z \mapsto P(\xi', z)$

Lemma 7.12 *There exists a Borel-measurable function*

$$\Phi : \mathbb{R}^{n-1} \to [-m, m]$$

such that for every $\xi' \in \mathbb{R}^{n-1}$,

$$\min_{j=1,\ldots,m} |\Phi(\xi') - \operatorname{Im} \lambda_j(\xi')| \geq 1.$$

Proof. The underlying idea is simple (*pigeonhole principle*):

The interval $[-m-1, m+1]$ has length $2m+2$. Thus, for each $\xi' \in \mathbb{R}^{n-1}$, there must be an open subinterval of length 2 that does not contain any of the m points $\operatorname{Im}\lambda_1(\xi'), \dots, \operatorname{Im}\lambda_m(\xi')$, and we can choose the midpoint of this interval for $\Phi(\xi')$. We only need to show that this can be done in a measurable way with respect to ξ'. To this end, we set

$$\begin{aligned} &\mu_0(\xi') := -m-1,\ \mu_{m+1}(\xi') := m+1, \\ &\mu_j(\xi') := \max\left(\{\min\left(\operatorname{Im}\lambda_j(\xi'), m+1\right)\}, -m-1\right), \qquad j = 1, \dots, m, \end{aligned}$$

i.e., $\mu_j(\xi') = \operatorname{Im}\lambda_j(\xi')$ if $\operatorname{Im}\lambda_j(\xi') \in [-m-1, m+1]$, $\mu_j(\xi') = m+1$ if $\operatorname{Im}\lambda_j(\xi') > m+1$, and $\mu_j(\xi') = -m-1$ if $\operatorname{Im}\lambda_j(\xi') < -m-1$. The functions μ_j are continuous by Lemma 7.11, so the sets

$$V_j := \{\xi' \in \mathbb{R}^{n-1} \mid \mu_{j+1}(\xi') - \mu_j(\xi') \geq 2\}, \quad j = 0, \dots, m,$$

are Borel-measurable. Since

$$-m-1 = \mu_0(\xi') \leq \mu_1(\xi') \leq \cdots \leq \mu_{m+1}(\xi') = m+1,$$

the sets V_j cover all of $\mathbb{R}^{n-1}$ (cf. the preceding remark). Thus, we can construct disjoint Borel sets $W_j \subset V_j$ that also cover $\mathbb{R}^{n-1}$.

Then, the function

$$\Phi(\xi') := \frac{1}{2}(\mu_{j+1}(\xi') + \mu_j(\xi')), \quad \text{if } \xi' \in W_j,$$

clearly has the desired properties. □

Theorem 7.13 (Malgrange–Ehrenpreis) *Every nontrivial linear differential operator $P(D)$ with constant coefficients has a fundamental solution.*

Proof. For $\varphi \in \mathcal{D}(\mathbb{R}^n)$, let $\mathcal{FL}\varphi$ denote the Fourier–Laplace transform of φ, which, by Theorem 6.8, is an entire holomorphic function on $\mathbb{C}^n$. We may assume that $P(\xi)$ has the form (7.29) and choose a measurable function Φ according to Lemma 7.12. We then define a linear functional K on $\mathcal{D}(\mathbb{R}^n)$ by

$$\langle K, \varphi\rangle := (2\pi)^{-n} \int\limits_{\mathbb{R}^{n-1}} \int\limits_{\operatorname{Im}\zeta_n = \Phi(\xi')} \frac{\mathcal{FL}\varphi(-\xi', -\zeta_n)}{P(\xi', \zeta_n)}\, \mathrm{d}\zeta_n\, \mathrm{d}\xi', \quad \varphi \in \mathcal{D}(\mathbb{R}^n). \tag{7.30}$$

The integral (7.30) converges absolutely because, according to (7.29),

$$P(\xi', \zeta_n) = (\zeta_n - \lambda_1(\xi')) \cdots (\zeta_n - \lambda_m(\xi')),$$

so by Lemma 7.12,

$$|P(\xi', \zeta_n)| \geq \prod_{j=1}^{m} |\operatorname{Im}\zeta_n - \operatorname{Im}\lambda_j(\xi')| \geq 1,$$

if $\operatorname{Im}\zeta_n = \Phi(\xi')$. For $\varphi \in \mathcal{D}_{\overline{B}_r}$, the Paley–Wiener Theorem 6.10 further ensures that for each $N \in \mathbb{N}$,

$$|\mathcal{FL}\varphi(-\xi', -\zeta_n)| \le C_N r^n \|\varphi\|_N (1 + |(\xi', t)|)^{-N} \mathrm{e}^{mr}, \tag{7.31}$$

if $\zeta_n = t + is$, $t, s \in \mathbb{R}$ with $|s| \le m$. Thus,

$$\begin{aligned}
&\int_{\mathbb{R}^{n-1}} \int_{-\infty}^{\infty} \left| \frac{\mathcal{FL}\varphi(-\xi', -t - i\Phi(\xi'))}{P(\xi', t + i\Phi(\xi'))} \right| \mathrm{d}t \, \mathrm{d}\xi' \\
&\quad \le C_N r^n \mathrm{e}^{mr} \|\varphi\|_N \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} (1 + |(\xi', t)|)^{-N} \mathrm{d}t \, \mathrm{d}\xi' \\
&\quad \le C'_N \mathrm{e}^{(m+1)r} \|\varphi\|_N,
\end{aligned}$$

if $N \ge n + 1$. This shows that (7.30) indeed defines a distribution $K \in \mathcal{D}'(\mathbb{R}^n)$. Finally,

$$\begin{aligned}
\langle P(D)K, \varphi \rangle &= \langle K, P(-D)\varphi \rangle \\
&= (2\pi)^{-n} \int_{\mathbb{R}^{n-1}} \int_{\operatorname{Im}\zeta_n = \Phi(\xi')} \mathcal{FL}\varphi(-\xi', -\zeta_n) \, \mathrm{d}\zeta_n \, \mathrm{d}\xi',
\end{aligned} \tag{7.32}$$

since

$$\mathcal{FL}(P(-D)\varphi)(-\zeta) = P(\zeta)\mathcal{FL}\varphi(-\zeta).$$

The inequality (7.31) allows us to use Cauchy's integral theorem to replace integration over the contour $\{\operatorname{Im}\zeta_n = \Phi(\xi')\}$ in (7.32) with integration over the real axis $\{\operatorname{Im}\zeta_n = 0\}$. Applying the Fourier inversion formula, we obtain

$$\langle P(D)K, \varphi \rangle = (2\pi)^{-n} \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} \hat{\varphi}(-\xi', -\xi_n) \, \mathrm{d}\xi_n \mathrm{d}\xi' = \varphi(0) = \langle \delta, \varphi \rangle,$$

i.e., $P(D)K = \delta$. □

Remarks 7.14 (a) Lojasiewicz [30] and Hörmander [20] have independently shown that one can even always find a tempered fundamental solution $K \in \mathcal{S}'(\mathbb{R}^n)$ for $P(D)$. However, the proofs in these works use deeper methods, particularly Hironaka's theorem on the resolution of singularities of algebraic functions.

(b) There exist linear differential operators (with non-constant coefficients) that are not locally solvable! The historically first example of this kind was described by Hans Lewy in 1957. It concerns the "complex vector field"

$$Z := \frac{\partial}{\partial x} + i\frac{\partial}{\partial y} + \frac{i}{2}(x + iy)\frac{\partial}{\partial t}$$

on $\mathbb{R}^3$ with coordinates (x, y, t) (see, e.g., Hörmander [22] and Folland [11]). The Lewy operator Z also exhibits interesting connections to the Heisenberg commutator relation (2.33) (see Stein [39, Ch. III.4]).

7.6 Exercises

7.1 Determine *all* fundamental solutions for $\frac{\mathrm{d}}{\mathrm{d}x} - a$ ($a \in \mathbb{C}$). Show that there is always a tempered one among them.

7.2 (Fundamental Solution for the Helmholtz Operator)
Consider for $c \in \mathbb{C}$ the **Helmholtz operator**

$$-\Delta + c^2 \quad \text{on } \mathbb{R}^3.$$

a) Proceed similarly to the proof of Lemma 7.6 to show that for $c > 0$, the function

$$\widehat{K_c}(\xi) := \frac{1}{|\xi|^2 + c^2}$$

defines a fundamental solution $K_c \in \mathcal{S}'(\mathbb{R}^3)$ for $-\Delta + c^2$, which can be represented by

$$\langle K_c, \varphi \rangle := \int_0^\infty \int_{\mathbb{R}^3} (4\pi t)^{-3/2} \mathrm{e}^{-\frac{|x|^2}{4t} - tc^2} \varphi(x)\, \mathrm{d}x\, \mathrm{d}t, \qquad \varphi \in \mathcal{D}(\mathbb{R}^3).$$

b) Show using suitable changes of variables that for every $a > 0$

$$\int_0^\infty \mathrm{e}^{-(\frac{a^2}{t} + t)} t^{-3/2}\, \mathrm{d}t = \sqrt{\pi}\, \frac{\mathrm{e}^{-2a}}{a}. \tag{I}$$

c) Conclude that for all $c > 0$, the distribution K_c is explicitly given as the integrable function

$$K_c(x) = \frac{\mathrm{e}^{-c|x|}}{4\pi|x|}. \tag{II}$$

d) Show using analytic continuation that (II) defines a locally integrable fundamental solution for $-\Delta + c^2$ for every $c \in \mathbb{C}$.

7.3 (Fractional Derivatives) For $\operatorname{Re}\alpha > -1$, the distribution χ_+^α on $\mathbb{R}$ is defined as the tempered function

$$\chi_+^\alpha(x) := \begin{cases} \frac{x^\alpha}{\Gamma(\alpha+1)}, & \text{if } x > 0, \\ 0, & \text{if } x \le 0. \end{cases}$$

a) Prove that $\{\chi_+^\alpha\}_{\operatorname{Re}\alpha > -1}$ forms an analytic family of tempered distributions.
b) Show that for every $k \in \mathbb{N}$, in the sense of distributions,

$$\frac{\mathrm{d}^k}{\mathrm{d}x^k} \chi_+^{\alpha+k} = \chi_+^\alpha, \qquad \text{if } \operatorname{Re}\alpha > -1. \tag{III}$$

By *defining* χ_+^α for $\operatorname{Re}\alpha > -k - 1$ by

$$\chi_+^\alpha := \frac{\mathrm{d}^k}{\mathrm{d}x^k} \chi_+^{\alpha+k},$$

one obtains an analytic continuation of the above family to the domain $\operatorname{Re}\alpha > -k-1$. Since $k \in \mathbb{N}$ is arbitrary, this constructs an analytic family of tempered distributions $\{\chi_+^\alpha\}_{\alpha\in\mathbb{C}}$ on all of $\mathbb{C}$.

c) We then have that

$$\chi_+^{\alpha-1} * \chi_+^{\beta-1} = \chi_+^{\alpha+\beta-1} \quad \text{for } \operatorname{Re}\alpha, \operatorname{Re}\beta > 0. \tag{IV}$$

Conclude that the convolution product $\chi_+^{\alpha-1} * \chi_+^{\beta-1}$ can be defined for arbitrary $\alpha, \beta \in \mathbb{C}$ in such a way that formula (IV) remains valid.

d) For every $k \in \mathbb{N}$, we have

$$\chi_+^{-k-1} = \delta^{(k)} := \frac{\mathrm{d}^k}{\mathrm{d}x^k}\delta.$$

e) Let J_α be the operator

$$J_\alpha\varphi := \varphi * \chi_+^{-\alpha-1}, \quad \varphi \in \mathcal{S}(\mathbb{R}).$$

Then, in particular, $J_k\varphi = \varphi^{(k)}$, $k \in \mathbb{N}$, and

$$J_{-1}\varphi(x) = \int_{-\infty}^{x} \varphi(y)\,\mathrm{d}y.$$

In view of these results, J_α for $\alpha \geq 0$ is called the **fractional derivative** of order α, and for $\alpha < 0$, the **fractional integration** of order $|\alpha|$.

Chapter 8
On the Regularity Theory of Linear Partial Differential Equations: The Singular Support and Hypoellipticity

8.1 Some Fundamental Concepts and Properties

Definitions. Let

$$L = \sum_{|\alpha| \le m} a_\alpha(x) D^\alpha$$

be a linear partial differential operator with smooth coefficients $a_\alpha \in C^\infty(\Omega)$ in a non-empty open subset $\Omega \subset \mathbb{R}^n$.

(a) L is called **hypoelliptic** if, for every distribution $u \in \mathcal{D}'(\Omega)$ and every open subset V of Ω, the following holds: If $(Lu)|_V \in C^\infty(V)$, then $u|_V \in C^\infty(V)$.
(b) Let $u \in \mathcal{D}'(\Omega)$. The **singular support** $\operatorname{sing\,supp} u$ of u is the set of all points in Ω that do not have an open neighborhood in which the restriction of u is a C^∞-function.

Using arguments very similar to those in the proof of Corollary 5.8, one shows that $\operatorname{sing\,supp} u$ *is the complement of the largest open subset of* Ω *in which the restriction of* u *is a* C^∞*-function*.

Clearly, $\operatorname{sing\,supp} u$ is closed in Ω, and we have

$$\operatorname{sing\,supp} u \subset \operatorname{supp} u \tag{8.1}$$

Examples 8.1 (a) If K_m denotes the fundamental solution for Δ^m from Theorem 7.8, then $\operatorname{sing\,supp} K_m = \{0\}$, but $\operatorname{supp} K_m = \mathbb{R}^n$.
(b) $\operatorname{sing\,supp} (p.v.\frac{1}{x}) = \{0\}$.
(c) If σ_S denotes the surface measure of a smooth compact hypersurface S in $\mathbb{R}^n$, then $\operatorname{sing\,supp} \sigma_S = S = \operatorname{supp} \sigma_S$.

Lemma 8.2 *Let* $L = \sum_{|\alpha| \le m} a_\alpha(x) D^\alpha$ *be a linear partial differential operator with smooth coefficients in* Ω*. Then:*

(a) $\operatorname{sing\,supp} (Lu) \subset \operatorname{sing\,supp} u$ *for every* $u \in \mathcal{D}'(\Omega)$.
(b) L *is hypoelliptic if and only if*

$$\operatorname{sing\,supp} (Lu) = \operatorname{sing\,supp} u \quad \text{for every } u \in \mathcal{D}'(\Omega). \tag{8.2}$$

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Proof. (a) is obvious.

(b) If L is hypoelliptic, consider for $u \in \mathcal{D}'(\Omega)$ the set $V := \Omega \setminus \operatorname{sing\,supp}(Lu)$. Then $(Lu)|_V \in C^\infty(V)$, and consequently, $u|_V \in C^\infty(V)$, i.e., $V \subset \Omega \setminus \operatorname{sing\,supp} u$. Together with (a), this implies (8.2).

Conversely, if (8.2) holds and $V \subset \Omega$ is an open subset such that $(Lu)|_V \in C^\infty(V)$, then $V \subset \Omega \setminus \operatorname{sing\,supp}(Lu) = \Omega \setminus \operatorname{sing\,supp} u$. Hence, $u|_V \in C^\infty(V)$, and it follows that L is hypoelliptic. □

Lemma 8.3 *If L is hypoelliptic in the open subset Ω of $\mathbb{R}^n$, then L is also hypoelliptic in every open subset of Ω.*

Proof. Let Ω_1 be an open subset of Ω, and let $u \in \mathcal{D}'(\Omega_1)$ such that $(Lu)|_V$ is smooth on the open subset V of Ω_1. Let $x_0 \in V$. Choose a cutoff function $\chi \in \mathcal{D}(\Omega_1)$ that is identically one on a neighborhood U of x_0 in V, and set $\tilde{u} := \chi u$. Then $\tilde{u} \in \mathcal{E}'(\Omega_1) \subset \mathcal{E}'(\Omega)$, and $(L\tilde{u})|_U \in C^\infty(U)$. Since L is hypoelliptic in Ω, it follows that $\tilde{u}|_U \in C^\infty(U)$, and hence $u|_U \in C^\infty(U)$. Thus, u is smooth on V. □

We now return to non-trivial linear partial differential operators with constant coefficients $P(D)$.

Definition. A distribution $G \in \mathcal{D}'(\mathbb{R}^n)$ is called a **parametrix** for $P(D)$ if

$$P(D)G - \delta \in C^\infty(\mathbb{R}^n).$$

Note, in particular, that then

$$\operatorname{sing\,supp} P(D)G = \{0\}.$$

Clearly, every fundamental solution of $P(D)$ is a parametrix. However, parametrices are generally much easier to construct than fundamental solutions and, as the following theorem shows, often play an equally important role in questions concerning the regularity of solutions to partial differential equations.

Proposition 8.4 *The following statements are equivalent:*

(i) *$P(D)$ is hypoelliptic on $\mathbb{R}^n$.*
(ii) *There exists a parametrix G for $P(D)$ such that* $\operatorname{sing\,supp} G \subset \{0\}$.
(iii) *For every parametrix G for $P(D)$, we have* $\operatorname{sing\,supp} G \subset \{0\}$.

Proof. (i) $\Longrightarrow$ (iii): Assume $P(D)$ is hypoelliptic, and let G be a parametrix for $P(D)$. Then, by Lemma 8.2, $\operatorname{sing\,supp} G = \operatorname{sing\,supp} P(D)G = \{0\}$.

(iii) $\Longrightarrow$ (ii): Choose G as a fundamental solution K of $P(D)$. Such a solution exists by the Malgrange–Ehrenpreis theorem (Theorem 7.13), and by (iii), we have $\operatorname{sing\,supp} G \subset \{0\}$.

(ii) $\Longrightarrow$ (i): Let $G \in \mathcal{D}'$ be a parametrix for $P(D)$ with $\operatorname{sing\,supp} G \subset \{0\}$, and let $u \in \mathcal{D}'(\mathbb{R}^n)$. We define $f := P(D)u$, and aim to show:

If $V \subset \mathbb{R}^n$ is open and $f|_V \in C^\infty(V)$, then $u|_V \in C^\infty(V)$.

Fix $x_0 \in V$, and choose $\varepsilon > 0$ such that $\overline{B}_{8\varepsilon}(x_0) \subset V$. It suffices to show that

$$u|_{B_\varepsilon(x_0)} \in C^\infty(B_\varepsilon(x_0)). \tag{8.3}$$

Now choose $\chi \in \mathcal{D}(\mathbb{R}^n)$ such that $\chi \equiv 1$ in a neighborhood of $\overline{B}_1(0)$, $\operatorname{supp}\chi \subset B_2(0)$, and define

$$\tilde{u} := \chi\left(\frac{\cdot - x_0}{4\varepsilon}\right) u \in \mathcal{E}'(\mathbb{R}^n).$$

Then $\tilde{u}$ coincides with u in a neighborhood of $\overline{B}_{4\varepsilon}(x_0)$, so that for $\tilde{f} := P(D)\tilde{u} \in \mathcal{E}'$, we have:

$$\tilde{f}|_{B_{4\varepsilon}(x_0)} = f|_{B_{4\varepsilon}(x_0)} \in C^\infty(B_{4\varepsilon}(x_0)).$$

By considering $\tilde{u}$ in place of u, it therefore suffices to prove the following:

If $u \in \mathcal{E}'(\mathbb{R}^n)$ and $f := P(D)u$ is smooth on $B_{4\varepsilon}(x_0)$, then (8.3) holds.

To this end, we may assume that our parametrix G is supported in $\overline{B}_\varepsilon(0)$. To see this, choose $\psi \in \mathcal{D}$ such that $\psi \equiv 1$ in a neighborhood of 0 and $\operatorname{supp}\psi \subset \overline{B}_\varepsilon(0)$. Then, using Leibniz's rule, we obtain

$$P(D)(\psi G) = \psi\, P(D)G + g = \psi(\delta + h) + g = \delta + \psi h + g,$$

with C^∞-functions g and h. Note here that every derivative of strictly positive order of ψ vanishes in a neighborhood of 0, and that G is smooth outside $\{0\}$.

Thus, ψG is also a parametrix for $P(D)$, but with support in $\overline{B}_\varepsilon(0)$.

So we may assume $\operatorname{supp} G \subset \overline{B}_\varepsilon(0)$, and let $u \in \mathcal{E}'$ be such that $f := P(D)u$ is smooth on $B_{4\varepsilon}(x_0)$. Then we first obtain

$$\begin{aligned} f * G &= (P(D)u) * G = u * P(D)G \\ &= u * (\delta + h) = u + u * h, \end{aligned}$$

where $h \in \mathcal{D}$, so that $u * h \in \mathcal{D}$ as well. To prove (8.3), it thus suffices to show that

$$(f * G)|_{B_\varepsilon(x_0)} \in C^\infty(B_\varepsilon(x_0)). \tag{8.4}$$

To this end, we decompose f as $f = f_1 + f_2$, with

$$f_1(x) := \chi\left(\frac{x - x_0}{2\varepsilon}\right) f(x), \quad f_2(x) := (1 - \chi)\left(\frac{x - x_0}{2\varepsilon}\right) f(x).$$

Clearly, $f_1 \in \mathcal{D}$ and $f_2 \in \mathcal{E}'$, where

$$\operatorname{supp} f_2 \subset \{x \in \mathbb{R}^n : |x - x_0| \geq 2\varepsilon\}.$$

Thus, $f * G = f_1 * G + f_2 * G$, where $f_1 * G \in \mathcal{D}$, and

$$\operatorname{supp} f_2 * G \subset \operatorname{supp} f_2 + \operatorname{supp} G \subset \{x \in \mathbb{R}^n : |x - x_0| \geq \varepsilon\}.$$

It follows that

$$(f * G)|_{B_\varepsilon(x_0)} = (f_1 * G)|_{B_\varepsilon(x_0)} \in C^\infty(B_\varepsilon(x_0)). \qquad \square$$

Corollary 8.5 (Weyl's Lemma) *The Laplace operator Δ is hypoelliptic. In particular, every harmonic distribution $u \in \mathcal{D}'(\Omega)$ is a smooth function, i.e., if $\Delta u = 0$ in Ω, then $u \in C^\infty(\Omega)$.*

Proof. The Newtonian potential $N = N_n$ on $\mathbb{R}^n$ is, as we have seen in Section 7.3, a fundamental solution for the Laplace operator, and clearly, sing supp $N = \{0\}$ (see also Example 8.1 (a)).

Thus, by Proposition 8.4, Δ is hypoelliptic on all of $\mathbb{R}^n$, and in particular, in every open subset Ω. $\square$

8.2 Elliptic Linear Differential Operators with Constant Coefficients

The above example can be significantly generalized.

Definition. Let $P(D) = \sum_{|\alpha| \le m} a_\alpha D^\alpha$ be a linear partial differential operator of order m with constant coefficients, and let

$$P_m(\xi) := \sum_{|\alpha| = m} a_\alpha \xi^\alpha, \qquad \xi \in \mathbb{R}^n,$$

be its principal symbol. $P(D)$ is called **elliptic** if

$$P_m(\xi) \neq 0 \quad \text{for all } \xi \in \mathbb{R}^n \setminus \{0\}. \tag{8.5}$$

Theorem 8.6 *Let $P(D)$ be an elliptic differential operator with constant coefficients. Then $P(D)$ is hypoelliptic.*

Proof. Since P_m is continuous, it follows from (8.5) that

$$a := \inf_{|\xi|=1} |P_m(\xi)| > 0.$$

Thus, for any $\xi \in \mathbb{R}^n \setminus \{0\}$, we obtain

$$|P_m(\xi)| = |\xi|^m \left| P_m\left(\frac{\xi}{|\xi|}\right) \right| \ge a|\xi|^m. \tag{8.6}$$

Furthermore, since $P - P_m$ is a polynomial of degree $\le m - 1$, there exists a constant $A \ge 0$ such that

$$|P(\xi) - P_m(\xi)| \le A|\xi|^{m-1}, \quad \text{for } |\xi| \ge 1.$$

From these two inequalities, it follows easily that there exists an $R \geq 1$ such that

$$|P(\xi)| \geq \frac{a}{2}|\xi|^m \quad \text{for all } \xi \text{ with } |\xi| \geq R. \tag{8.7}$$

Now let $\chi \in \mathcal{D}(\mathbb{R}^n)$ be such that $\chi \equiv 1$ on a neighborhood of $\overline{B}_R(0)$, and $\operatorname{supp}\chi \subset B_{2R}(0)$. Then, by (8.7), the function

$$g(\xi) := \begin{cases} (1-\chi(\xi))/P(\xi), & \text{if } |\xi| \geq R, \\ 0, & \text{otherwise}, \end{cases}$$

defines a bounded C^∞ function g on $\mathbb{R}^n$. In particular, $g \in \mathcal{S}'$. Define

$$G := \mathcal{F}^{-1}g \in \mathcal{S}'.$$

Then G is a parametrix for $P(D)$, since

$$(P(D)G)^\wedge = Pg = 1 - \chi,$$

i.e.,

$$P(D)G = \delta - \mathcal{F}^{-1}\chi,$$

where $\mathcal{F}^{-1}\chi \in \mathcal{S}$. By Proposition 8.4, it remains to show that

$$\operatorname{sing\,supp} G \subset \{0\}. \tag{8.8}$$

To this end, we first observe that for all $\alpha \in \mathbb{N}^n$, we have (on the set where $P(\xi) \neq 0$)

$$D^\alpha\left(\frac{1}{P}\right) = \frac{q_\alpha}{P^{|\alpha|+1}},$$

where q_α is a polynomial of degree $\leq |\alpha|(m-1)$. This follows immediately by induction on the order $|\alpha|$ of the derivative. From (8.7), we then obtain the estimate

$$\left|D^\alpha\left(\frac{1}{P}\right)(\xi)\right| \leq C_\alpha \frac{|\xi|^{|\alpha|(m-1)}}{|\xi|^{m(|\alpha|+1)}} = C_\alpha|\xi|^{-m-|\alpha|}, \quad \text{for } |\xi| \geq R,$$

and thus, for every $\alpha \in \mathbb{N}^n$,

$$D^\alpha g(\xi) = O(|\xi|^{-m-|\alpha|}), \qquad \xi \in \mathbb{R}^n. \tag{8.9}$$

In particular, for $|\alpha| \geq -m + n + 1$, we clearly have $D^\alpha g \in L^1(\mathbb{R}^n)$, and for such α, the inverse Fourier transform of $D^\alpha g$ is given pointwise by

$$\mathcal{F}^{-1}(D^\alpha g)(x) = (2\pi)^{-n}\int D^\alpha g(\xi)e^{ix\cdot\xi}\,d\xi.$$

More generally, if $k \in \mathbb{N}$ and $|\alpha| \geq -m+n+1+k$, then the theorem on differentiation of parameter-dependent integrals, together with (8.9) (see Proposition 2.18), shows that $\mathcal{F}^{-1}(D^\alpha g) \in C^k(\mathbb{R}^n)$, and that

$$D^\beta\big(\mathcal{F}^{-1}(D^\alpha g)\big)(x) = (2\pi)^{-n} \int \xi^\beta D^\alpha g(\xi) \mathrm{e}^{ix\cdot\xi}\, \mathrm{d}\xi, \qquad |\beta| \leq k.$$

Furthermore, in the sense of distributions,

$$\mathcal{F}^{-1}(D^\alpha g) = (-1)^{|\alpha|} x^\alpha G,$$

so that

$$x^\alpha G \in C^k(\mathbb{R}^n), \quad \text{if } |\alpha| \geq -m+n+1+k.$$

In particular, for $2\ell \geq -m+n+1+k$, we have $|\cdot|^{2\ell} G \in C^k(\mathbb{R}^n)$. Since $|\cdot|^{-2\ell}$ is smooth in $\mathbb{R}^n \setminus \{0\}$, we conclude that $G|_{\mathbb{R}^n\setminus\{0\}} \in C^k(\mathbb{R}^n \setminus \{0\})$ for every $k \in \mathbb{N}$. Thus, $G|_{\mathbb{R}^n\setminus\{0\}}$ is smooth, i.e., (8.8) holds. □

Remarks 8.7 Theorem 8.5 has numerous generalizations. For example, using the calculus of pseudodifferential operators, one can construct generalized parametrices even for "elliptic" linear differential operators with variable coefficients and thereby establish hypoellipticity. The fundamental idea remains the same as in the proof of Theorem 8.5 (see, e.g., Folland [12]). In the case of linear differential operators with constant coefficients, Hörmander has also provided a complete characterization of hypoellipticity based on properties of the associated symbol (see Hörmander [22], as well as [24] for a somewhat more modern approach).

Examples 8.8 (a) The **heat operator** $\frac{\partial}{\partial t} - \Delta_x$ on $\mathbb{R} \times \mathbb{R}^n$ is the prototype of a "*parabolic*" differential operator. In Theorem 7.5, we constructed a fundamental solution W for this operator, where, according to (7.8), the singular support of W is given by $\operatorname{sing\,supp} E = \{(0,0)\}$. Hence, by Theorem 8.4, $\frac{\partial}{\partial t} - \Delta_x$ is also hypoelliptic.

(b) In contrast, the **d'Alembertian** or **wave operator** $\Box := \frac{\partial^2}{\partial t^2} - \Delta_x$ on $\mathbb{R} \times \mathbb{R}^n$, which serves as the prototype of a "*hyperbolic*" differential operator, is not hypoelliptic: For instance, if w is an arbitrary function in $L^1_{\mathrm{loc}}(\mathbb{R})$, and if we set

$$u(t, x_1, \ldots, x_n) := w(x_1 - t),$$

then it is easy to see that $\Box u = 0$ (for C^2 functions w, this is immediately clear; in the general case, one can use a Friedrichs mollification of w). However, this solution u will generally not be smooth.

In the theory of hyperbolic differential equations, questions concerning the *propagation of singularities* play an important role. If, in the above example, the function w has a singularity at the origin 0, then the solution is singular at time $t = 0$ along the hyperplane $x_1 = 0$, and this singularity propagates over time t so that at time t, the solution is singular along the parallel affine hyperplane $\{x_1 = t\}$. In studying such questions, the concept of the wave front set of a distribution has proven to be particularly significant, which we will briefly discuss in the next section.

8.3 On the Spectral Analysis of Singularities: The Wavefront Set*

A more precise description of the singularities of a distribution is enabled by the concept of the "wavefront set". This not only allows for the localization of singularities but also provides a certain spectral localization of the same.

In particular, in the theory of "Lagrangian distributions" and associated "Fourier integral operators" established by Hörmander (see, e.g., Hörmander [21], Duistermaat [7], and Sogge [38]), the concept of the wavefront set plays a significant role. We will not delve into these deeper theories here and will instead limit ourselves to briefly introducing this concept and discussing a few examples.

As a preliminary observation, note that a distribution $u \in \mathcal{D}'(\Omega)$ (where again $\emptyset \neq \Omega \subset \mathbb{R}^n$ is open) is smooth near a point $y \in \Omega$ (in the sense that it can be represented by a C^∞-function on a neighborhood of y) if and only if there exists an open neighborhood U of y in Ω such that following holds:

$$\chi u \in C^\infty(\mathbb{R}^n) \qquad \text{for every } \chi \in C_0^\infty(U).$$

Since χu has compact support, this is again equivalent to the validity of estimates of the form

$$|\widehat{\chi u}(\xi)| \leq C_N (1 + |\xi|)^{-N} \qquad \text{for all } N \in \mathbb{N}, \xi \in \mathbb{R}^n. \tag{8.10}$$

Definition. Let $u \in \mathcal{D}'(\Omega)$. The **wavefront set** $WF(u) \subset \Omega \times (\mathbb{R}^n \setminus \{0\})$ of u is defined as follows:

A point $(y, \eta) \in \Omega \times (\mathbb{R}^n \setminus \{0\})$ does *not* belong to $WF(u)$ if and only if there exists a neighborhood U of y in Ω and a conic neighborhood Γ of η in $\mathbb{R}^n \setminus \{0\}$ such that for all functions $\chi \in C_0^\infty(U)$, the following estimates hold:

$$|\widehat{\chi u}(\xi)| \leq C_N (1 + |\xi|)^{-N} \qquad \text{for all } N \in \mathbb{N}, \xi \in \Gamma. \tag{8.11}$$

A subset $\Gamma \subset \mathbb{R}^n \setminus \{0\}$ is called **conic** if

$$r\xi \in \Gamma \qquad \text{for all } r > 0, \xi \in \Gamma.$$

It is clear that the complement of $WF(u)$ in $\Omega \times (\mathbb{R}^n \setminus \{0\})$ is open, so that $WF(u)$ is closed in $\Omega \times (\mathbb{R}^n \setminus \{0\})$.

From the definition, together with (8.10), it follows directly that

$$\operatorname{singsupp} u = \{x \in \Omega : \text{there exists } \xi \in \mathbb{R}^n \setminus \{0\} \text{ with } (x, \xi) \in WF(u)\} \tag{8.12}$$

(just look at the complements of the sets above), i.e., the singular support of u is precisely the image of the wavefront set $WF(u)$ under the canonical projection

$$\pi : \Omega \times (\mathbb{R}^n \setminus \{0\}) \to \Omega, \ (x, \xi) \mapsto x$$

onto the "spatial domain" Ω.

In fact, $\Omega \times \mathbb{R}^n$ can be naturally identified with the cotangent bundle of Ω, since wavefront sets transform like subsets of the cotangent bundle under coordinate transformations. This allows us to define the wavefront sets even for distributions on any smooth manifold (see, e.g., Hörmander [23]).

Examples 8.9 For a better understanding of the concept of wavefront set, let us consider two examples:

(a) For the Newtonian potential N_n on $\mathbb{R}^n$ from Section 7.3 we have:

$$WF(N_n) = \{0\} \times (\mathbb{R}^n \setminus \{0\}).$$

Proof. Using (8.12), we see that $WF(N_n) \subset \{0\} \times (\mathbb{R}^n \setminus \{0\})$, since $\operatorname{sing\,supp} N_n = \{0\}$.

Conversely, let $(0, \eta) \in \{0\} \times (\mathbb{R}^n \setminus \{0\})$, and assume that $(0, \eta) \notin WF(N_n)$. Then we can choose a radial function $\chi \in C_0^\infty(\mathbb{R}^n)$ with $\chi(0) \neq 0$ such that in a conical neighborhood Γ of η,

$$|\widehat{\chi N_n}(\xi)| \leq C_M (1 + |\xi|)^{-M} \qquad \text{for all } M \in \mathbb{N}, \xi \in \Gamma.$$

Since χN_n is a radial function, $\widehat{\chi N_n}$ is also radial, meaning that the above estimates hold not only on Γ but even on the whole of $\mathbb{R}^n$. Thus, as we have seen, χN_n would have to be smooth, which leads to a contradiction. Consequently, $(0, \eta)$ must lie in $WF(N_n)$. □

(b) Let $E \subset \mathbb{R}^n$ be the k-dimensional linear subspace $E := \mathbb{R}^k \times \{0\}$, where we assume $0 \leq k < n$. Furthermore, let $\mathrm{d}\sigma := \mathrm{d}x_1 \cdots \mathrm{d}x_k \otimes \delta_0$ be the Riemannian surface measure of E, and let $\psi \in C_0^\infty(E)$. Then for the distribution $u := \psi\sigma$ supported in E, we have:

$$WF(u) = \operatorname{supp} u \times (E^\perp \setminus \{0\}) \subset E \times (E^\perp \setminus \{0\}), \tag{8.13}$$

where $E^\perp = \{(x', x'') \in \mathbb{R}^n = \mathbb{R}^k \times \mathbb{R}^{n-k} : x' = 0\}$ is the orthogonal complement of E with respect to the canonical Euclidean inner product on $\mathbb{R}^n$.

Proof. Since $k < n$, it is clear that $\operatorname{sing\,supp} u = \operatorname{supp} u$. With $x = (x', x'')$, where $x' = (x_1, \ldots, x_k) \in \mathbb{R}^k$ and $x'' = (x_{k+1}, \ldots, x_n) \in \mathbb{R}^{n-k}$, we also have for $\chi \in C_0^\infty(\mathbb{R}^n)$,

$$\widehat{\chi u}(\xi) = \int_{\mathbb{R}^k} \mathrm{e}^{-i\xi' \cdot x'} \chi(x', 0) \psi(x')\, \mathrm{d}x', \tag{8.14}$$

where we similarly decompose $\xi \in \mathbb{R}^n$ as $\xi = (\xi', \xi'')$. Hence, $\widehat{\chi u}(\xi) = \widehat{\chi u}(\xi', \xi'')$ decays rapidly in ξ' but is independent of ξ''.

Now let $(y, \eta) \in \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\})$. If $\eta' \neq 0$, from (8.14) we quickly see that $(y, \eta) \notin WF(u)$. Together with (8.12), it follows that

$$WF(u) \subset \operatorname{supp} u \times (E^\perp \setminus \{0\}).$$

To show the reverse inclusion, let $(y,\eta) \in \operatorname{supp} u \times (E^\perp \setminus \{0\})$, i.e., $y \in \operatorname{supp} u$ and $\eta = (0,\eta'')$ with $\eta'' \neq 0$. First, we make the following observation:

If $\chi(y,0) \neq 0$, the amplitude in (8.14) does not vanish identically, and hence there exists a $\zeta' \neq 0$ such that

$$\widehat{\chi u}(\zeta',\xi'') = \widehat{\chi u}(\zeta',0) \neq 0 \quad \text{for all } \xi'' \in \mathbb{R}^{n-k}.$$

This implies that there cannot exist a conic neighborhood Γ of $(0,\eta'')$ where $\widehat{\chi u}$ decays rapidly, and consequently, $(y,\eta) \in WF(u)$. □

8.4 Exercises

8.1 a) Show that for all functions $f, g \in L^1_{\mathrm{loc}}(\mathbb{R})$, the expressions

$$u_+(t,x) := f(x+ct) \quad \text{and} \quad u_-(t,x) := g(x-ct), \qquad (t,x) \in \mathbb{R}^2,$$

define solutions (in the sense of distributions) of the one-dimensional homogeneous wave equation

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$$

with a given propagation speed $c > 0$. In particular, this shows that there exist classical solutions (i.e., solutions in $C^2(\mathbb{R}^2)$) that are not smooth.

b) Every classical solution $u \in C^2(\mathbb{R}^2)$ can be written as a superposition $u = u_+ + u_-$, with C^2-functions f and g.

8.2 Let $u \in \mathcal{S}'(\mathbb{R}^n)$ be a homogeneous distribution (cf. Exercise 6.5) that is smooth outside the origin. Then

$$WF(u) = \{(0,\xi) : \xi \in \operatorname{supp} \hat{u} \setminus \{0\}\}.$$

Hints: If $\chi \in C_0^\infty(\mathbb{R}^n)$ with $\chi(0) = 1$, let $\phi := \hat{\chi}$. Since, by Exercise 6.5 (i), $\hat{u}$ is also homogeneous, there exists a $\beta \in \mathbb{C}$ such that

$$\hat{u} * \phi_{1/R}(\xi) = R^\beta \widehat{\chi u}(R\xi) \qquad \text{for all } R > 0, \xi \in \mathbb{R}^n,$$

where $\{(2\pi)^{-n}\phi_{1/R}\}_R$ forms a Dirac family as $R \to \infty$. Consider in particular $R \gg 1$ and $|\xi| \sim 1$.

Chapter 9
The Schwartz Kernel Theorem, and the Tensor Product of Distributions

The Schwartz Kernel Theorem provides a very elegant and useful characterization of continuous linear operators

$$T : \mathcal{S}(\mathbb{R}^m) \to \mathcal{S}'(\mathbb{R}^n) \quad (\text{respectively } T : \mathcal{D}(\Omega_1) \to \mathcal{D}'(\Omega_2)) :$$

all these operators are, in a generalized sense, "integral operators", whose integral kernels, however, are suitable distributions on the product space $\mathbb{R}^n \times \mathbb{R}^m$ (respectively $\Omega_2 \times \Omega_1$).

By utilizing the results from Chapter 2.8 and Proposition 9.3 from the next section, we will present here an alternative and conceptually more elementary approach to this result than is usually found in the literature. The key idea, which has become a kind of mathematical "folklore", will be to expand Schwartz functions into Hermite series. It has been used for instance (possibly for the first time) by R. Howe in [25, Cor. 1.7.2].

9.1 A Characterization of Schwartz Functions via Fourier–Hermite Coefficients

Let $\mathcal{S} := \mathcal{S}(\mathbb{R}^n)$. In generalizing the operators Z and Z^* from Chapter 2.8, we set

$$Z_j := X_j + iD_j = x_j + \frac{\mathrm{d}}{\mathrm{d}x_j}, \qquad Z_j^* := X_j - iD_j = x_j - \frac{\mathrm{d}}{\mathrm{d}x_j}, \qquad j = 1, \dots, n.$$

For these, the following commutator relations are evident:

$$[Z_j, Z_k^*] = 2\delta_{jk} I. \tag{9.1}$$

Furthermore, one easily computes that

$$\sum_{k=1}^{n} Z_k^* Z_k = \mathcal{H}_n - nI, \quad \sum_{k=1}^{n} Z_k Z_k^* = \mathcal{H}_n + nI, \tag{9.2}$$

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where

$$\mathcal{H}_n := \sum_{j=1}^{n} \left(-\frac{\partial^2}{\partial x_j^2} + x_j^2 \right) = -\Delta + |x|^2$$

again denotes the Hermite operator from Chapter 2.8. From (9.1) and (9.2), we immediately deduce that

$$[Z_j, \mathcal{H}_n] = 2Z_j, \qquad [\mathcal{H}_n, Z_j^*] = 2Z_j^*, \qquad j = 1, \dots, n. \tag{9.3}$$

We set

$$\mathcal{A} := \{Z_1, \dots, Z_n, Z_1^*, \dots, Z_n^*\}.$$

Let $T_1, \dots, T_m \in \mathcal{A}$. Then, as a generalization of the last two identities in (2.37), we have

$$\begin{aligned} [\mathcal{H}_n, T_1 \cdots T_m] &= \sum_{j=1}^{m} T_1 \cdots T_{j-1} [\mathcal{H}_n, T_j] T_{j+1} \cdots T_m \\ &= \sum_{j=1}^{m} 2\epsilon(T_j) T_1 \cdots T_{j-1} T_j T_{j+1} \cdots T_m, \end{aligned}$$

where

$$\epsilon(T_i) := \begin{cases} -1, & \text{if } T_j = Z_j, \\ 1, & \text{if } T_j = Z_j^*. \end{cases}$$

The first identity follows by induction on the number m of factors from the product rule (2.32), and the second from (9.3). Consequently,

$$[\mathcal{H}_n, T_1 \cdots T_m] = aT_1 \cdots T_m, \qquad \text{with } a \in \{-2m, \dots, 2m\}. \tag{9.4}$$

We also need the following:

Lemma 9.1 *For every $\varphi \in \mathcal{S}$, we have*

$$\|\varphi\|_2 \le \|\mathcal{H}_n \varphi\|_2.$$

Moreover, for every $k = 1, \dots, n$,

$$\|Z_k \varphi\|_2 \le \|\mathcal{H}_n \varphi\|_2, \quad \|Z_k^* \varphi\|_2 \le (n+1)^{\frac{1}{2}} \|\mathcal{H}_n \varphi\|_2.$$

Proof. Let $\check{\varphi}(\mathbf{k})$ be the Fourier–Hermite coefficient of order $\mathbf{k} \in \mathbb{N}^n$ of φ, and let $\psi := \mathcal{H}_n \varphi$. From the formula (2.42) for the eigenvalues of $\mathcal{H}_n$, it follows that

$$\check{\varphi}(\mathbf{k}) = (\varphi, h_{\mathbf{k}}) = (\varphi, \tfrac{\mathcal{H}_n h_{\mathbf{k}}}{n+2|\mathbf{k}|}) = \frac{1}{n+2|\mathbf{k}|} (\varphi, \mathcal{H}_n h_{\mathbf{k}}) = \frac{1}{n+2|\mathbf{k}|} \check{\psi}(\mathbf{k}). \tag{9.5}$$

In particular, $|\check{\varphi}(\mathbf{k})| \le |\check{\psi}(\mathbf{k})|$. The first inequality $\|\varphi\|_2 \le \|\mathcal{H}_n \varphi\|_2$ now follows immediately from the Plancherel–Hermite identity (2.30).

The third inequality follows from (9.2):

$$\begin{aligned}\|Z_k^*\varphi\|_2^2 &\le \sum_{j=1}^n \|Z_j^*\varphi\|_2^2 = (\sum_{j=1}^n Z_j Z_j^*\varphi, \varphi) = ((\mathcal{H}_n + nI)\varphi, \varphi)\\ &\le \|\mathcal{H}_n\varphi\|_2\|\varphi\|_2 + n\|\varphi\|_2^2 \le (n+1)\|\mathcal{H}_n\varphi\|_2^2,\end{aligned}$$

where we used the Cauchy–Schwarz inequality and, in the last step, the previously proven first inequality.

The proof of the second inequality is similar and slightly simpler. □

Lemma 9.2 *For every $m \in \mathbb{N}$, there exists a constant $C_m > 0$ such that for all $\varphi \in \mathcal{S}$ and $T_1, \dots, T_m \in \mathcal{A}$,*

$$\|T_m \cdots T_1\varphi\|_2 \le C_m\|\mathcal{H}_n^m\varphi\|_2.$$

Proof. The proof is by induction on m, starting with the trivial case $m = 0$.

Assume that the above inequality holds for some $m \in \mathbb{N}$, and let $T_1, \dots, T_{m+1} \in \mathcal{A}$. Set $\psi := T_m \cdots T_1\varphi$. Then, by Lemma 9.1,

$$\|T_{m+1} \cdots T_1\varphi\|_2^2 = \|T_{m+1}\psi\|_2^2 \le (n+1)\|\mathcal{H}_n\psi\|_2^2.$$

Furthermore, by (9.4),

$$\mathcal{H}_n\psi = \mathcal{H}_n T_m \cdots T_1\varphi = T_m \cdots T_1(\mathcal{H}_n\varphi) + aT_m \cdots T_1\varphi,$$

so that

$$\begin{aligned}\|\mathcal{H}_n\psi\|_2 &\le \|T_m \cdots T_1(\mathcal{H}_n\varphi)\|_2 + |a|\|T_m \cdots T_1\varphi\|_2\\ &\le C_m\|\mathcal{H}_n^m(\mathcal{H}_n\varphi)\|_2 + C_m 2m\|\mathcal{H}_n^m\varphi\|_2\\ &\le C_m(1+2m)\|\mathcal{H}_n^{m+1}\varphi\|_2.\end{aligned}$$

Here, we used the induction hypothesis in the second-to-last inequality and Lemma 9.1 in the last one.

Thus, we conclude that

$$\|T_{m+1} \cdots T_1\varphi\|_2 \le C_{m+1}\|\mathcal{H}_n^{m+1}\varphi\|_2,$$

with $C_{m+1} := (n+1)^{1/2}C_m(1+2m)$. □

We now denote by $\ell^2(\mathbb{N}^n)$ the Hilbert space of all square-summable mappings $c : \mathbb{N}^n \to \mathbb{C}$, endowed with the norm

$$\|c\|_2 := \big(\sum_{\mathbf{k}\in\mathbb{N}^n} |c(\mathbf{k})|^2\big)^{1/2},$$

and by $\mathcal{F}_H$ the isometric linear mapping

$$\mathcal{F}_H : L^2(\mathbb{R}^n) \to \ell^2(\mathbb{N}^n), \quad f \mapsto \check{f},$$

where again $\check{f}(\mathbf{k}) = (f, h_{\mathbf{k}})$ denotes the Fourier–Hermite coefficient of order $\mathbf{k}$ of f. We shall call the mapping $\mathcal{F}_H$ the **Fourier–Hermite transform** (cf. (2.30)).

Its restriction to the subspace $\mathcal{S} = \mathcal{S}(\mathbb{R}^n)$ is also denoted by $\mathcal{F}_H$ for simplicity. The inverse mapping $\mathcal{F}_H^{-1}$ is evidently given by

$$\mathcal{F}_H^{-1}(c) = \sum_{\mathbf{k}\in\mathbb{N}^n} c(\mathbf{k}) h_{\mathbf{k}}, \qquad c \in \ell^2(\mathbb{N}^n).$$

A function $c : \mathbb{N}^n \to \mathbb{C}$ is called **rapidly decaying** if, for every $N \in \mathbb{N}$, there exists a constant C_N such that

$$|c(\mathbf{k})| \le C_N (1 + |\mathbf{k}|)^{-N} \quad \text{for every} \quad \mathbf{k} \in \mathbb{N}^n.$$

The set of all such functions clearly forms a linear subspace $\mathfrak{s} = \mathfrak{s}(\mathbb{N}^n)$ of $\ell^2(\mathbb{N}^n)$, which we endow, in analogy with the topology on $\mathcal{S}$, with the locally convex topology induced by the following family of norms:

$$\|c\|_{(N)} := \sup_{\mathbf{k}\in\mathbb{N}^n} (1 + |\mathbf{k}|)^N |c(\mathbf{k})|, \qquad N \in \mathbb{N}.$$

Similar to $\mathcal{S}$, one can show that $\mathfrak{s}$ is a Fréchet space (the proof is even somewhat simpler).

In view of Lemma 9.2, however, it is more convenient here to work on $\mathcal{S}$ with the L^2-based Schwartz norms

$$\|\varphi\|_{(2,N)} := \max_{|\alpha|,|\beta|\le N} \|x^\alpha D^\beta \varphi\|_2, \qquad \varphi \in \mathcal{S}(\mathbb{R}^n),$$

from Exercise 4.12, rather than with the family of norms $\{\|\cdot\|_{(N)}\}_N$.

According to Exercise 4.12, these two families of seminorms on $\mathcal{S}$ are equivalent.

For the sake of completeness, let me prove this directly here. In the case $p = 2$, one can even argue more quickly using Plancherel's theorem:

Let $m \in \mathbb{N}$ be even with $m > n/2$. We show that for all $\varphi \in \mathcal{S}$,

$$\|\varphi\|_{(2,N)} \le C_n \|\varphi\|_{(N+m)}, \tag{9.6}$$

$$\|\varphi\|_{(N)} \le C_n \|\varphi\|_{(2,N+m)}. \tag{9.7}$$

From

$$|x^\alpha D^\beta \varphi(x)| = |\langle x\rangle^m x^\alpha D^\beta \varphi(x)|\, \langle x\rangle^{-m}$$

it follows immediately that

$$\|x^\alpha D^\beta \varphi\|_2 \le C \|\langle x\rangle^m x^\alpha D^\beta \varphi\|_\infty,$$

which yields (9.6). Conversely, by the Fourier inversion formula and Hölder's inequality,

$$|x^\alpha D^\beta \varphi(x)| \le C \|\widehat{x^\alpha D^\beta \varphi}\|_1 \le C' \|\langle \xi\rangle^m \widehat{x^\alpha D^\beta \varphi}\|_2.$$

Applying Plancherel's theorem, we obtain

$$\|x^\alpha D^\beta \varphi\|_\infty \le C\|(1-\Delta)^{m/2} x^\alpha D^\beta \varphi\|_2 \le C_n \|\varphi\|_{(2,N+m)},$$

which gives (9.7) as well.

Proposition 9.3 *If $\varphi \in \mathcal{S} = \mathcal{S}(\mathbb{R}^n)$ is a Schwartz function, then its Fourier–Hermite transform $\check{\varphi}$ is rapidly decaying, i.e., $\check{\varphi} \in \mathfrak{s} = \mathfrak{s}(\mathbb{N}^n)$. In fact, the linear mapping defined by*

$$\mathcal{F}_H : \mathcal{S} \to \mathfrak{s}, \quad f \mapsto \check{f},$$

is an isomorphism of the Fréchet space $\mathcal{S}$ onto the Fréchet space $\mathfrak{s}$. In particular, every Schwartz function $\varphi \in \mathcal{S}$ can be expanded into the series

$$\varphi = \sum_{\mathbf{k}\in\mathbb{N}^n} \check{\varphi}(\mathbf{k}) h_{\mathbf{k}}$$

converging in $\mathcal{S}$.

Proof. Let $\varphi \in \mathcal{S}$. Iterating the argument in (9.5), we obtain

$$\check{\varphi}(\mathbf{k}) = (\varphi, h_{\mathbf{k}}) = (n+2|\mathbf{k}|)^{-N} (\varphi, \mathcal{H}_n^N h_{\mathbf{k}}) = (n+2|\mathbf{k}|)^{-N} (\mathcal{H}_n^N \varphi, h_{\mathbf{k}}),$$

so that

$$\begin{aligned} |\check{\varphi}(\mathbf{k})| &\le (n+2|\mathbf{k}|)^{-N} \|\mathcal{H}_n^N \varphi\|_2 \\ &\le C_N (n+2|\mathbf{k}|)^{-N} \|\varphi\|_{(2,2N)}. \end{aligned}$$

Thus, $\check{\varphi}$ is rapidly decaying, and we also see that

$$\|\check{\varphi}\|_{(N)} \le C'_N \|\varphi\|_{(2,2N)},$$

with constants C'_N depending only on N (and n). This shows that $\mathcal{F}_H$ continuously maps $\mathcal{S}$ into $\mathfrak{s}$.

Next, we observe the following: since $X_j = (Z_j + Z_j^*)/2$ and $D_j = (Z_j - Z_j^*)/2i$, we immediately obtain from Lemma 9.2 and Lemma 9.1 the estimates

$$\|\varphi\|_{(2,N)} \le C'_N \|\mathcal{H}_n^{2N} \varphi\|_2, \qquad N \in \mathbb{N}.$$

In particular, this implies together with (2.42) that

$$\|h_{\mathbf{k}}\|_{(2,N)} \le C'_N (n+2|\mathbf{k}|)^{2N}. \tag{9.8}$$

Now, if $c \in \mathfrak{s}$, it follows that

$$\sum_{\mathbf{k}\in\mathbb{N}^n} \|c(\mathbf{k}) h_{\mathbf{k}}\|_{(2,N)} \le C'_N \sum_{\mathbf{k}\in\mathbb{N}^n} (n+2|\mathbf{k}|)^{2N} |c(\mathbf{k})| \le C''_N \|c\|_{(2N+n+1)} < \infty.$$

The series $\sum_{\mathbf{k}\in\mathbb{N}^n} c(\mathbf{k})h_{\mathbf{k}}$ thus converges absolutely with respect to each of the seminorms $\|\cdot\|_{(2,N)}$, and hence also in $\mathcal{S}$, since $\mathcal{S}$ is complete. Since it also converges in L^2 to $\mathcal{F}_H^{-1}c$, we see that $\mathcal{F}_H^{-1}c$ lies in $\mathcal{S}$ and that

$$\|\mathcal{F}_H^{-1}c\|_{(2,N)} \le C_N''\|c\|_{(2N+n+1)}.$$

Thus, $\mathcal{F}_H^{-1} : \mathfrak{s} \to \mathcal{S}$ is also continuous. □

9.2 The Kernel Theorem for Tempered Distributions

We shall use the abbreviations $X := \mathbb{R}^n, Y := \mathbb{R}^m$ throughout this section.

To begin with, let us recall the following result on integral operators, also called *Schur's test* (see Exercise 2.7):

Assume that $\mathcal{K} : X \times Y \to \mathbb{C}$ is a measurable function (or "kernel"), and that there exists a constant $C \ge 0$ such that

$$\sup_{x\in X}\int_Y |\mathcal{K}(x,y)|\,\mathrm{d}y \le C \quad \text{and} \quad \sup_{y\in Y}\int_X |\mathcal{K}(x,y)|\,\mathrm{d}x \le C. \tag{9.9}$$

Then, for $f \in L^p(Y)$, the integral

$$T_{\mathcal{K}}f(x) := \int_Y \mathcal{K}(x,y)f(y)\,\mathrm{d}y$$

converges for almost every $x \in X$, and the mapping $f \mapsto T_{\mathcal{K}}f$ defines a bounded linear **kernel operator**

$$T_{\mathcal{K}} : L^p(Y) \to L^p(X),$$

with integral kernel $\mathcal{K}$.

A significantly larger class of operators is given by the space of all continuous linear operators

$$T : \mathcal{S}(Y) \to \mathcal{S}'(X).$$

Note that this continuity condition is a rather weak assumption.

The kernel theorem of L. Schwartz states that even such operators can be represented, in a generalized sense, by "kernels" $\mathcal{K}$, which, however, will generally only be tempered distributions $\mathcal{K} \in \mathcal{S}'(X \times Y)$.

Indeed, if $\mathcal{K} \in \mathcal{S}'(X \times Y)$, we can define a linear operator $T_{\mathcal{K}} : \mathcal{S}(Y) \to \mathcal{S}'(X)$ by setting

$$\langle T_{\mathcal{K}}(\varphi), \psi\rangle := \langle \mathcal{K}, \psi \otimes \varphi\rangle \qquad \text{for all } \varphi \in \mathcal{S}(Y), \psi \in \mathcal{S}(X). \tag{9.10}$$

Here, as usual, $\psi \otimes \varphi$ denotes the Schwartz function on $X \times Y$ defined by

$$\psi \otimes \varphi(x,y) := \psi(x)\varphi(y), \qquad (x,y) \in X \times Y.$$

This operator $T_{\mathcal{K}}$ is clearly continuous from $\mathcal{S}(Y)$ to $\mathcal{S}'(X)$, since $\mathcal{S}'(X)$ is endowed with the weak*-topology with respect to $\mathcal{S}(X)$. The converse also holds:

Theorem 9.4 *Let $T : \mathcal{S}(Y) \to \mathcal{S}'(X)$ be a continuous linear operator. Then there exists a unique tempered distribution $\mathcal{K}_T \in \mathcal{S}'(X \times Y)$ such that T is represented by $\mathcal{K}_T$ as in (9.10), i.e.,*

$$\langle T(\varphi), \psi \rangle = \langle \mathcal{K}_T, \psi \otimes \varphi \rangle \qquad \text{for all } \varphi \in \mathcal{S}(Y), \psi \in \mathcal{S}(X). \tag{9.11}$$

Proof. We must show that (9.11) uniquely determines a tempered distribution $\mathcal{K}_T$ on $X \times Y$.

We begin with uniqueness. Suppose $\mathcal{K}_T \in \mathcal{S}'(X \times Y)$ is a distribution satisfying (9.11), and let $\phi \in \mathcal{S}(X \times Y)$.

Let $\{h_{\mathbf{j}}\}_{\mathbf{j} \in \mathbb{N}^n}$ be the Hermite basis of $L^2(X)$, and $\{h_{\mathbf{k}}\}_{\mathbf{k} \in \mathbb{N}^m}$ the Hermite basis of $L^2(Y)$. Then the set $\{h_{\mathbf{j}} \otimes h_{\mathbf{k}}\}_{(\mathbf{j},\mathbf{k}) \in \mathbb{N}^n \times \mathbb{N}^m}$ forms the Hermite basis of $L^2(X \times Y)$, and according to Proposition 9.3 we can expand ϕ in the $\mathcal{S}$-convergent series

$$\phi = \sum_{(\mathbf{j},\mathbf{k}) \in \mathbb{N}^n \times \mathbb{N}^m} \check{\phi}(\mathbf{j}, \mathbf{k}) h_{\mathbf{j}} \otimes h_{\mathbf{k}}. \tag{9.12}$$

Since $\mathcal{K}_T$ is continuous on $\mathcal{S}$, we obtain

$$\langle \mathcal{K}_T, \phi \rangle = \sum_{(\mathbf{j},\mathbf{k}) \in \mathbb{N}^n \times \mathbb{N}^m} \check{\phi}(\mathbf{j}, \mathbf{k}) \langle \mathcal{K}_T, h_{\mathbf{j}} \otimes h_{\mathbf{k}} \rangle, \tag{9.13}$$

and thus, together with (9.11),

$$\langle \mathcal{K}_T, \phi \rangle = \sum_{(\mathbf{j},\mathbf{k}) \in \mathbb{N}^n \times \mathbb{N}^m} \check{\phi}(\mathbf{j}, \mathbf{k}) \langle T(h_{\mathbf{k}}), h_{\mathbf{j}} \rangle. \tag{9.14}$$

This establishes the uniqueness of $\mathcal{K}_T$.

To complete the proof, suppose we can show that the series on the right-hand side of (9.14) is convergent. This would allow us to *define* a tempered distribution $\mathcal{K}_T$ by this series, which then would evidently have the desired properties.

To this end, let us consider the following bilinear form on $\mathcal{S}(Y) \times \mathcal{S}(X)$:

$$\mathcal{B}(\varphi, \psi) := \langle T(\varphi), \psi \rangle, \qquad \varphi \in \mathcal{S}(Y), \psi \in \mathcal{S}(X).$$

This form is separately continuous, meaning it is continuous in φ for fixed ψ (since $T : \mathcal{S}(Y) \to \mathcal{S}'(X)$ is continuous), and continuous in ψ for fixed φ (since $T(\varphi) \in \mathcal{S}'(X)$).

Since $\mathcal{S}(Y)$ and $\mathcal{S}(X)$ are Fréchet spaces, the Baire category theorem (see Proposition A.8 in Appendix A) implies that $\mathcal{B}$ is jointly continuous. And, by Proposition A.7, this in return is equivalent to the existence of constants $C \geq 0$ and $M, N \in \mathbb{N}$ such that

$$|\mathcal{B}(\varphi, \psi)| \leq C \|\varphi\|_{(2,M)} \|\psi\|_{(2,N)} \quad \text{for all } \varphi \in \mathcal{S}(Y), \psi \in \mathcal{S}(X).$$

Using (9.8), we conclude that

$$|\langle T(h_{\mathbf{k}}), h_{\mathbf{j}}\rangle| \le C_{N,M}(n+2|\mathbf{j}|)^{2N}(n+2|\mathbf{k}|)^{2M} \le C'(1+|(\mathbf{j},\mathbf{k})|)^{2N+2M}.$$

Since $\check{\phi}$ is rapidly decaying, this ensures the convergence of (9.14). More precisely, we even have an estimate

$$|\langle \mathcal{K}_T, \phi\rangle| \le C\|\check{\phi}\|_{2N+2M+n+m+1} \le C'\|\phi\|_{(2,\tilde{N})}$$

for a suitable $\tilde{N} \in \mathbb{N}$ (in the last inequality, we used the fact that, according to Proposition 9.3, the Fourier–Hermite transform is continuous on $\mathcal{S}$). This shows that $\mathcal{K}_T$ is indeed a tempered distribution. □

9.3 The Tensor Product of Tempered Distributions

Using Theorem 9.4, the tensor product of two arbitrary tempered distributions can now be easily defined. Let again $X := \mathbb{R}^n, Y := \mathbb{R}^m$.

Corollary 9.5 *Given two tempered distributions $u \in \mathcal{S}'(X)$ and $v \in \mathcal{S}'(Y)$, there exists a unique tempered distribution $u \otimes v \in \mathcal{S}'(X \times Y)$ such that*

$$\langle u \otimes v, \psi \otimes \varphi\rangle = \langle u, \psi\rangle \langle v, \varphi\rangle \qquad \text{for all } \psi \in \mathcal{S}(X), \varphi \in \mathcal{S}(Y).$$

Proof. Consider the linear operator $T = T_{u,v} : \mathcal{S}(Y) \to \mathcal{S}'(X)$ defined by

$$T(\varphi) := \langle v, \varphi\rangle\, u, \qquad \varphi \in \mathcal{S}(Y).$$

This operator is clearly continuous, so by Theorem 9.4, there exists a unique tempered distribution $\mathcal{K}_T \in \mathcal{S}'(X \times Y)$ such that

$$\langle T(\varphi), \psi\rangle = \langle \mathcal{K}_T, \psi \otimes \varphi\rangle,$$

i.e.,

$$\langle \mathcal{K}_T, \psi \otimes \varphi\rangle = \langle u, \psi\rangle \langle v, \varphi\rangle \qquad \text{for all } \psi \in \mathcal{S}(X), \varphi \in \mathcal{S}(Y).$$

Thus, $u \otimes v := \mathcal{K}_{T_{u,v}}$ has the desired properties. □

The tempered distribution $u \otimes v$ defined in Corollary 9.5 is called the **tensor product** of the tempered distribution u with the tempered distribution v.

The following proposition is useful for applications. Again, let $u \in \mathcal{S}'(X)$ and $v \in \mathcal{S}'(Y)$. For $\phi \in \mathcal{S}(X \times Y)$, define

$$\phi_v(x) := \langle v, \phi(x, \cdot)\rangle,\ x \in X; \qquad {}_u\phi(y) := \langle u, \phi(\cdot, y)\rangle,\ y \in Y. \tag{9.15}$$

Proposition 9.6 *If $u \in \mathcal{S}'(X)$ and $v \in \mathcal{S}'(Y)$, and if $\phi \in \mathcal{S}(X \times Y)$, then $\phi_v \in \mathcal{S}(X)$ and ${}_u\phi \in \mathcal{S}(Y)$, and the following holds:*

$$\langle u \otimes v, \phi \rangle = \langle u, \phi_v \rangle = \langle v, {}_u\phi \rangle. \tag{9.16}$$

Proof. Expanding

$$\phi = \sum_{(\mathbf{j},\mathbf{k}) \in \mathbb{N}^n \times \mathbb{N}^m} \check{\phi}(\mathbf{j}, \mathbf{k}) h_{\mathbf{j}} \otimes h_{\mathbf{k}}$$

as in (9.12), we obtain for fixed $x \in X$, using the continuity of $v \in \mathcal{S}'(Y)$,

$$\phi_v(x) = \sum_{(\mathbf{j},\mathbf{k}) \in \mathbb{N}^n \times \mathbb{N}^m} \langle v, h_{\mathbf{k}} \rangle \check{\phi}(\mathbf{j}, \mathbf{k})\, h_{\mathbf{j}}(x) = \sum_{\mathbf{j} \in \mathbb{N}^n} c_{\mathbf{j}}\, h_{\mathbf{j}}(x),$$

with $c_{\mathbf{j}} := \sum_{\mathbf{k} \in \mathbb{N}^m} \langle v, h_{\mathbf{k}} \rangle \check{\phi}(\mathbf{j}, \mathbf{k})$. From (9.8), it follows immediately that there exist constants $C \geq 0$ and $N \in \mathbb{N}$ such that

$$|\langle v, h_{\mathbf{k}} \rangle| \leq C(1 + |\mathbf{k}|)^N \quad \text{for all } \mathbf{k} \in \mathbb{N}^m.$$

From this, one easily deduces with Proposition 9.3 that the sequence of coefficients $c_{\mathbf{j}}$ decays rapidly, so that the function ϕ_v belongs to $\mathcal{S}(X)$ by Proposition 9.3. Thus,

$$\begin{aligned} \langle u, \phi_v \rangle &= \sum_{\mathbf{j} \in \mathbb{N}^n} c_{\mathbf{j}} \langle u, h_{\mathbf{j}} \rangle = \sum_{(\mathbf{j},\mathbf{k}) \in \mathbb{N}^n \times \mathbb{N}^m} \check{\phi}(\mathbf{j}, \mathbf{k}) \langle v, h_{\mathbf{k}} \rangle \langle u, h_{\mathbf{j}} \rangle \\ &= \sum_{(\mathbf{j},\mathbf{k}) \in \mathbb{N}^n \times \mathbb{N}^m} \check{\phi}(\mathbf{j}, \mathbf{k}) \langle u \otimes v, h_{\mathbf{j}} \otimes h_{\mathbf{k}} \rangle = \langle u \otimes v, \phi \rangle. \end{aligned}$$

This proves the statement about ϕ_v as well as the first identity in (9.16).

The corresponding statement for ${}_u\phi$ and the second identity in (9.16) are shown analogously by swapping the roles of x and y in the above argument. □

9.4 The Kernel Theorem for Distributions in Open Sets

We now consider the situation where $X := \Omega_2 \subset \mathbb{R}^n$ is a non-empty open subset of $\mathbb{R}^n$, and $Y := \Omega_1 \subset \mathbb{R}^m$ is a non-empty open subset of $\mathbb{R}^m$.

If $\mathcal{K} \in \mathcal{D}'(X \times Y)$ is a distribution on $X \times Y$, we can define a linear operator

$$T_{\mathcal{K}} : \mathcal{D}(Y) \to \mathcal{D}'(X)$$

analogously by setting

$$\langle T_{\mathcal{K}}(\varphi), \psi \rangle := \langle \mathcal{K}, \psi \otimes \varphi \rangle \qquad \text{for all } \varphi \in \mathcal{D}(Y), \psi \in \mathcal{D}(X). \tag{9.17}$$

This operator is then continuous from $\mathcal{D}(Y)$ to $\mathcal{D}'(X)$, since $\mathcal{D}'(X)$ is endowed with the weak*-topology with respect to $\mathcal{D}(X)$. The converse also holds:

Theorem 9.7 *Let $T : \mathcal{D}(Y) \to \mathcal{D}'(X)$ be a continuous linear operator. Then there exists a unique distribution $\mathcal{K}_T \in \mathcal{D}'(X \times Y)$ such that T is represented by $\mathcal{K}_T$ as in (9.17), i.e.,*

$$\langle T(\varphi), \psi \rangle = \langle \mathcal{K}_T, \psi \otimes \varphi \rangle \qquad \text{for all } \varphi \in \mathcal{D}(Y), \psi \in \mathcal{D}(X). \tag{9.18}$$

Proof. We reduce this to Theorem 9.4 by localization.

Let $\{K_j^X\}_{j\in\mathbb{N}}$ and $\{K_j^Y\}_{j\in\mathbb{N}}$ be compact exhaustions of X and Y, respectively, and let $\chi_j^X \in \mathcal{D}(X)$ and $\chi_j^Y \in \mathcal{D}(Y)$ satisfy $\chi_j^X \equiv 1$ on a neighborhood of K_j^X and $\operatorname{supp} \chi_j^X \subset K_{j+1}^X$, with χ_j^Y having analogous properties with respect to $\{K_j^Y\}$.

Then the sequence of compacta $K_j := K_j^X \times K_j^Y$ forms a compact exhaustion of $X \times Y$, and setting $\chi_j := \chi_j^X \otimes \chi_j^Y$, we have $\chi_j \equiv 1$ on a neighborhood of K_j, and $\operatorname{supp} \chi_j \subset K_{j+1}$.

We first prove the uniqueness of $\mathcal{K}_T$. Suppose $\mathcal{K}_1, \mathcal{K}_2$ are distributions in $X \times Y$ satisfying $\langle T(\varphi), \psi \rangle = \langle \mathcal{K}_\iota, \psi \otimes \varphi \rangle$ for all $\varphi \in \mathcal{D}(Y), \psi \in \mathcal{D}(X)$, $\iota = 1, 2$. Then for $u := \mathcal{K}_1 - \mathcal{K}_2$,

$$\langle u, \psi \otimes \varphi \rangle = 0 \qquad \text{for all } \varphi \in \mathcal{D}(Y), \psi \in \mathcal{D}(X).$$

For each $j \in \mathbb{N}$, let u_j be the tempered distribution $u_j := \widetilde{u\chi_j}$ on $\mathbb{R}^n \times \mathbb{R}^m$. Then

$$\langle u_j, \Psi \otimes \Phi \rangle = \langle u, (\Psi \chi_j^X)|_X \otimes (\Phi \chi_j^Y)|_Y \rangle = 0 \qquad \text{for all } \Phi \in \mathcal{S}(\mathbb{R}^m), \Psi \in \mathcal{S}(\mathbb{R}^n).$$

By the continuity of u_j on $\mathcal{S}(\mathbb{R}^n \times \mathbb{R}^m)$, it follows (see (9.13)) that $u_j = 0$. Since $\chi_j = 1$ on K_j, we see that u vanishes on the interior K_j^0 of K_j for each j, hence on all of $X \times Y$. Thus, $\mathcal{K}_1 = \mathcal{K}_2$.

To prove existence, we define for each $j \in \mathbb{N}$ the continuous linear operator $\tilde{T}_j : \mathcal{S}(\mathbb{R}^m) \to \mathcal{S}'(\mathbb{R}^n)$ by

$$\tilde{T}_j(\Phi) := [\chi_j^X T(\chi_j^Y \Phi|_Y)]\tilde{}, \quad \text{i.e.,} \quad \langle \tilde{T}_j(\Phi), \Psi \rangle = \langle T(\chi_j^Y \Phi|_Y), \chi_j^X \Psi|_X \rangle,$$

for $\Phi \in \mathcal{S}(\mathbb{R}^m), \Psi \in \mathcal{S}(\mathbb{R}^n)$. By Theorem 9.4, there exists a unique tempered distribution $\tilde{\mathcal{K}}_j \in \mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^m)$ such that

$$\langle \tilde{T}_j(\Phi), \Psi \rangle = \langle \tilde{\mathcal{K}}_j, \Psi \otimes \Phi \rangle, \qquad \Phi \in \mathcal{S}(\mathbb{R}^m), \Psi \in \mathcal{S}(\mathbb{R}^n).$$

We then define $\mathcal{K}_j := \tilde{\mathcal{K}}_j|_{\mathcal{D}(K_j^0)} \in \mathcal{D}'(K_j^0)$.

For $j_2 \geq j_1$, we then have

$$\mathcal{K}_{j_2}|_{K_{j_1}^0} = \mathcal{K}_{j_1}, \tag{9.19}$$

for if $\varphi \in \mathcal{D}((K_{j_1}^Y)^0)$ and $\psi \in \mathcal{D}((K_{j_1}^X)^0)$, then, using the usual embedding of test functions in open subsets into the respective space of Schwartz functions:

$$\langle \mathcal{K}_{j_2}, \psi \otimes \varphi \rangle = \langle \tilde{T}_{j_2}(\varphi), \psi \rangle = \langle T(\chi^Y_{j_2}\varphi|_Y), \chi^X_{j_2}\psi|_X \rangle = \langle T(\varphi|_Y), \psi|_X \rangle$$
$$= \langle T(\chi^Y_{j_1}\varphi|_Y), \chi^X_{j_1}\psi|_X \rangle = \langle \mathcal{K}_{j_1}, \psi \otimes \varphi \rangle,$$

that is, $\langle \mathcal{K}_{j_2}, \psi \otimes \varphi \rangle = \langle \mathcal{K}_{j_1}, \psi \otimes \varphi \rangle$.

But, as we have already seen in the first part of the proof concerning uniqueness, this indeed implies (9.19).

We now set for $\phi \in \mathcal{D}(X \times Y)$

$$\langle \mathcal{K}_T, \phi \rangle := \langle \mathcal{K}_j, \phi \rangle, \qquad \text{if} \quad \operatorname{supp} \phi \subset K^0_j.$$

By (9.19), it is easily seen that in this way a linear functional $\mathcal{K}_T$ is well-defined on the space $\mathcal{D}(X \times Y)$. Furthermore, from Theorem 5.3, we immediately see that it is continuous, since its restriction to each of the subspaces $\mathcal{D}_{K_{j-1}}$ is continuous.

The distribution $\mathcal{K}_T \in \mathcal{D}'(X \times Y)$ defined in this way then also satisfies (9.18) – just choose j sufficiently large so that $\operatorname{supp} \psi \otimes \varphi \subset K^0_j$. □

9.5 The Tensor Product of Distributions in Open Sets

Let $X := \Omega_2 \subset \mathbb{R}^n$ be a non-empty open subset of $\mathbb{R}^n$ and $Y := \Omega_1 \subset \mathbb{R}^m$ a non-empty open subset of $\mathbb{R}^m$. Using Theorem 9.7 instead of Theorem 9.4, we can now prove the following result analogously to Corollary 9.5:

Corollary 9.8 *Given two distributions $u \in \mathcal{D}'(X)$ and $v \in \mathcal{D}'(Y)$, there exists a unique distribution $u \otimes v \in \mathcal{D}'(X \times Y)$ such that*

$$\langle u \otimes v, \psi \otimes \varphi \rangle = \langle u, \psi \rangle \, \langle v, \varphi \rangle \qquad \textit{for all } \psi \in \mathcal{D}(X), \varphi \in \mathcal{D}(Y).$$

We refer to the distribution $u \otimes v \in \mathcal{D}'(\Omega_2 \times \Omega_1)$ defined in Corollary 9.8 as the **tensor product** of the distribution $u \in \mathcal{D}'(\Omega_2)$ with the distribution $v \in \mathcal{D}'(\Omega_1)$.

By localization, we also obtain the following analogue of Proposition 9.6, where for $\phi \in \mathcal{D}(X \times Y)$, the functions ϕ_v and ${}_u\phi$ are defined analogously to (9.15), i.e.,

$$\phi_v(x) := \langle v, \phi(x, \cdot) \rangle, \; x \in X; \qquad {}_u\phi(y) := \langle u, \phi(\cdot, y) \rangle, \; y \in Y.$$

Proposition 9.9 *If $u \in \mathcal{D}'(X)$ and $v \in \mathcal{D}'(Y)$, and if $\phi \in \mathcal{D}(X \times Y)$, then $\phi_v \in \mathcal{D}(X)$ and ${}_u\phi \in \mathcal{D}(Y)$, and we have:*

$$\langle u \otimes v, \phi \rangle = \langle u, \phi_v \rangle = \langle v, {}_u\phi \rangle. \tag{9.20}$$

Proof. Similar to the proof of Theorem 9.7, we choose functions $\chi^X \in \mathcal{D}(X)$ and $\chi^Y \in \mathcal{D}(Y)$ such that $\phi = (\chi^X \otimes \chi^Y)\phi$, and consider the tempered distributions

$$\tilde{u} := (u\chi^X)^\sim \in \mathcal{S}'(\mathbb{R}^n), \; \tilde{v} := (v\chi^Y)^\sim \in \mathcal{S}'(\mathbb{R}^m), \text{ and}$$
$$[(u \otimes v)(\chi^X \otimes \chi^Y)]^\sim \in \mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^m).$$

Clearly, we then have

$$[(u \otimes v)(\chi^X \otimes \chi^Y)]\tilde{} = (u\chi^X)\tilde{} \otimes (v\chi^Y)\tilde{} = \tilde{u} \otimes \tilde{v} \tag{9.21}$$

(it suffices to test this on tensor products $\Psi \otimes \Phi$ of Schwartz functions!).

Thus, if $\phi \in \mathcal{D}(X \times Y)$, with trivial extension $\tilde{\phi} \in \mathcal{S}(\mathbb{R}^n \times \mathbb{R}^n)$, then we have for $x \in X$:

$$\phi_v(x) = \langle v, [(\chi^X \otimes \chi^Y)\phi](x, \cdot)\rangle = \langle (v\chi^Y)\tilde{}, \tilde{\phi}(x, \cdot)\rangle = \langle \tilde{v}, \tilde{\phi}(x, \cdot)\rangle = \tilde{\phi}_{\tilde{v}}(x),$$

so that $\tilde{\phi}_{\tilde{v}}|_X = \phi_v$.

Together with Proposition 9.6, this shows in particular that ϕ_v is smooth, and thus $\phi_v \in \mathcal{D}(X)$. Furthermore, using (9.21) and Proposition 9.6, we obtain

$$\langle u \otimes v, \phi\rangle = \langle (u \otimes v)(\chi^X \otimes \chi^Y), \phi\rangle = \langle \tilde{u} \otimes \tilde{v}, \tilde{\phi}\rangle = \langle \tilde{u}, \tilde{\phi}_{\tilde{v}}\rangle = \langle u, \phi_v\rangle.$$

The statement about ${}_u\phi$ as well as the second identity in (9.20) follow analogously by swapping the roles of x and y in the above argument. □

9.6 Exercises

9.1 (Translation-Invariant Distributions) Let $X := \mathbb{R}^m, Y := \mathbb{R}^n$, and let $U \in \mathcal{D}'(X \times Y)$ (respectively $U \in \mathcal{S}'(X \times Y)$).

For $z \in \mathbb{R}^m$, let λ_z denote the operator of translation by z in X, i.e., for test functions $\phi \in \mathcal{S}(X \times Y)$, we define λ_z by

$$\lambda_z\phi(x, y) := \phi(x - z, y).$$

By means of our scheme from Section 4.3 respectively 5.3, we can also apply λ_z to the distribution U.

Assume that U is translation-invariant in X, i.e., that

$$\lambda_z U = U \qquad \text{for all } z \in X.$$

Show that there exists a unique distribution $u \in \mathcal{D}'(Y)$ (respectively $u \in \mathcal{S}'(Y)$) such that

$$U = 1 \otimes u,$$

i.e., that for every $\phi \in \mathcal{D}(X \times Y)$ (respectively $\phi \in \mathcal{S}(X \times Y)$):

$$\langle U, \phi\rangle = \langle u, \int_X \phi(x, \cdot)\, \mathrm{d}x\rangle.$$

Hints:

(i) Using induction on $m = \dim X$, it is sufficient to prove the case $m = 1$, i.e., $X = \mathbb{R}$. Furthermore, the case $U \in \mathcal{S}'(X \times Y)$ can easily be reduced to the case $U \in \mathcal{D}'(X \times Y)$ since $\mathcal{D}$ is dense in $\mathcal{S}$.

(ii) Choose $\chi \in \mathcal{D}(\mathbb{R})$ with $\int \chi(x)\,\mathrm{d}x = 1$, and set for $\phi \in \mathcal{D}(X \times Y)$

$$\psi_1(x, y) := \int_{-\infty}^{x} (\phi(x', y) - \chi(x')\phi_1(y))\,\mathrm{d}x',$$

with $\phi_1(y) := \int_{\mathbb{R}} \phi(x, y)\,\mathrm{d}x$. Then $\psi_1 \in \mathcal{D}(X \times Y)$, and

$$\phi(x, y) = \chi(x)\phi_1(y) + \frac{\partial}{\partial x}\psi_1(x, y).$$

Define u by

$$\langle u, \psi \rangle := \langle U, \chi \otimes \psi \rangle, \qquad \psi \in \mathcal{D}(Y).$$

9.2 (Continuous Translation-Invariant Operators) Prove the following:

Theorem 9.10 *Let $T : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ be a continuous and translation-invariant linear operator (i.e., $T(\lambda_x \varphi) = \lambda_x(T(\varphi))$ for all $x \in \mathbb{R}^n$, $\varphi \in \mathcal{S}(\mathbb{R}^n)$). Then there exists a unique tempered distribution $u \in \mathcal{S}'(\mathbb{R}^n)$ such that*

$$T\varphi = u * \varphi \qquad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^n).$$

An analogous statement also holds for $\mathcal{D}(\mathbb{R}^n)$ and $\mathcal{D}'(\mathbb{R}^n)$ in place of $\mathcal{S}(\mathbb{R}^n)$ and $\mathcal{S}'(\mathbb{R}^n)$.

Hints: Let $X := Y := \mathbb{R}^n$, and let $\mathcal{K} = \mathcal{K}_T$ be the distributional Schwartz kernel of T according to Theorem 9.4 respectively Theorem 9.7. Show that this kernel is invariant under the family of simultaneous translations $\sigma_z : (x, y) \mapsto (x - z, y - z), z \in X$. Then use the linear transformation

$$A : X \times Y \to X \times Y, \quad (x, y) \mapsto \left(\frac{x + y}{2}, \frac{x - y}{2}\right),$$

to reduce the result to Exercise 9.1.

An example of such a translation-invariant operator is the Hilbert transform $\mathcal{H}\varphi = \varphi * H = H * \varphi$ on $\mathbb{R}$ (see Exercises 4.14, 4.15). Here, $u := H = p.v.(\frac{1}{\pi x})$.

Appendix A
The Baire Category Theorem and the Banach–Steinhaus Theorem*

A.1 The Baire Category Theorem

Definitions. Let X be a topological space. A subset $M \subset X$ is called **nowhere dense** if its closure $\overline{M}$ in X contains no interior point.

M is said to be **of the first category** if M can be represented as a countable union of nowhere dense subsets of X.

A more suggestive term for sets of the first category would be **meager**, which is occasionally used. However, here we shall keep with the more standard term "of the first category" introduced by Baire, since it has become widely established in the literature.

Subsets of X that are not of the first category are called **sets of the second category** (or occasionally also **non-meager**).

Example. The (dense) subset $\mathbb{Q}$ of $\mathbb{R}$ is of the first category.

The following observations are often helpful:

(a) *M is nowhere dense in X if and only if the complement $\overline{M}^c = X \setminus \overline{M}$ of $\overline{M}$ is dense in X.*
(b) *Every subset of a set of the first category is of the first category.*
(c) *Every countable union of sets of the first category is of the first category.*

(a) follows immediately from the easily seen fact that the open set $U := \overline{M}^c$ is dense in X if and only if its complement $U^c = \overline{M}$ contains no interior point, and (b) and (c) are obvious.

Theorem A.1 (Baire's Theorem for Metric Spaces) *In a complete metric space, every intersection of a countably collection of dense open subsets is dense in X.*

The following two corollaries will be proven jointly with Theorem A.1:

Corollary A.2 *Let X be a complete metric space, and let $M \subset X$ be a set of the first category. Then M has no interior point. In particular, X is of the second category.*

Corollary A.3 *Let X be a complete metric space, and let A_k, $k \in \mathbb{N}$, be closed subsets of X. If their union $\bigcup_k A_k$ has an interior point, then there exists some $k_0 \in \mathbb{N}$ such that A_{k_0} has an interior point.*

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Remarks A.4 Baire's theorem also holds, with a similar proof, in locally compact Hausdorff spaces (see, e.g., Rudin [34]).

Proof. Let (X, d) be a complete metric space, and let U_j, $j \geq 1$, be a sequence of open dense subsets of X. We show that $\bigcap_j U_j$ is dense in X.

To this end, let x_0 be an arbitrary point in X, let $\varepsilon_0 > 0$, and let $B_0 := B_{\varepsilon_0}(x_0) = \{x \in X : d(x, x_0) < \varepsilon_0\}$ be the open ball with center x_0 and radius ε_0. We construct recursively balls $B_j := B_{\varepsilon_j}(x_j)$ with

$$\overline{B_j} \subset U_j \cap B_{j-1} \quad \text{and} \quad \varepsilon_j < \frac{\varepsilon_0}{2^j}, \qquad j = 1, 2, 3, \ldots .$$

as follows:

First, since $U_1 \cap B_0 \neq \emptyset$, there exists a ball $B_1 = B_{\varepsilon_1}(x_1)$ with $\varepsilon_1 < \varepsilon_0/2$ and $\overline{B_1} \subset U_1 \cap B_0$. And, assuming that we have already chosen balls $B_0, B_1, \ldots, B_n$ with the above properties, then $U_{n+1} \cap B_n \neq \emptyset$, so that we can similarly find a ball B_{n+1} with the desired properties.

Since $d(x_j, x_{j-1}) < \varepsilon_0/2^{j-1}$, the sequence $\{x_j\}_j$ then forms a Cauchy sequence in X, and since X is complete, there exists a limit $x := \lim x_j \in X$. For each $k \in \mathbb{N}$, we then have $x_j \in \overline{B_k}$ whenever $j \geq k$. By passing to the limit, we obtain:

$$x \in \bigcap_{k \geq 0} \overline{B_k} \subset \left(\bigcap_{k \geq 0} U_k \right) \cap B_{\varepsilon_0}(x_0).$$

Since $\varepsilon_0 > 0$ was arbitrary, it follows that $x_0 \in \overline{\bigcap_k U_k}$. This proves Theorem A.1.

Corollary A.2 follows from this by taking complements: Let $M = \bigcup_j M_j$, where $\{M_j\}_j$ is a sequence of nowhere dense subsets of X. The sets $U_j := \overline{M_j}^c$ are then open and dense in X by (a), so that by Baire's theorem, their intersection $\bigcap_j U_j$ is dense in X. Thus, the complement

$$\left(\bigcap_j U_j \right)^c = \bigcup_j \overline{M_j}$$

has no interior point. In particular, this then also applies to $M = \bigcup_j M_j$.

Corollary A.3 follows immediately from Corollary A.2. □

A.2 The Banach–Steinhaus Theorem

In the following, as in Section 4.2, let E and F always denote locally convex vector spaces over the field $\mathbb{K}$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$. The topologies on E and F are assumed to be generated by the separating families of seminorms $\mathcal{N}_E = \{\|\cdot\|_{E,i}\}_{i \in I}$ on E and $\mathcal{N}_F = \{\|\cdot\|_{F,j}\}_{j \in J}$ on F, respectively.

Definitions. A subset $B \subset F$ is called **bounded** if, for every zero neighborhood U, i.e. a neighborhood of 0 in F, there exists an $r > 0$ such that $B \subset rU$.

Assuming without loss of generality that the zero neighborhoods U are generalized balls as defined in Section 4.2, we see that B is bounded if and only if each of the seminorms $\|\cdot\|_{F,j}$, $j \in J$, is bounded on B.

Denoting by $L(E,F)$ the linear space of all continuous linear operators $T : E \to F$, a subset $\Gamma \subset L(E,F)$ of such operators is said to be **pointwise bounded** if, for every $v \in E$, the set

$$\Gamma(v) := \{Tv : T \in \Gamma\}$$

is bounded in F.

Γ is called **uniformly bounded** if, for every zero neighborhood U in F, there exists a zero neighborhood V in E such that $T(V) \subset U$ for all $T \in \Gamma$. Essentially, this property means that the family Γ of continuous linear mappings T is "**equicontinuous**". Using arguments similar to those in the proof of Proposition 4.8, one easily shows the following equivalent characterization:

(d) $\Gamma \subset L(E,F)$ *is uniformly bounded if and only if for every* $j \in J$, *there exists a finite index set* $\{i_1, \dots, i_m\} \subset I$ *and a constant* $C \geq 0$ *such that*

$$\|Tv\|_{F,j} \leq C \sum_{k=1,\dots,m} \|v\|_{E,i_k} \quad \text{for all } v \in E, T \in \Gamma. \tag{A.1}$$

It follows immediately that uniformly bounded sets $\Gamma \subset L(E,F)$ are in particular **pointwise bounded**.

Surprisingly, under suitable conditions on E and F, the converse also holds, as shown by the following theorem of Banach and Steinhaus:

Theorem A.5 (The Principle of Uniform Boundedness) *Let E be a Fréchet space and F a locally convex vector space. Then every pointwise bounded subset $\Gamma \subset L(E,F)$ is uniformly bounded.*

In particular, if $(E, \|\cdot\|_E)$ is a Banach space and $(F, \|\cdot\|_F)$ a normed space, this means the following:

If for every $v \in E$, there exists a constant C_v such that $\|Tv\|_F \leq C_v$ for every $T \in \Gamma$, then there even exists a constant C such that

$$\|Tv\|_F \leq C\|v\|_E \quad \text{for all} \quad T \in \Gamma, v \in E.$$

Proof of Theorem A.5. Let U be a zero neighborhood in F. Then there exists a generalized ball of the form

$$B_{F,f,\varepsilon}(0) = \{w \in F : \|w\|_{F,j_l} < \varepsilon \quad \text{for } l = 1, \dots, L\}$$

with $f = \{j_1, \dots, j_L\} \subset J$ and $\varepsilon > 0$, such that $B_{F,f,\varepsilon}(0) \subset U$. Set $U' := B_{F,f,\varepsilon/4}(0)$, and define

$$A := \bigcap_{T \in \Gamma} T^{-1}(\overline{U'}) = \{v \in E : Tv \in \overline{U'} \text{ for all } T \in \Gamma\}.$$

Since each operator $T \in \Gamma$ is continuous, A is closed.

Let $v \in E$. By assumption, $\Gamma(v)$ is bounded, i.e., there exists an $n = n(v) \in \mathbb{N}_{\geq 1}$, such that $\Gamma(v) \subset n\,U'$. Thus, $v \in n\,T^{-1}(U')$ for all T in Γ, so that $v \in n\,A$. It follows that

$$E = \bigcup_{n=1}^{\infty} n\,A.$$

Since E as a Fréchet space is metrizable, Baire's theorem (specifically Corollary A.3) implies that there exists an $m \in \mathbb{N}_{\geq 1}$ such that $(m\,A)^0 \neq \emptyset$. Since $(m\,A)^0 = m\,A^0$, it follows that $A^0 \neq \emptyset$.

Fix some $v_0 \in A^0$, and set $V := -v_0 + A^0$. Then V is a zero neighborhood in E, and for every $T \in \Gamma$, we have:

$$T(V) = -Tv_0 + T(A^0) \subset -T(A) + T(A) \subset -\overline{U'} + \overline{U'} \subset B_{F,f,\varepsilon}(0) \subset U. \qquad \square$$

Corollary A.6 *Let $\{T_n\}_n \subset L(E,F)$ be a sequence of continuous linear operators on a Fréchet space E with values in a locally convex space F. If for every $v \in E$, the limit $Tv := \lim_{n\to\infty} T_n v \in F$ exists, then the family*

$$\Gamma := \{T_n : n \in \mathbb{N}\} \subset L(E,F)$$

is uniformly bounded, and the resulting linear operator $T : E \to F$ is continuous.

Proof. The linearity of T follows immediately from the linearity of the T_n. Furthermore, the set $\Gamma(v) := \{T_n v : n \in \mathbb{N}\}$ is clearly bounded for every $v \in E$ since convergent sequences in F are always bounded. By the Banach–Steinhaus theorem (Theorem A.5), the family Γ is therefore uniformly bounded.

Now let U be a zero neighborhood in F. As before, we choose a zero neighborhood U' in F such that $\overline{U'} \subset U$. Then there exists a zero neighborhood V in E such that $T_n(V) \subset U'$ for all $n \in \mathbb{N}$. It follows that $T(V) \subset \overline{U'} \subset U$. This shows that T is continuous at $0 \in E$, and thus continuous on E by Proposition 4.8. $\square$

A.3 Continuity of Bilinear Mappings

Let E, F, and Z be locally convex vector spaces, and let

$$\mathcal{B} : E \times F \to Z$$

be a bilinear mapping.

The topologies on E, F, and Z are assumed to be generated by the separating families of seminorms $\mathcal{N}_E = \{\|\cdot\|_{E,i}\}_{i\in I}$ on E, $\mathcal{N}_F = \{\|\cdot\|_{F,j}\}_{j\in J}$ on F, and $\mathcal{N}_Z = \{\|\cdot\|_{Z,k}\}_{k\in K}$ on Z.

Without loss of generality, we may and shall assume again that the maximum of finitely many seminorms from $\mathcal{N}_E$, $\mathcal{N}_F$, and $\mathcal{N}_Z$ is again one of our seminorms (cf. Example 4.7 (a)), to simplify the formulation of the following theorem.

Analogously to Proposition 4.8, the continuity of $\mathcal{B}$ can then be characterized as follows:

Proposition A.7 *Under the above assumptions, the following statements are equivalent:*

(i) $\mathcal{B}$ *is continuous on* $E \times F$.
(ii) $\mathcal{B}$ *is continuous at* $(0,0) \in E \times F$.
(iii) *For every* $k \in K$, *there exist indices* $i \in I$ *and* $j \in J$ *and a constant* $C \geq 0$, *such that*

$$\|\mathcal{B}(v,w)\|_{Z,k} \leq C\|v\|_{E,i}\|w\|_{F,j} \quad \text{for all } v \in E, w \in F. \tag{A.1}$$

Proof. $(i) \implies (ii)$ is clear.

$(ii) \implies (iii)$: If $\mathcal{B}$ is continuous at $(0,0)$, and if $k \in K$, then we aim to prove (A.1).

Since the ball $B_{Z,k,1}(0)$ is a zero neighborhood in Z, its preimage $\mathcal{B}^{-1}(B_{Z,k,1}(0))$ is a zero neighborhood in $E \times F$ due to the continuity of $\mathcal{B}$ at $(0,0)$. Thus, there exist balls $B_{E,i,r_1}(0)$ in E and $B_{F,j,r_2}(0)$ in F such that

$$\mathcal{B}(B_{E,i,r_1}(0) \times B_{F,j,r_2}(0)) \subset B_{Z,k,1}(0).$$

Thus, if $\|v\|_{E,i} < r_1$ and $\|w\|_{F,j} < r_2$, then $\|\mathcal{B}(v,w)\|_{Z,k} < 1$.

Now, making use of the bilinearity of $\mathcal{B}$ and applying the scaling argument used in the proof of Proposition 4.8 analogously to $v \in E$, and to $w \in F$, we quickly obtain the estimate (A.1), with the constant $C := 4/(r_1 r_2)$.

$(iii) \implies (i)$: Let $(\tilde{v}, \tilde{w}) \in E \times F$ be given. To show the continuity of $\mathcal{B}$ at this point, let $\varepsilon > 0$, and choose indices $i \in I$ and $j \in J$ as well as a constant $C \geq 0$ such that (A.1) holds for a given $k \in K$. Since

$$\mathcal{B}(v,w) - \mathcal{B}(\tilde{v},\tilde{w}) = \mathcal{B}(v-\tilde{v},w) + \mathcal{B}(\tilde{v},w-\tilde{w}),$$

we obtain

$$\begin{aligned}\|\mathcal{B}(v,w) - \mathcal{B}(\tilde{v},\tilde{w})\|_{Z,k} &\leq \|\mathcal{B}(v-\tilde{v},w)\|_{Z,k} + \|\mathcal{B}(\tilde{v},w-\tilde{w})\|_{Z,k} \\ &\leq C\|v-\tilde{v}\|_{E,i}\|w\|_{F,j} + C\|\tilde{v}\|_{E,i}\|w-\tilde{w}\|_{F,j}.\end{aligned}$$

Let now $0 < r_1, r_2 \leq 1$ and $\|v-\tilde{v}\|_{E,i} < r_1, \|w-\tilde{w}\|_{F,j} < r_2$. Then we have in particular that

$$\|w\|_{F,j} \leq \|w-\tilde{w}\|_{F,j} + \|\tilde{w}\|_{F,j} \leq 1 + \|\tilde{w}\|_{F,j},$$

and thus

$$\|\mathcal{B}(v,w) - \mathcal{B}(\tilde{v},\tilde{w})\|_{Z,k} \leq C(1+\|\tilde{w}\|_{F,j})\|v-\tilde{v}\|_{E,i} + C\|\tilde{v}\|_{E,i}\|w-\tilde{w}\|_{F,j}.$$

Thus, by choosing $r_1, r_2 > 0$ small enough, we see that $\|\mathcal{B}(v,w) - \mathcal{B}(\tilde{v},\tilde{w})\|_{Z,k} < \varepsilon$, provided $\|v-\tilde{v}\|_{E,i} < r_1$ and $\|w-\tilde{w}\|_{F,j} < r_2$.

This proves the continuity of $\mathcal{B}$ at the point$(\tilde{v},\tilde{w})$. □

For $v \in E, w \in F$, define

$${}_v\mathcal{B}(w) = \mathcal{B}_w(v) := \mathcal{B}(v,w).$$

Definition. The bilinear mapping $\mathcal{B}$ is called *separately continuous* if the mappings

$$ {}_v\mathcal{B} : F \to Z \quad \text{and} \quad \mathcal{B}_w : E \to Z $$

are continuous for all $v \in E$ respectively $w \in F$.

Proposition A.8 *Let $\mathcal{B} : E \times F \to Z$ be a bilinear and separately continuous mapping. If E is a Fréchet space and F is metrizable, then $\mathcal{B}$ is continuous.*

Proof. By Theorem A.7, it suffices to prove the continuity of $\mathcal{B}$ at the origin $(0,0) \in E \times F$. Since both E and F are metrizable, it is enough to prove:

If $\{(v_n, w_n)\}_n$ is a sequence in $E \times F$ converging to $(0,0) \in E \times F$, then

$$ \mathcal{B}(v_n, w_n) \to 0 \quad \text{in} \quad Z. \tag{A.2} $$

To this end, define

$$ T_n(v) := \mathcal{B}(v, w_n), \quad v \in E, n \in \mathbb{N}. $$

The linear operators $T_n : E \to Z$ are continuous since $T_n(v) = \mathcal{B}_{w_n}(v)$ is continuous in v. Moreover, since ${}_v\mathcal{B}$ is continuous, we have for every $v \in E$:

$$ \lim_{n\to\infty} T_n(v) = \mathcal{B}(v, 0) = 0. $$

By the Banach–Steinhaus theorem (specifically, Corollary A.6), the sequence $\{T_n\}_n$ is uniformly bounded in $L(E, Z)$. Thus, for each $k \in K$, there exists $i \in I$ and $C \geq 0$ such that

$$ \|\mathcal{B}(v, w_n)\|_{Z,k} = \|T_n(v)\|_{Z,k} \leq C\|v\|_{E,i}, \quad \text{for all } n \in \mathbb{N}, v \in E. $$

In particular,

$$ \|\mathcal{B}(v_n, w_n)\|_{Z,k} \leq C\|v_n\|_{E,i}, \quad \text{for all } n \in \mathbb{N}, $$

which implies $\|\mathcal{B}(v_n, w_n)\|_{Z,k} \to 0$ for each $k \in K$. Thus, (A.2) follows. □

A.4 Exercises

A.1 Let $\mathcal{F} \subset C(\mathbb{R})$ be a pointwise bounded family of continuous functions. Show that there exists a non-empty open interval I on which $\mathcal{F}$ is uniformly bounded, i.e., that there exists a constant C such that

$$ |f(t)| \leq C $$

for all $f \in \mathcal{F}$ and all $t \in I$.

A.2 (*) Let $\{a_k\}_{k\in\mathbb{N}}$ be a sequence of complex numbers such that for every convergent sequence $\{x_k\}_{k\in\mathbb{N}}$ in $\mathbb{C}$, the series

$$\sum_{k=0}^{\infty} a_k x_k$$

is convergent. Show that this implies

$$\sum_{k=0}^{\infty} |a_k| < \infty.$$

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